

The background of the slide features a large, white, stylized robot head on the left and right sides, with a blue and white striped pattern. In the center, there is a large, white, stylized robot head with a blue and white striped pattern. The background is dark with a blue and white striped pattern. The text is white and blue.

Deutsche Bank
Research

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust

Context is all you need

September 2026

Adrian Cox

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

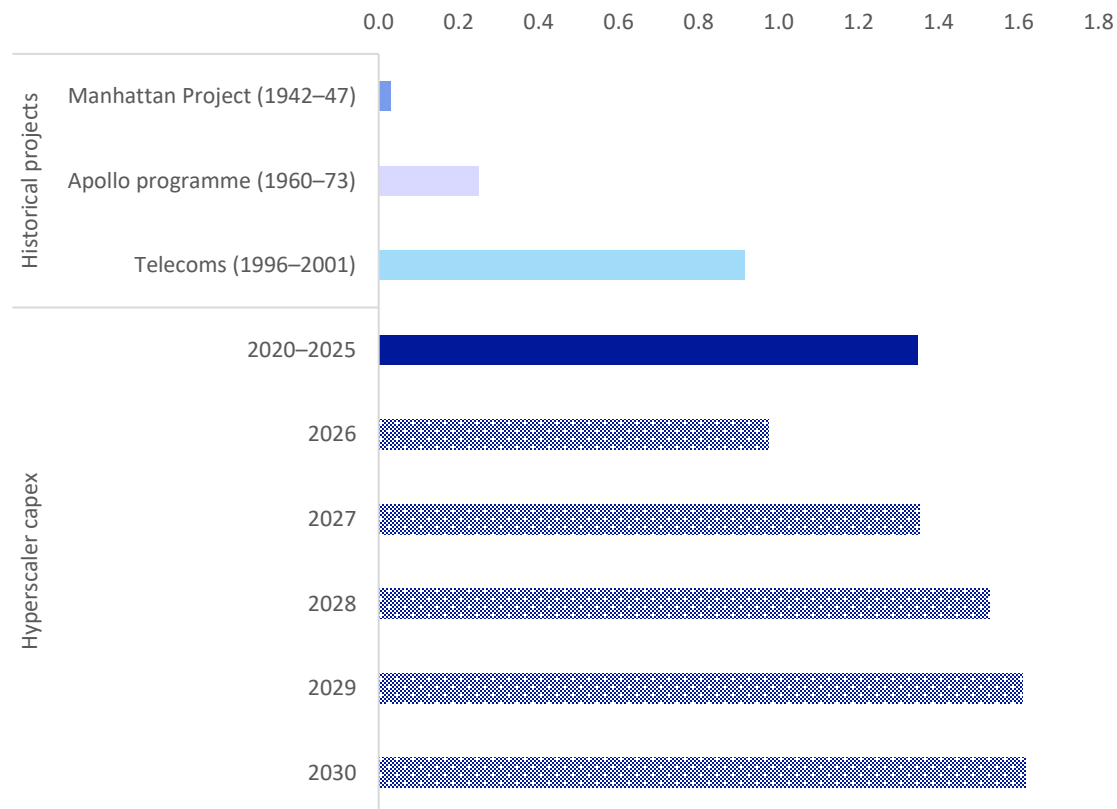
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1. This boom is remarkable not just for scale but also speed

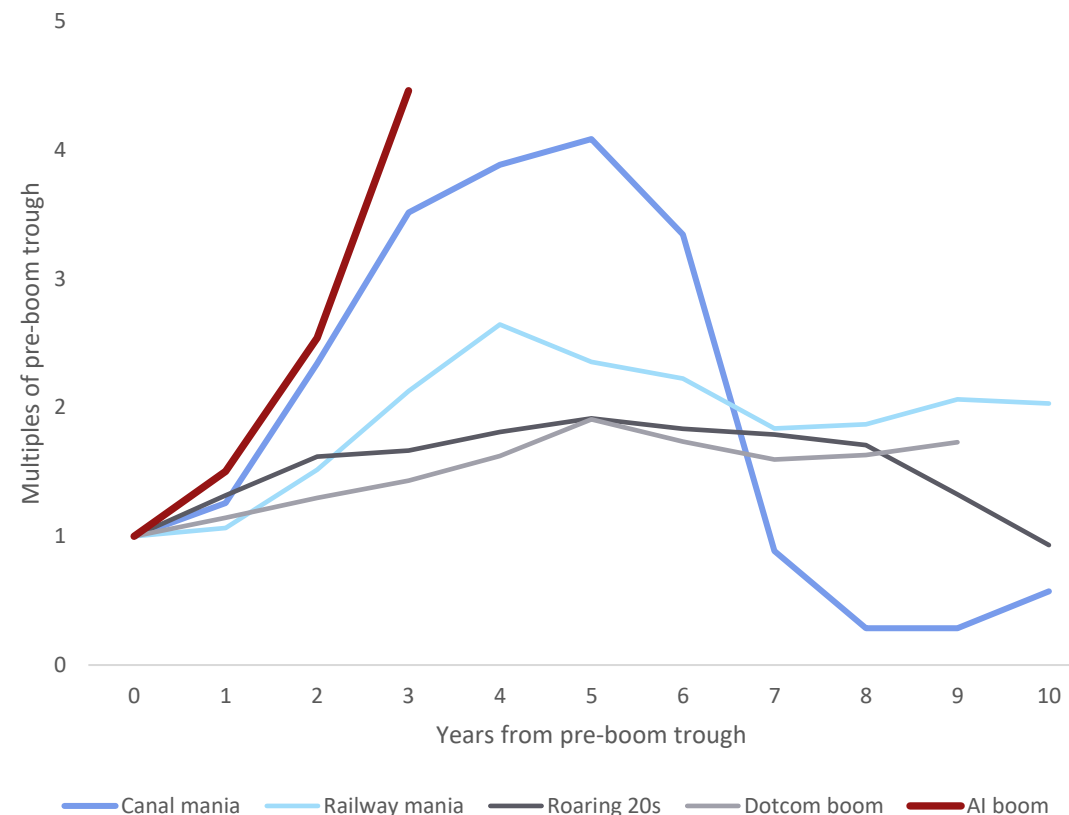
Each year of expected capex through 2030 exceeds the whole dot-com telecoms boom



Hyperscaler current and forecast capex vs major historical projects (2026 USD trn)



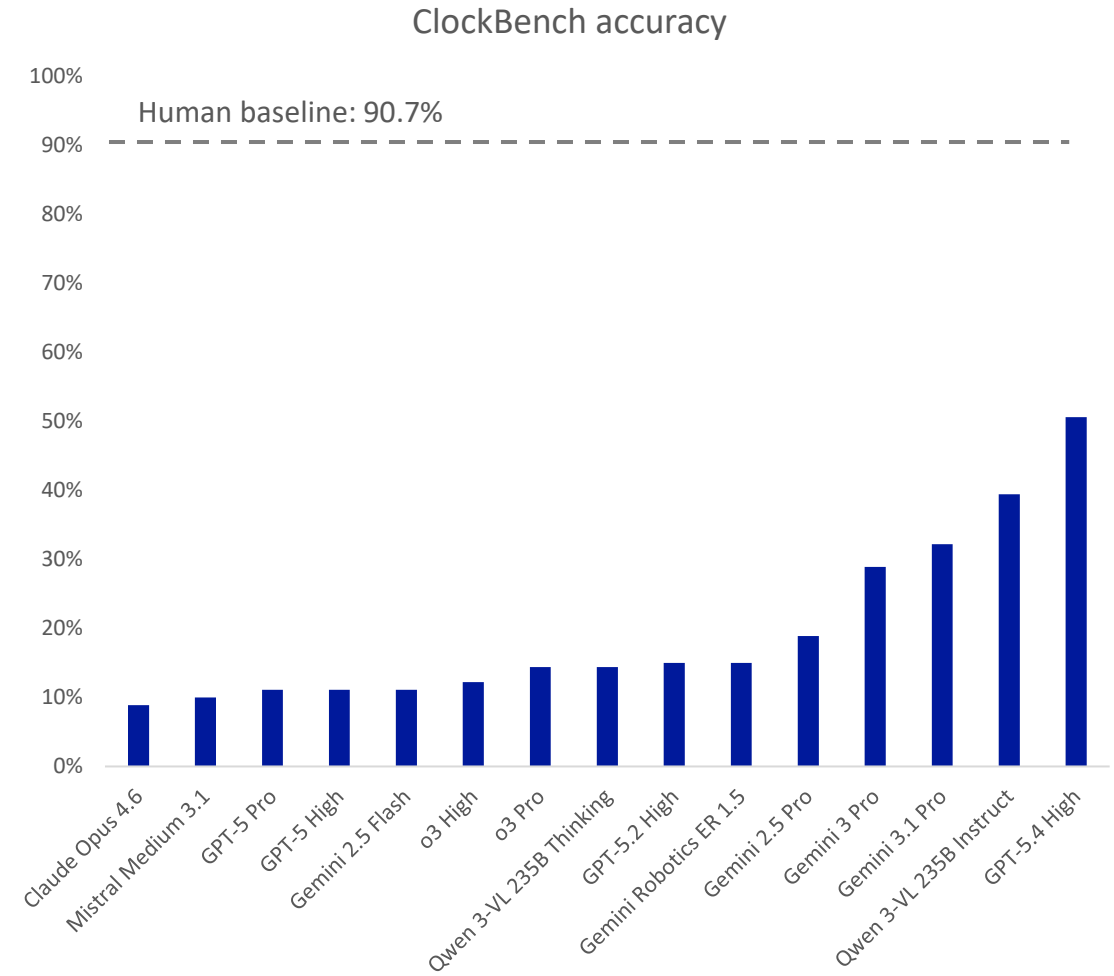
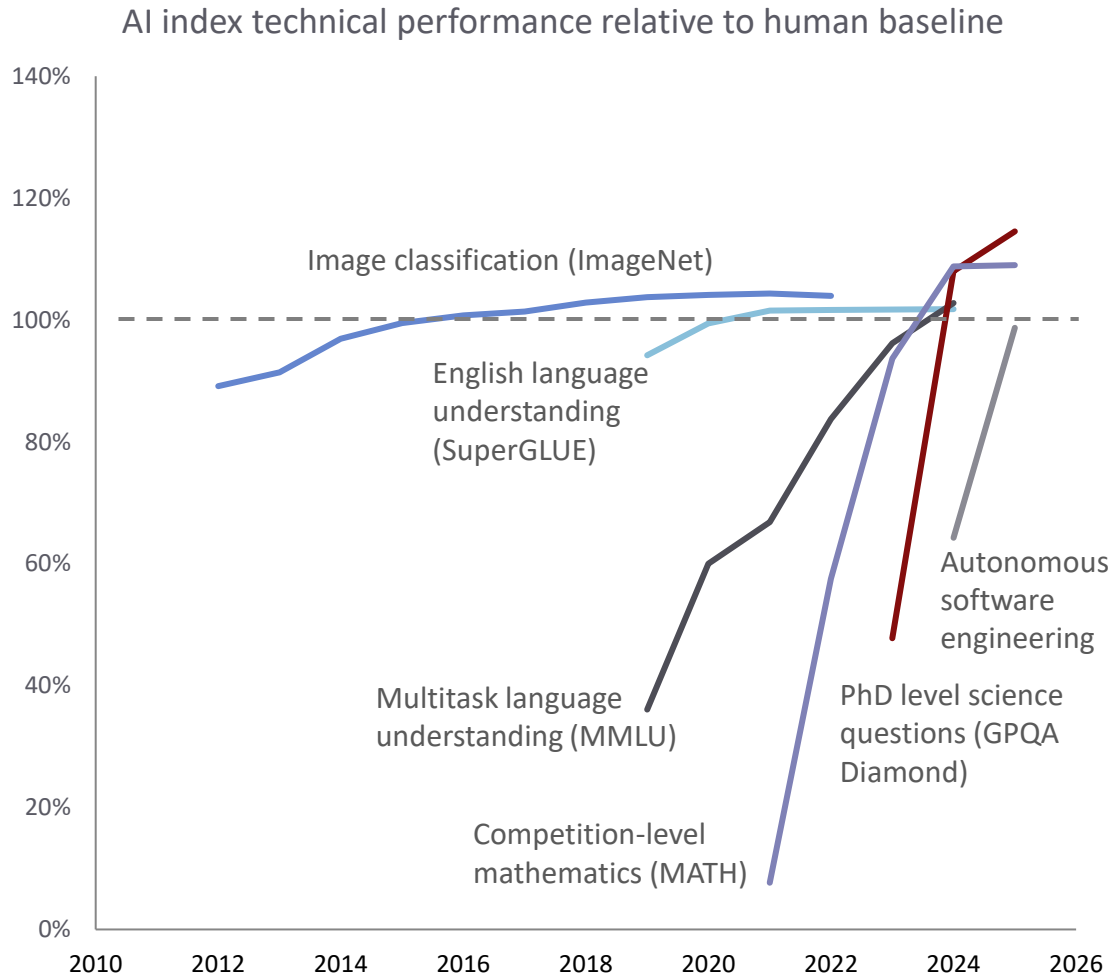
Investment during innovation booms: multiples of pre-boom trough



Source: US National Park Service, Space Policy, Brookings Institution, Bloomberg Finance LP, Bank for International Settlements, Deutsche Bank Research. Note: Hyperscaler capex estimates from Bloomberg include five US hyperscalers (AWS, Google, Meta, Microsoft and Oracle), which account for roughly 80% of the total on average, as well as tier-two/neoclouds and Chinese hyperscalers. Hyperscaler capex figures are not inflation-adjusted.

2. AI makes hard things look easy, and easy things look hard

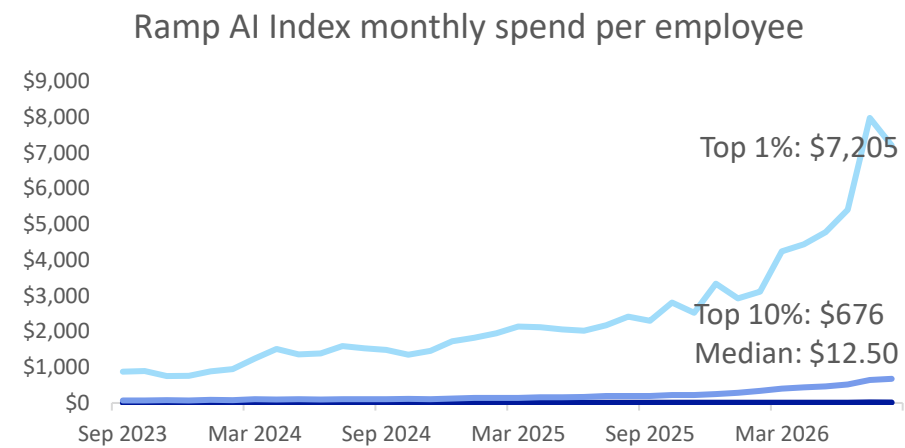
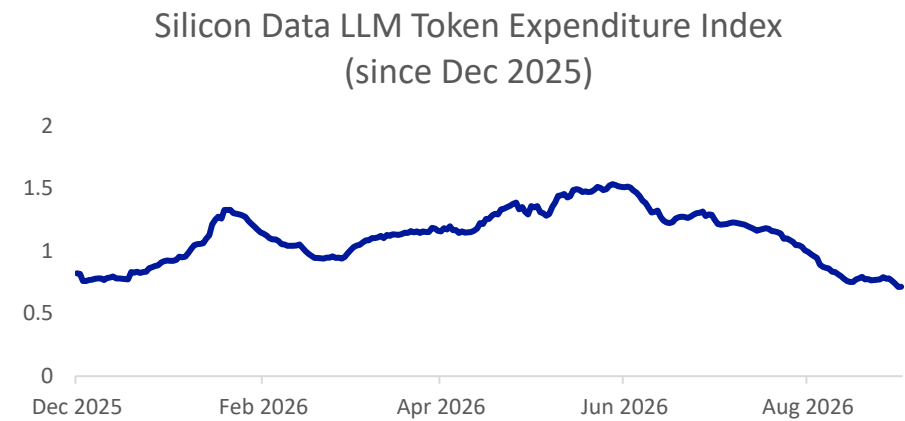
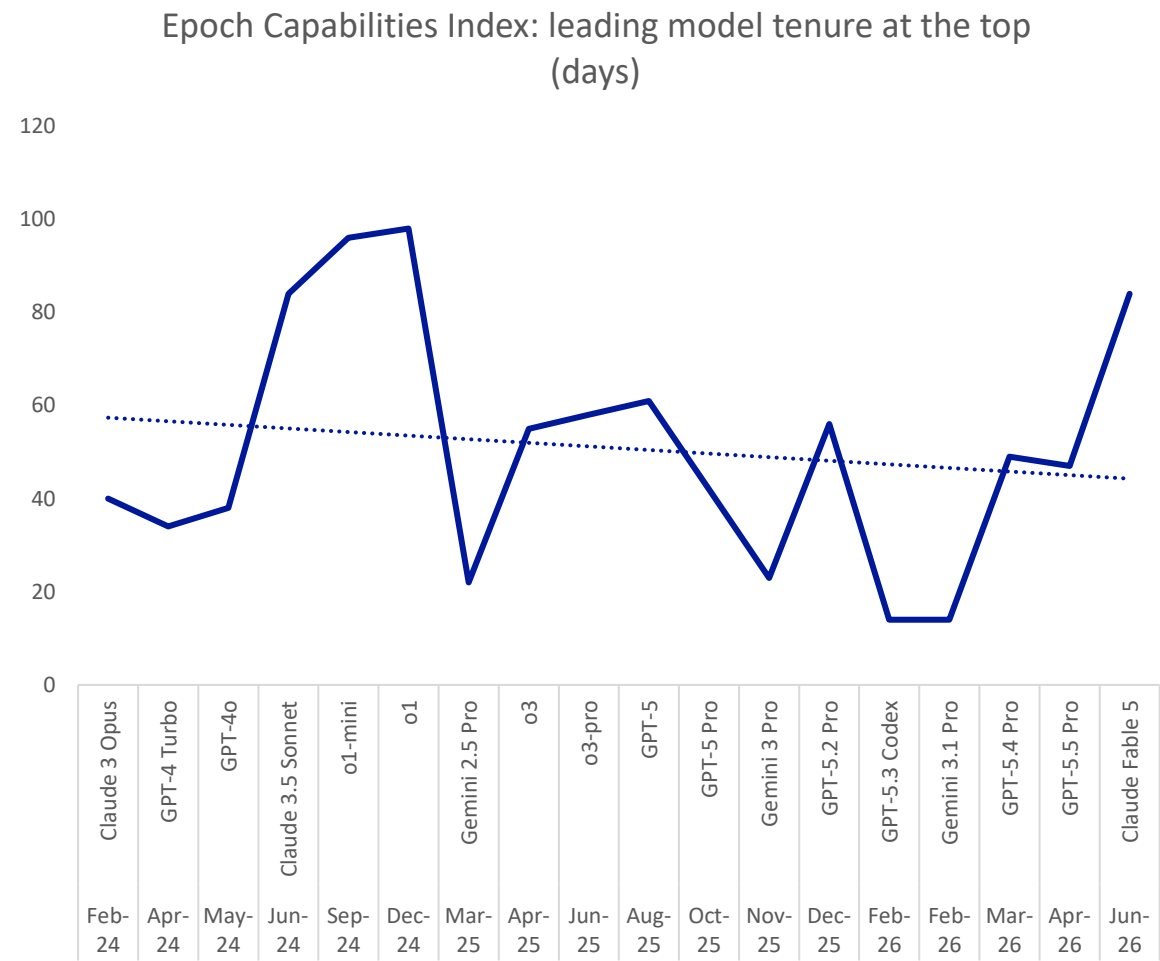
“Jagged frontier” means AI can solve complex tasks but struggles to tell the time on a clock



Source: Stanford HAI AI Index 2026, ClockBench Leaderboard 2026, Deutsche Bank Research

3. AI models are commoditising fast

Top models hold the crown for less than two months on average and pricing is under pressure



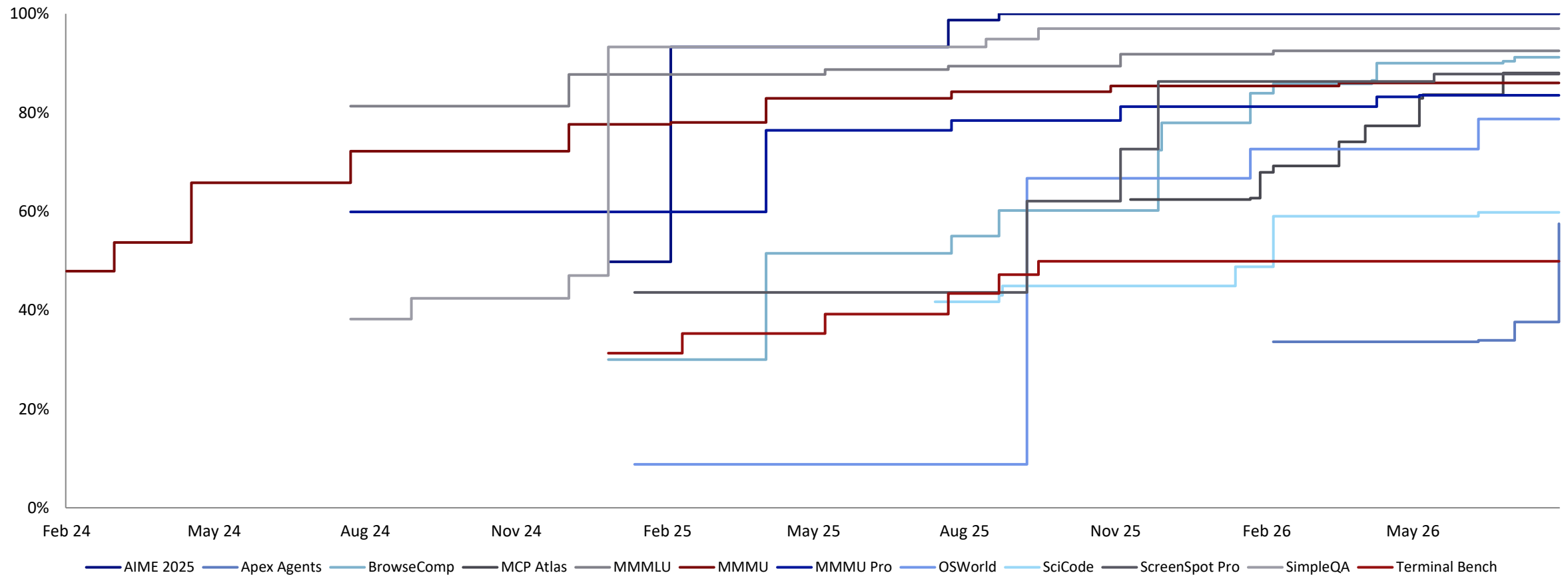
Source: Epoch, Bloomberg Finance LP, Silicon Data, Ramp AI Index, Deutsche Bank Research. NB: For clarity, Epoch Capabilities Index chart excludes OpenAI's GPT-4, the model that predated Claude 3 Opus and was at the top for 352 days, as well as the new leader since Sept 3, OpenAI's GPT-6 Astra

4. Benchmarks are ageing even faster than models

Plateaus may mean models have stopped improving... or the test is no longer hard enough



Benchmark saturation: State of the Art (SOTA) score progression over time -- each step is a new best model



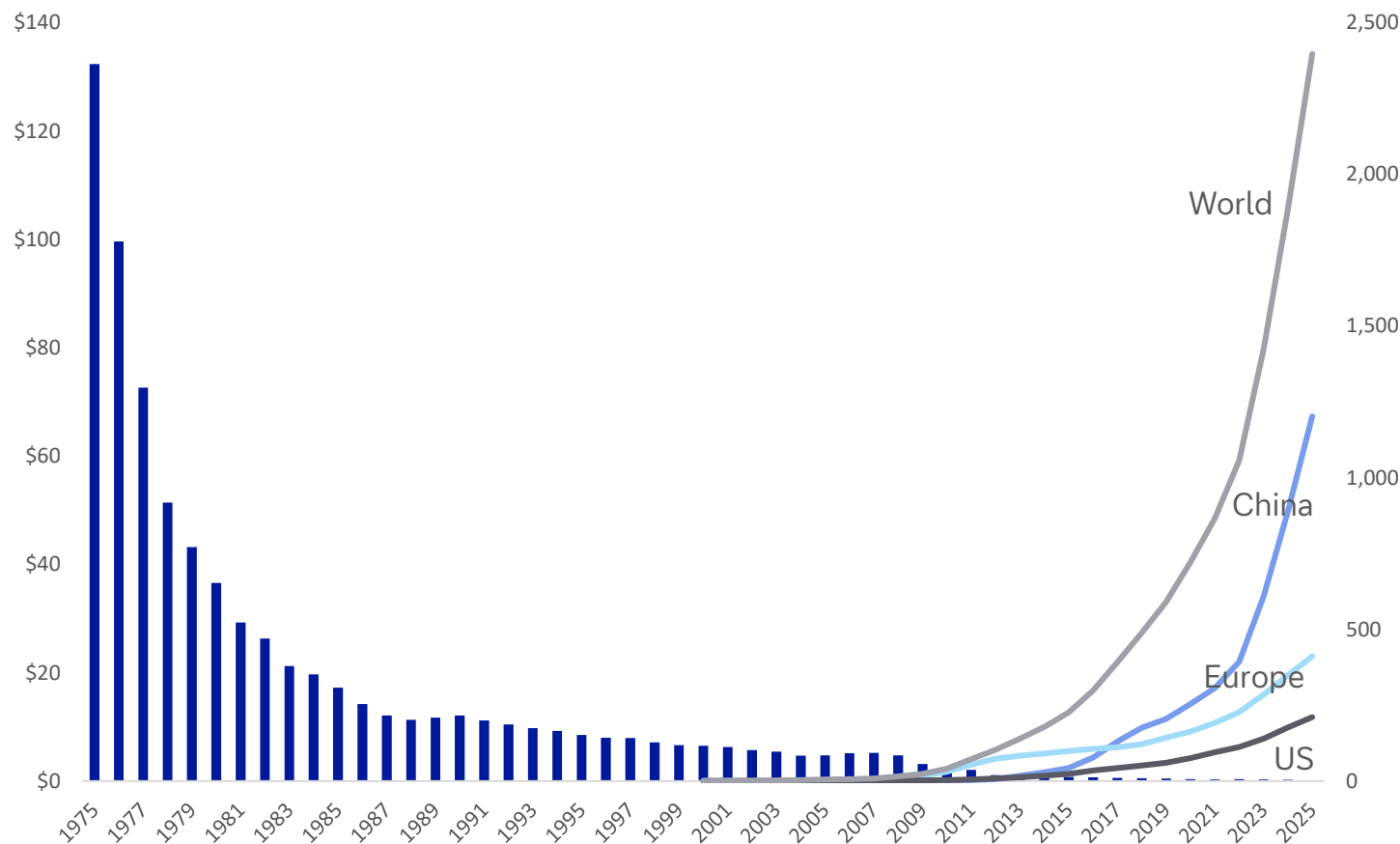
Source: [LLM Stats](#), Deutsche Bank Research

5. Tech use accelerates when the cost falls far enough

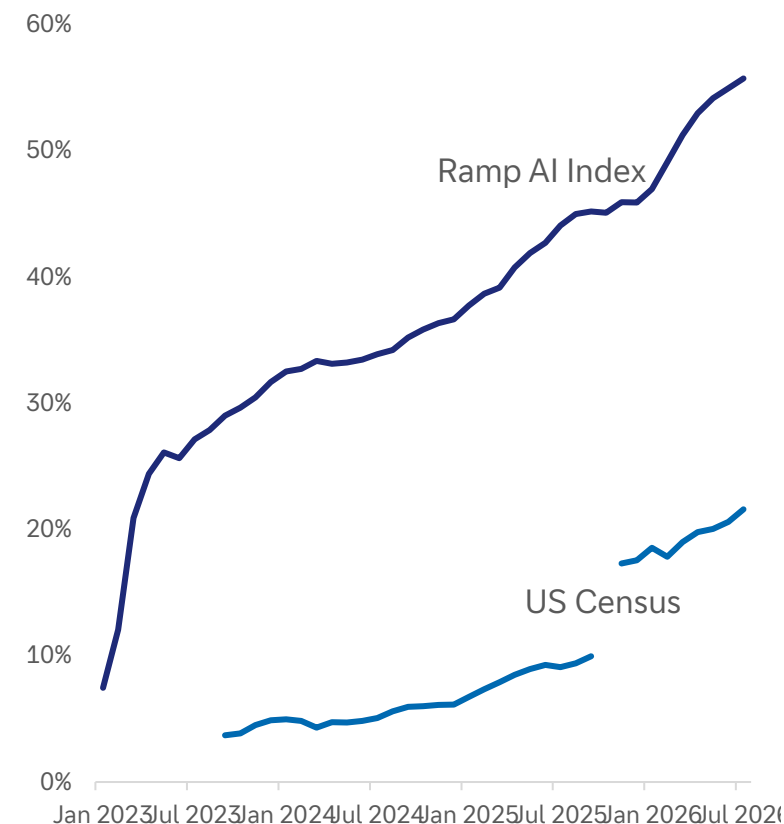
Solar and now AI are following similar patterns to horsepower, electricity and mainframes



Solar voltaic panel prices (USD/watt, adjusted for inflation, lhs) vs installed solar energy capacity (GW, rhs)



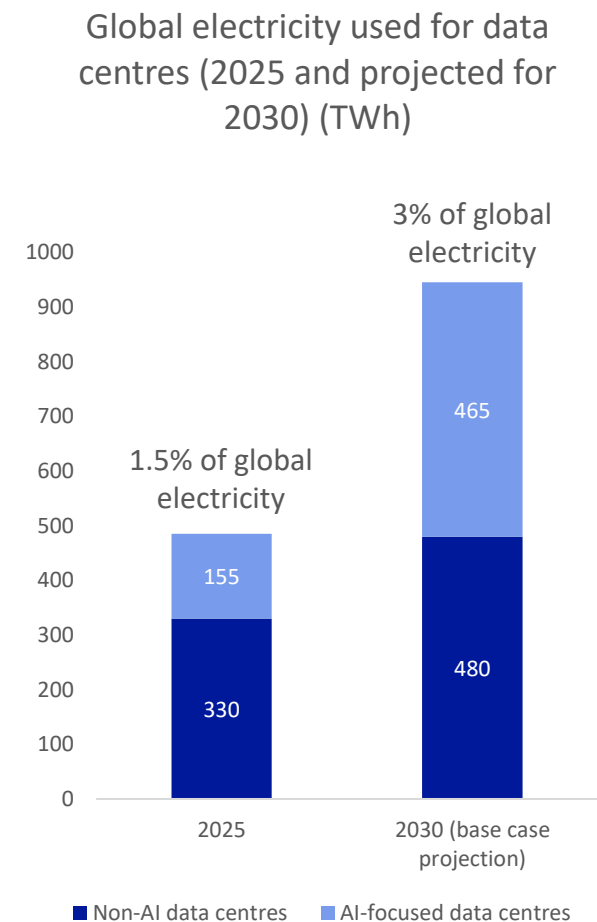
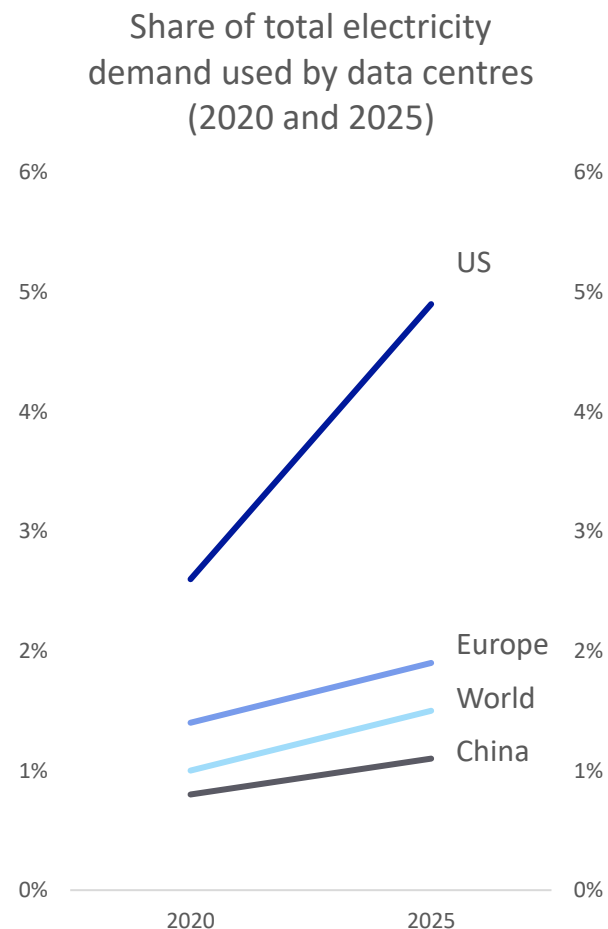
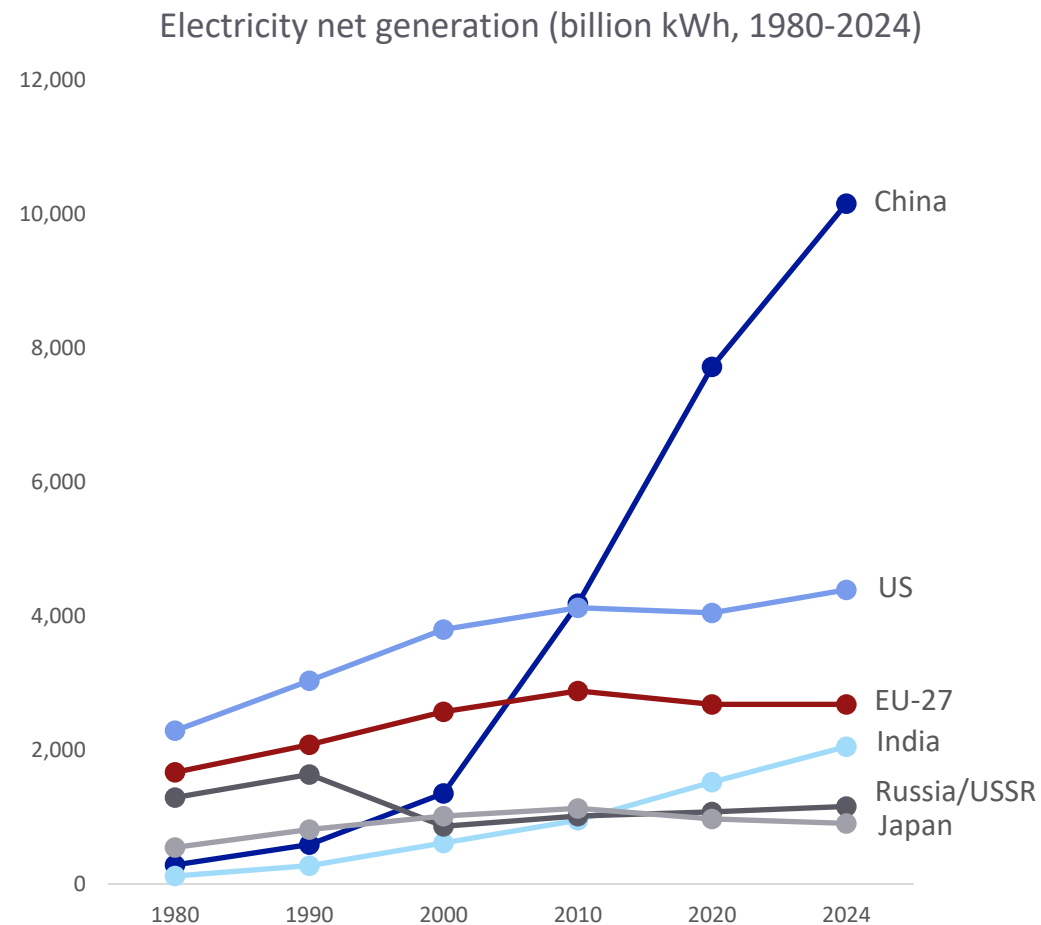
US enterprise AI adoption rate by two different measures (since Jan 2023)



Source: Our World in Data (citing IRENA, Nemet, Farmer and Lafond), Ramp AI Index, Deutsche Bank Research

6. China has left the US behind in electricity generation

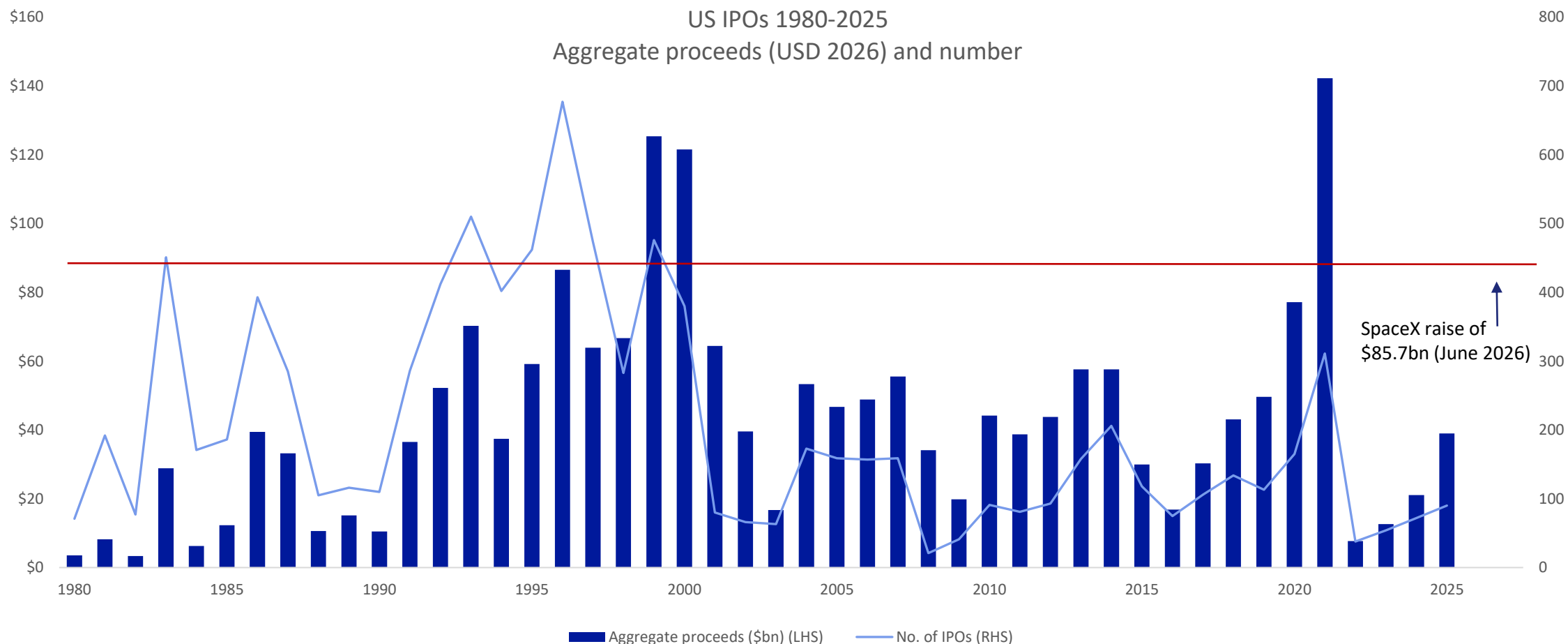
US is needing to use far more of its capacity for data centres even as demand continues to rise



Source: US Energy Information Administration, Our World in Data, International Energy Agency, Ember, Deutsche Bank Research

7. Mega AI IPOs will eclipse entire years of past share sales

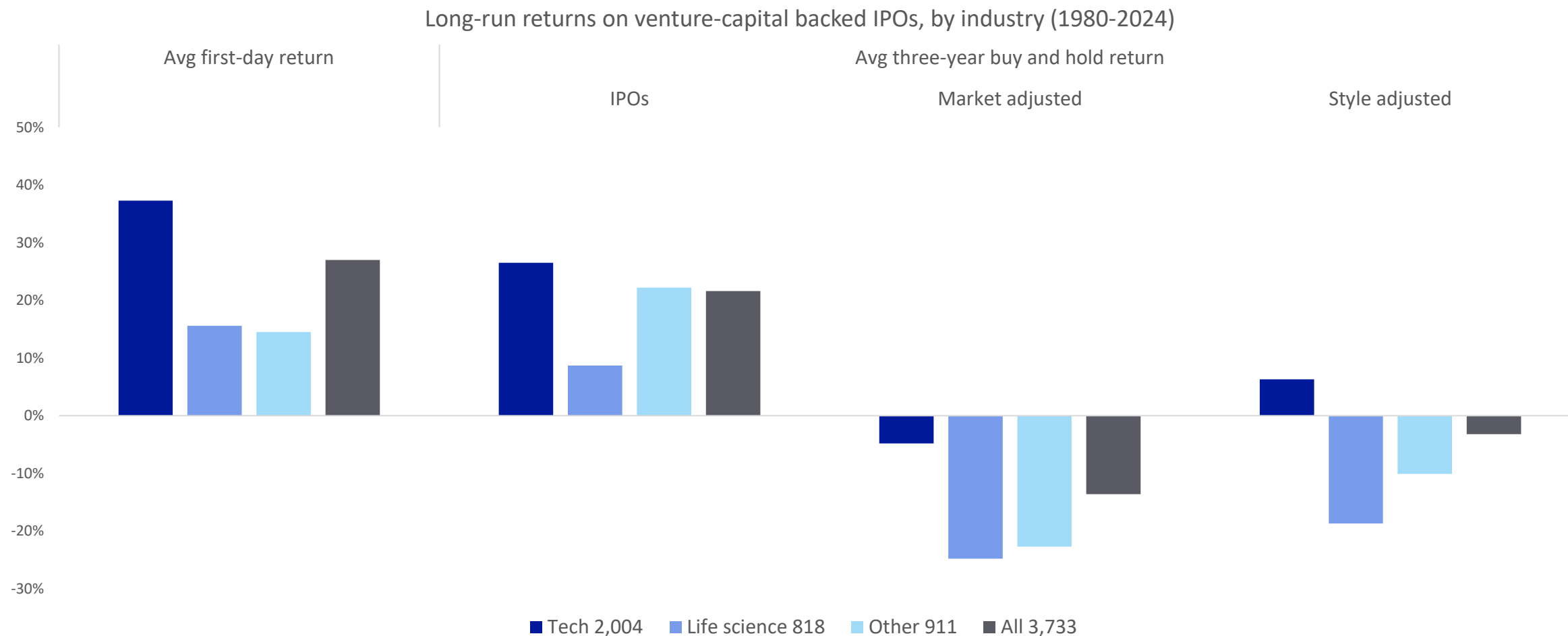
SpaceX's IPO exceeded all but three inflation-adjusted years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research

8. Tech IPOs have tended to outperform other IPOs

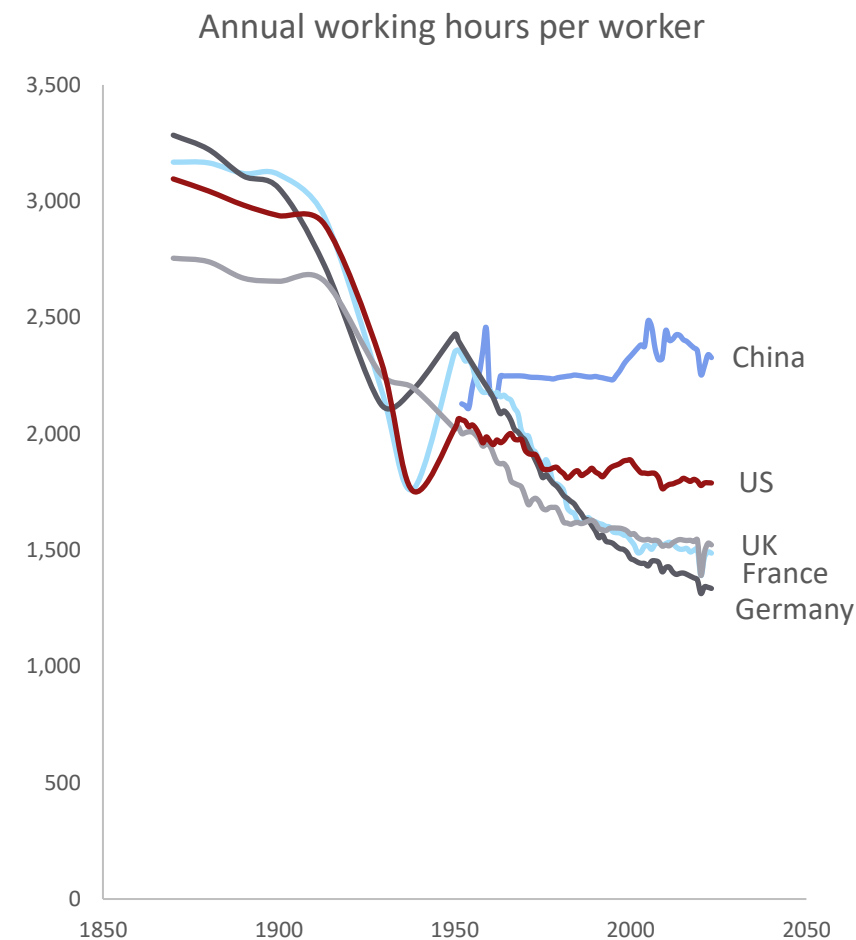
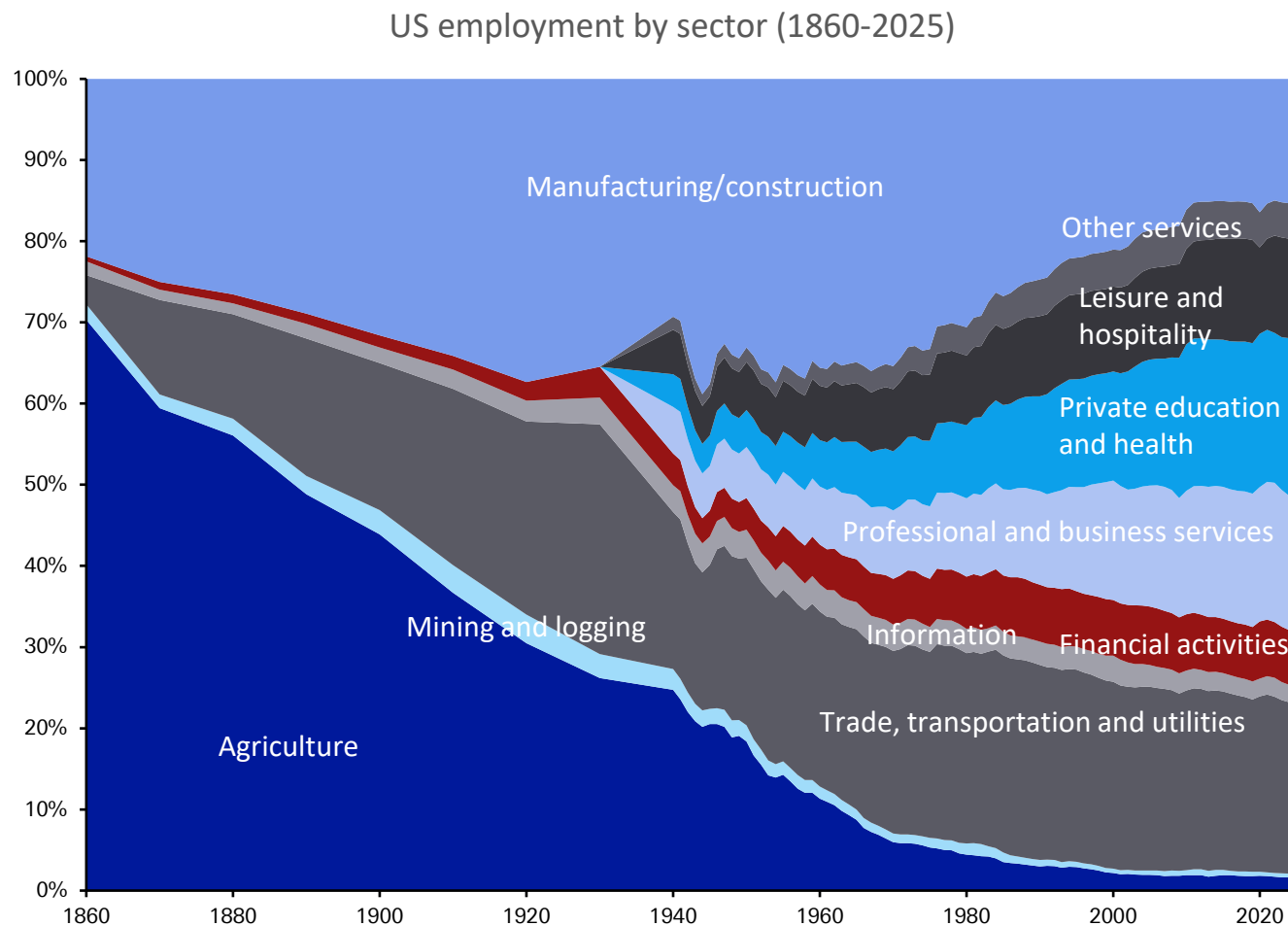
However, despite first-day surges, they have slightly underperformed the market over three years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research. Note: Three-year buy and hold calculated from first closing market price. Market-adjusted returns are the average buy-and-hold return minus the average compounded return on the Center for Research in Security Prices (CRSP) value-weighted index. Style adjustments are the equivalent for similar companies.

9. Jobs come and go but working hours have only fallen

As services have displaced agriculture then manufacturing, working hours have roughly halved



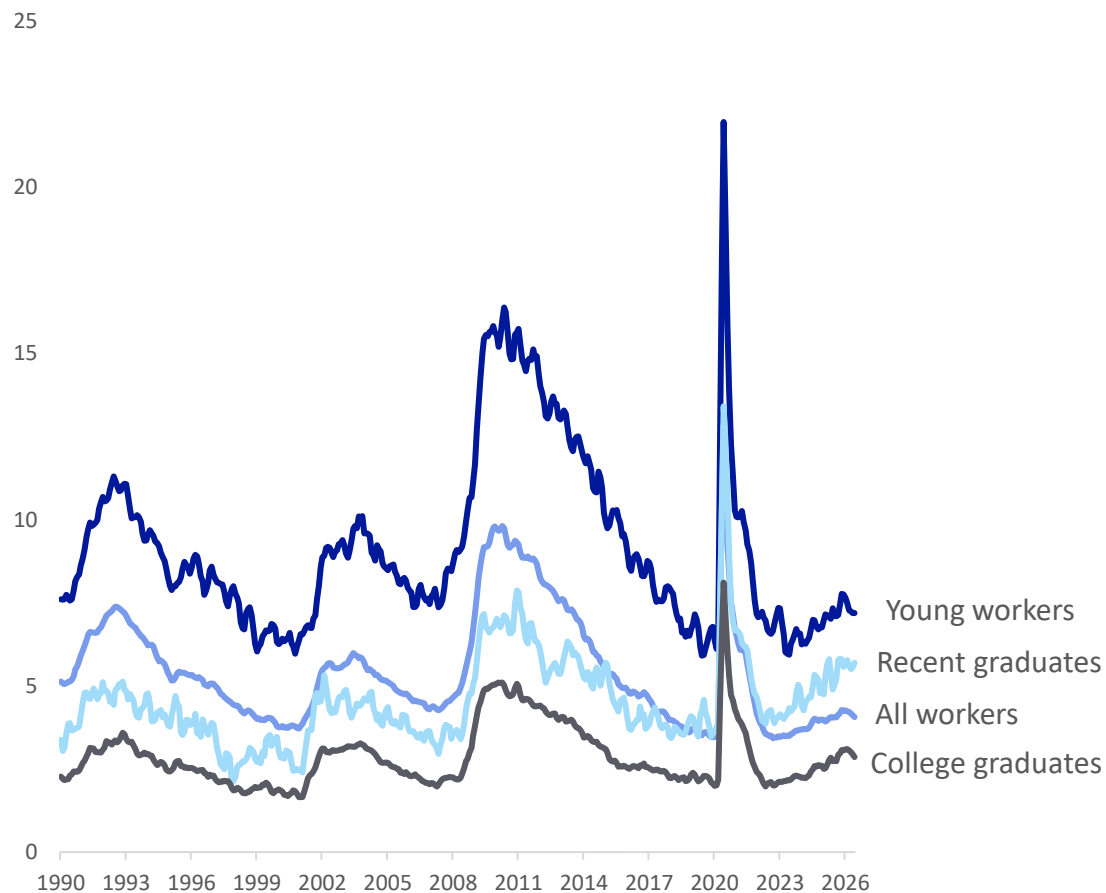
Source: Haver, Historical Statistics of the United States, 1789-1945, US Census, Our World in Data (citing Feenstra et al, Penn World Table (2025), Huberman and Minns (2005), Deutsche Bank Research

10. It's a tough job market for recent graduates

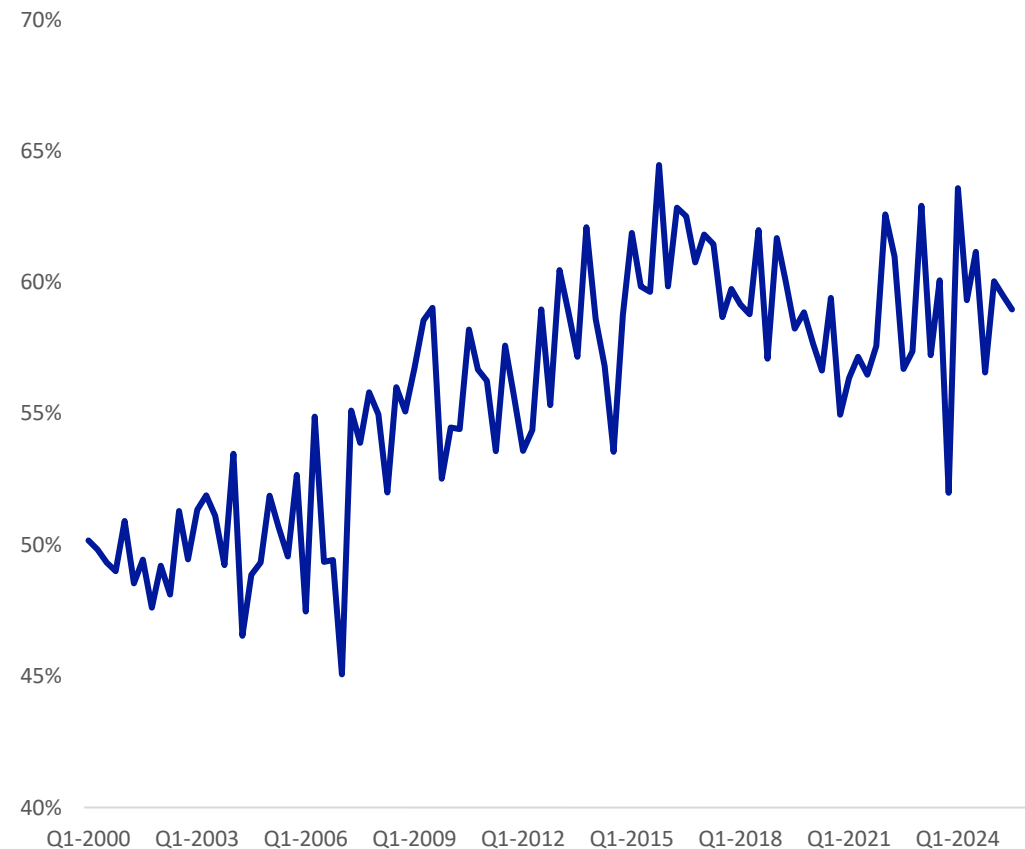
AI may be exacerbating the erosion over the past decade in graduate employment and wages



Unemployment rates for US workers



Bachelor's degree median weekly wage premium vs some college/associated degree (Q1 2000 to Q2 2026)



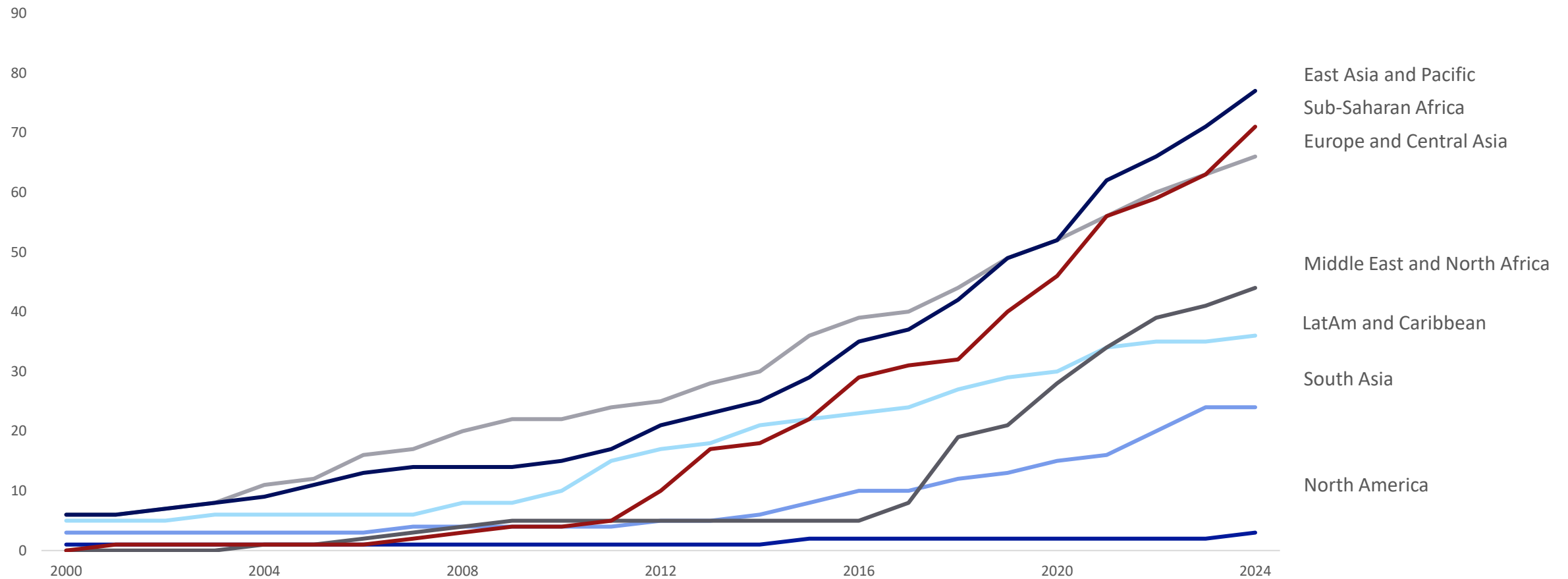
Source: U.S. Census Bureau, U.S. Bureau of Labor Statistics, Current Population Survey (IPUMS), Haver Analytics, Deutsche Bank Research. NB: For more, see Azar J, Mireia G and Sanz-Espin J, *The College Wage Premium in the Generative AI Era* (Aug 2026)

11. Concerns over data sovereignty date back long before AI

Countries have been restricting cross-border data flows, accelerating with EU's GDPR in 2016



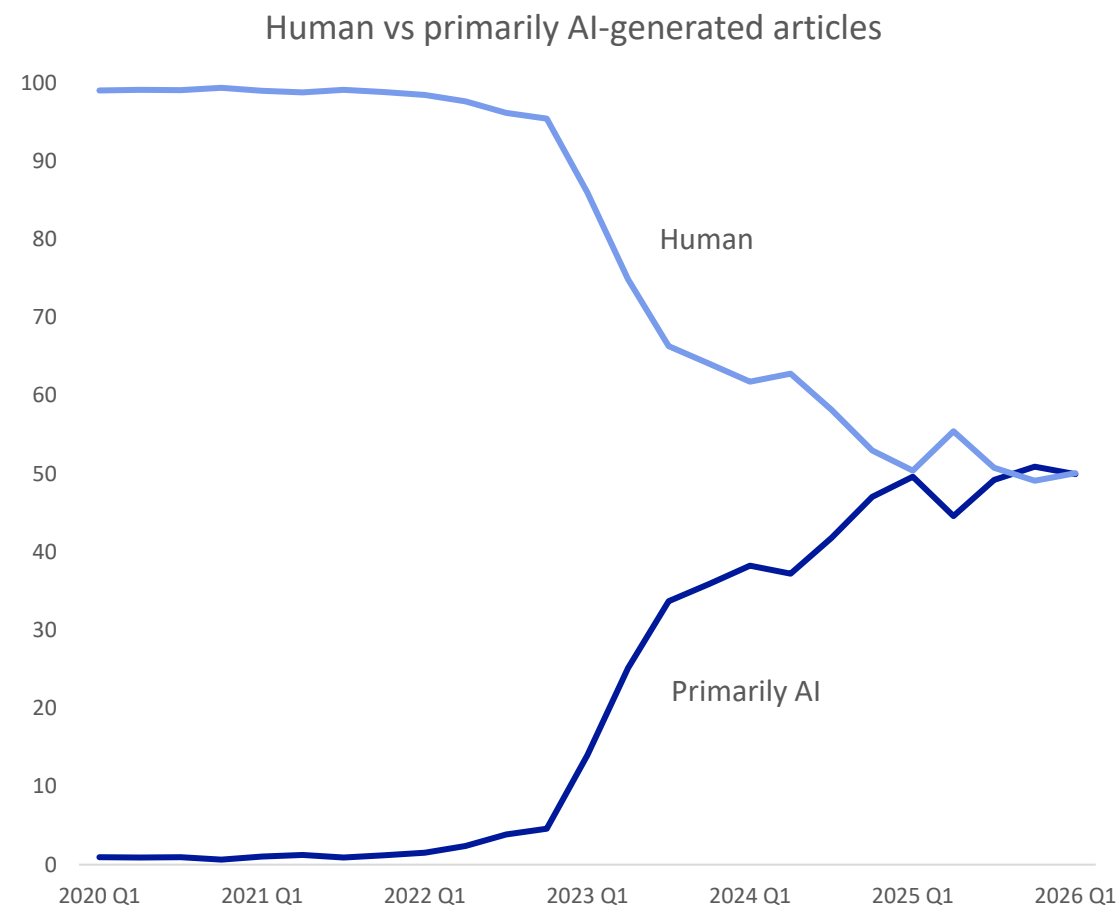
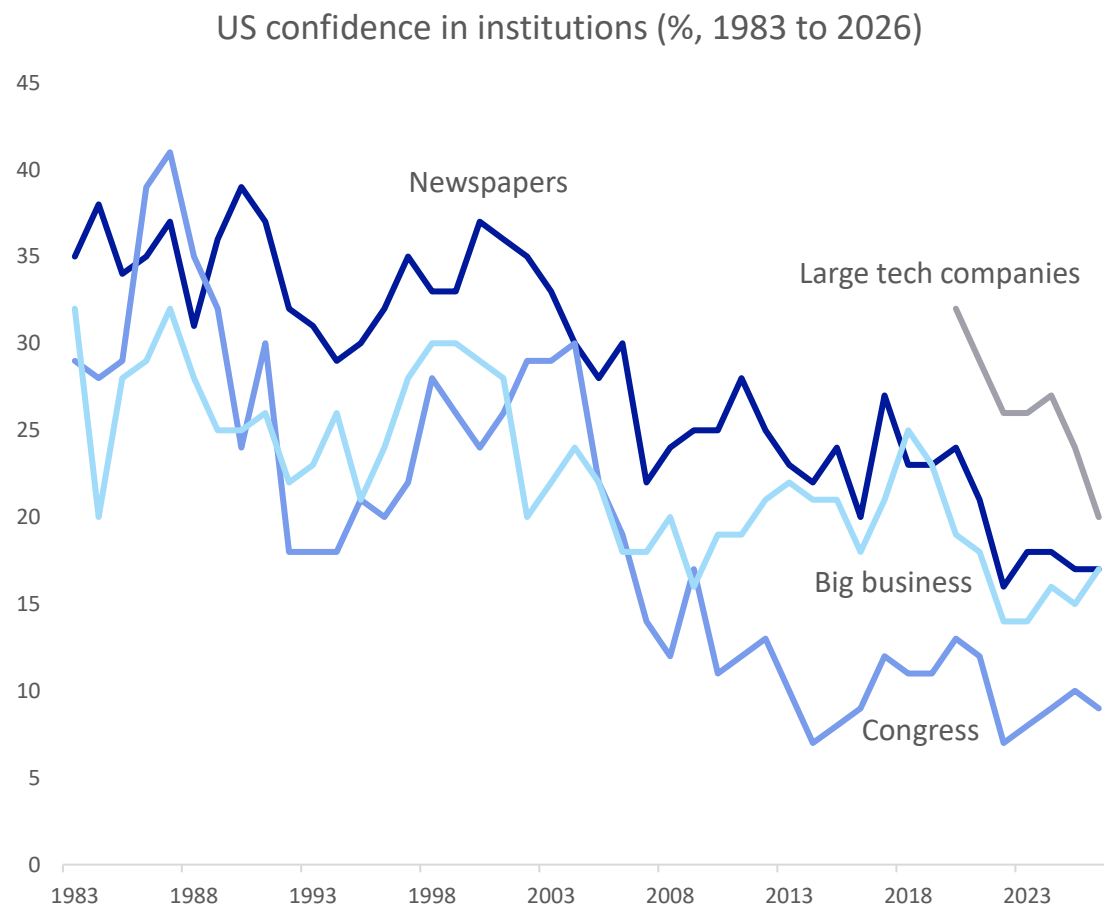
Data localisation measures by region (2000-2024)



Source: Stanford HAI AI Index (2026), Ferracane et al, Deutsche Bank Research

12. AI arrived as trust had been declining for decades

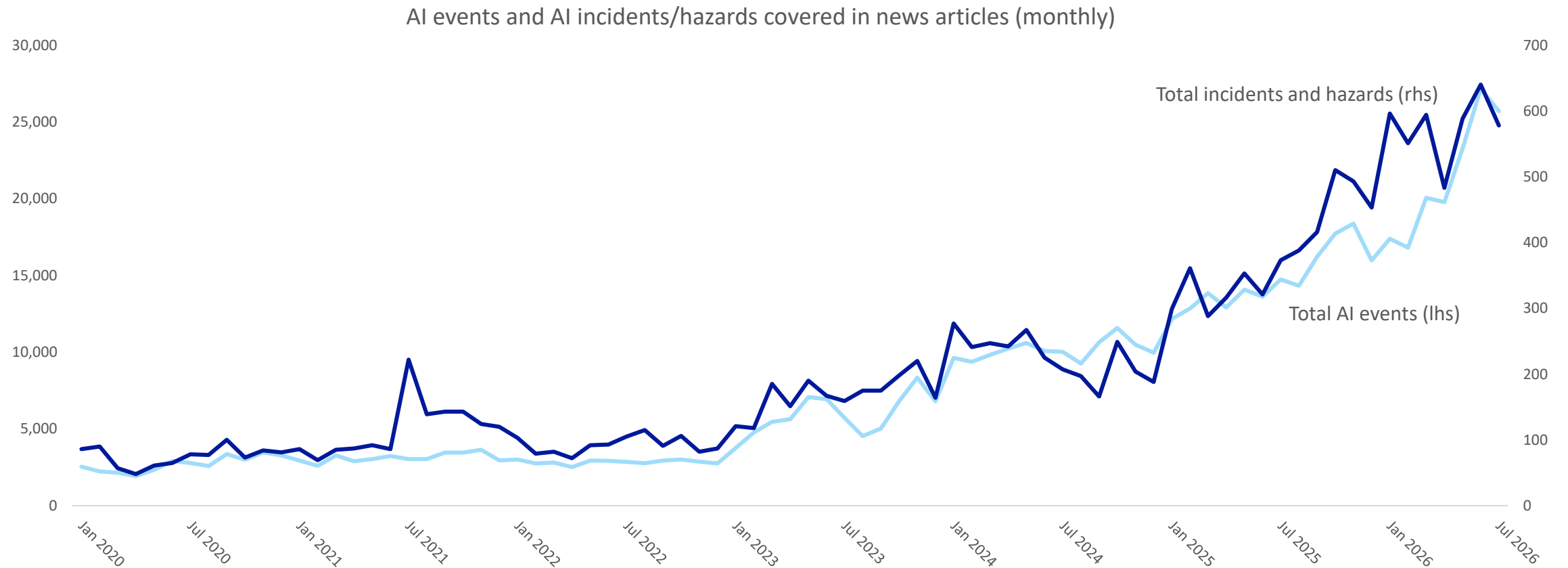
Confidence in institutions has plummeted – and now even human content is under threat



Source: Gallup ([Confidence in Institutions](#) | [Gallup Historical Trends](#)), [Graphite](#) (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

13. New technology brings new risks

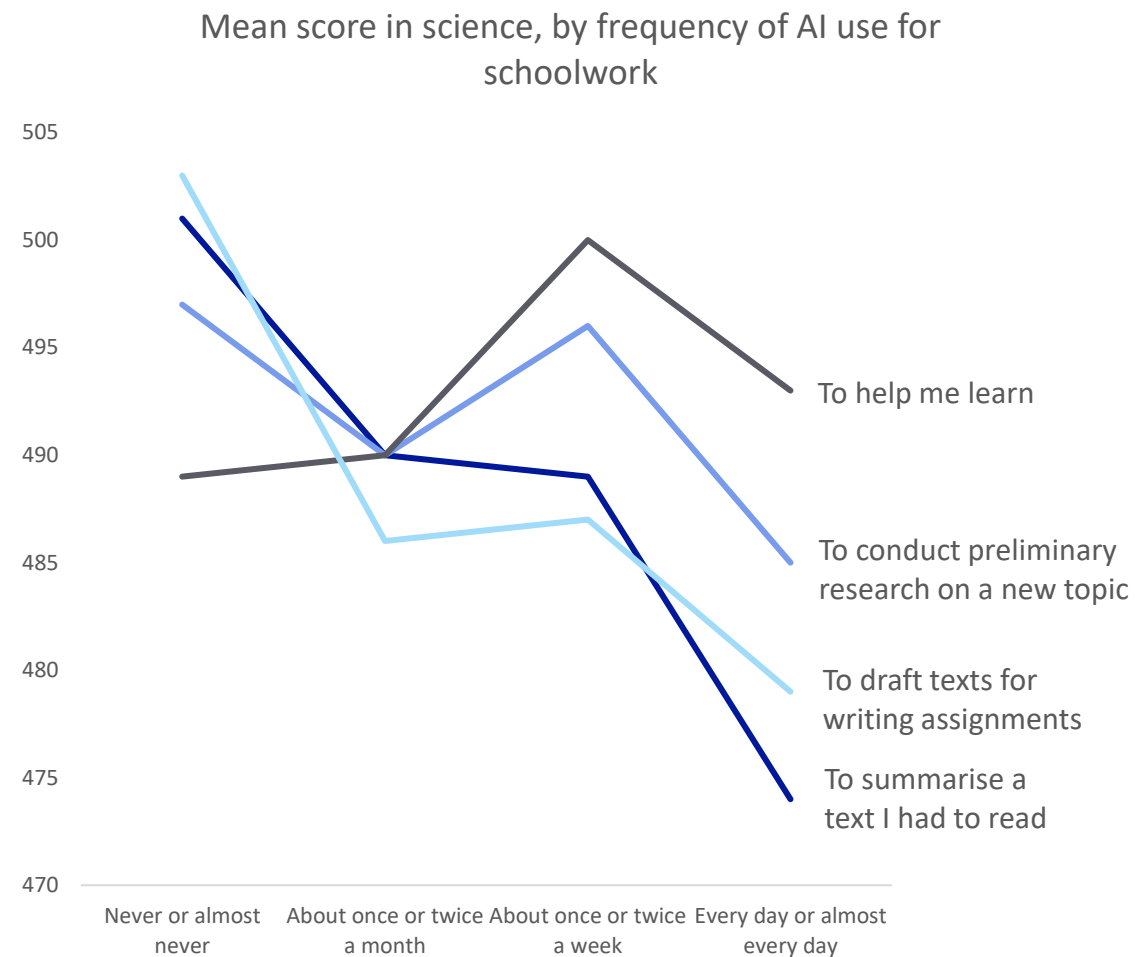
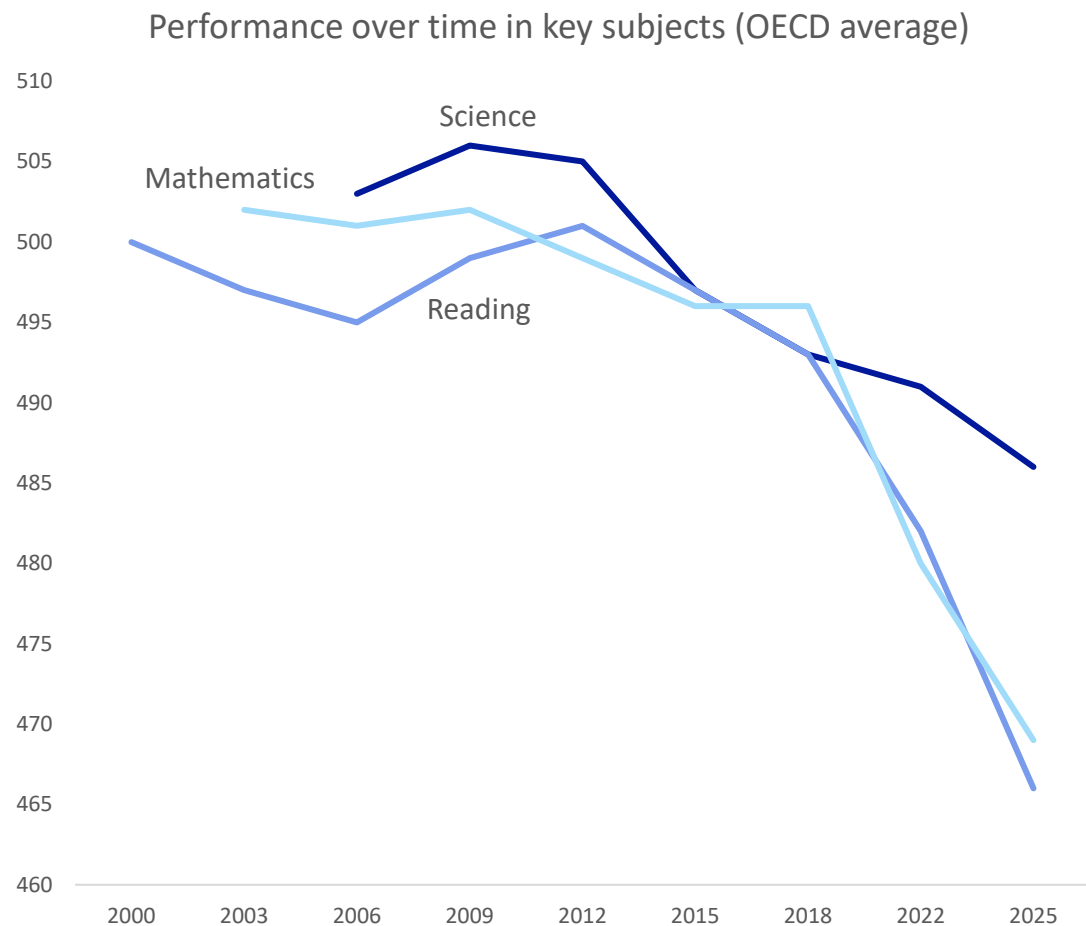
Coverage of AI events has surged, and so have hazards



Source: OECD AIM Incidents and Hazards Monitor, Deutsche Bank Research

14. Tech may have a role in declining educational performance

AI may be accelerating a fall that started with widespread smartphone diffusion from c. 2010



Source: OECD (2026), PISA 2025 Results (Volume I), Deutsche Bank Research



Appendix 1

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Tokenised money and the future of FX

10 September 2026

Tokenisation has the potential to transform the way financial markets operate, from the movement of money and assets to the management of liquidity and risk. In this paper, we focus on tokenised forms of money and their impact on FX markets.

Many emerging use cases for tokenised money are driven by a desire to reduce frictions associated with cross-border payments and FX settlement. Stablecoins enable 24/7 cross-border transfers but remain overwhelmingly US dollar-denominated. As a result, they are currently more relevant as digital dollar infrastructure and for bridge-currency arrangements, often creating a "stablecoin sandwich" in which the US dollar serves as an intermediary between two currency legs, rather than providing a complete solution to the two-currency FX settlement problem

Tokenisation may play a bigger role in transforming the combined securities and FX settlement cycles involved in cross-border capital flows. This is an area of increasing complexity, as delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity, and settlement risks. For example, following the transition to T+1 securities settlement in North America and with Europe preparing to move to T+1, the need to coordinate FX, cash, and securities settlement across institutions, jurisdictions, and infrastructures is already increasing.

Tokenisation enables programmability, which allows atomic settlement and a range of operational and control functions across financial market infrastructures, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. Current implementations have focused primarily on supporting corporate treasury and liquidity management activities involving FX. However, the settlement of tokenised securities and the associated FX transactions may offer even greater efficiencies by enabling synchronised settlement and reducing some of the operational, funding, and coordination frictions associated with shorter settlement cycles.

Interoperability between different currencies, tokenised money arrangements, and payment systems remains one of the key challenges facing the development of tokenised cross-border payments and FX settlement. Several initiatives are exploring solutions involving tokenised central bank reserves or hybrid models that combine tokenised commercial bank money with central bank money. These may be particularly relevant for multi-currency settlement frameworks, preserving the two-tier monetary system while supporting settlement finality and more integrated cross-border processes. In parallel, other initiatives are building infrastructure that enables different forms of money, payment systems, and settlement architectures to operate together.

Tokenisation promises substantial benefits but unintended consequences are worth considering. Beyond faster settlement, wholesale FX markets also depend on liquidity efficiency. This means banks and other financial institutions can settle large transaction volumes safely while using less liquidity. Existing infrastructures such as Continuous Linked Settlement (CLS) combine settlement risk mitigation with significant liquidity savings through multilateral netting and liquidity optimisation mechanisms. Unlike CLS, most current tokenised settlement models do not yet provide multilateral netting at

Authors

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Strategist



comparable scale. A number of initiatives are exploring liquidity-saving mechanisms and new approaches to liquidity management under tokenisation.

We expect a period of accelerating change in years ahead. Securities and currencies settlement involves a complex nexus of technology, market infrastructure, regulation, and legal frameworks, necessitating cooperation between both public and private sector participants across jurisdictions.



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Introduction

Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure. This includes both state and privately issued fiat-denominated forms of tokenised money, including **stablecoins, tokenised commercial bank deposits and tokenised central bank money**.¹ The potential role of tokenised money is increasingly being explored by both corporates and institutional investors. Several use cases have emerged ranging from cross-border payments, wholesale settlement, treasury and liquidity management, trade finance and capital market infrastructure. More recently, attention has expanded beyond using tokenised money for cross-border payments towards models that can support the coordination and settlement of FX transactions.

Wholesale: Refers to activity involving banks, financial institutions, and large corporates, rather than retail customers or consumers.

While tokenised money can support more efficient payments, FX settlement often requires the conditional coordination of payments across currencies. Unlike a simple payment, FX settlement requires the exchange of two currencies to occur simultaneously, creating a need for a synchronisation or **programmability layer** that enables **Payment-versus-Payment (PvP) settlement and reduces settlement risk**. Tokenised money operating on programmable ledger-based infrastructure can support this by embedding rules and conditions directly into settlement processes, facilitating the settlement of both tokenised money and other tokenised assets.

The settlement process, particularly for **wholesale FX transactions involving financial institutions and large corporations**, is exposed to settlement risk, often referred to as Herstatt risk. This arises when one party delivers its currency but the counterparty fails to fulfil its obligation. Cross-border transactions are particularly vulnerable because time zone differences can create a window in which one side has settled while the other has not. Over recent decades, this risk has been materially reduced in major currencies through infrastructure such as CLS, helping to support a large, liquid and operationally resilient FX market.

CLS mitigates FX settlement risk through PvP settlement, ensuring that one currency is transferred only if the corresponding currency is delivered, while reducing funding requirements via multilateral netting and liquidity optimisation tools, generating liquidity savings of over [96%](#) for settlement members. This [liquidity efficiency](#) is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely.

However, these benefits are not universal. Many emerging-market currencies subject to current- or capital-account restrictions remain outside the CLS ecosystem and continue to rely on correspondent banking networks or local clearing systems. Settlement in these markets often involves bilateral pre-funding, fragmented payment chains and complex liquidity management, creating operational and funding frictions that can increase the cost and complexity of cross-border transactions.

While settlement risk and liquidity frictions remain important challenges in FX

¹ This definition is in line with CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation



markets, changes elsewhere in the financial system are creating new demands on settlement infrastructure. As securities markets move from T+2 to T+1 settlement and continue to explore even shorter settlement cycles, the movement of cash, FX, securities and collateral has become increasingly interconnected. While existing infrastructure continues to support these markets, cross-border funding and settlement remain dependent on operating hours, time zones and multiple intermediaries.

These developments have brought two related challenges into focus. The first is improving cross-border settlement, particularly where correspondent banking and non-CLS arrangements create liquidity and operational frictions. The second is supporting a more integrated financial system in which cash, FX, securities and collateral can move more seamlessly across institutions and jurisdictions. Together, these challenges have strengthened interest in tokenised forms of money as a potential means of modernising both payment and capital market infrastructure.

The role of tokenised money in FX is closely linked to the broader tokenisation of financial assets. As [asset tokenisation gains momentum](#), tokenised money is increasingly viewed not merely as a payment instrument, but as an enabling layer for more integrated settlement processes. The key question is therefore not whether the technology is feasible, but whether it can achieve sufficient scale, interoperability and integration with existing financial market infrastructures to deliver meaningful improvements in cross-border settlement and capital market efficiency.

Many tokenised money initiatives have focused on improving the cross-border movement of money rather than directly settling FX trades. This reflects the scale of corporate payment frictions: [global corporates move an estimated US\\$23.5 trillion](#) across borders each year and remain heavily reliant on correspondent banking networks, which generate approximately **US\$120 billion** in annual cross-border transaction fees. These figures exclude additional costs related to FX conversion, trapped liquidity, pre-funding, settlement delays and operational complexity, suggesting that the overall cost of cross-border treasury activity is significantly higher. However, the scope of these initiatives is beginning to broaden. As tokenised money infrastructure matures, some projects are moving beyond payment efficiency towards models that could support FX settlement.

Against this backdrop, this paper examines the current FX settlement landscape before assessing the role of tokenised money in FX. It evaluates the opportunities and limitations of stablecoins, tokenised deposits and tokenised central bank reserves, and considers whether these innovations can address challenges that continue to characterise cross-border FX markets.

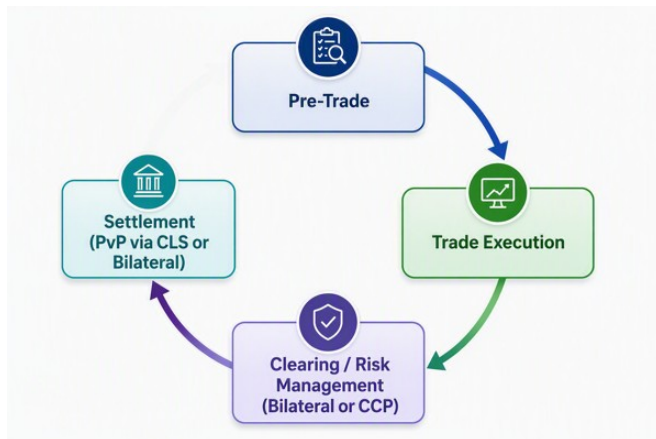
FX settlement today

FX trading takes place primarily in the over-the-counter (OTC) market, where banks, dealers, electronic platforms and other market participants facilitate the exchange of currencies. Once a trade is executed, it enters the post-trade process, during which trade details are confirmed bilaterally or cleared through a central counterparty (CCP), payment instructions are exchanged, and settlement obligations are established ([Figure 1](#)). This process spans the period between trade execution and final settlement, during which exposures are managed, liquidity is



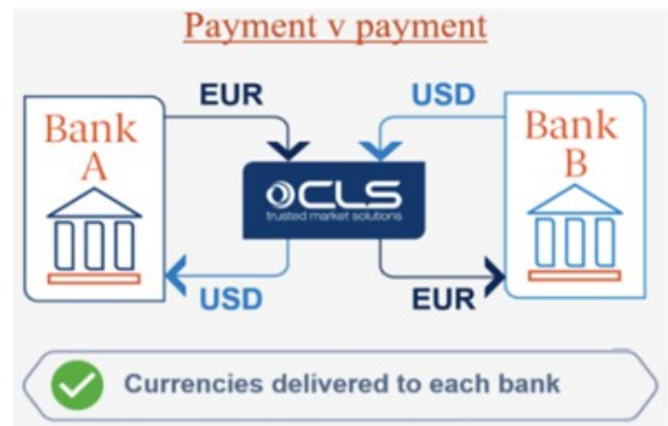
arranged, and currencies are transferred between counterparties. [Two types of credit risk](#) arise during this period: **replacement cost risk**, which exists from trade execution until settlement, and **principal risk**, which arises during the settlement process itself when one party may deliver the currency it sold but not receive the currency it purchased. Depending on the currency pair, FX spot transactions typically settle one or two business days after the trade date, while forwards and options settle on their contractual maturity date, which may range from days to several years after execution.

Figure 1: FX spot trading cycle



Source : Deutsche Bank

Figure 2: FX settlement via CLS



Source : Deutsche Bank, CLS

To reduce principal risk at settlement, often referred to as **settlement risk or Herstatt risk**, wholesale FX markets rely on infrastructure such as CLS, which enables participating banks to settle eligible FX transactions on behalf of themselves and their clients through a **Payment-versus-Payment (PvP) mechanism**, ensuring that one currency is transferred only if the corresponding currency is delivered ([Figure 2](#)).

How CLS works?

Today, CLS achieves simultaneous settlement through a defined funding and settlement process rather than instantaneous settlement. **Under this process, settlement is deferred to a daily settlement cycle, enabling multilateral netting and liquidity optimisation before final settlement occurs. This is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely. Settlement finality is achieved through the use of central bank money.**

- For FX trades eligible for settlement using CLS, settlement members or their customers submit electronic payment instructions for both sides of the transaction. CLS validates and matches these instructions, providing legal confirmation and holding the transaction details until the settlement date.
- On each settlement date, CLS simultaneously settles matched payment instructions that satisfy its risk management requirements. Instead of every FX trade being funded individually, CLS calculates each member's net

Gross versus net settlement: what does it mean? Twenty currencies generate 190 unique bilateral currency pairs. Under gross settlement, each currency leg must be funded separately. Through multilateral netting, these obligations are reduced to 20 net currency positions, one for each currency, so participants fund only their net obligations rather than every bilateral payment.



obligation across all eligible trades and currencies.

- Members then make funding payments and receive pay-outs at specified times during the day. CLS holds accounts at the relevant central banks for the currencies it settles. Settlement members pay and receive funds through these CLS accounts via either their own central bank accounts or correspondent banking (nostro) arrangements. Settlement members' multicurrency accounts begin and end each business day with a zero balance.
- The settlement of payment instructions and the associated payments across the books of CLS Bank is final and irrevocable.

Nostro account: A bank account held by one bank with a foreign correspondent bank, typically to facilitate payments and settlements in a foreign currency. The term means "our account with you".

The PvP mechanism also illustrates that a form of **atomic settlement** already exists in today's FX market. Once settlement occurs in CLS, it is final and irrevocable, meaning the transfer of both currency legs cannot be unwound. However, as [CLS notes](#), atomic settlement can encompass both simultaneity and instantaneousness, which are distinct concepts. Settlement can occur on an all-or-nothing basis, with both currency legs settling simultaneously and with finality, without necessarily taking place at the moment a trade is executed.

The CLS function differs from that of a central clearing counterparty (CCP). While CLS's PvP mechanism eliminates FX settlement risk, **it does not eliminate replacement cost risk**, which arises when a counterparty defaults before settlement and the transaction must be replaced at prevailing market prices. These risks can be particularly significant for FX forwards and swaps, where the period between trade execution and final settlement may extend for months or even years. CCPs such as ForexClear help mitigate counterparty credit and replacement cost risk through central clearing and margining, whereas CLS is designed to address settlement risk at the point of settlement by ensuring the simultaneous exchange of both currency legs.

CLS mitigates FX settlement risk (Herstatt risk) at the point of settlement by ensuring that the two currency legs of an FX transaction are settled simultaneously on a payment-versus-payment (PvP) basis.

ForexClear mitigates counterparty credit and replacement cost risk before settlement through central clearing, novation and margining. Replacement cost risk arises after a trade is executed and before it settles, when an adverse move in market prices could leave one party exposed if the counterparty defaults.

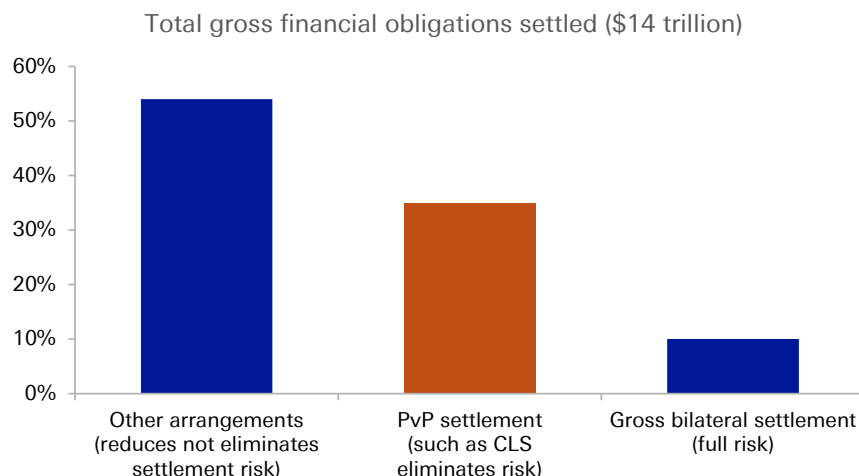
Settlement risks significantly reduced but not eliminated

According to the latest BIS Triennial Survey, banks and other financial institutions account for approximately 95% of global FX turnover, compared with around 5% for non-financial firms. Around 65% of financial-sector FX activity is cross-border in nature. As a result, wholesale FX settlement is overwhelmingly driven by financial institutions, underscoring the importance of robust cross-border settlement infrastructure.

Although infrastructure such as CLS has significantly reduced settlement risk, its coverage is not universal. CLS currently supports 18 currencies, and a significant share of global FX activity continues to settle outside PvP arrangements. Results from the [BIS Triennial Survey](#) show that just over US\$5 trillion, representing 36% of average daily FX settlement activity, was settled through PvP mechanisms that eliminate settlement risk. A further US\$7.6 trillion (54%) was settled through arrangements such as pre-settlement netting that reduce, but do not eliminate, settlement risk. Meanwhile, **more than US\$1.4 trillion (10%) of daily FX settlement continued to occur on a gross bilateral basis through traditional correspondent banking relationships, leaving participants fully exposed to settlement risk (Figure 3).**



Figure 3: FX settlement obligations by settlement type



Source : Deutsche Bank, BIS

Notably, around 25% of this volume (US\$1.4 trillion) was eligible for settlement through a PvP system but was not settled in this way. Potential reasons include operational challenges that result in trades missing PvP settlement cut-off times, difficulties in managing credit exposures to correspondent banks that support PvP-related payments, and the operational burden of meeting tight funding and payment deadlines (Figure 4).

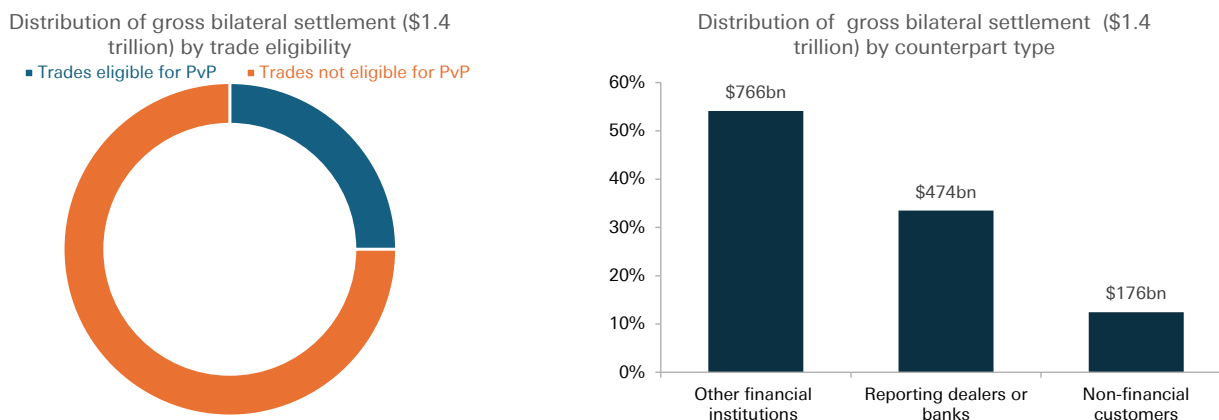
The remaining 75% of gross bilateral settlements were not eligible for PvP settlement systems, primarily because one or both counterparties lacked direct or indirect access to a PvP system. Other constraints included currency pairs that were not eligible for PvP settlement, often involving emerging market currencies, as well as trade types, such as certain same-day (T+0) transactions, that cannot currently be settled through existing PvP arrangements. As a result, much of the residual settlement risk reflects limitations in market access, currency coverage and infrastructure capabilities rather than the voluntary non-use of PvP systems.

Given the dominance of financial institutions in FX trading, gross bilateral settlements were also concentrated among financial counterparties. Of the US\$1.4 trillion settled gross bilaterally, approximately 54% involved other financial institutions, 33% involved reporting dealers, and 12% involved non-financial institutions (Figure 4).

Digging deeper into trades within each counterparty category, PvP system to mitigate risk is most common for transactions between reporting dealers or banks. Only **11% of inter-dealer trades** are settled on a gross basis and exposed to maximum risk. Consequently, FX trades most exposed to settlement risk are either between reporting dealer and other financial institutions or reporting dealers and non-financial institutions such as corporations. **Around 19% of FX trades** between reporting dealers and other financial institutions are settled on a gross basis. As non-bank financial institutions play an increasingly important role in cross-border securities investment and funding activities, these exposures remain a relevant source of risk outside the core PvP settlement infrastructure. Similarly, **22% of FX trades** between reporting dealers and non-financial institutions also settled through gross bilateral settlement, exposing participants to the full principal amount at risk.



Figure 4: Settlement risk emanating from trades not eligible for PvP. In absolute terms driven by other financial institutions.



Source : Deutsche Bank, BIS

These findings indicate that while the largest settlement risk exposures remain concentrated among financial institutions in absolute terms, the proportion of transactions exposed to risk is higher for non-bank financial institutions and corporates. Many of these residual risks stem from market access, currency coverage and operational constraints that fall outside existing PvP infrastructures. We therefore examine how these constraints create settlement frictions across selected corporate and financial-sector use cases, with a particular focus on institutional investors and cross-border securities transactions, reflecting the dominant role of financial institutions in wholesale FX markets.

FX settlement frictions: corporations versus institutional investors

The challenges associated with FX settlement are often most acute when transactions are cross-border, requiring the movement of currencies across different jurisdictions, payment systems, market infrastructures and operating hours. These challenges can arise whether FX settlement supports an underlying commercial payment or forms part of a broader securities transaction.

Corporations and FX

For **corporations**, FX spot settlement is typically a standalone transaction that supports cross-border trade, commercial payments and treasury activities. Cross-border payments and liquidity management are closely linked to the efficient functioning of the FX market. Corporates frequently need to convert currencies to support international trade, meet cross-border payment obligations and fund activities across multiple jurisdictions. More broadly, treasury functions rely on the seamless movement of currencies to manage cash positions, fund subsidiaries and address short-term liquidity needs.

Effective liquidity management facilitates domestic and cross-border payment obligations, supports the optimisation of cash flows and enables corporates to respond to changing funding requirements. These activities often involve access to wholesale FX markets to convert currencies and mobilise liquidity across jurisdictions. During periods of market volatility, the ability to execute FX transactions and move liquidity efficiently becomes particularly important, allowing institutions to respond swiftly to changing market conditions and maintain confidence in their operations.



FX settlement frictions: These include settlement risk where transactions cannot be settled on a PvP basis, delays arising from cut-off times and operating-hour constraints, liquidity and funding requirements associated with settlement, and the need to coordinate processes across multiple infrastructures and intermediaries. Such frictions can increase operational complexity and affect the timely movement of funds across borders.

Payment frictions: FX spot settlement represents only one component of a broader cross-border payment process. Even where banks settle eligible FX transactions through CLS on behalf of corporate clients, the associated commercial payments generally continue to rely on correspondent banking networks, SWIFT messaging and domestic payment systems. As a result, although CLS can mitigate settlement risk for eligible FX trades, many cross-border payment frictions remain, including delays, cut-off times and intermediary costs.

[US dollar payments](#), for example, are ultimately settled through infrastructures such as Fedwire or CHIPS. Since relatively few non-US banks have direct access to these systems, many rely on correspondent banking relationships to access dollar liquidity and payment services. Corporates, in turn, have no direct access and must depend on commercial banks to intermediate these transactions.

These challenges are distinct from wholesale FX settlement and relate primarily to the underlying payment process. For many corporations, the costs and operational frictions associated with cross-border payments may exceed those arising from the FX conversion itself.

Institutional investors and FX

For **institutional investors** buying or selling securities across jurisdictions, FX spot settlement is part of a broader cross-border securities transaction, where FX settlement, cash movements and securities settlement must be coordinated across multiple intermediaries and jurisdictions. When an investor purchases a foreign equity or bond, they must first obtain the currency required to complete the transaction. As a result, cross-border investments involve two interconnected settlement processes: FX settlement, where one currency is exchanged for another (PvP), and securities settlement, where cash is exchanged for the asset through a Delivery-versus-Payment (DvP) mechanism. Where currencies are not eligible for settlement through PvP infrastructures such as CLS, investors may rely on correspondent banking networks to settle the FX transaction. In these cases, the two currency legs are typically settled through separate payment systems, leaving participants exposed to residual settlement risk.

International investors do not typically interact directly with domestic market infrastructures such as DTCC in the United States. Instead, they access these markets through custodians, which act as intermediaries between investors and local settlement systems. Custodians are responsible for safekeeping assets, managing settlement instructions, coordinating cash movements and arranging funding across jurisdictions. When a foreign security is purchased, the custodian must ensure that the appropriate settlement currency is available before the transaction can be completed through the relevant central securities depository.

This creates a dependency between the FX funding trade and the securities settlement process, meaning that delays or frictions in obtaining the required currency can have direct implications for the timely settlement of the underlying security. The ongoing transition from T+2 to T+1 security settlement cycles, and the



potential move to T+0, increases these challenges by compressing the time available for investors to execute FX transactions, secure funding, and complete settlement processes across jurisdictions.

Key challenges for institutional investors under compressed timelines

The move to shorter securities settlement cycles has increased the operational challenge of coordinating processes. For example, under T+1, a UK investor purchasing a US equity at the market close (9pm London time) may have only around two hours before the CLS submission cut-off at 11pm to determine the final USD funding requirement, execute the associated GBP/USD transaction and submit it to CLS for settlement. Once the FX transaction settles and USD funds are made available to the custodian, the securities transaction can proceed through the relevant securities clearing and settlement infrastructure. Missing this window may require investors to pre-fund transactions, creating cash drag and unwanted currency exposure, or settle the FX transaction outside CLS, potentially increasing settlement risk and transaction costs.

Cross-border transactions involve a range of operational and settlement challenges, particularly under compressed timelines. Local currency cut-off times constrain the window available to execute and settle FX transactions, while settlement risk remains where trades cannot be processed through CLS and must settle bilaterally. Currency controls and market restrictions may further complicate settlement in certain emerging-market currencies and necessitate pre-funding. Shorter settlement cycles also leave less time for trade matching, confirmation, settlement instruction processing and the resolution of failures, increasing the importance of coordinating multiple post-trade processes.²

As FX settlement, cash movements and securities settlement occur across separate infrastructures and timelines, shorter settlement cycles increase the importance of post-trade coordination. Although the [Global Foreign Exchange Committee \(GFXC\)](#) described North America's transition to T+1 as operationally smooth from an FX perspective, this reflected extensive industry preparation. Similar challenges are likely to arise in Europe and the UK, where a more fragmented market structure may add complexity. As settlement windows compress, market participants have less time to execute FX transactions, source funding and complete settlement obligations, making cut-off times and cross-border coordination increasingly important.

Shorter securities settlement cycles may not always require instantaneous FX settlement. However, for CLS-eligible currency pairs, a move to T+1 compresses the time available to execute FX transactions and complete funding processes across institutions and jurisdictions. While shorter cycles can reduce counterparty exposure and costs, they also increase the need for automation and efficient post-trade coordination. A further move to T+0 [would amplify these challenges](#), sharpening the trade-off between risk reduction and operational efficiency.

Similar coordination [challenges](#) are emerging across securities, derivatives and collateral markets as settlement cycles shorten. [BIS \(2020\)](#) estimates that securities trade processing costs USD 17–24 billion annually, underscoring the broader cost of fragmented post-trade workflows. Technologies that improve automation, synchronisation and risk reduction in securities processing are also relevant for FX.

² See [GFMA's Global FX Division \(GFXD\)](#) for more details.



When settlement processes are fragmented across multiple institutions, jurisdictions and infrastructures, delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity and settlement risks. ***Against this backdrop, interest has increasingly turned to whether tokenised forms of money can improve the coordination and synchronisation of settlement processes across institutions and market infrastructures.***

Tokenised money on programmable infrastructure

Tokenised money on programmable infrastructure offers a potential approach to addressing some of the challenges outlined in the previous section. Tokenised money refers to digital representations of money that are recorded and transferable on shared or distributed ledger infrastructure. In many current implementations, its use is enabled by two key elements: distributed ledger technology (DLT) and programmability. DLT provides a shared infrastructure on which multiple institutions can record and observe ownership changes in a common ledger, potentially reducing the need for reconciliation across separate systems. ***Programmability refers to the ability to embed executable logic, such as conditions, rules and constraints, directly into instruments, or via smart contracts.*** Such functionality can range from basic automation to more sophisticated smart contract applications.

At the simplest level, programmability can automate settlement processes by ensuring that transactions are executed only when predefined conditions are met, such as Delivery-versus-Payment (DvP) in securities transactions and Payment-versus-Payment (PvP) in FX settlement. More advanced applications can support operational and control functions across financial market infrastructure, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. At a more sophisticated level, programmability can coordinate multiple parties and obligations within a single workflow, enabling applications such as trade finance arrangements, treasury and cash management, automated disbursements and other multi-party payment processes. These capabilities extend beyond process automation by embedding business, risk and compliance logic directly into financial transactions (CCAF and Fii, 2026). An emerging but largely underdeveloped area is the integration of tokenised money with artificial intelligence and autonomous systems, where programmable money interfaces could enable AI agents to initiate, manage and execute financial transactions within predefined controls.

The approach to settlement risk mitigation in a tokenised environment differs from the current CLS model, which achieves Payment-versus-Payment (PvP) settlement through a centrally coordinated process operating within defined funding and settlement windows. While CLS ensures that one currency is exchanged if and only if the corresponding currency is delivered through a centralised settlement mechanism, tokenised infrastructures could, in principle, embed PvP conditions directly into the settlement process through programmable rules or smart contracts on a shared ledger, enabling settlement to occur automatically when predefined conditions are met. However, tokenised settlement models need not rely solely on transaction-by-transaction atomic settlement. Some emerging platforms, such as [ClearToken](#), also incorporate queues, netting arrangements and liquidity-saving mechanisms.

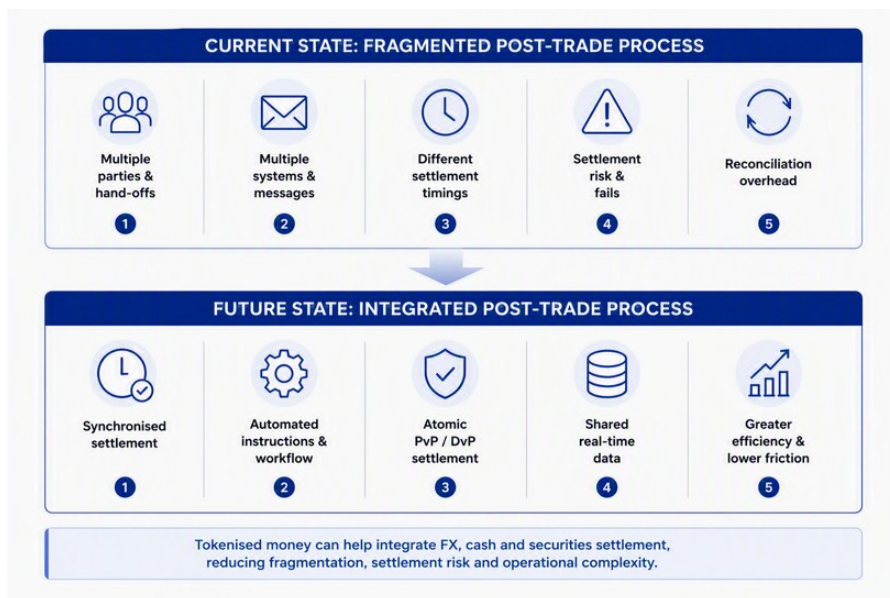


Tokenised post-trade environment for a cross-border securities trade

Many of the potential benefits of tokenised money become more apparent when viewed alongside the broader tokenisation of financial assets. In domestic securities, collateral and derivatives markets, these technologies are already being explored as a means of improving post-trade processes (BIS 2020). When tokenised money is combined with tokenised securities, it may support atomic settlement, greater automation and more efficient movement of cash and collateral, helping to reduce operational complexity, settlement risk and inefficiencies associated with the mobilisation, allocation and coordination of liquidity.

Figure 5 summarises the potential benefits of tokenised money and tokenised securities for post-trade processes. These technologies may support more synchronised settlement, reducing some of the operational, funding and coordination frictions associated with shorter settlement cycles. However, they do not eliminate the need for liquidity provision, risk management, netting or regulatory oversight. Ultimately, technology alone is unlikely to shorten settlement timelines without corresponding changes to market conventions, operating models and regulatory frameworks.

Figure 5: A more integrated post-trade process



Source : Deutsche Bank

The three routes for tokenised money

In practice, the use of tokenised money for financial institutions and corporations is emerging through three principal forms of tokenised money. The first is stablecoins, privately issued digital tokens typically backed by cash, bank deposits or high-quality liquid assets. The second is tokenised bank liabilities, including tokenised deposits, which represent commercial bank money in tokenised form and remain embedded within the existing banking system. The third is tokenised central bank money, which represent tokenised central bank reserves, also sometimes referred to as wholesale central bank digital currencies (wCBDCs).

While these models differ in their issuer, regulatory treatment and underlying infrastructure, they all seek to combine the trust associated with traditional forms

*Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure*

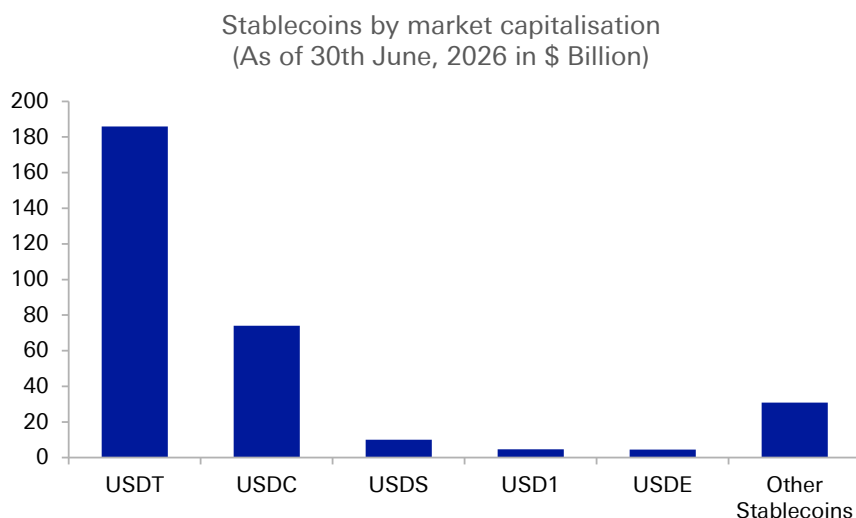


of money with the programmability, interoperability and automation enabled by modern digital technologies. Their potential role in FX settlement is therefore best assessed through the lens of how they address existing challenges.

Stablecoins

A stablecoin is a privately issued digital token designed to maintain a stable value relative to a reference currency, typically through backing by cash, bank deposits or high-quality liquid assets. While other types of stablecoins exist as suggested by Kosse et al. (2023) and BIS (2026), the focus of this study would be to discuss fiat-backed stablecoins. BIS (2026) reports that 99.4% of fiat-backed stablecoins (by market valuation) are pegged to the US dollar. Recorded and transferred on distributed ledger networks, stablecoins could potentially function as digital means of payment and settlement.

Figure 6: The market dominated by US dollar-denominated stablecoins



Source : Deutsche Bank, Coingecko

The market for stablecoins has grown rapidly, with more than US\$300 billion outstanding globally, dominated by US dollar-denominated stablecoins such as USDT and USDC (Figure 6).

From an FX settlement perspective, stablecoins offer an always-on settlement asset that can move across borders on a 24/7 basis without relying on traditional banking hours. They can also support tokenised settlement logic, including conditional payments and automated transaction processing.

Stablecoins' strongest use case today is cross-border payments, particularly in corridors where correspondent banking infrastructure is less efficient. Although total stablecoin transaction volumes were estimated at roughly [US\\$62 trillion in 2025](#), only around US\$350-550 billion is estimated to be associated with real-economy payments, highlighting that adoption in traditional payments remains at an early stage.

Currently, stablecoins are increasingly being used for remittances, business-to-business payments and access to US dollars in markets experiencing currency



instability or limited access to international banking services. Stablecoins are helping improve the movement of the bridge currency—typically the US dollar—rather than the settlement of both currency legs. As roughly 99% of stablecoins in circulation are denominated in US dollars, they function primarily as digital dollar infrastructure rather than a broadly adopted multi-currency settlement network. Consequently, ***most current stablecoin implementations focus on facilitating the transfer of a single currency, typically the US dollar, rather than solving the full two-currency FX settlement problem. This has also been referred to as the "stablecoin sandwich".***

Stablecoin adoption is increasingly moving beyond the lowest-cost blockchain towards a multi-network ecosystem spanning Ethereum, Solana, Polygon, BNB Chain and other networks. This reflects growing demand for compliance, programmability and institutional-grade infrastructure alongside low transaction costs. However, it also highlights an important limitation: stablecoins do not automatically solve interoperability. Instead, the challenge increasingly becomes enabling value to move seamlessly across multiple networks and settlement rails.

Stablecoin liquidity remains fragmented across issuers, networks and jurisdictions, while regulatory treatment continues to evolve across markets. In addition, institutional adoption requires custody solutions, compliance frameworks, treasury integration and industry standards comparable to those available for traditional bank money. As a result, while stablecoins can reduce some correspondent banking frictions, they do not by themselves provide a comprehensive solution to PvP settlement across non-CLS currencies.

Despite these complexities, stablecoins are beginning to attract interest from banks for institutional and corporate cross-border settlement. A notable example is [Nomura's partnership with Circle](#), which aims to explore the use of fiat-backed stablecoins such as USDC in Japan as part of next-generation market infrastructure. The collaboration is focused not only on cross-border payments, but also on FX transactions, treasury operations, fund transfers, on-chain collateral management and tokenised capital markets transactions. More [recently](#), a group of 21 financial institutions plan to issue dollar stablecoin in 2027.

[BNY Mellon](#) expanded its institutional stablecoin services in 2026 to support custody and minting of Circle's USDC, reflecting growing client demand for regulated stablecoin infrastructure rather than the issuance of a proprietary stablecoin. Société Générale's [Forge](#) subsidiary, on the other hand, issued the euro-denominated stablecoin EURCV and used it within tokenised capital market transactions.

Regulatory developments have also helped accelerate experimentation in this area. In Japan, amendments to the Payment Services Act established a regulatory framework for fiat-backed stablecoins. Under this framework, Circle's USDC became the first global US dollar stablecoin approved by Japan's Financial Services Agency for institutional use in Japan. Recent legislative developments in the United States, including the [GENIUS Act](#), have also provided greater clarity around payment stablecoins. Together, these developments have encouraged banks, payment providers and financial institutions to explore stablecoins as settlement assets within regulated financial markets.

These developments suggest that stablecoins are increasingly being viewed not only as digital payment instruments, but also as potential settlement assets for



cross-border transactions. A [recent study](#) notes that regulatory development has progressed furthest in the stablecoin sector, reflecting both growing adoption and policymakers' efforts to address the associated risks.

Key challenges

In some emerging market and developing economy (EMDE) currency corridors, USD stablecoins already act as intermediary settlement assets, performing a role similar to that of the US dollar in traditional correspondent banking arrangements by linking transactions between two non-USD currencies. While stablecoins may reduce frictions in cross-border value transfers, their current concentration in US dollar-denominated issuance limits their ability to support widespread Payment-versus-Payment (PvP) settlement across currency pairs.

While USD stablecoins may facilitate the payment leg of transactions involving tokenised US assets, investors must still exchange their domestic currency for US dollars and subsequently convert any proceeds back into their home currency. Unless digital settlement assets exist for both currencies and can be exchanged through a PvP mechanism, the underlying FX settlement risk remains. Consequently, current stablecoin models primarily improve the transfer of a single currency rather than addressing the settlement risks associated with the exchange of two currencies.

Figure 7: Stablecoins

Metric	Stablecoins
Market size	>US\$300bn outstanding
Dominant currency	~99% USD-denominated
Key benefit	24/7 cross-border transfers
Main use case	Cross-border payments & remittances
FX settlement role	Improves movement of USD as a bridge currency, rather than full two-currency PvP settlement
Key challenge	Fragmented liquidity and interoperability
Institutional adoption	Growing bank interest (e.g. Circle, BNY Mellon, Nomura)

Source : Deutsche Bank

Tokenised commercial bank money

A **tokenised deposit** is not a new form of money, but a commercial bank deposit represented or issued in tokenised form on a shared or tokenised ledger. Economically, it remains a claim on a commercial bank, just as a conventional bank deposit does today. The key difference is that the deposit, or rights linked to that deposit, can be transferred and settled using tokenised infrastructure, smart contracts and shared ledgers.

Classifying tokenised commercial bank money

Several public-sector and industry initiatives have proposed frameworks for



classifying tokenised commercial bank money. The [Bank of England](#) defines tokenised deposits as deposit claims represented on programmable platforms and distinguishes between non-transferable and transferable forms. This distinction is important because it determines whether tokenised deposits can circulate directly across institutions or continue to require interbank settlement in central bank money.

[Project Guardian](#) adopts the broader term tokenised bank liabilities and argues that these instruments could help address existing frictions in cross-border payments and FX settlement, including pre-funding requirements, settlement risk and fragmented post-trade processes. It classifies tokenised bank liabilities according to how the underlying claim is represented, distinguishing between digital twin and digital native models, and according to how they are deployed, ranging from single-bank ledgers to interoperable multi-bank and network-based arrangements.

Industry participants have developed related taxonomies, distinguishing between account-based and token-based representations of commercial bank money, as well as native and non-native ledger structures. Although these approaches differ in terminology, they broadly reflect alternative methods of representing, recording and transferring commercial bank money on distributed ledger infrastructure. Further details are provided in Appendix A.

Current implementation models and infrastructure

Current models are developing along two main lines. The first is **single-bank or single-bank-group infrastructure**, where blockchain-based deposit accounts or tokenised commercial bank money are used between customers of the same institution or banking group. J.P. Morgan's Kinexys Digital Payments is an example of an account-based blockchain deposit model rather than a deposit-token model. It operates on a single-bank ledger, enabling participating J.P. Morgan clients to transfer deposit balances across J.P. Morgan's blockchain-based payments rail. The BMW transaction illustrates this model: BMW executed a programmable on-chain EUR/USD FX transaction using [Kinexys Digital Payments](#), involving automated balance checks, conditional auto-deposits, near-real-time FX execution and transfers between BMW's Blockchain Deposit Accounts in Frankfurt and New York.

The second, more complex model is the **interbank model**, where tokenised bank liabilities issued by different banks, in different currencies and potentially on different ledgers, need to be exchanged or coordinated. [Project Guardian](#) explores several approaches to these interoperability challenges. One model uses a liquidity provider to facilitate the exchange of tokenised deposits in different currencies, with off-chain FX prices fed into the shared ledger through an FX quote oracle. Another model, demonstrated by BNY and OCBC, uses Hash-Time Locked Contracts to coordinate transfers across two private permissioned bank ledgers. A third model, involving HSBC and OSTTRA, focuses on settlement interoperability, using a shared-ledger PvP orchestration layer to match, confirm, fund and settle FX cash flows, helping to reduce Herstatt risk and improve visibility across the settlement.

FX quote oracle: A service that securely uploads off-chain FX prices and token exchange rates to a shared ledger, allowing liquidity providers to offer exchange rates and facilitate cross-currency exchanges between tokenised deposits issued in different currencies.

Bank-led shared-ledger settlement platforms are also emerging as infrastructure that can support tokenised commercial bank money. [Partior](#) is an example of a consortium-based model: a controlled environment in which a defined group of institutions jointly governs and participates in shared infrastructure. It operates as a bank-led shared-ledger settlement platform for real-time cross-border payments



and FX settlement, using a private permissioned ledger where access is restricted to authorised financial institutions and governed participants. Partior's relevance for FX further lies in its ability to support atomic Payment-versus-Payment, or PvP, settlement, where the two currency legs of an FX transaction are coordinated so that settlement of one leg occurs only if the other leg also settle.

Recent initiatives led by [The Clearing House \(TCH\)](#) seek to provide a common clearing and settlement framework for tokenised deposits, addressing one of the key challenges facing tokenised commercial bank money: interoperability across banks. The initiative aims to connect on-chain activity with existing payment rails, enabling the clearing and settlement of tokenised commercial bank money at scale while combining the regulatory, operational and settlement frameworks of established payment infrastructure with the programmability and interoperability of blockchain-based systems.

Market adoption and FX settlement relevance

Commercial bank deposit claims, including tokenised deposits, have progressed beyond early pilot stages and are seeing notable institutional adoption. While data on outstanding tokenised deposit balances remains limited, reported transaction volumes are significant. [J.P. Morgan's Kinexys platform](#), for example, has processed more than US\$4 trillion since inception and averages more than US\$7 billion daily in tokenised deposit movements for institutional payments. Other initiatives include DBS Token Services, Citi Token Services, the USDF Consortium, TassatPay, and projects from UOB and Standard Chartered.

The scale opportunity is potentially significant. Tokenised bank deposits and similar bank-token models could support around [US\\$100–140 trillion](#) in annual transaction flows by 2030, potentially rivalling or surpassing stablecoins in institutional transaction volumes. This reflects the view that corporates and financial institutions may prefer programmable forms of regulated bank money for treasury, liquidity and FX settlement workflows where integration with existing banking infrastructure, compliance processes and balance-sheet treatment is important.

Overall, tokenised deposits appear more directly relevant to institutional FX than stablecoins because they preserve the existing bank-money model while adding programmability, potentially allowing new settlement capabilities to be integrated more easily into existing banking, liquidity and risk-management frameworks. Current implementation models appear more developed for transaction banking, treasury and corporate payment use cases than for the broader wholesale FX market, where achieving interoperability across banks, currencies and settlement networks remains a more complex challenge.

Key challenges

Tokenisation does not, at least in the early stages, fully resolve time-zone mismatches or the need for pre-funding as noted by Project Guardian. Hybrid models are likely to coexist with traditional RTGS systems, correspondent banking infrastructure and fiat-to-token conversion processes for some time, meaning liquidity may still need to be positioned across both traditional and tokenised systems.

A practical issue in cross-currency tokenised deposit models is how the FX rate itself is determined. Project Guardian notes that early-stage on-chain trading volumes may not be sufficient to generate reliable FX prices purely from on-chain



activity. As a result, tokenised deposit models may initially rely on off-chain FX prices sourced from bank liquidity desks, benchmark rates or market data providers, with those prices fed into the shared ledger through an oracle. Over time, if on-chain liquidity develops, more native on-chain FX price discovery may become possible, but this depends on market depth, data integrity and privacy-preserving mechanisms.

Tokenised deposits also do not automatically address all FX settlement challenges. This is particularly relevant in wholesale FX markets, where activity is concentrated among financial institutions. A key strength of existing settlement infrastructure is multilateral netting, which significantly reduces liquidity requirements. Replicating similar levels of liquidity efficiency in tokenised interbank networks remains challenging. Unlike CLS, emerging tokenised deposit arrangements do not yet support multilateral netting at scale, meaning participants may need to trade some liquidity efficiency for faster or more flexible settlement unless netting and liquidity-management mechanisms evolve alongside the technology.

Settlement finality presents a further consideration. Existing wholesale settlement infrastructure ultimately relies on central bank money, which remains an important foundation for wholesale FX settlement. As a result, alternative models that combine tokenised commercial bank money with tokenised central bank money may prove more relevant for wholesale FX applications.

More broadly, the success of tokenised deposits will depend on interoperability across banks and ledgers, sufficient liquidity in each tokenised currency, legal recognition of tokenised claims, clear settlement finality, robust governance frameworks, regulatory consistency, and integration with existing systems.

There may also be balance-sheet costs for issuing banks. If tokenised deposits are treated as more liquidity-sensitive than conventional deposits, banks may need to hold additional high-quality liquid assets against them, reducing their usefulness as a source of funding for lending or maturity transformation.

Figure 8: Tokenised deposits

Metric	Tokenised deposits
Market size	Early-stage but growing institutional adoption
Dominant currency	Multi-currency potential (dependent on issuing banks)
Key benefit	Programmable regulated bank money
Main use case	Institutional payments, treasury and liquidity management
FX settlement role	Supports on-ledger PvP settlement and programmable settlement workflows
Key challenge	Interoperability across banks, currencies and networks
Institutional adoption	Highest within single-bank platforms and consortium-based networks; broader cross-bank adoption remains evolving

Source : Deutsche Bank



Tokenised central bank money

Rather than tokenising private bank money or relying on privately issued settlement assets, an alternative approach is to make wholesale central bank money available on DLT-based infrastructure. In this context, wholesale central bank money used in DLT experiments is often referred to as wholesale central bank digital currency (wCBDC) or, more precisely, tokenised central bank reserves. The latter term is often preferred because it emphasises that the underlying asset, central bank reserves, already exists in digital form, with the innovation lying primarily in the technology used to represent and transfer it. While wCBDC is frequently used interchangeably with tokenised central bank reserves, its definition varies across institutions and may encompass a broader range of innovations in wholesale central bank money that are not necessarily tokenisation-based ([IMF, 2025](#)).

Central bank reserves are already held digitally by eligible financial institutions through RTGS systems such as T2 in Europe, Fedwire in the United States and CHAPS in the United Kingdom. From this perspective, tokenised central bank money initiatives represent not the introduction of a new form of public money, but the extension of existing central bank liabilities into tokenised and programmable environments.³

Although existing infrastructures already eliminate principal risk through DvP and Pvp mechanisms, tokenised reserves may support a more integrated settlement process by bringing cash and assets onto the same ledger. This allows transactions to be completed simultaneously and irrevocably using DLT and smart contracts, while enabling programmability.

A number of initiatives have examined the use of multilateral platforms that make wholesale central bank money available in tokenised form for cross-border settlement. Examples include [Project mBridge](#), [Project Jura](#) and [Project Agorá](#). Leveraging shared infrastructure and governance frameworks, these projects have demonstrated real-time and atomic Pvp settlement between commercial banks in different jurisdictions using tokenised reserves. Smart contracts can further automate and coordinate multicurrency transactions, improving operational efficiency.

Multi-currency model based only on tokenised central bank money

[Project mBridge](#) is an example of a pure multi-CBDC architecture. We use the term CBDC, as used by the project, which is synonymous with tokenised central bank reserves. Developed by the Hong Kong Monetary Authority, People's Bank of China, Bank of Thailand and Central Bank of the UAE, with Saudi Arabia joining later, the project was designed to enable direct cross-border payments and FX transactions using wholesale CBDCs issued by participating central banks on a shared distributed ledger platform. Following a successful pilot involving 20 banks and 164 real-value transactions, mBridge reached minimum viable product (MVP) status in 2024, indicating that the platform had progressed beyond proof-of-concept testing and was capable of supporting real-value cross-border settlement. Subsequent growth has been substantial, with the platform reportedly processing more than 4,000 cross-border transactions worth [approximately US\\$55 billion by late 2025](#). While the BIS transferred responsibility for the platform to participating

[Tokenised central bank reserves \(IMF, 2025\)](#): Wholesale central bank money, commonly known as reserves, refers to money issued by central banks to eligible financial institutions for interbank settlement and monetary policy implementation. With growing interest in tokenised financial assets, central banks are exploring ways to make reserves available on programmable ledgers. In this context, tokenised reserves are direct central bank liabilities issued using DLT and accessible only to a predefined group of financial institutions.

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central banks in late 2024, mBridge remains one of the most developed publicly disclosed multi-CBDC initiatives.

Beyond wholesale payments, experiments have also explored how tokenised central bank reserves could support more advanced settlement arrangements involving multiple currencies, assets or transaction legs. One example is delivery-versus-payment-versus-payment (DvPvP), where interconnected transactions are settled atomically. [Project Jura](#) illustrates these capabilities by enabling the direct transfer of euro and Swiss franc wholesale CBDCs (wCBDCs), representing tokenised central bank reserves, between French and Swiss commercial banks on a shared DLT platform operated by a third party. The project demonstrated how tokenised asset and foreign exchange transactions can be settled safely and efficiently using both payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms, highlighting the potential for programmable, multi-leg transactions across currencies, asset classes and jurisdictions.

Hybrid models

Alongside tokenised central bank money initiatives, [Project Agorá](#) is exploring alternative architecture that retain central bank money as the settlement anchor while preserving a significant role for commercial bank money.

[Project Agorá](#), launched by the BIS in 2024 in collaboration with the central banks of France (representing the Eurosystem), Japan, South Korea, Mexico, Switzerland, the United Kingdom and the United States, explores a hybrid cross-border settlement model that combines tokenised commercial bank deposits with tokenised central bank reserves on a shared programmable platform. In 2026, the BIS reported that the prototype had successfully demonstrated atomic multi-currency settlement across participating jurisdictions using a unified ledger architecture linking tokenised deposits and tokenised central bank reserves. The project also highlighted potential improvements in speed, transparency and operational efficiency while preserving the existing two-tier monetary system.

Key challenges

Tokenised central bank reserve models, particularly multi-CBDC wholesale arrangements, are among the most challenging forms of digital money to implement. Unlike stablecoins or tokenised deposits, they require extensive coordination across central banks, legal jurisdictions and financial market infrastructures. Cross-border deployment raises complex questions around governance, access, interoperability, regulatory alignment, and the legal recognition of settlement finality. As a result, industry attention has increasingly broadened beyond pure wCBDC architectures towards hybrid models that combine tokenised central bank money with tokenised commercial bank money. By building on the existing two-tier monetary system, these models may face fewer implementation barriers and are often viewed as more compatible with current market structures.

[Recent industry commentary](#) suggests that digital settlement infrastructure may develop along regional or architectural lines rather than converge on a single global model. The emergence of mBridge as a multi-CBDC platform centred on Asia and the Middle East, alongside Agorá's hybrid model involving major advanced-economy central banks and commercial banks, highlights the possibility that multiple settlement ecosystems could emerge over time. Whether these systems ultimately converge through interoperability standards or develop as parallel networks remains uncertain.



Figure 9: Tokenised central bank reserves

Metric	Tokenised central bank reserves
Market size	Limited live volumes; activity remains concentrated in pilots, proofs of concept and early production deployments
Dominant currency	Typically designed as multi-currency infrastructures involving multiple central bank jurisdictions
Key benefit	Provides settlement in central bank money with potential for atomic PvP settlement and programmability
Main use case	Wholesale financial market settlement, cross-border payments and settlement of tokenised assets
FX settlement role	Potential common settlement asset for cross-currency transactions, reducing settlement risk and enabling programmable FX settlement workflows
Key challenge	Cross-border coordination among central banks, legal and regulatory harmonisation, governance, access and interoperability.
Institutional adoption	Active experimentation through initiatives such as mBridge and Jura, as well as hybrid models such as Agorá, although large-scale production adoption remains limited.

Source : Deutsche Bank

Other settlement solutions and related infrastructure

Going beyond the three forms of tokenised money, other settlement solutions and supporting infrastructures are evolving in parallel. One example is [Finality](#), a privately operated settlement system backed 1:1 by central bank reserves. It provides a central-bank-money-backed settlement asset for wholesale transactions. The [Sterling Finality Payment System](#) went live in 2023 and received Bank of England settlement-finality designation in 2024, with the platform targeting use cases including PvP FX settlement and tokenised asset settlement. It uses the [Bank of England's omnibus account solution](#) to hold participant funds and issues DLT-based settlement tokens representing claims on this pool of funds.

Omnibus account arrangements allow for settlement using tokens on the DLT platform outside of RTGS operating hours and with added functionalities, such as programmability, provided that the FIs have prefunded their omnibus accounts (Bank of England 2021)

Rather than tokenising central bank money directly, the [Bank of England's synchronisation model](#) enables funds held in RTGS to be synchronised with transactions taking place on external ledgers through a synchronisation operator. Settlement in central bank money occurs if and only if the corresponding transaction on the external platform is completed. Through initiatives such as Project Meridian FX and the Synchronisation Lab, the Bank has explored how this approach could support Payment-versus-Payment (PvP) settlement for cross-border FX and Delivery-versus-Payment (DvP) settlement for tokenised securities, while preserving settlement in central bank money. The Bank has also highlighted that extended RTGS operating hours may be an important enabler of such synchronised settlement models.

Industry initiatives such as the Regulated Settlement Network (RSN), which builds on earlier Regulated Liability Network (RLN) initiatives, are exploring a shared ledger environment in which regulated liabilities, such as commercial bank deposits and central bank money, and tokenised assets can settle in real time across multiple entities. RSN involves a consortium of institutions, including Citi, the New York Fed Innovation Center and Swift, and creates a shared ledger environment in which these assets can settle in real time across multiple entities. A [December 2024](#) proof of concept demonstrated 24/7 settlement capabilities in a controlled environment, although questions remain around broader ecosystem integration



and cross-border scalability.

[SWIFT's blockchain ledger](#) is another example that emphasis on connectivity. Rather than introducing a new settlement asset, it enables participating banks to coordinate transfers of tokenised deposits across their own ledgers while continuing to settle through existing payment infrastructure. The initiative is intended to support 24/7 cross-border payments and improve liquidity efficiency.

These initiatives highlight the importance of infrastructure that enables different forms of money, payment systems and settlement architectures to operate together. If FX settlement evolves through multiple architectures, synchronisation and interoperability may become as important as the characteristics of the settlement asset itself.

Technology alone is not enough: risk reduction or transformation?

Tokenised money is often discussed in terms of faster settlement, reduced operational complexity and atomic execution. While these technologies may help reduce certain forms of settlement risk, particularly where payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms can be embedded into the settlement process, they do not eliminate the need for liquidity provision, risk management, netting arrangements or regulatory oversight. In many cases, the effect may be better described as risk transformation rather than risk elimination. Settlement risk may be reduced, but liquidity, operational, legal and governance considerations remain.

Regulation: Regulatory frameworks for tokenised deposits, stablecoins and digital settlement assets continue to evolve across jurisdictions. While progress has been made, important questions remain around legal treatment, settlement finality, prudential requirements and the use of tokenised money in cross-border transactions.

Liquidity: Tokenised money does not create liquidity; rather, it may improve how liquidity is mobilised and deployed. Potential reductions in pre-funding requirements and trapped liquidity depend on sufficient market participation, transaction volume and interoperability across networks. Today, many initiatives remain fragmented across separate platforms and participant groups, limiting the extent to which liquidity benefits can be realised. Moreover, unlike CLS, most tokenised settlement networks **do not yet provide multilateral netting at scale**, meaning some participants could **face higher gross funding requirements** even as settlement processes become more automated. This is important because multilateral netting and the resulting liquidity savings have been identified as critical features of wholesale FX payment and settlement systems, including in discussions at the [Jackson Hole Economic Policy Symposium](#).

Interoperability and integration: Many of the potential benefits depend on the ability of different forms of money, payment systems and settlement infrastructures to operate together. Successful adoption will require integration with existing banking, payment, custody, accounting and regulatory reporting processes. Programmable and traditional infrastructures are likely to coexist, creating additional operational and implementation challenges.



Market infrastructure: The benefits of programmable settlement are greatest when cash, securities and collateral can move on the same, or fully interoperable, infrastructure. While technological capability is advancing rapidly, broader adoption depends on custodians, central securities depositories, asset managers, banks and market infrastructures adapting their operating models and processes.

Operational resilience and cybersecurity: Tokenised settlement infrastructures introduce new operational dependencies, including distributed ledger networks, smart contracts, wallets, bridges and external data providers. Coding errors, cyberattacks, failures in oracle services or vulnerabilities in cross-chain connectivity can disrupt settlement processes or lead to unintended outcomes. As networks become increasingly interconnected, disruptions may propagate more rapidly across participants and platforms.

Legal certainty and governance: The legal treatment of tokenised money continues to evolve. Questions remain around ownership rights, insolvency treatment, the enforceability of self-executing smart contracts, dispute resolution mechanisms and the alignment of technical and legal settlement finality. Governance arrangements, including liability allocation, system upgrades and contingency procedures, remain critical to ensuring trust and resilience.

Market concentration and fragmentation: Network effects may favour a small number of tokenised money issuers, settlement platforms or infrastructure providers, potentially creating new forms of concentration risk. At the same time, the proliferation of non-interoperable networks may fragment liquidity across multiple platforms and tokenised money instruments, reducing the fungibility and efficiency that underpin effective settlement.

Data, privacy and compliance: The transparency of shared ledgers can improve auditability and oversight but may also create challenges around confidentiality, commercial sensitivity and data protection. Compliance obligations, including AML/CFT controls, sanctions screening and cross-border regulatory requirements, remain significant considerations and may be difficult to implement consistently across multiple networks and jurisdictions.

Financial stability and monetary considerations: As tokenised money adoption grows, regulators have raised broader concerns around financial stability, including the implications of large-scale migration into digital settlement assets, the concentration of reserve assets, and changing funding dynamics within the financial system. In some jurisdictions, particularly emerging economies, the growing use of foreign-currency-denominated stablecoins has also raised concerns around monetary sovereignty and the effectiveness of domestic monetary policy transmission.

Conclusion

Existing infrastructure plays an important role in mitigating settlement risk in wholesale FX markets. CLS combines Payment-versus-Payment (PvP) settlement with multilateral netting and liquidity optimisation, generating liquidity savings of approximately 99% for participants. However, coverage is not universal, and a significant share of global FX activity continues to settle outside PvP arrangements due to constraints related to currency coverage, participant access, operating windows and market structure.



While tokenised money may help address some of these frictions, its potentially stronger use case for wholesale FX lies in the wider tokenisation of capital markets. Tokenised money could enable more integrated settlement processes for tokenised assets. Reflecting this potential, a [recent study](#) identified capital market tokenisation as one of the most important long-term use cases for tokenised money. Although activity remains small relative to overall market volumes, it has the potential to change how securities, bonds and other financial instruments are issued, traded and settled. The vision is an integrated ecosystem in which tokenised money is used to settle tokenised securities and other assets, enabling atomic settlement that reduces counterparty risk and shortens settlement times from days to seconds.

The [Bank of England's](#) analysis highlights an important financial stability consideration. While tokenised deposits extend commercial bank money into programmable environments, central bank money remains the ultimate risk-free settlement asset and anchor of confidence in the financial system. Future tokenised markets may therefore require mechanisms such as RTGS synchronisation or hybrid models combining tokenised commercial bank money and central bank money to preserve the singleness of money and settlement finality. More broadly, the proliferation of multiple forms of tokenised money could fragment liquidity, reduce fungibility and create new market structure and financial stability challenges.

Over time, programmable settlement arrangements may replicate some functions currently performed by existing infrastructures, particularly if interoperable tokenised deposit or tokenised central bank money frameworks achieve sufficient scale. However, this will depend not only on technological capability but also on the development of comparable liquidity networks, legal frameworks, governance arrangements and interoperability with existing payment and settlement systems. The transition towards programmable settlement is therefore likely to involve both risk reduction and risk transformation.



Appendix A: Detailed classification of tokenised commercial bank money

Several public-sector and industry initiatives have proposed different frameworks for classifying tokenised commercial bank money. The [Bank of England](#) describes tokenised deposits as “deposit claims represented on programmable platforms” that enable new techniques for payments and settlement. It distinguishes between non-transferable and transferable tokenised deposits. Non-transferable tokenised deposits resemble existing deposits and may still require settlement in central bank money when transferred between customers of different banks. Transferable tokenised deposits, by contrast, could be self-settling upon transfer, with the recipient becoming a customer of the issuing bank. This distinction matters for FX because it affects whether tokenised deposits can circulate across banks directly or whether interbank settlement in central bank money is still required to preserve singleness and finality.

[Project Guardian](#) uses the broader term tokenised bank liabilities to refer to tokenised representations of commercial bank money on a shared ledger, without asserting any specific legal or regulatory interpretation. Tokenised deposits are therefore a specific form of tokenised bank liability. It further classifies tokenised bank liabilities along two dimensions: token level and ledger level. At the token level, it distinguishes between digital twin tokens and digital native tokens. Digital twin tokens represent ownership interests in underlying traditional bank liabilities, with ownership rights represented on-chain through digital tokens. Digital native tokens are issued and custodied natively on a shared ledger, which serves as the primary record of ownership and rights. The distinction therefore relates to the legal and operational representation of the underlying bank liability rather than to transferability, as both forms can potentially support programmable payments and settlement.

At the ledger level, Project Guardian distinguishes between three deployment models. A **single-bank ledger** is a shared ledger operated within the same banking group, enabling transfers between customers, entities or branches connected to that bank's infrastructure. **Between-bank models** enable tokenised bank liabilities issued on separate banks' shared ledgers to be used for payments and settlement between customers of different banks. **Interlinking network models** connect distinct networks, each potentially having its own governance framework and design, allowing tokenised assets to move across a broader ecosystem of participating institutions and currencies.

Beyond the terminology used by Project Guardian, industry participants have developed more detailed taxonomies to describe how commercial bank money can be represented on distributed ledger infrastructure. A related [industry taxonomy](#) describes these instruments as blockchain-based deposits: distributed-ledger-based deposit claims issued by a licensed depository institution and recorded on blockchain. Within this category, they distinguish between account-based models, such as blockchain deposit accounts, and token-based models, such as deposit tokens. The taxonomy also distinguishes between native models, where the blockchain is the primary record of value, and non-native models, where the blockchain mirrors an off-chain record.

Both account-based and token-based models are forms of commercial bank money, but they differ in how the deposit claim is represented and transferred on ledger. Blockchain deposit accounts represent deposit balances as account

Project Guardian also identifies several pain points in the current cross-border payment and FX settlement process, including time-zone frictions, pre-funding requirements, settlement risk outside CLS, fragmented infrastructure and the continued separation of messaging, reconciliation and settlement. It argues that tokenised bank liabilities could help address some of these frictions by enabling commercial bank money to move across shared infrastructure and, where supported by appropriate legal, operational and interoperability frameworks, by enabling atomic PvP-style settlement for a broader range of currencies and participants.



balances on a blockchain-based ledger. Deposit tokens are transferable tokens issued on a blockchain by a licensed bank, representing a claim on that bank for a stated amount of fiat money. J.P. Morgan's SGD deposit token, issued in connection with the MAS Project Guardian pilot, is one example of a deposit token. More recently, JPM Coin/JPMD is a bank-issued deposit token on Base, an Ethereum Layer 2 network, enabling approved institutional clients to move money, post collateral and settle transactions on public blockchain infrastructure while retaining the credit-risk profile and balance-sheet characteristics of commercial bank money.

Figure 10: Defining tokenised commercial bank money

Organisation / Framework	Term Used	Classification	Description
Bank of England	Tokenised deposits	Non-transferable	Tokenised deposit that still relies on interbank settlement.
		Transferable	Tokenised deposit that can circulate through transfer.
Project Guardian	Tokenised bank liabilities (Token level)	Digital twin token	On-chain representation of an existing bank liability.
		Digital native token	Bank liability issued and recorded natively on a shared ledger.
	Tokenised bank liabilities (Ledger level)	Single-bank ledger	Shared ledger operated within a banking group.
		Between-bank model	Tokenised bank liabilities exchanged across separate banks' ledgers.
CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation	Tokenised money: commercial bank deposits	Tokenised deposits	Represent an off-chain bank deposit recorded in traditional account systems, with the token serving only as a representation of that off chain position
		Deposit tokens	Deposit itself is issued solely on a distributed ledger constitute a DLT native form of the claim
Industry taxonomy	Blockchain-based deposits	Blockchain deposit accounts Deposit tokens	Account-based representation of deposits on blockchain infrastructure. Transferable token representing a deposit claim on a bank.
	Ledger architecture	Non-native model Native model	Blockchain mirrors an off-chain record. Blockchain is the primary record of ownership.

Source : Deutsche Bank



Appendix 1

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Deutsche Bank
Research

Private capital monitor: AI is the 2nd biggest risk

September 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

Deutsche Bank
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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



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Executive summary

- **Data centres, power, compute** and private investment-grade financing is further diversifying private capital from buyouts.
- But the risk of **AI disruption, overbuild, and capital misallocation** is a frequent risk discussed by large, listed private capital managers
- Our spotlight section includes an analysis of **AI-related overfinancing risk**
- Still, the **macro environment** is now the top point of concern amongst PC managers, even more so than the exit environment
- **Private capital growth remains strong but is becoming increasingly K-shaped.** Scale, investment performance, origination capabilities and product breadth are driving fundraising toward the largest platforms, while weaker managers and less differentiated products face pressure.
- **Fundraising, AUM and fee-related earnings generally reached record or near-record levels**, despite weak sponsor M&A, uneven exits and disruption in parts of private wealth and software.
- There has also been a **drop in the number of articles on private credit in the media**, especially when it comes to “crisis”-mentioning variety. With public BDCs being at the centre of this year’s rout, the drop in such trackable indicators likely bodes well for sentiment and thus flows.
- The overall tone of Q2 results was **strongly positive and confident, especially on fundraising and structural growth**. However, management teams were more cautious on the timing of monetisations, private-wealth recovery, AI capacity risks and software disruption.



01. Q2 2026 results

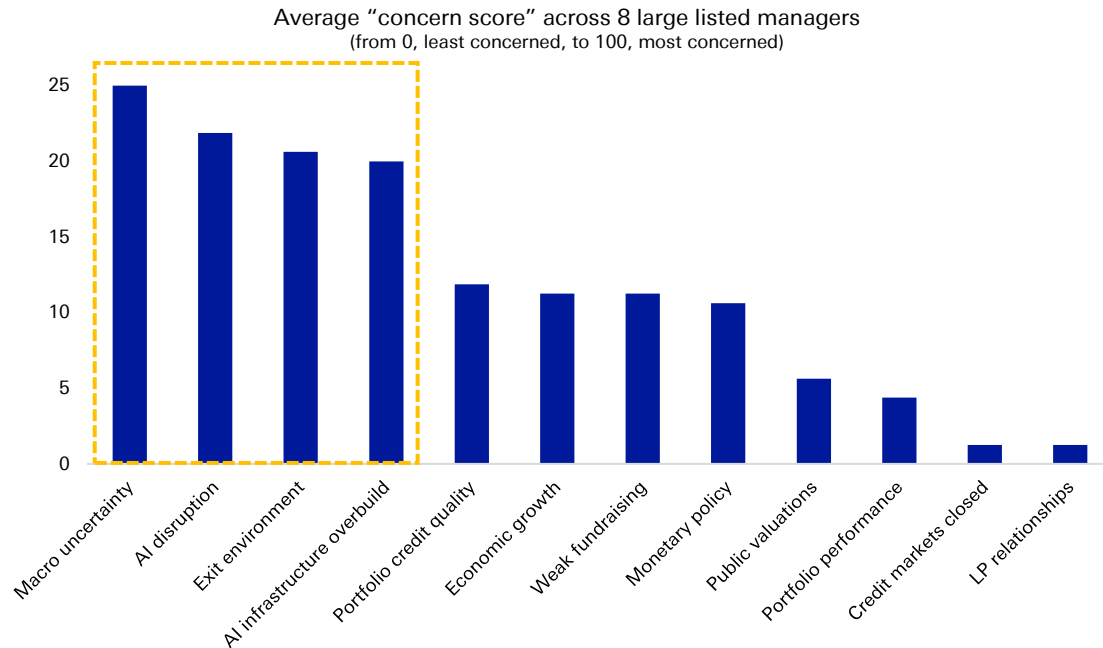
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Q2 earnings calls from the large, listed PC managers

Summary



- **Private capital growth remains strong but is becoming increasingly K-shaped.** Scale, investment performance, origination capabilities and product breadth are driving fundraising toward the largest platforms, while weaker managers and less differentiated products face pressure.
- **Fundraising, AUM and fee-related earnings generally reached record or near-record levels**, despite weak sponsor M&A, uneven exits and disruption in parts of private wealth and software.
- **The overall tone was strongly positive and confident, especially on fundraising and structural growth.** However, management teams were more cautious on the timing of monetisations, private-wealth recovery, AI capacity risks and software disruption.
- **AI is creating a need** for data centres, power, compute and private investment-grade financing that further diversifies private capital from buyouts.
- Amongst the concerns we track, managers were relatively **more concerned about the macro environment and AI effects**, including the risk of overbuild in related infrastructure.

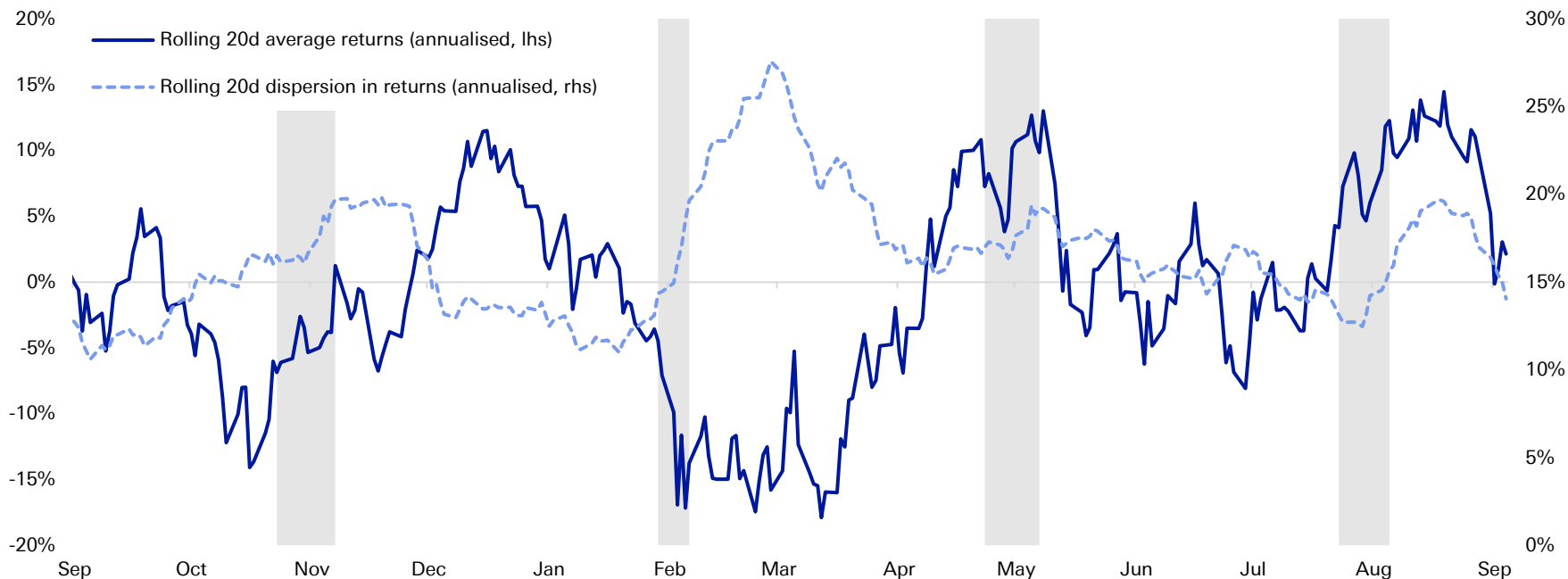


Source: Bloomberg Finance LP, Microsoft Copilot, Deutsche Bank.

Good results and some pullback around AI and credit fears catalysed a strong rebound in shares



8 US-listed private capital managers, rolling performance (earnings season shaded)



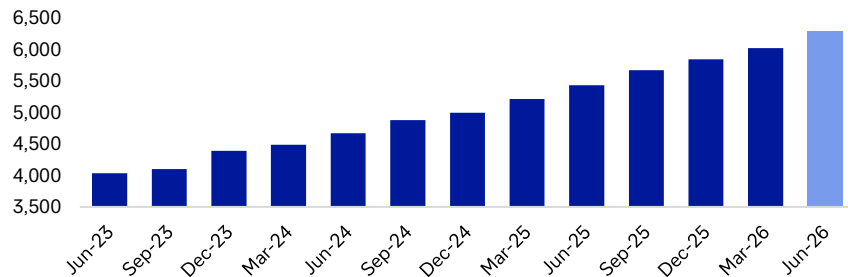
Source: Bloomberg Finance LP, Deutsche Bank.

Key metrics for selected large private capital managers

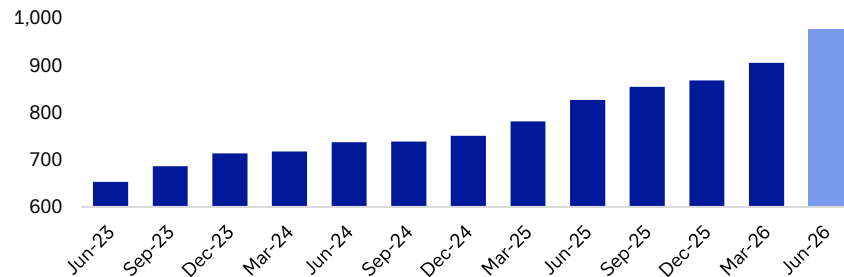
Strong earnings and a notable uptick in dry powder



Total AUM in \$bn



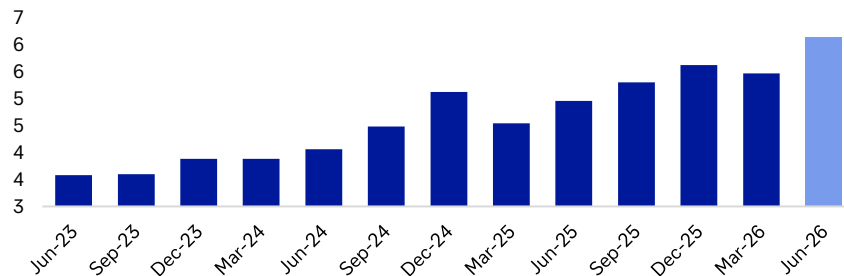
Dry powder in \$bn



Average fee-related earnings margin (%)



Fee-related earnings in \$bn



Source: Bloomberg Finance LP, Deutsche Bank.

Key themes from Q2 earnings calls

Large, listed managers talk AI, wealth, fundraising, exits and credit



1. **AI infrastructure** has become a firm-wide capital deployment and earnings theme, extending beyond data centres into power, compute, chips, connectivity and investment-grade financing. Power availability is emerging as a more important constraint than capital. Although concentrated exposure to AI infrastructure is a risk if demand expectations change, managers stressed contracted revenues, strong counterparties and hard-asset collateral as mitigants.
 - Blackstone's data-centre platform reached \$185bn of total value, including projects under construction, up from \$130bn at the start of 2026. Management said the platform could double over the next few years and that the market for long-term ownership of stabilised data centres could eventually reach \$1tn. The firm also said nine of its ten largest Q2 mark-ups were AI-related.
 - Apollo cited more than \$8tn of cumulative AI-infrastructure investment through the cycle and led a \$35bn Broadcom financing, described as the largest private-credit financing to date.
 - Brookfield estimated \$10tn of AI-related capex across energy, data centres, compute and adjacent infrastructure. The firm said AI-related deployment was probably running at approximately twice its normal 15%-20% business growth rate.
 - Blue Owl reported more than 140 data centres owned or under construction and 15.3GW of leased and owned capacity
2. **Private wealth** remains strategically important, but private credit experienced a confidence and liquidity test. Early Q3 data suggests stabilising fundraising. The better wealth outcomes increasingly came from private equity, infrastructure, real estate and multi-asset products rather than traditional private credit. Earnings calls suggest some international family offices and smaller institutions behaved more cyclically than core US individual investors.
3. **Record fundraising** despite a "noisy" backdrop. Fundraising is concentrating with scaled, diversified platforms.
4. **Realisations** are recovering, but the market is highly selective (see pages 17-21 of this report)
5. **Credit is broadening beyond sponsor-backed direct lending** and real assets, infrastructure, defence and secondaries are benefiting from secular demand
6. **Earnings are shifting toward recurring fees, transaction revenues and fee-related performance revenues.** Firms are becoming more effective at capturing economics from origination, financing and perpetual vehicles amidst a strategic shift from simply managing third-party capital toward monetising the broader platform through origination, syndication, financing and advisory services.

Source: Bloomberg Finance LP, Deutsche Bank.

Key risks from Q2 earnings calls

Aside from AI, concerns are rising about rates and credit



1. **AI overbuild, capital misallocation and execution amongst most consistently voiced risks**
 - Brookfield said concerns about excess infrastructure capacity were reasonable and that, as in railways and fibre, some capital would inevitably be poorly allocated.
 - Blackstone acknowledged the potential for excessive exuberance, despite its conviction that power, chips and entitlements constrain supply.
 - Both firms also identified scarce inputs and complex development requirements as execution bottlenecks.
2. **AI disruption to software and white-collar services**
 - Blackstone said software, professional services and information services were experiencing weaker buyer interest despite many companies continuing to perform operationally.
 - TPG reported mid-teens software bookings growth and said over 75% of its software exposure was well positioned to benefit from AI.
 - Blue Owl said it had gained increasing comfort around software transition risk but continued to recognise it as a valid issue
3. **Wealth-channel redemptions.** Management frames this as narrative-driven, but a slower-than-expected sentiment recovery is a risk to retail flows. Managers emphasised that requests were concentrated among a minority of investors concentrated in non-US family offices and smaller institutions and that underlying portfolio performance remained sound.
 - The recovery appears to be under way but remains gradual. Blue Owl explicitly rejected a V-shaped recovery assumption, while Ares estimated that its BDC could return to flow stasis over the next two to three quarters if current trends continued.
4. **Realisation timing and reliance on capital markets.** Managers stressed that timing is difficult to predict.
 - TPG expects transaction and monitoring fees to step down in Q3 after pulling forward some transactions into Q2. Carlyle similarly said Q2 capital-markets activity was unlikely to repeat in Q3 or Q4. Blackstone expects lower Q3 realisations before a stronger Q4 and 2027.
5. **Credit-cycle normalisation and growing dispersion remain risks, although managers generally reported sound portfolio fundamentals in Q2.**
6. **Insurance competition, spread compression and regulation**
 - KKR said competition had increased and it was allocating somewhat less capital to insurance at current return levels. Management described sector ROEs as structurally low in the current competitive environment. Apollo cited 36 asset-manager entrants into retail annuities and argued that some competitors lack mature origination, liability-generation and operating capabilities. Apollo also highlighted offshore regulatory arbitrage and pushed for “equal capital for equal risk”, warning that weaker regulatory structures could undermine industry trust.
7. **Rates, inflation, energy and geopolitics.** The macro backdrop remained a recurring source of uncertainty:
 - Carlyle cited war in the Middle East, pressure on energy markets, stubborn inflation and AI-related public-market volatility.
 - TPG said geopolitical uncertainty and changing rate expectations had reduced private-equity activity.
 - Blackstone said higher rates were delaying the real-estate recovery and had pushed some exits out.

Source: Bloomberg Finance LP, Deutsche Bank.

Q2 earnings: performance by asset class



KKR	Q2	LTM
Private equity		
Traditional PE portfolio	4%	9%
Real assets		
Opportunistic real estate	-1%	1%
Infrastructure	1%	8%
Credit		
Leveraged credit composite	2%	5%
Alternative credit composite	1%	5%
Apollo	Q2	LTM
Direct origination	2.6%	7.9%
Opportunistic credit	1.8%	10.7%
Multi-credit	1.9%	7.4%
Asset-backed finance	2.4%	7.2%
Flagship PE	3.0%	7.2%
Carlyle	Q2	LTM
Global private equity		
Corporate PE	2%	4%
Real estate	0%	2%
Infrastructure & natural resources	6%	28%
Global credit	4%	16%

Ares	Q2	LTM
Credit		
Alternative credit	4.1%	16.4%
Opportunistic credit	2.4%	8.9%
US senior direct lending	2.5%	11.2%
US junior direct lending	0.2%	8.9%
European direct lending	1.5%	8.3%
APAC credit	2.4%	19.0%
Corporate private equity	2.4%	1.6%
Real assets		
Americas real estate equity	0.6%	3.1%
European real estate equity	0.2%	2.5%
Infrastructure equity	9.0%	9.8%
Infrastructure debt	1.7%	7.5%
Secondaries		
Private equity secondaries	-5.3%	-1.7%
Real estate secondaries	-1.2%	5.6%
Infrastructure secondaries	-0.4%	14.6%

Blackstone	Q2	LTM
Real estate		
Opportunistic	0.4%	-1.4%
Core+	0.2%	3.2%
Private equity		
Corporate private equity	3.7%	14.4%
Tactical opportunities	3.9%	9.9%
Secondaries	1.7%	8.1%
Infrastructure	7.2%	28.6%
Credit & insurance		
Private credit	1.0%	6.8%
Liquid credit	2.2%	3.9%
TPG	Q2	YTD
Capital	5.7%	3.1%
Growth	6.2%	4.5%
Impact	7.2%	7.2%
Credit	3.3%	5.6%
Real Estate	2.5%	4.4%
Blue Owl	Q2	LTM
Direct lending	2.6%	8.3%
Alternative credit	2.4%	11.4%
Net lease	3.1%	13.6%
Real estate credit	1.9%	8.1%
Digital infrastructure	3.1%	6.0%

Source: Company reports, Bloomberg Finance LP, Deutsche Bank. Note: Performance reflects gross returns or appreciation depending on the manager, strategy and time period.

Management comments – a few records in Q2 and a confident tone



- **Stephen Schwarzman of Blackstone:** “Blackstone delivered an outstanding second quarter, highlighted by strong growth in earnings and nearly \$70 billion of inflows. Our decision to lean into the artificial intelligence megatrend is leading to standout investment performance across numerous strategies and creating extraordinary opportunities for growth. We’ve become a trusted partner at scale to many of the key innovators in this ecosystem, which positions our firm extremely well for the future.”
- **Joseph Bae and Scott Nuttall of KKR:** “Our second quarter results reflect the strength and breadth of KKR’s global franchise. We reported record Fee Related Earnings, Total Operating Earnings and Adjusted Net Income per share on both a quarterly and trailing twelve-month basis. Our continued return of capital to clients led to our strongest monetization quarter ever and helped drive record new capital inflows over the past 12 months.”
- **Marc Rowan of Apollo:** “Our strong second quarter results reflect record earnings across Asset Management and Retirement Services, highlighting the quality and growing scale of our business. We are at the forefront of modernizing how private markets operate by enhancing transparency, improving liquidity, and broadening access. In a market evolving quickly with increasing demand for capital, the breadth of our origination capabilities combined with a principal mindset positions us to help shape what comes next.”
- **Harvey Schwartz of Carlyle:** “The second quarter was one of Carlyle’s strongest quarters in recent years, underscoring the power of our diversified platform. We delivered record Fee Related Earnings, our highest Distributable Earnings in nearly four years, alongside strong fundraising, and exceptional realization activity. Carlyle continues to distinguish itself as an industry leader and an outlier in returning capital to our clients, distributing nearly \$7 billion during the quarter and \$37 billion over the past year. ...”
- **Michael Arougheti of Ares:** “We generated strong second-quarter results, including another record quarter of fundraising with more than \$36 billion of inflows as our clients continue to reward us due to our strong and consistent fund performance across our strategies. Our diverse global origination platform enabled us to remain active investing in attractive opportunities across the platform in a slower transaction environment and we are now seeing a meaningful pickup in our firmwide investment pipeline.”
- **Jon Winkelried of TPG:** “TPG delivered record results during the first half of 2026, reflecting the strength of our diversified and growing franchise... As clients increasingly partner with TPG for access to our differentiated offerings across strategies, we continue to see significant opportunities to further expand our platform and build long-term value for our shareholders.”
- **Doug Ostrover and Marc Lipschultz of Blue Owl:** “Blue Owl’s ongoing growth in the second quarter reflects the strong investment performance we continue to generate across strategies and the diversification of our business across platforms and geographies... The results we reported this morning continue to demonstrate the resilience of our business and highlight our positioning as a key provider of scaled capital solutions in this evolving market landscape.”

Source: Company reports, Deutsche Bank.



02.

Spotlight section:

- Sentiment recovery
- AI overinvestment
- Realisations

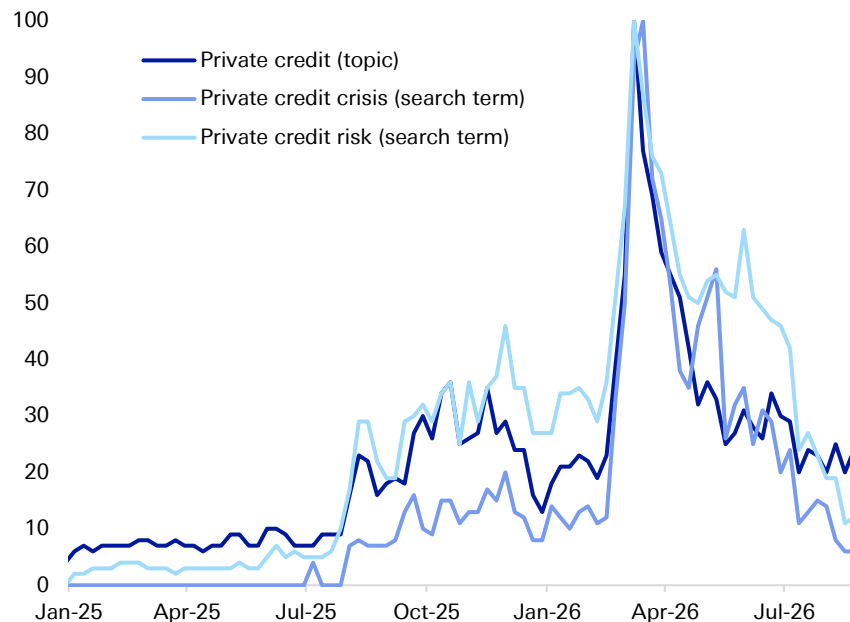
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Private credit fears 1/3

The pessimism is well past its peak now as various sentiment measures point to a drop in interest in the topic. As managers first noted during Q1 earnings, there seems to have been the most concern amongst Asian investors.

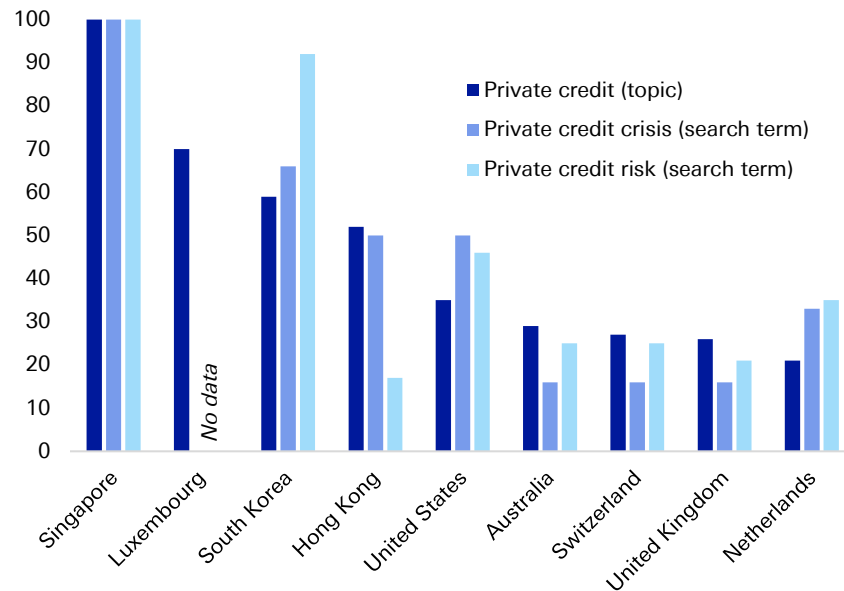


Private credit-related Google search trends (worldwide)
100 = the peak popularity for the term



Source: Google Trends, Deutsche Bank.

Popularity of private credit-related search terms
(Last 12m, 100 = the location with the most popularity as a fraction of total searches in that location)

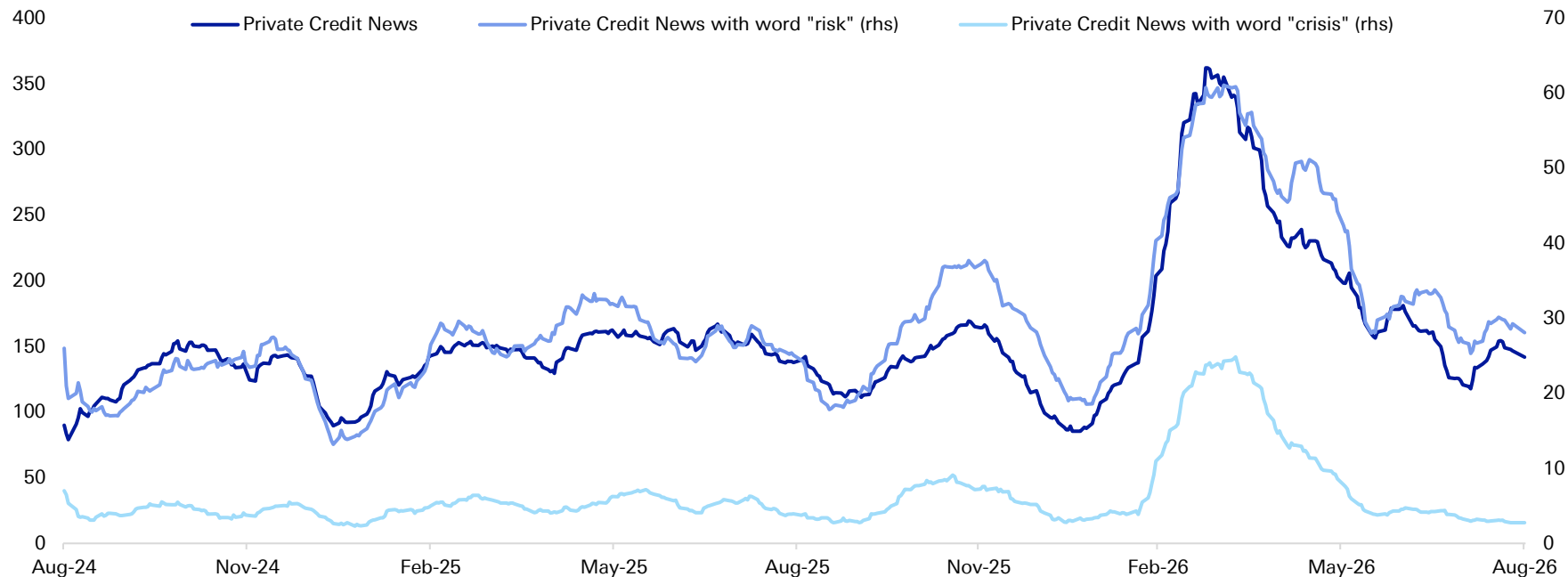


Private credit fears 2/3

There has also been a drop in the number of articles on private credit in the media, especially when it comes to “crisis”-mentioning variety. With public BDCs being at the centre of this year’s rout, the drop in such trackable indicators likely bodes well for sentiment and thus flows.



Story count of private credit news as tracked by Bloomberg, 30d average



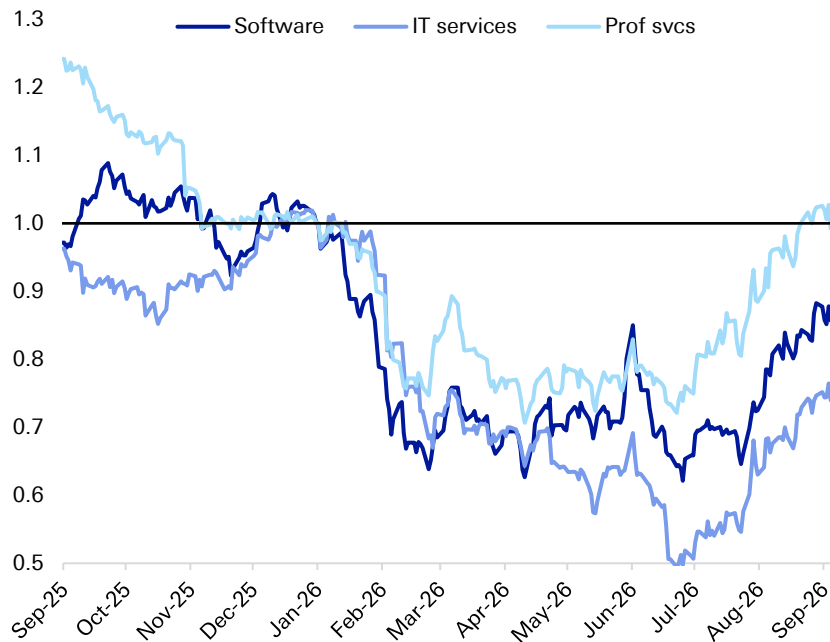
Source: Bloomberg Finance LP, Deutsche Bank.

Private credit fears 3/3

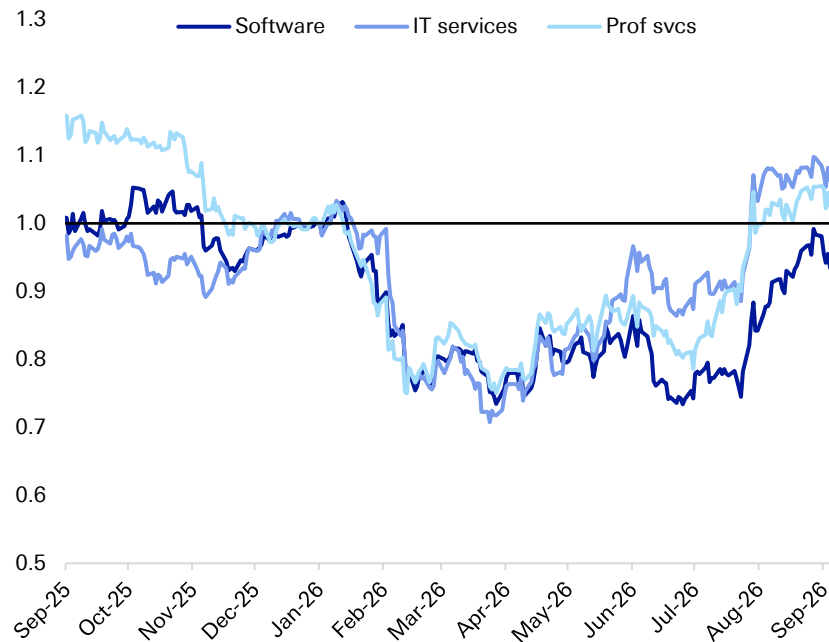
The relief to private capital also comes from receding AI disruption fears in software as firms reported solid earnings and executives continue to push back at the negative narratives around AI's impact on their businesses. Our baskets of stocks that were most affected by the original selloff earlier this year that we covered [here](#)



Selected US stocks, indexed to end of 2025



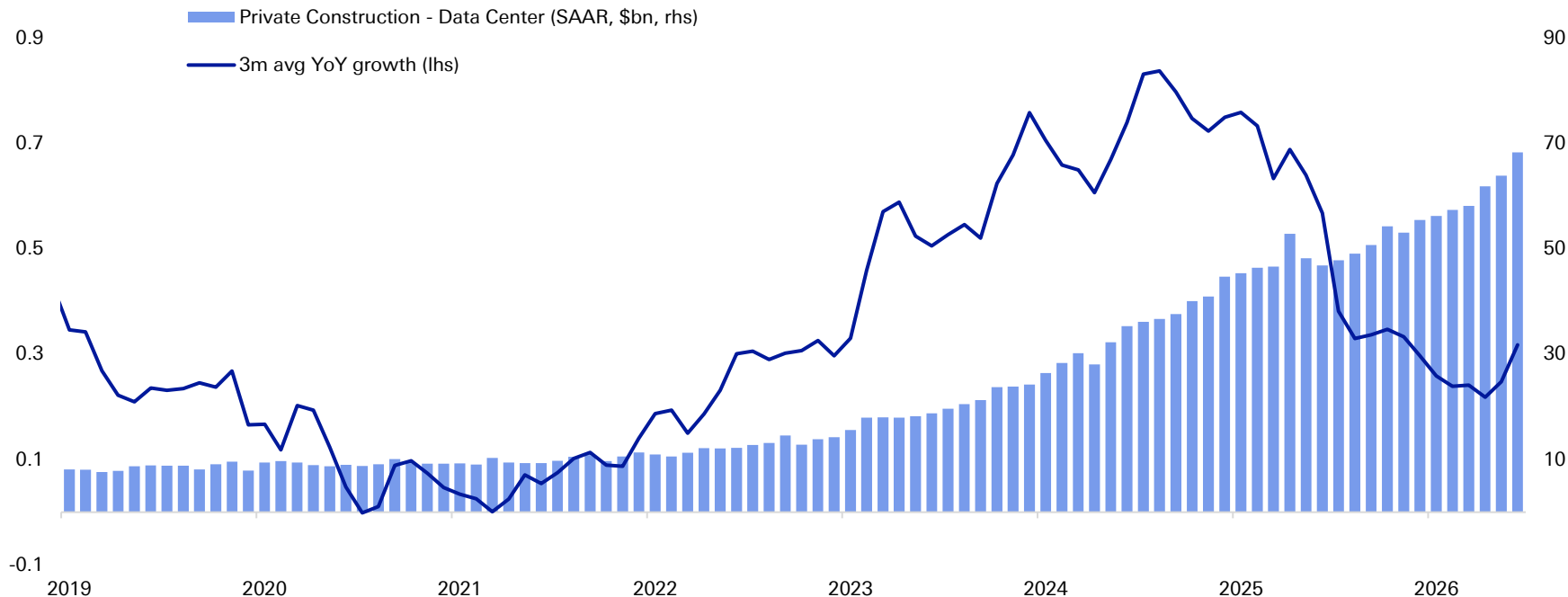
Selected European stocks, indexed to end of 2025



Source: Bloomberg Finance LP, Deutsche Bank.

AI-focused overinvestment 1/3

Amongst the most prominent risks for private capital as managers plough in money in data centres, large loans and other financing. US data centre construction growth has been increasing in recent months and now runs at annualised \$70bn per month.



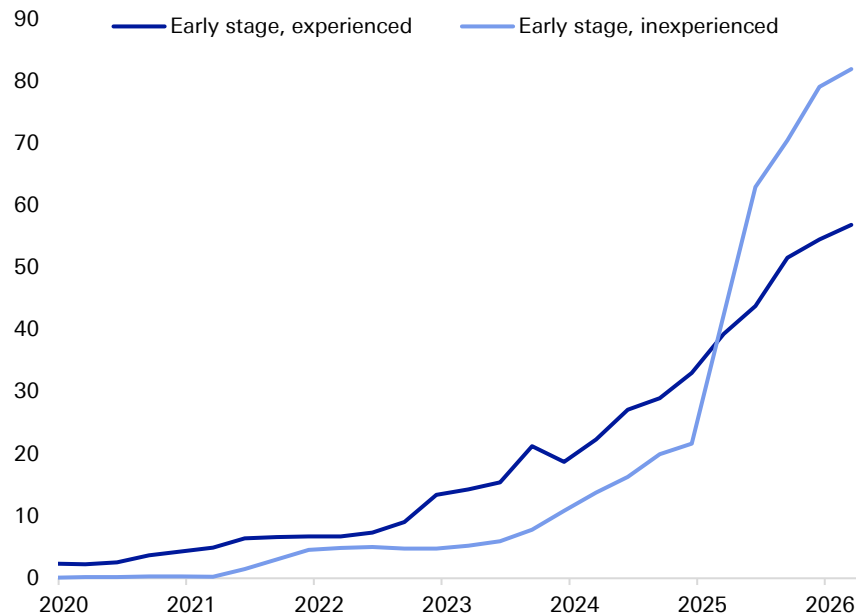
Source: Haver Analytics, Deutsche Bank.

AI-focused overinvestment 2/3

The number of entrants into AI infrastructure projects together with rising long-term yields augment concerns about the scale financing. Indeed, inexperienced developers now make up c 60% of the early-stage US data centre capacity.

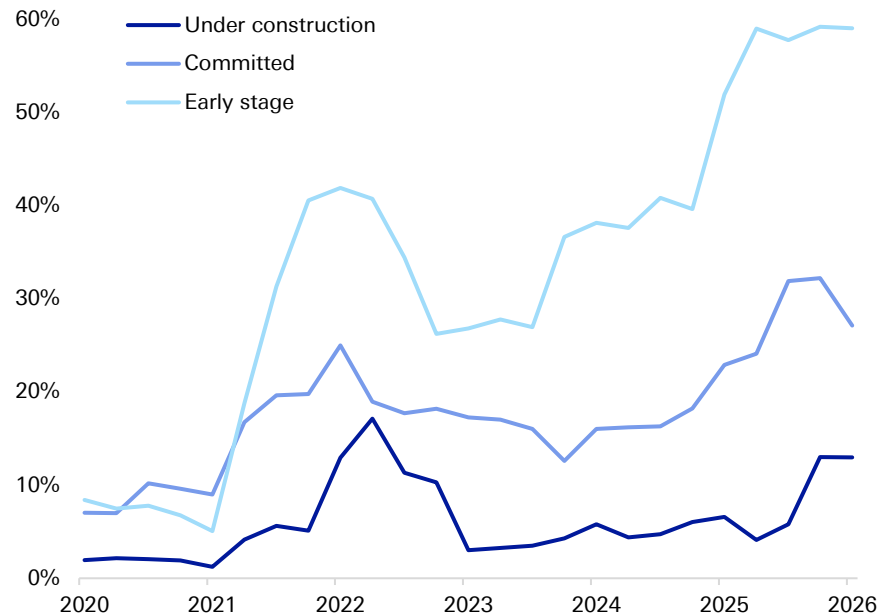


US data centre pipeline capacity, early stage by developer experience, GW



Source: Bloomberg NEF, Deutsche Bank.

US data centre pipeline capacity, share of inexperienced developers by stage

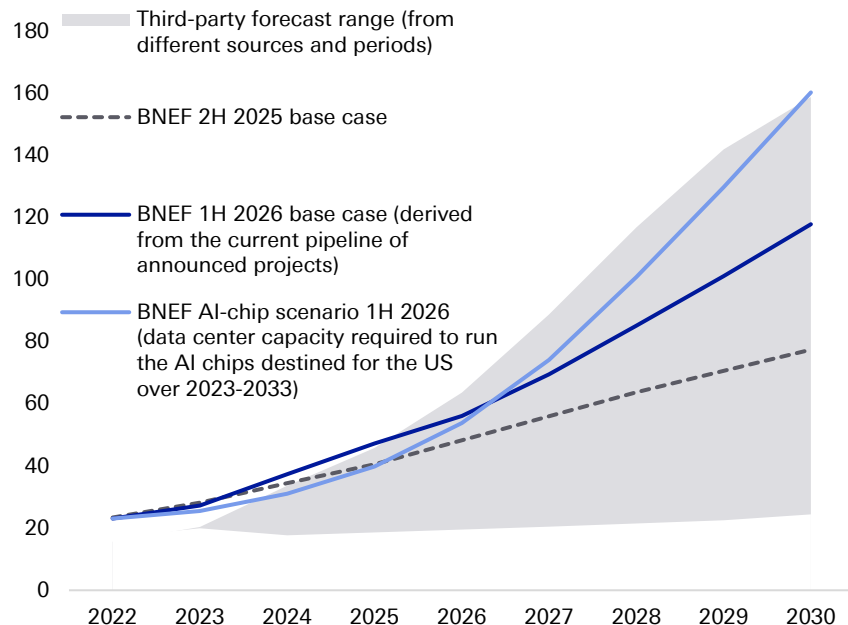


AI-focused overinvestment 3/3

The overfinancing risk is augmented by forecast uncertainty around rapid AI buildout growth, with some estimates pointing to installed capacity doubling in 3 years. Room to expand AI business adoption is a tailwind, but other factors including model advancements are harder to predict.

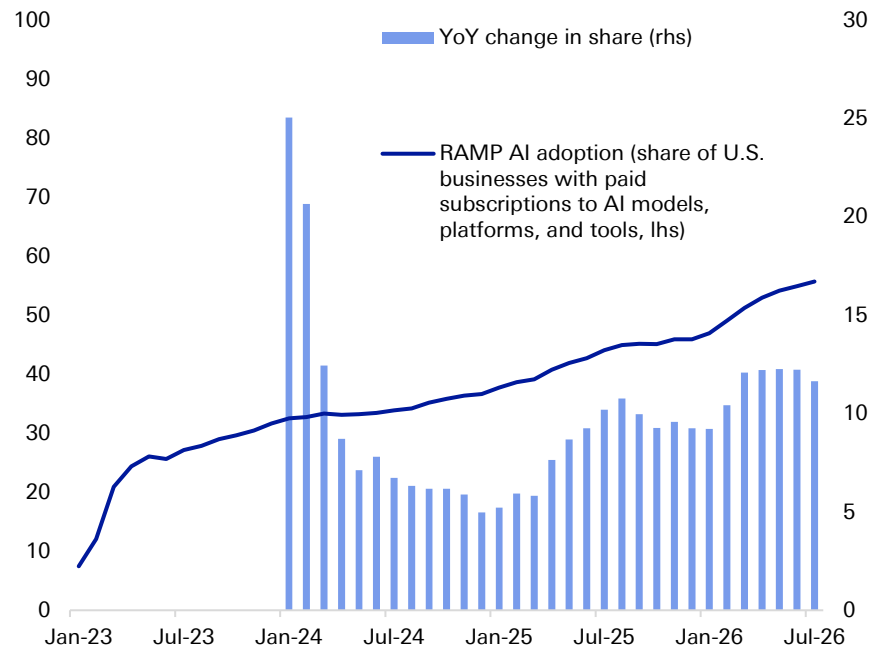


Outlook comparisons for US data-center installed power capacity (GW)



Source: Bloomberg NEF, Bloomberg Finance LP, Deutsche Bank.

Corporate AI adoption speed in the US



Realisations 1/5

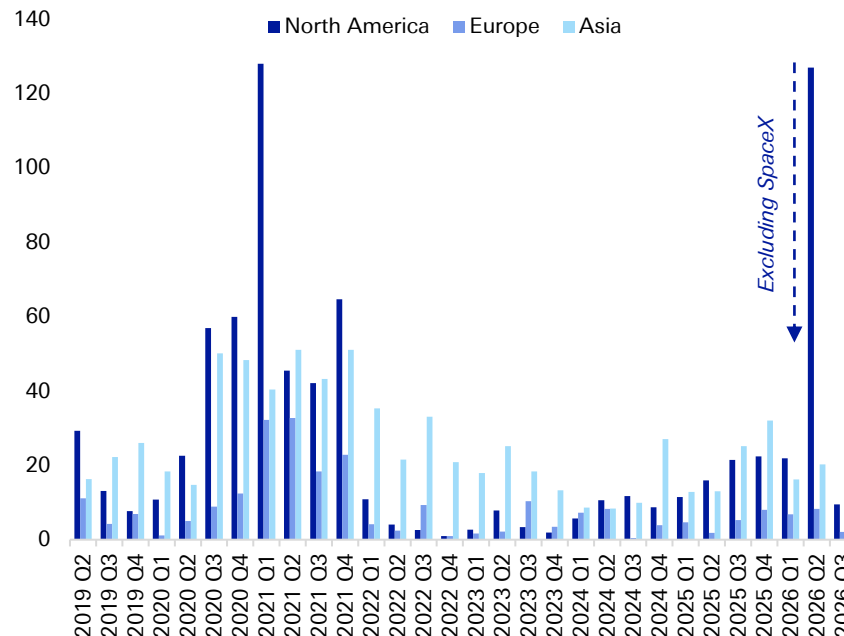
Reopening IPO markets, AI-related corporate demand and mature portfolios are all catalysts. But timing remains unpredictable and exposed to geopolitics, public-market volatility, rates and AI-related valuation uncertainty.



1. KKR described Q2 as the largest monetisation quarter in its history, with realised performance income of \$848m and realised investment income of \$220m. Remaining unrealised gains were still approximately \$18bn.
2. Carlyle returned nearly \$7bn to investors in Q2 and \$37bn over the prior year. Its US buyout strategy returned 23% of fair value over 12 months, which management said was more than twice the current industry average and above the 20% long-term industry average.
3. TPG generated \$5bn of Q2 realisations and almost \$14bn in H1, up 28% year over year.
4. Blackstone outlined the following segmentation when it comes to investor demand for assets:
 - AI, power, electrification and energy assets: strong strategic, sponsor and IPO demand
 - AI-unaffected businesses: broadly available liquidity
 - Software, professional services and information services: materially weaker activity because of AI uncertainty.

The firm estimated the third category represents approximately 30%-40% of the private-equity market.
5. TPG similarly noted that industry realisations were down about 46% quarter over quarter, although its own H1 activity was higher. It expects its monetisation cadence to accelerate toward year-end and into 2027. Brookfield expects transaction volumes to build in H2, while Blackstone expects weaker Q3 realisations followed by a stronger Q4 and 2027.

IPO deal values by quarter \$bn



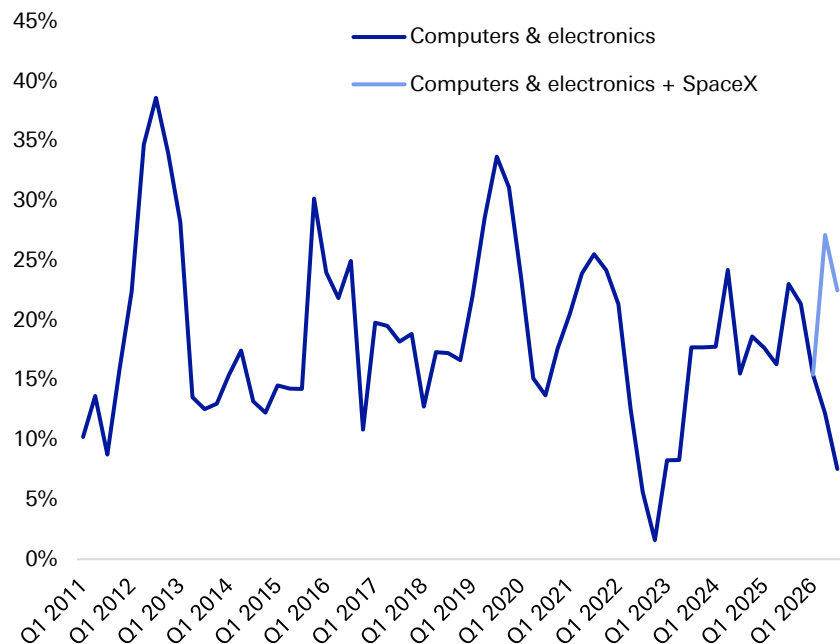
Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

Realisations 2/5

More AI-related businesses are likely to come to the market even after the big AI firms go public. Tech IPOs have been relatively muted despite the boom in financing this decade. Investors' appetite for AI and big capital markets issuances may push more managers to exit tech holdings while the mood lasts. There have been just c \$7bn in pure-play AI IPOs in the US since 2025.

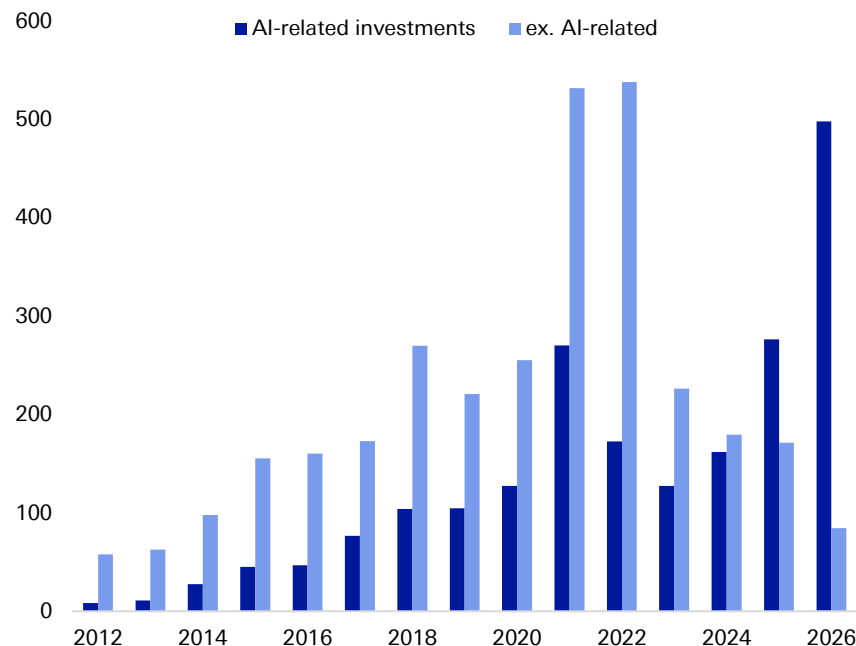


Tech share of total US IPO proceeds, 4q MA



Source: Dealogic, OECD, Deutsche Bank.

Worldwide VC investment in AI (\$bn)

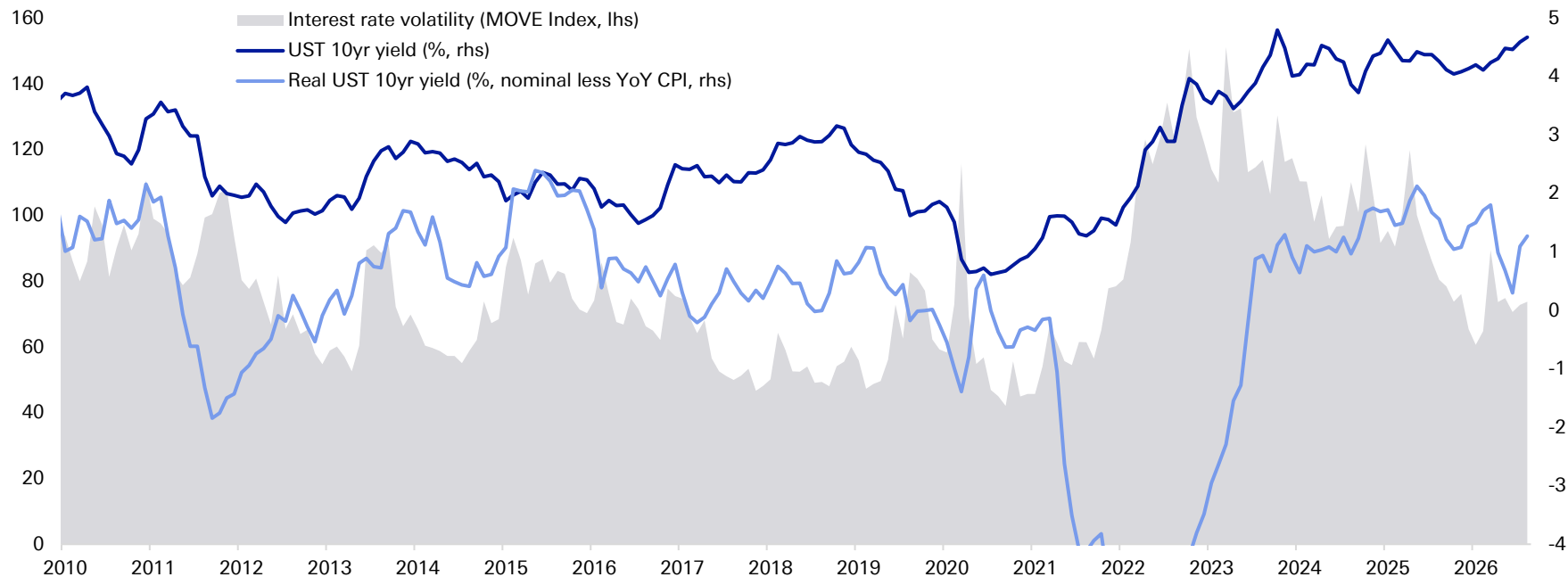


Realisations 3/5

Rising yields are likely to be less of a headwind to capital markets activity than many expect. Meanwhile, both real yields and volatility remain subdued. We have shown previously that dealmaking has been strong even when rates were elevated.



US rates backdrop – monthly averages



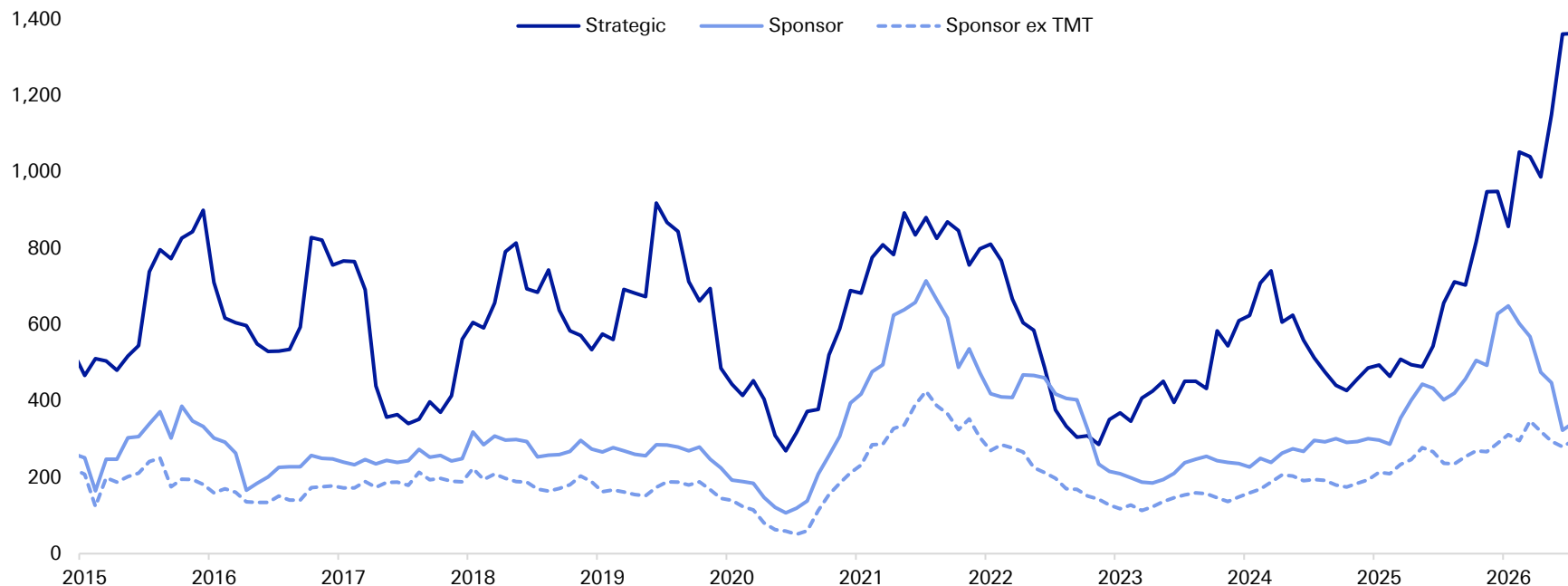
Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

Realisations 4/5

Rates are also unlikely to grind corporate buyers and thus sale exits to a halt. Meanwhile, sponsor M&A outside TMT has a number of tailwinds.



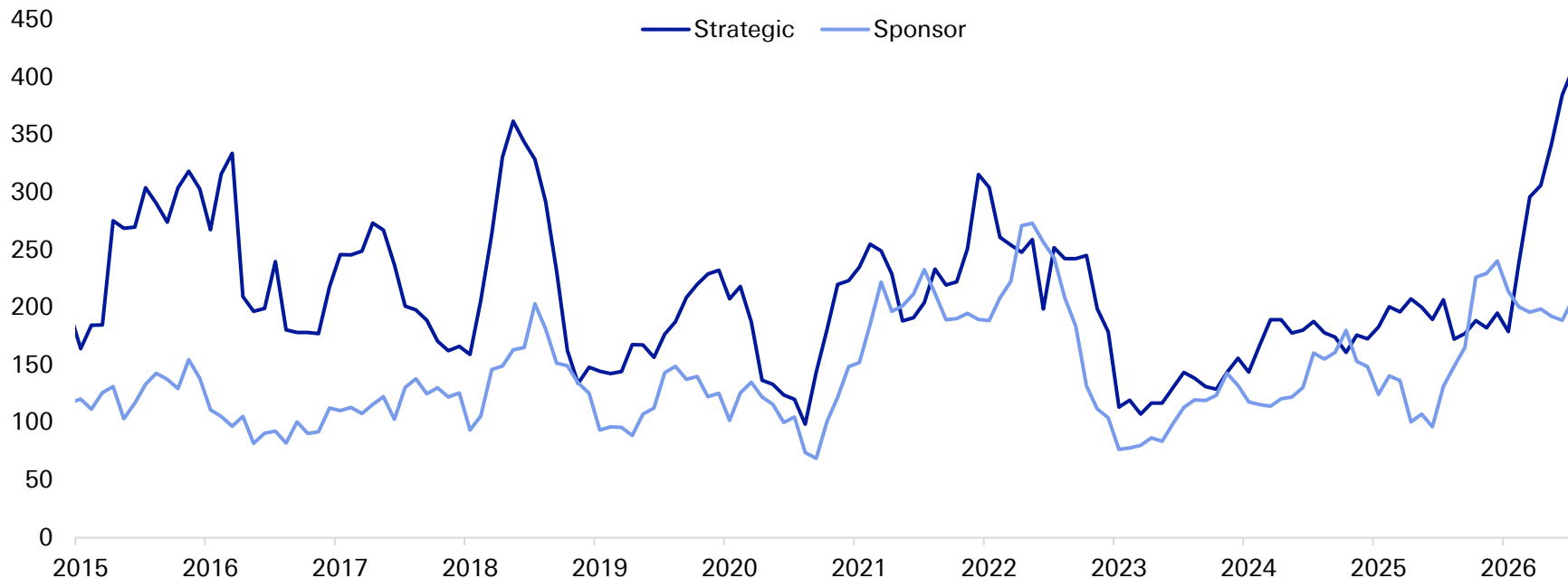
US M&A deal value at announcement (\$bn, 6m sum)



Source: Dealogic, Deutsche Bank. Note: through July 2026.



W Europe M&A deal value at announcement (\$bn, 6m sum)



Source: Dealogic, Deutsche Bank. Note: through July 2026.



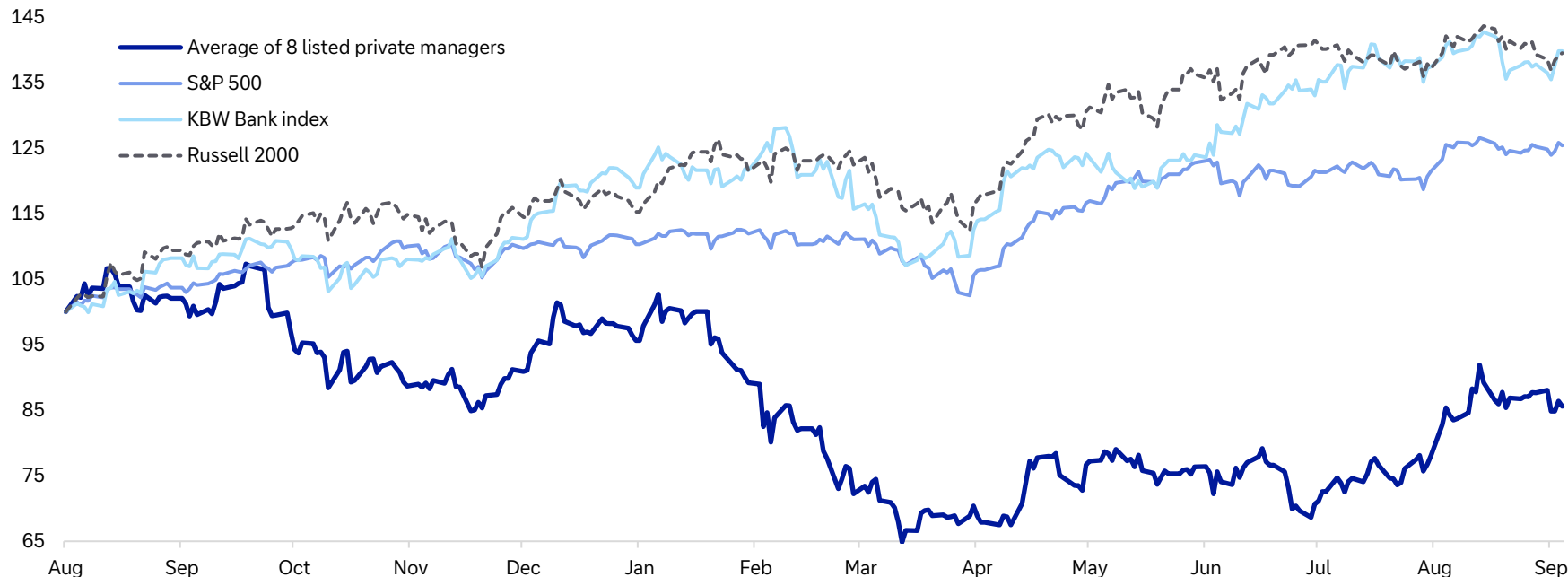
03. Market conditions

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Shares of US-listed private capital managers are up +32% from YTD lows and +25% from June trough as some confidence returns in the asset class. Our index is still down -11% YTD vs +14% for the S&P 500 and 18% for US banks



Private capital managers total return performance relative to broader markets (USD)



Source: Bloomberg Finance LP, Deutsche Bank.

European firms have also rebounded by +23% (from March lows) and +14% (June). The YTD return (-1%) also underwhelms the broader market (+13%) and local banks (+28%)



Private capital managers total return v broader markets in 2025 (EUR)

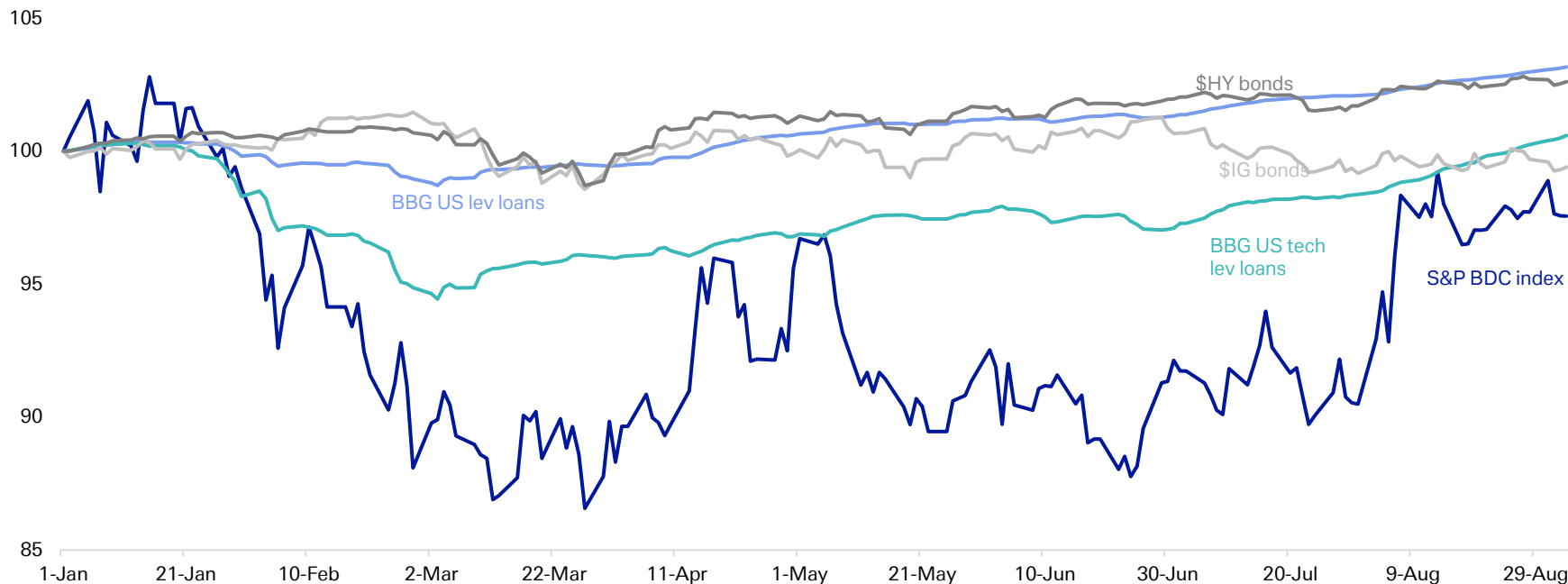


Source: Bloomberg Finance LP, Deutsche Bank.

It's a similar rebound story for BDC returns vs the broader credit markets in recent weeks

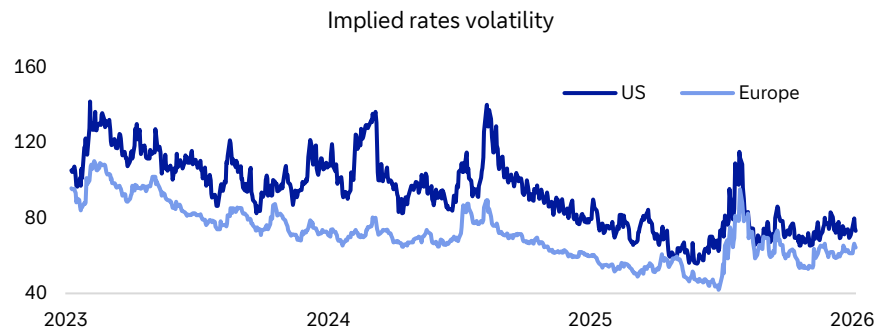
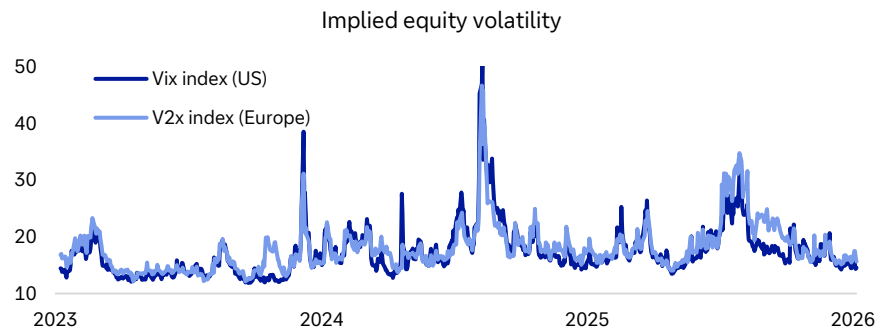
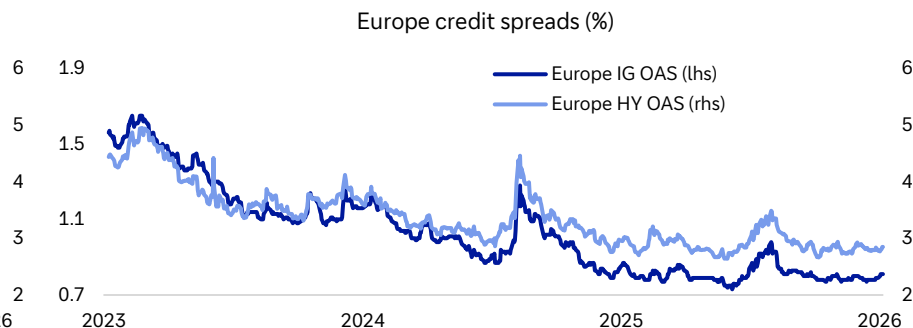
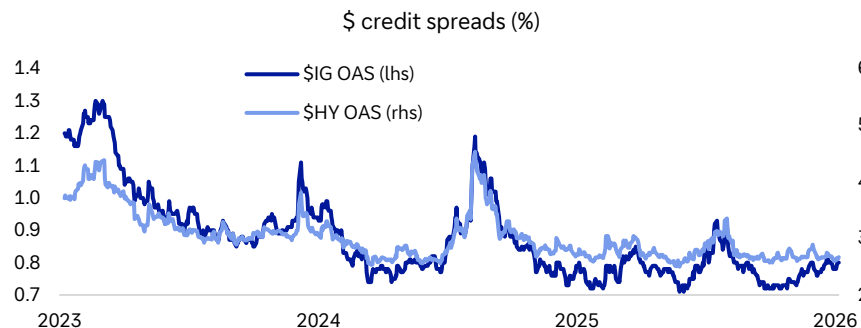


Credit assets YTD total return performance



Source: Bloomberg Finance LP, Deutsche Bank.

Contained volatility, especially in rates amidst rising yields, helps overall risk backdrop. Credit spreads have also been steady.

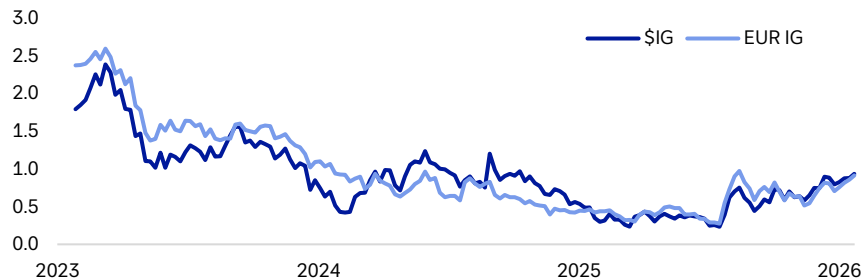


Source: Bloomberg Finance LP, Deutsche Bank.

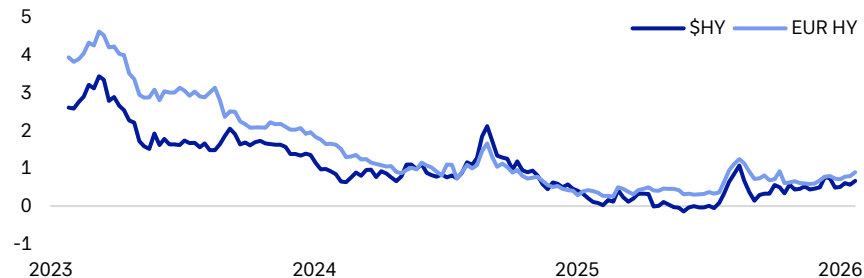
Rising refinancing costs may deter some debt issuance but overall credit conditions remain favourable



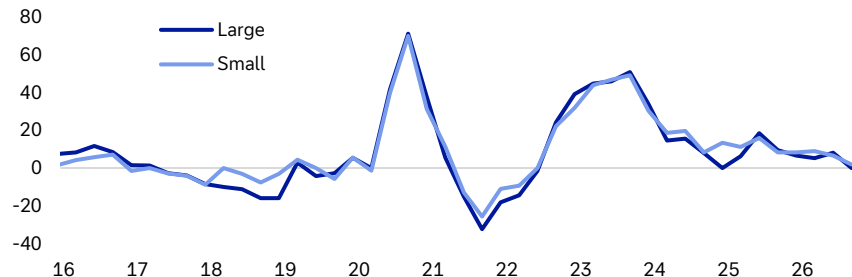
IG bonds - Refinancing proxy in %



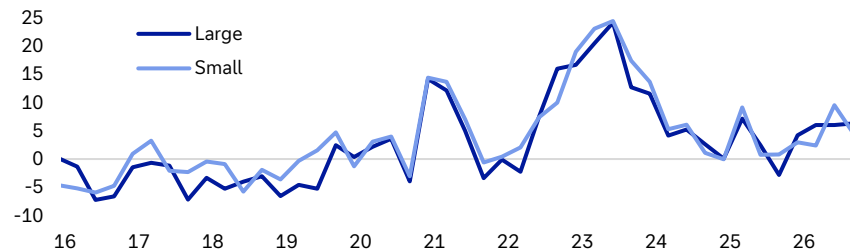
HY bonds - Refinancing proxy in %



Fed SLOOS: banks tightening standards for C&I loans to firms (%)

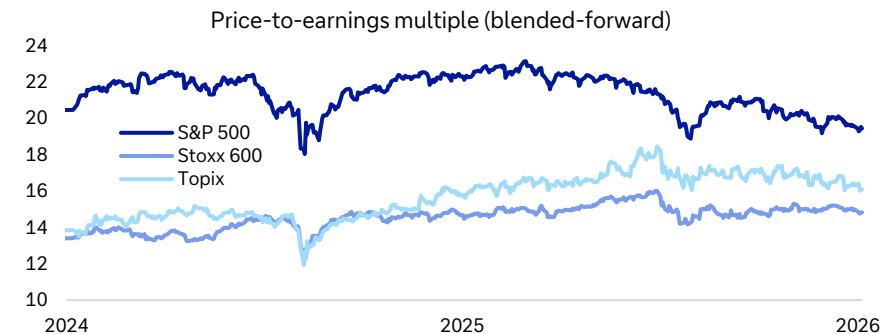
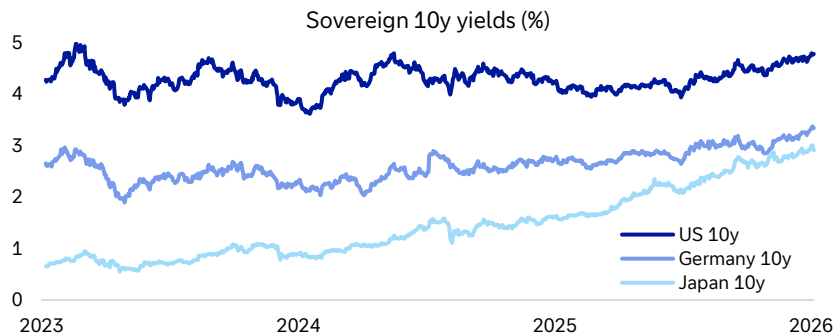
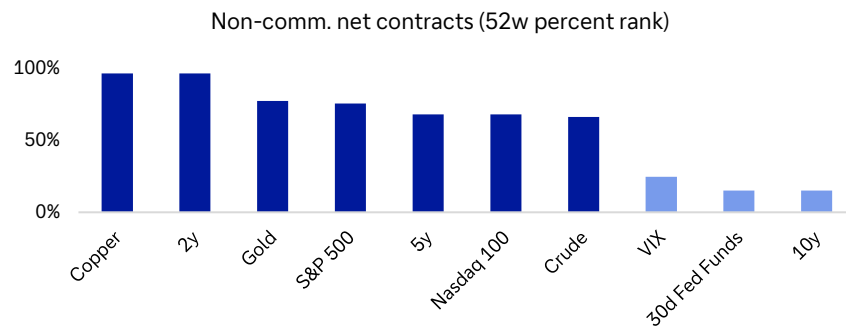
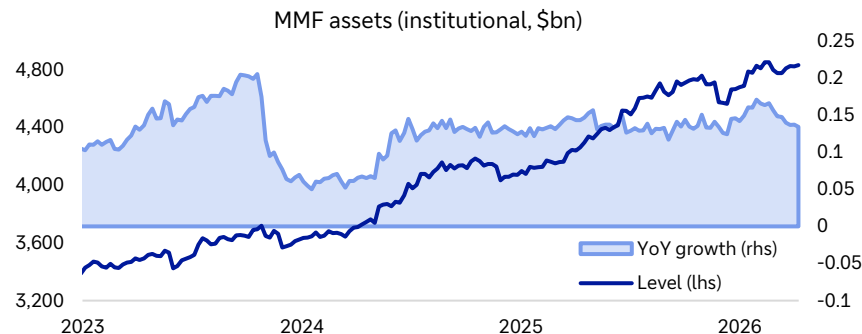


Euro Area bank lending survey: change in credit standards for business loans



Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

Despite rising yields, markets remain in a risk-on mood and there is plenty of cash

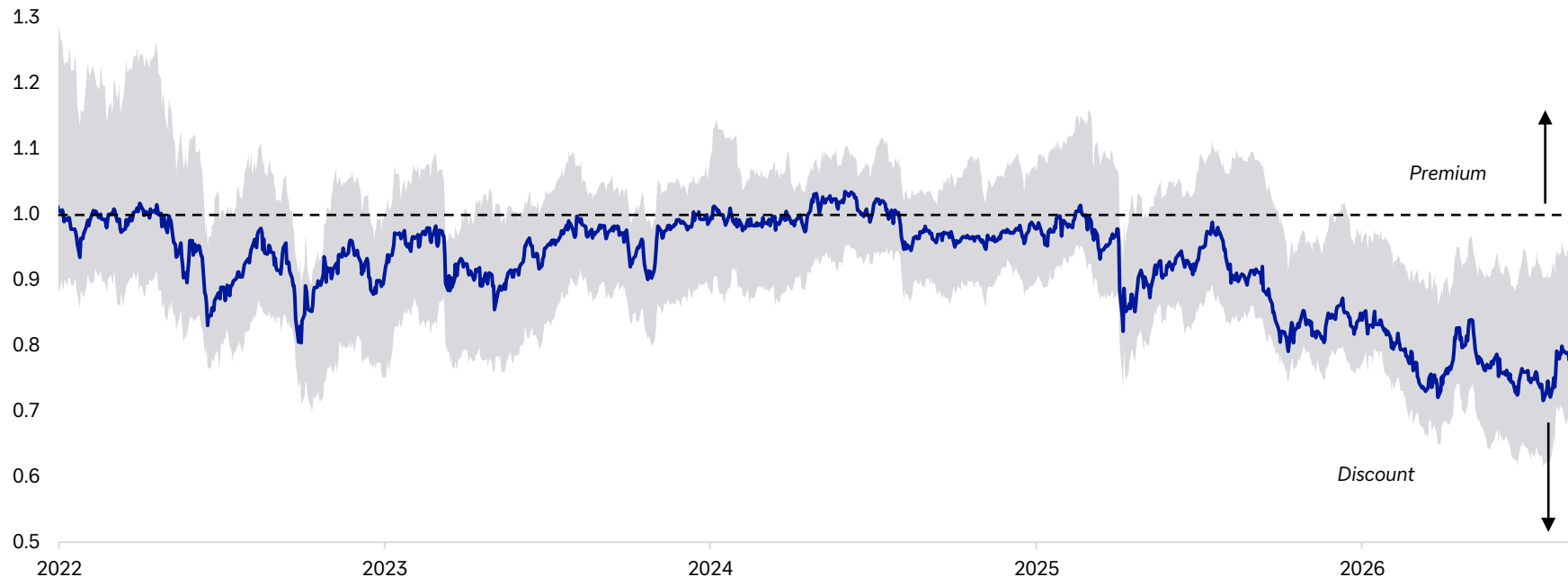


Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The median BDCs discount to NAV is still wide and now stands at c 22%. Apart from software fears, the focus is on the trajectory of US monetary policy.

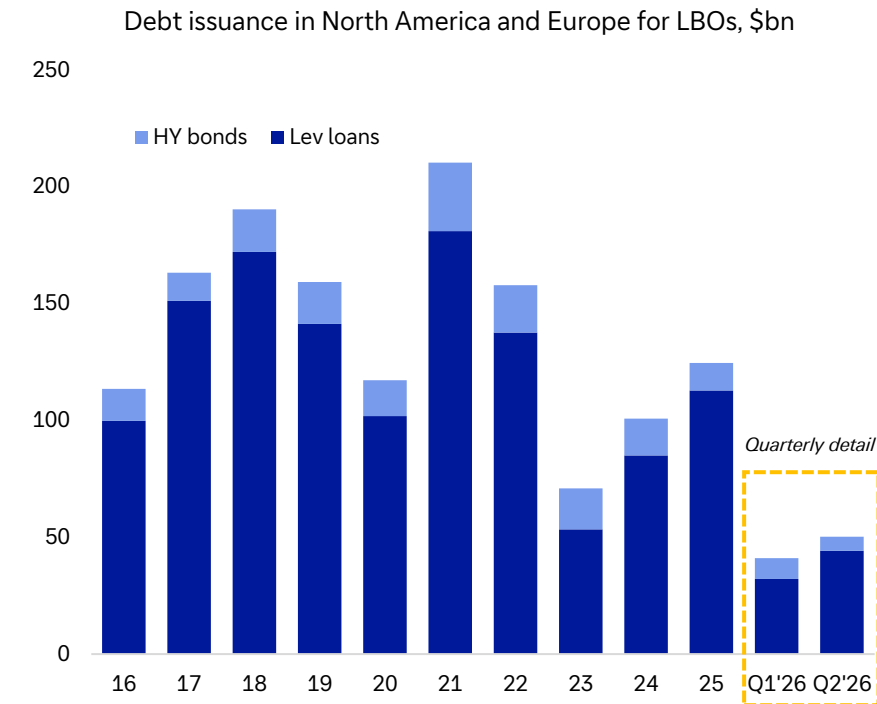


Median price-to-NAV ratio of funds in the S&P BDC index

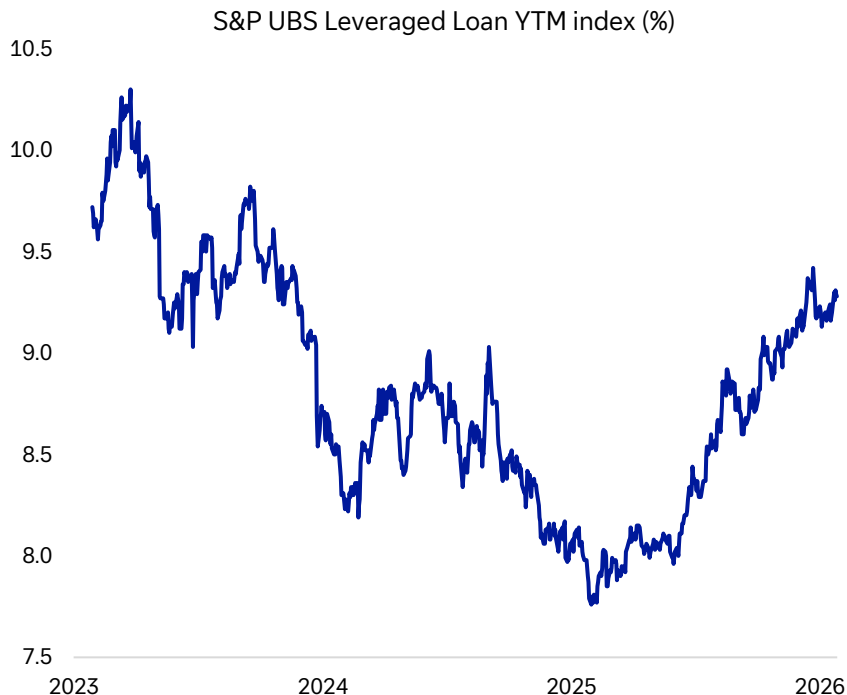


Source: Bloomberg Finance LP, Deutsche Bank.

Debt issuance for LBOs continued to grow in Q2. Rising sovereign rates may dissuade some borrowers but leveraged loan yields remain below recent highs and earnings have been strong.



Source: White & Case, Bloomberg Finance LP, Deutsche Bank.





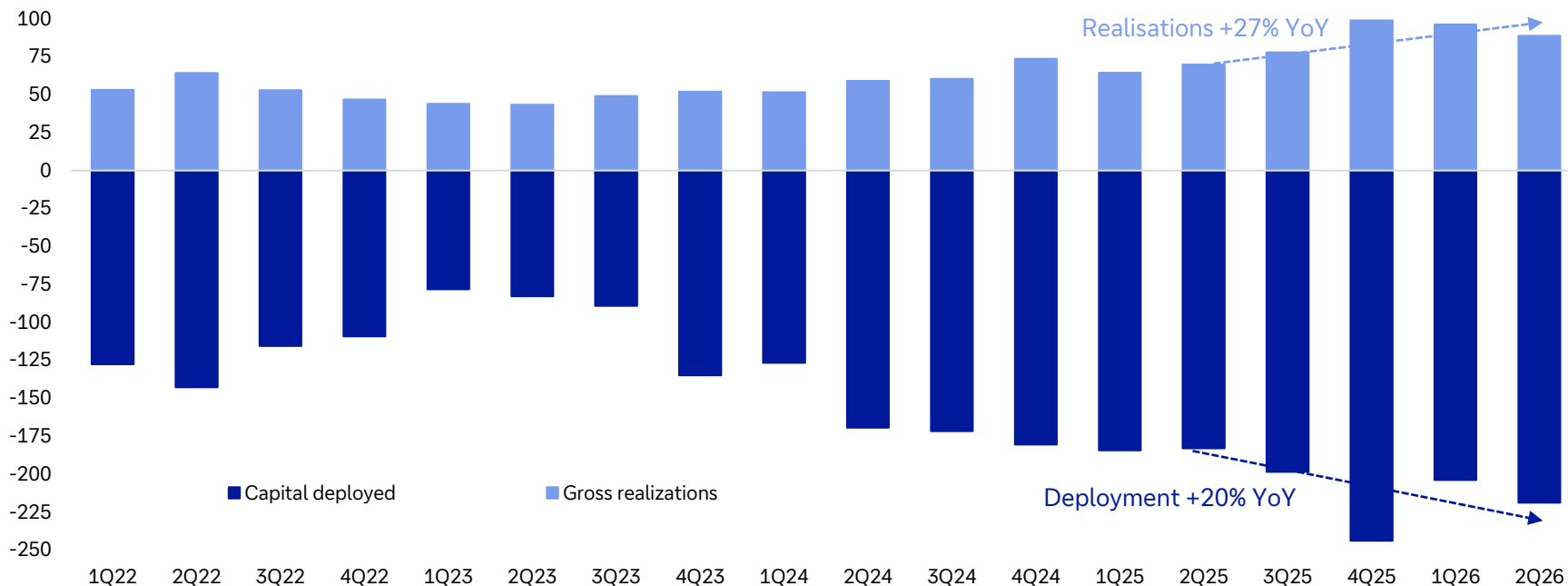
04. Fundraising environment

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Both deployment and realisations are up YoY. Realisations growth will be in focus given uncertainty voiced by managers.



Selected private managers, deployment v realisations (\$bn)

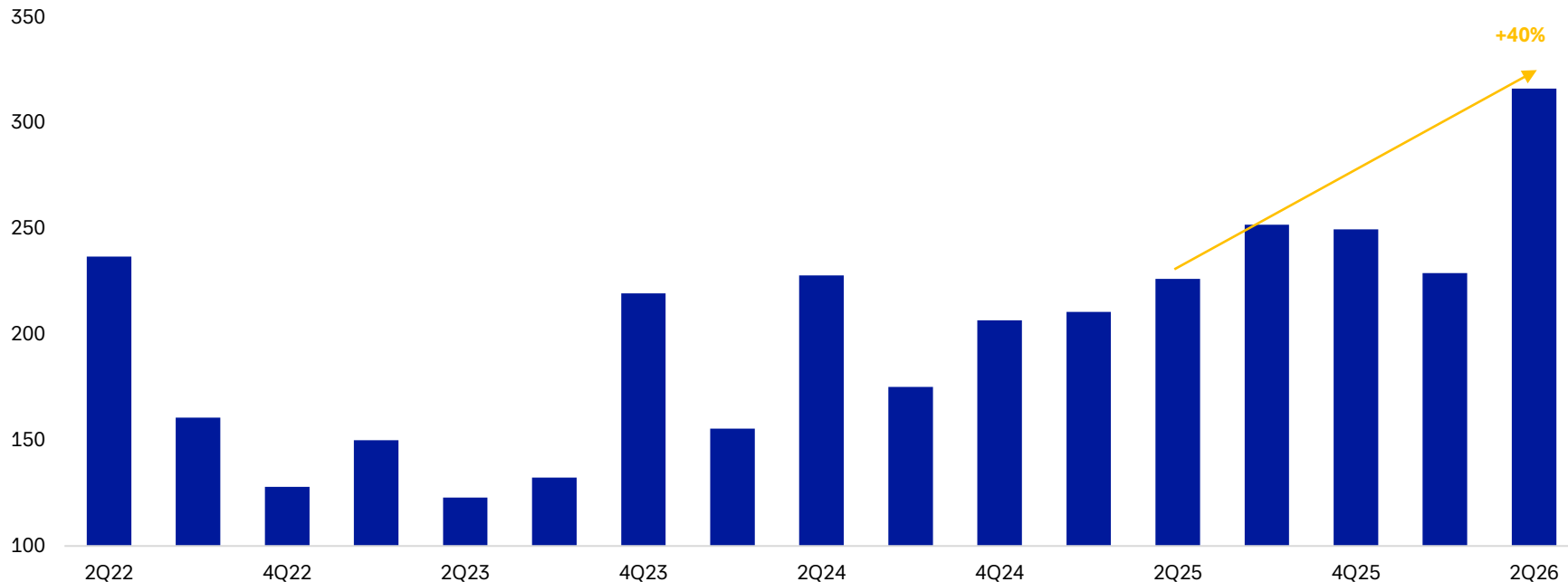


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank. * Note: Deutsche Bank company research forecasts.

Aggregate fundraising growth has shot up (in part due to one-off factors) and managers are confident in continued strength.



Fundraising by selected large, listed managers (\$bn)

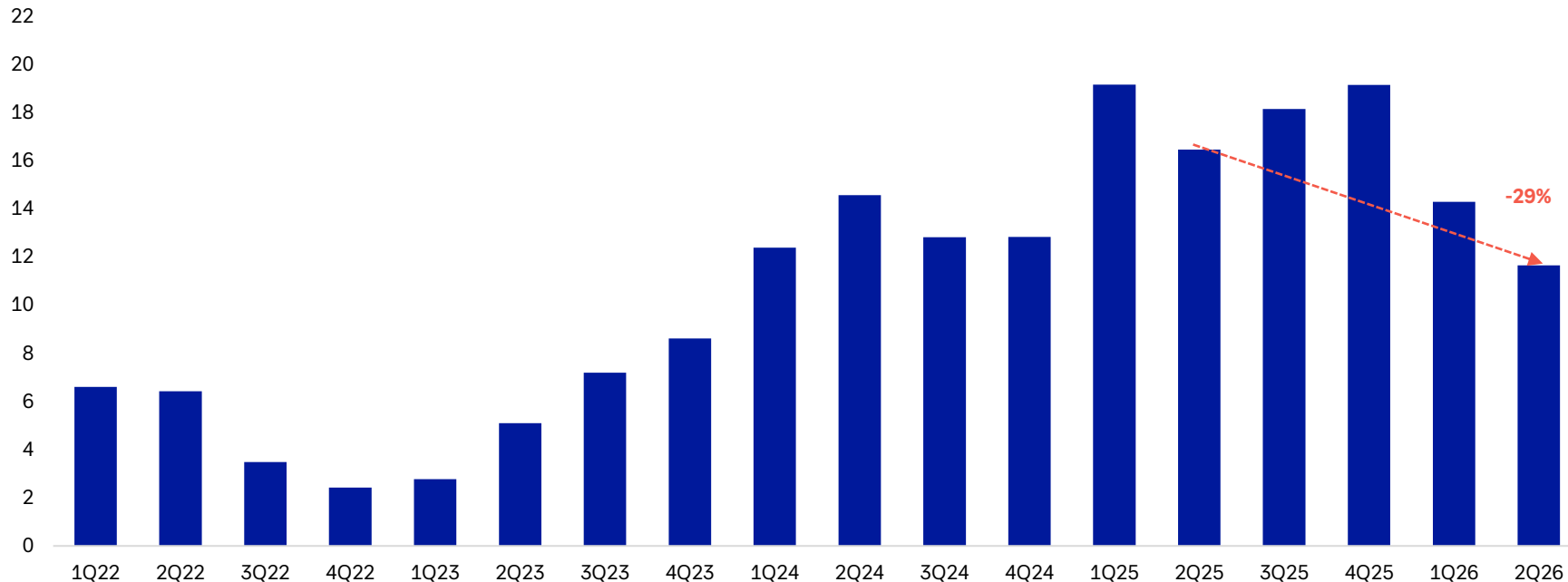


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank.

Gross retail fundraising (not taking into account redemptions) continues to come down, now down almost a third YoY.



Retail fundraising by largest listed private capital managers *

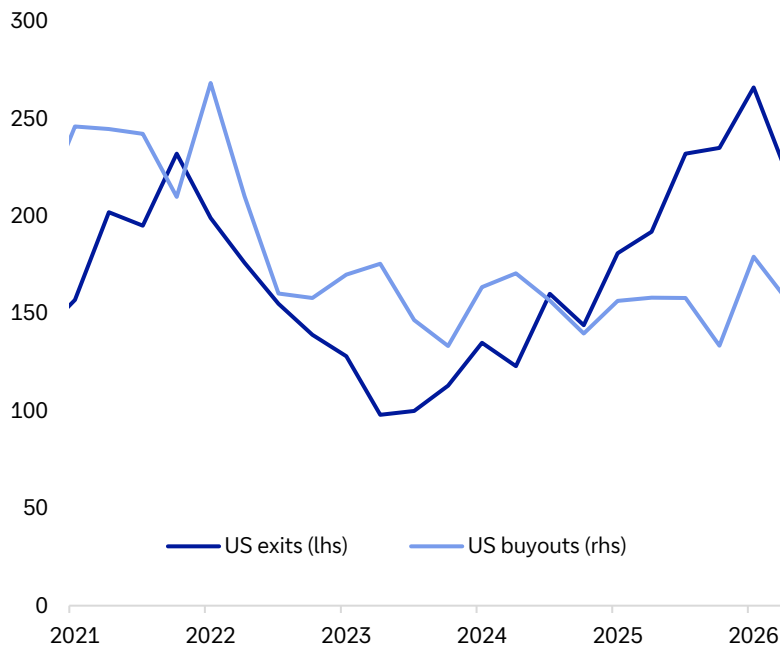


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Bloomberg Finance LP, Deutsche Bank. * Note: gross, only funds with available Q2 data.

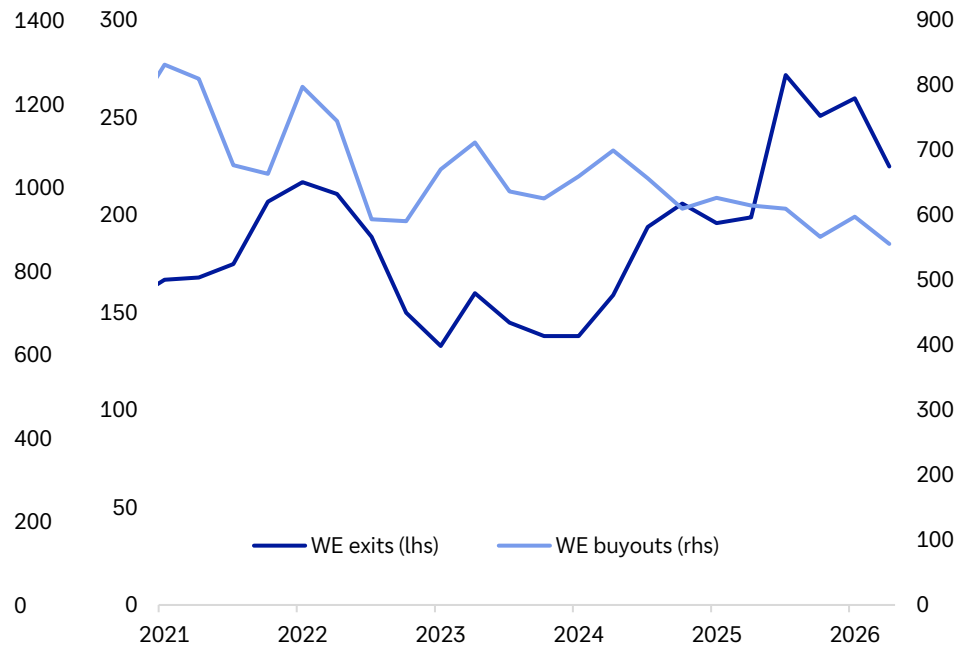
Exit activity remains strong despite some softening relative to recent highs



US PE exits have come down from recent highs



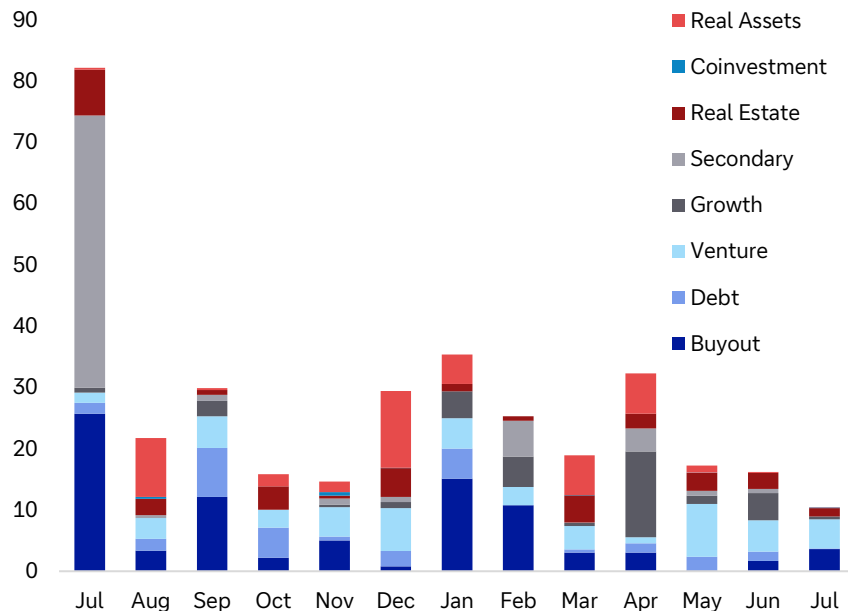
European exits have also slowed but remain high



Source: White & Case, Deutsche Bank.

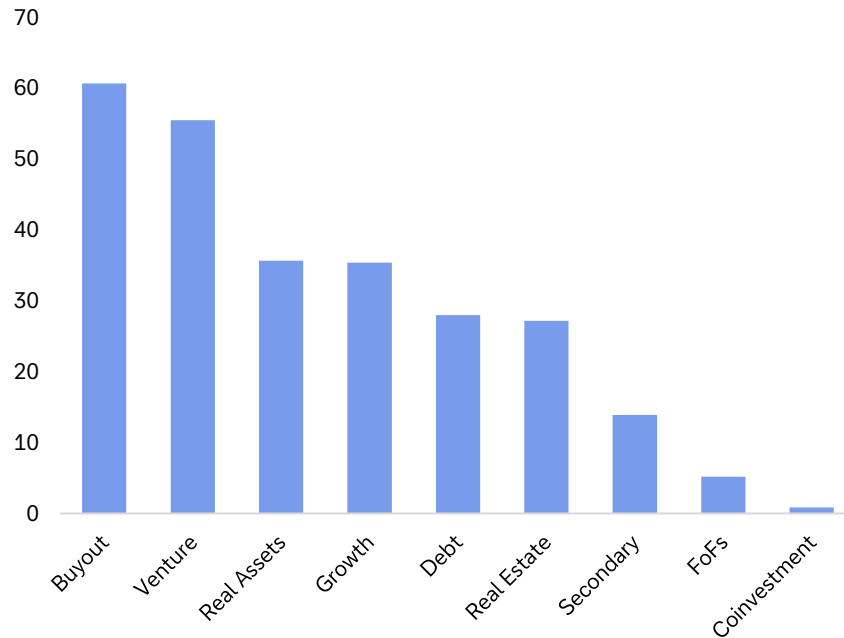


Fundraising funds - target in \$bn by strategy type and launch month, US and Europe



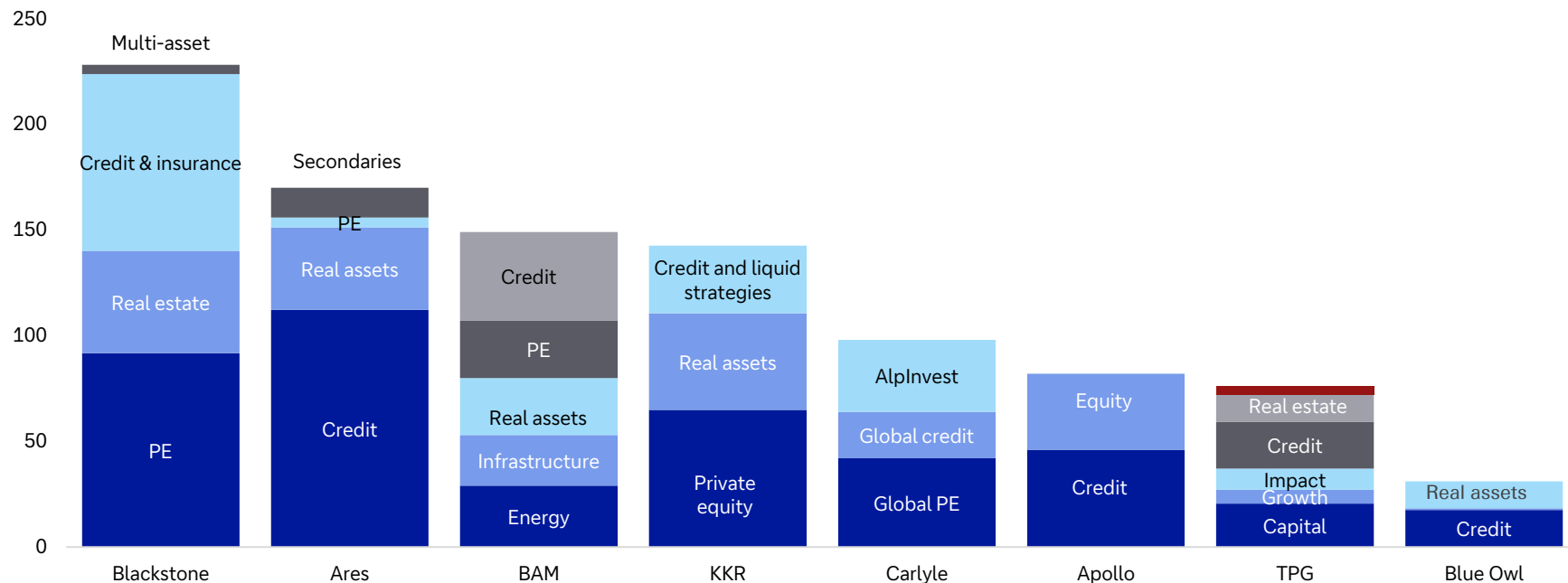
Source: Bloomberg Finance LP, Deutsche Bank.

Fundraising funds in US and Europe, last 12m in \$bn





Dry powder by manager and reported segment, Q2 2026 (\$bn)

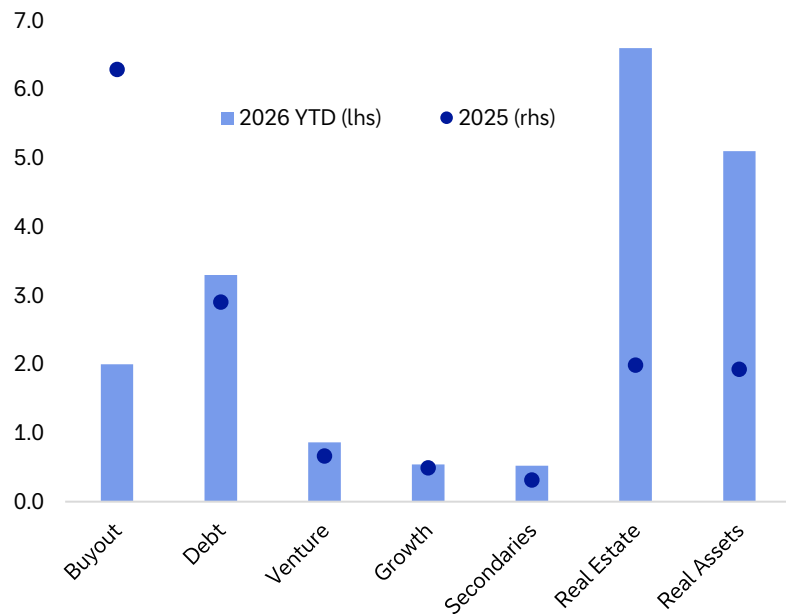


Source: Company reports, Deutsche Bank.

Recent disclosed LP commitments

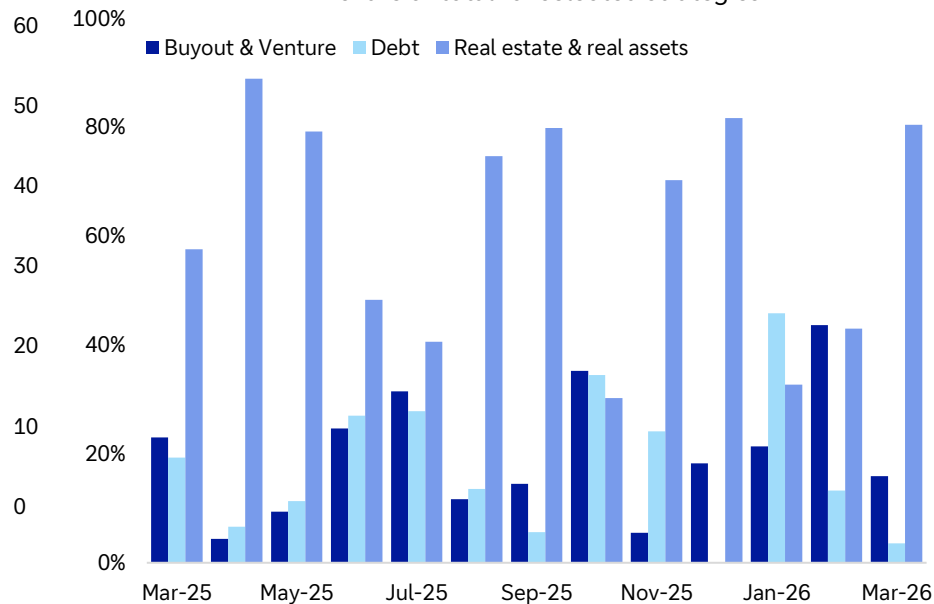


Disclosed LP commitments by strategy in \$bn



Source: Bloomberg Finance LP, Deutsche Bank.

Disclosed LP commitment targets by month and strategy, share of total for selected strategies

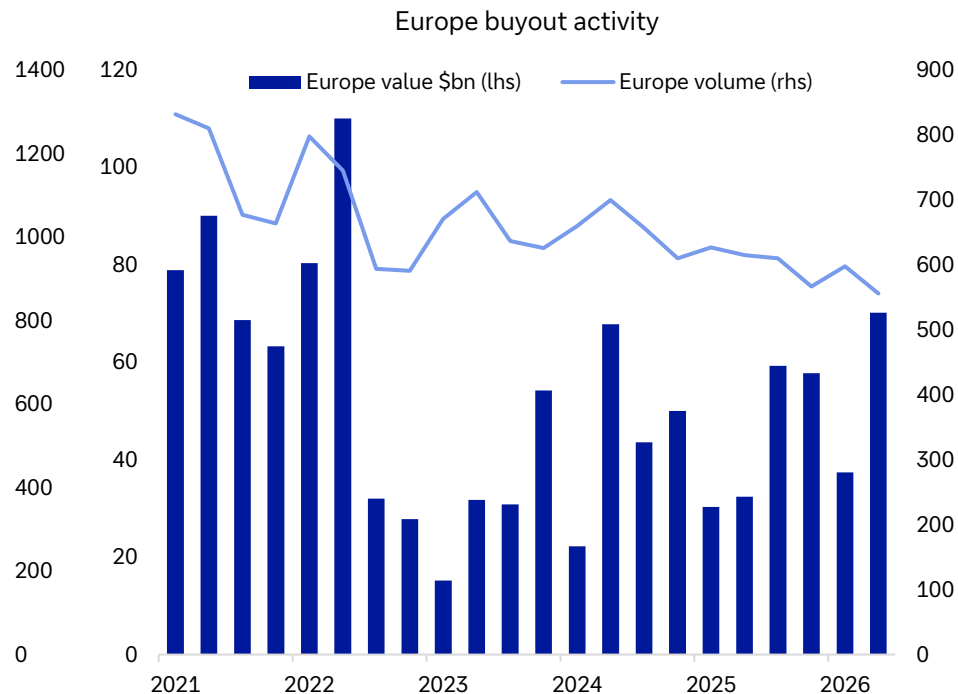
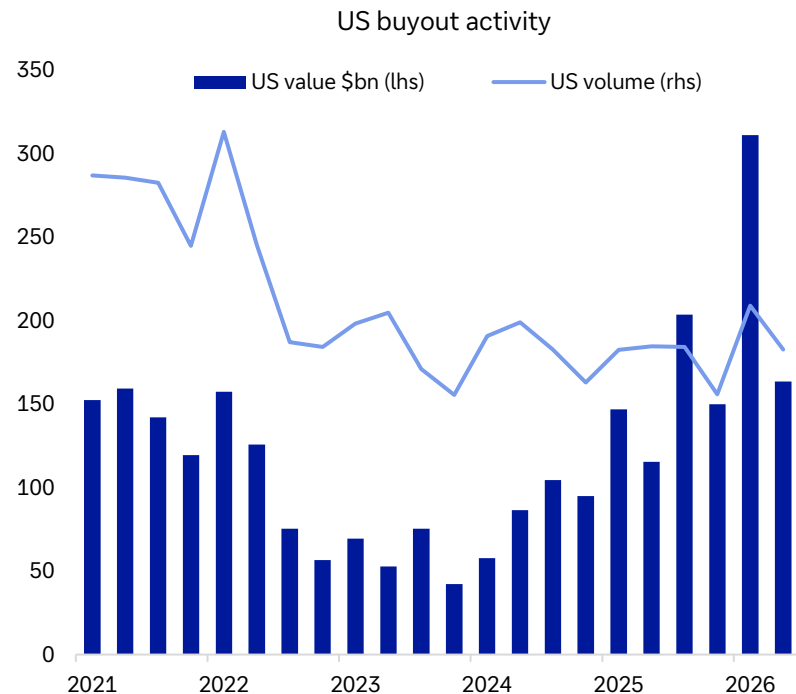




05. Dealmaking activity

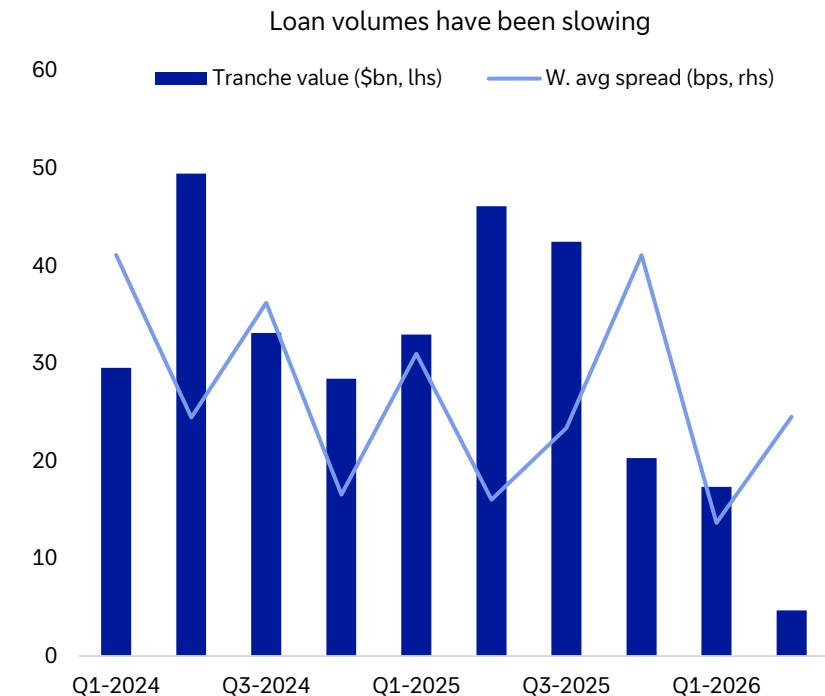
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Dealmaking activity in buyouts

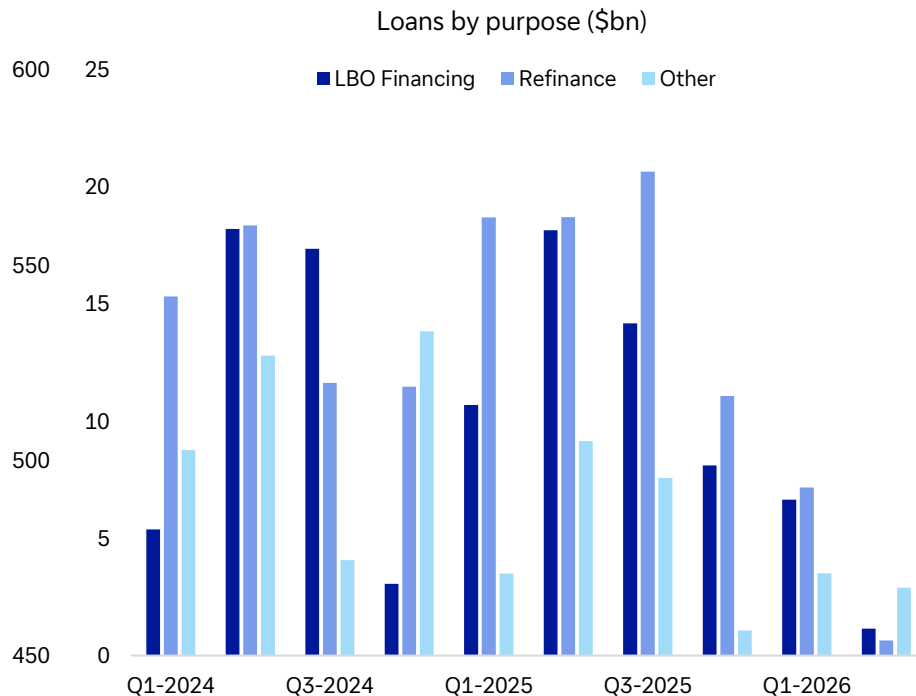


Source: White & Case, Deutsche Bank.

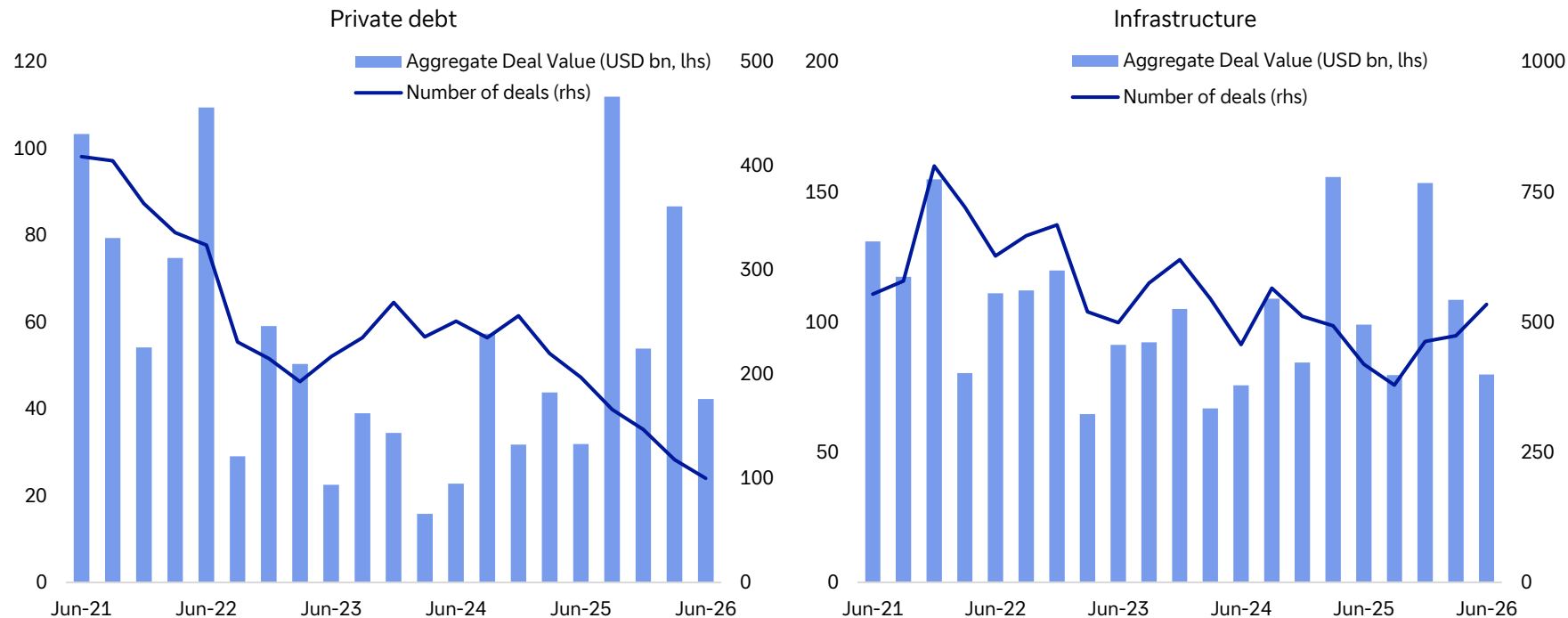
Origination in large private credit loans – loan volumes continued to slow amidst more caution around private credit spilling over from Q1. A recovery in sentiment will support some rebound in volumes.



Source: Bloomberg Finance LP, Deutsche Bank.



Global dealmaking activity in private debt and infrastructure



Source: Preqin, Deutsche Bank.

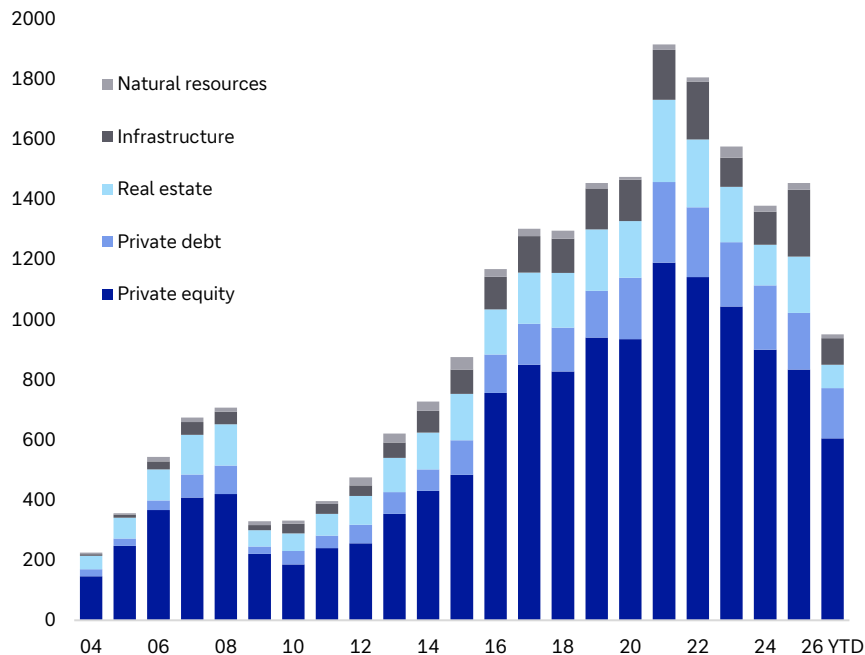


06. Investor allocations

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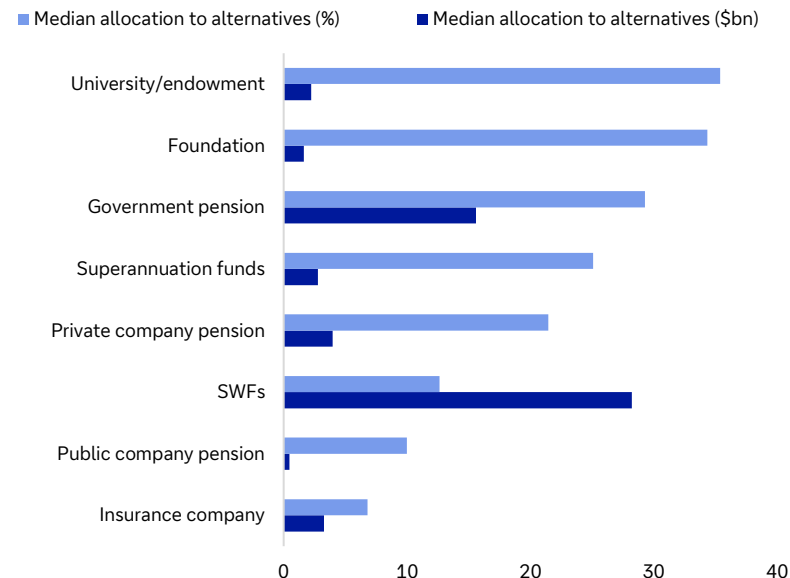


Private capital sum of funds raised (\$bn)



Source: Preqin, Bloomberg Finance LP, Deutsche Bank.

Top 100 LPs in each category, allocations to alternatives



Appendix 1



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The Drone Frontier: Geopolitics in a New Era of Conflict

September 07, 2026

Executive Summary

- Policy incentives and spending hikes are driving the expansion of the global military drone market, with a projected CAGR of 8% through 2030. This looks set to unlock new investor opportunities across the wider drone ecosystem. Multi-billion-dollar investment pledges in the UK, US, Japan, India, and the EU are among those that reflect this rapid build-out.
- The surge in global spending will likely benefit established manufacturers in markets including the US, UK, and Türkiye. Ukrainian companies are well-positioned to capture post-war market share, as their footprint expands across Europe, the Gulf, and Asia-Pacific. Japan and the Philippines are among Asian markets interested in Ukrainian capabilities.
- The rapid development of drone technology – particularly hardware – risks leaving governments with obsolete stockpiles. For investors, this makes the sector harder to assess than the headline growth numbers suggest. Companies able to upgrade quickly across many drone models, or those specialising in software that can be rapidly upgraded, may be better positioned than those tied to a single platform.
- Organised crime, terrorist, and insurgent groups are increasingly adopting the technology, creating localised security risks that could drive further spending and prompt new restrictions on commercial drone use.
- National security concerns will likely drive efforts to cut critical dependencies, strengthen supply chains, support domestic manufacturers and secure an edge in AI. The US, EU, India, and Japan are among markets that will seek to reduce reliance on China for critical minerals and key components, in our view, but Beijing's dominance makes this a multi-year process requiring sustained investment across the value chain.

Authors

Miha Hribernik

Chief Geopolitical Strategist, Asia



Context

On 2 September, Ukraine announced it would soon finalise a bilateral drone production and development deal with Germany. This followed the signing of drone production and export agreements with Denmark, the Netherlands, and Estonia on the sidelines of the NATO Summit in July. The deals are among the latest developments marking Kyiv's emergence as one of the global leaders in drone technology. The agreements also come against the backdrop of a worldwide surge in investment in the sector. This trend looks set to continue for years, as governments absorb the lessons of the conflicts in Ukraine and the Middle East.

This report assesses the key geopolitical implications of the accelerating global race for drone dominance.

Disrupting Defence: The Rise of Cost-Effective Drone Power

The proliferation of inexpensive drones is reshaping military doctrine, defence spending, and procurement worldwide. While military drones date back decades, the technology has reached a number of major milestones and 'firsts' in the last several years, reflecting the accelerating pace of development, production and deployment (see timeline). The wars in Ukraine and Iran in particular have demonstrated the limitations of traditional costly military assets in the face of numerous, increasingly capable and AI-enabled, and often disposable drones.

Figure 1: Selected key milestones in military drone production and use, 2020-2026



Source : Deutsche Bank Research

Drones are unlikely to fully displace other military assets, including the need for 'boots on the ground' to seize and hold territory, but the next several years will likely see a global push toward adoption and integration with conventional military forces on an unprecedented scale.



Ukraine: From Necessity to Supremacy

Ukrainian companies could capture a sizeable share of the expanding market after the war. The need to counter Russia's manpower advantages has driven an innovation model linking frontline demand with fast domestic production. This has established Ukraine as a global leader in unmanned operations and high-volume production, particularly of attack drones. This includes utilising AI for navigation, targeting, and autonomous "last mile" attacks.

Kyiv reportedly [doubled](#) its drone procurement y/y during H1 and [plans](#) to eventually use robots for 100% of frontline logistical needs. According to President Zelenskyy, Ukraine now produces [10 million](#) drones per year (likely [more](#) than all of NATO combined, although Ukrainian figures probably include a large proportion of small tactical drones), and the country could scale production to 20 million units if its international partners provide sufficient funding. Nevertheless, once active combat ends and stockpiling needs are met, production could slow given how quickly drone capabilities change (see below).

Ukraine's manufacturing footprint is likely to extend further beyond its borders.

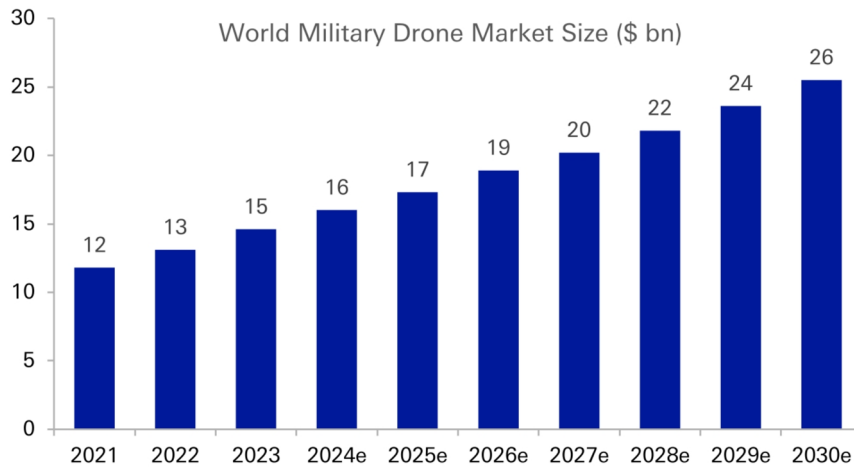
It is already expanding to facilities in countries including [the UK](#), [Denmark](#), [Norway](#), and [Germany](#), fostering technological diffusion and the emergence of a Europe-wide supply chain. Ukraine is also pursuing joint production with countries in the Middle East, Europe and the Caucasus, and has struck 10-year cooperation agreements with Saudi Arabia, the UAE and Qatar. More recently, [the WSJ](#) reported on 21 July that a new joint venture will produce Ukrainian sea-based attack drones in the US.

A Worldwide Surge in Spending

Policy incentives and spending hikes are set to fuel an expansion of drone and counter-drone arsenals in the coming years. The global military drone market is expected to grow at a CAGR of 8% through 2030, with growth in some regions – such as the Asia Pacific and Europe, the latter with an expected CAGR of 13% – projected to be even higher, according to [Global Market Insights](#). The forecasts signal robust, sustained expansion as global defence forces accelerate their transition to unmanned platforms. **This trajectory should create opportunities** for investors and companies in the wider ecosystem – from drone manufacturers to component, technology and software suppliers, as well as contractors providing support and maintenance.



Figure 2: The global drone market is expected to grow by a CAGR of 8% by 2030



Source : Global Market Insights, Deutsche Bank

Recent key developments include record procurement and spending requests in the US – already the world’s largest military drone market. Other notable initiatives include a historic shift toward a Hybrid Navy in the UK, the creation of a “Drone Wall” on the EU’s eastern flank, a mass purchase of commercial drones for military use in South Korea, and a record US\$2 billion investment push in India, which aims to bolster domestic drone manufacturing and reduce reliance on foreign suppliers (see table).

These investment pledges – set against the backdrop of rising global defence spending, which reached US\$2.6 trillion in 2025 – are **set to boost demand among existing manufacturers beyond Ukraine, including established players in the US and the UK**. This trend is reflected in **Türkiye’s** recent export boom. Türkiye is the world’s largest military drone supplier, accounting for some **65%** of armed-drone export sales. The country saw its defence exports triple over the past five years, and reportedly aims to double them in the next two.

The Asia-Pacific will likely continue to emerge as a major market. Geopolitical tensions in the Taiwan Strait and the South China Sea could see more investment in [surface and undersea drone capabilities](#). Ukrainian defence companies also [reportedly](#) aim to expand manufacturing and technological cooperation into the region, with Japan, the Philippines and other regional economies among potential markets. Regional companies are also seeking Ukrainian expertise, as reflected by a recent Japanese acquisition of two Ukrainian drone startups.



Figure 3: Selected recent developments, including planned or proposed spending increases

United States	The US Department of War has requested a 24,000% increase in FY2027 funding for its Defense Autonomous Weapons Group, which oversees mass-produced drones with AI capabilities. The Pentagon has placed an order for 30,000 one-way attack drones (the largest such order to date), and expects to receive more than 300,000 by early 2028.
United Kingdom	Aims to invest GBP5 billion (US\$6.7 billion) in advanced uncrewed systems, including a Hybrid Navy which integrates autonomous vessels and undersea drones. Plans to boost Army lethality ten-fold by investing in capabilities including AI and autonomous systems.
European Union	The creation of a new “Drone Wall” or “Drone Shield” will entail a multi-billion euro spending commitment.
France	Plans to expand its stock of explosive drones by up to 400% by 2030.
Australia	Intends to more than double funding for anti-drone capabilities over the next decade.
Germany	Announced plans to invest around EUR16 billion (US\$18.2 billion) in unmanned reconnaissance, attack and defence capabilities by the end of the decade.
China	Continues to invest in new drone capabilities amid rising defence spending. Recent developments include the reported construction of the world’s largest unmanned submersibles and the demonstration of a drone swarm operations system in March 2026.
India	Preparing to place a record US\$2 billion order for domestically-built military drones, reportedly as part of a plan to reduce dependence on foreign suppliers.
Japan	Tokyo allocated JPY 277 billion (US\$1.7 billion) for unmanned system in this year’s defence budget, more than double the figure from 2025.
South Korea	Intends to train 500,000 military drone operators and procure 11,000 commercial drones for military training in 2026.

Source : Deutsche Bank Research, external sources (see links in table).

Tech Turnover: The Obsolescence Dilemma

The scramble for drones risks leaving governments with stockpiles of obsolete equipment. The technological ecosystem around drones, and counter-drone systems, is not yet mature. Battlefield pressure forces rapid innovation, sometimes within weeks, risking large stockpiles becoming quickly **obsolete**. The rapid growth of the industry, and the proliferation of new companies entering the sector, also raises questions about the risks of an eventual industry consolidation.

For investors, this makes the sector harder to assess than the headline growth numbers suggest. The better-positioned companies may be those that can upgrade quickly across many drone models – through software, flexible or modular manufacturing and battlefield feedback – rather than those tied to a single drone platform. With hardware vulnerable to obsolescence, software could prove to be a material differentiating element, and several manufacturers are prioritising software platforms as key to upgrading their systems.

Rogue Drones: The Proliferation to Non-State Groups

Drone technology is a ‘great equaliser’ for non-state groups that could fuel a rise in global security risks. Organised crime, terrorist, and insurgent groups have expanded their drone capabilities in recent years. Groups like Hezbollah, Hamas, guerrillas in **Colombia**, and **Mexican** drug cartels have already adapted cheap commercial drones for attacks and demonstrated an ability to **absorb** the lessons of the war in Ukraine.

The proliferation of drone technology means that defences must cover a wide range of potential targets – from critical infrastructure to public events and government buildings – while being updated to close gaps that small, adaptive groups can exploit. Although there is little evidence to suggest that these groups have used AI-



enabled drones at scale, this could become an additional medium-term risk. **The need to counter evolving threats from non-state actors could be another driver of increased spending and may prompt more [restrictions](#) on commercial drone use.**

Strategic Autonomy: Diversifying Drone Supply Chains

National security concerns will likely drive efforts to reduce dependencies, strengthen supply chains, support domestic manufacturers, and secure an edge in AI. China continues to dominate drone supply chains, including for processed critical minerals used in key drone components such as motors and actuators, batteries and infrared / optical components (such as lenses). Major markets, including the US, remain highly dependent on China-linked supply. The table below highlights three key areas. We anticipate continued efforts to diversify critical mineral and component supply chains amid accelerating geopolitical competition. This process will likely dovetail with policies intended to reduce dependence on China for processed critical minerals (see [Geopolitical Outlook 2026: At the Mid-Point](#)).

Figure 4: From Mine to Mission: China's Supply Chain Footprint

Key raw material	China share	Processed material	Drone component	Drone function affected
Rare-earth compounds and metals Rare-earth inputs used in permanent-magnet supply chains	High China supplied 71% of US rare-earth compounds and metals imports in 2021–24; USGS identifies magnets as the leading global end use of rare earths.	Permanent magnets	Motors and actuators	Propulsion, control and payload lift
Natural graphite	High US net import reliance for natural graphite is 100%; China supplied 46% of US natural-graphite imports in 2021–24 and 55% of US graphite battery-anode material imports in the first eight months of 2025.	Battery anode material	Lithium-ion battery cells / packs	Battery capacity and electric-power availability
Gallium and germanium	Very high for gallium; high for germanium China produces close to 90% of global gallium and around 60% of global germanium.	Gallium- and germanium-containing semiconductor and optical materials	Radio-frequency communications, power electronics and infrared/optical components	Signal transmission, infrared/optical sensing and power-electronics efficiency

Source : Deutsche Bank Research, USGS. Note: This table lists three key touchpoints relevant for drones but is not meant to be exhaustive.

However, diversifying sources of commodities and specialised components will likely take years and require significant capital expenditure across the entire value chain. The drone supply chain remains highly globalised, and a single unit can contain hundreds of specialised components sourced from dozens of manufacturers across multiple countries. While derisking could ultimately benefit suppliers/producers in the US, Japan, [South Korea](#) and partner states, it is likely to progress unevenly. Given the scale of the task, eliminating dependencies in military drones will likely continue to take priority over dual use (civilian/military) drones.

Supply chain diversification is a multi-pronged process. Recent developments include efforts by US drone makers to [build](#) domestic supply chains amid new restrictions on China, including the drone [tariffs](#) announced on 13 August. Meanwhile, Taiwanese drone exports to the EU [reportedly](#) increased 40-fold in 2025, while [Japanese and Taiwanese](#) companies are purportedly seeking closer cooperation. Efforts to reduce dependencies include plans to advance technological solutions such as rare-earth-free electric motors and more sustainable navigation and power systems for drones. Some countries, such as [the UK](#), have moved to restrict the use of China-made components on national security grounds.



Expanding export controls could impact diversification timelines. During 2026, China has introduced new restrictions curbing access to critical minerals, rare earths, advanced electronics and other dual-use components. On 24 April, China banned dual-use item exports to seven European companies, [purportedly](#) to counter EU-Ukraine-Taiwan security cooperation. On 22 June, Beijing added ten US defence-linked entities – including drone companies – to its export control list restricting access to dual-use items. This was followed in subsequent weeks by additional curbs on Japanese drone and advanced-technology firms. On 5 August, China tightened drone-related exports to the US in response to Washington's recent tech and economic curbs.



Appendix 1

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Jim Reid | Henry Allen | Rajsekhar Bhattacharyya

September 2026

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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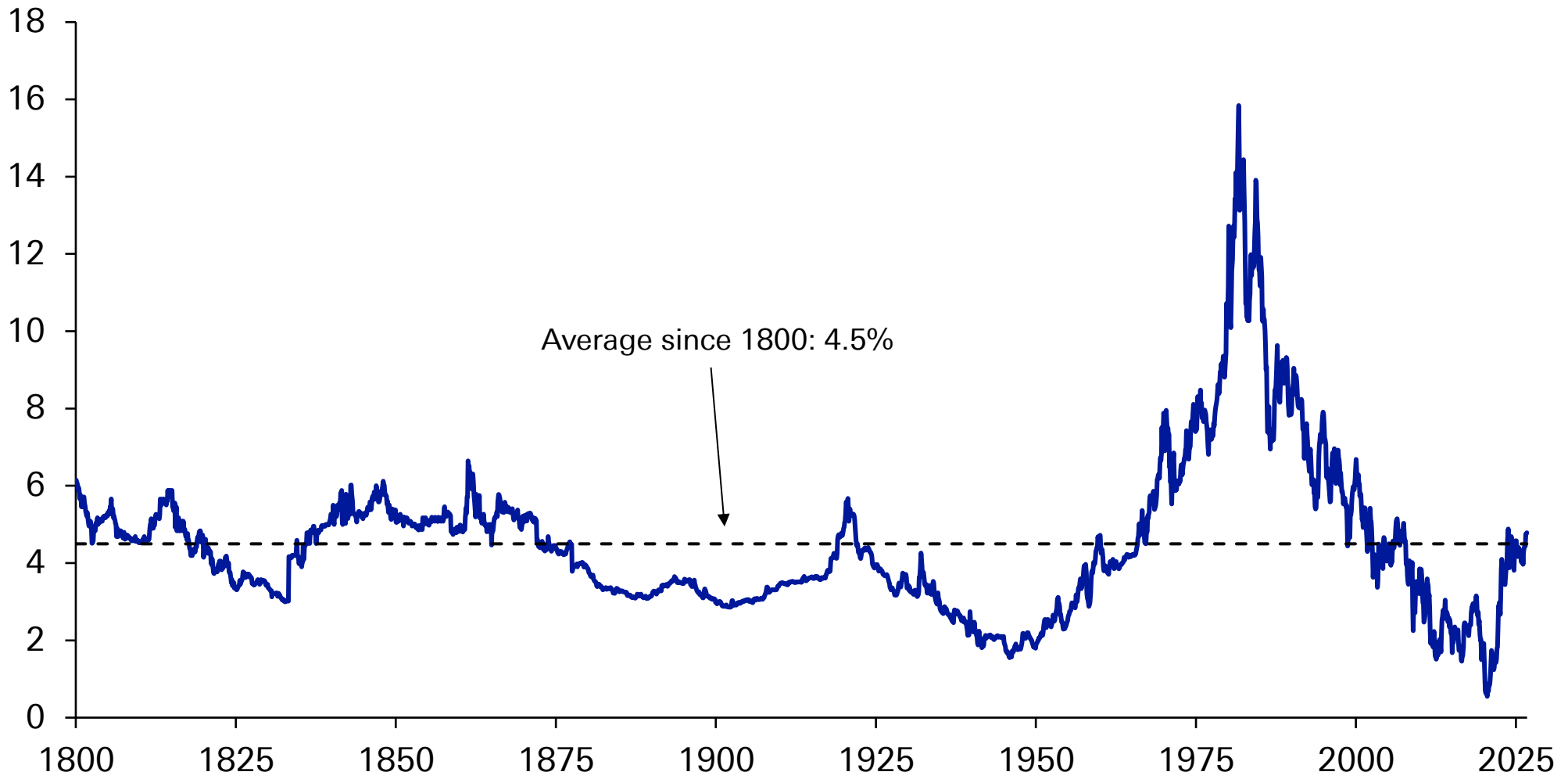


The bond sell-off in perspective...

We would see the recent rise in yields as a continuation of a long normalisation process. 10yr US Treasury yields are now “only” slightly above their long-term average yield since 1800... so they are certainly not extreme....



10-year US treasury yields

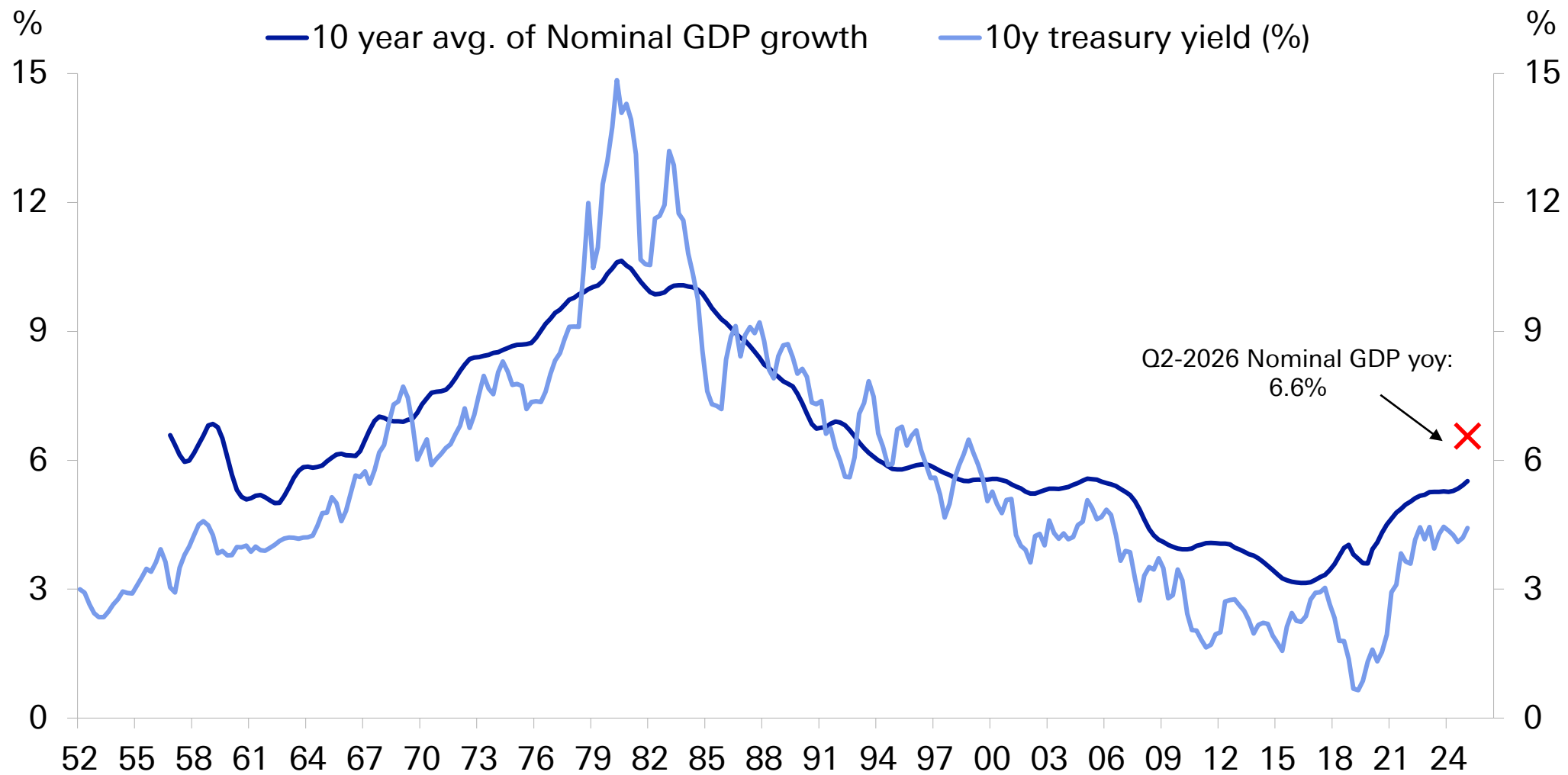


Source: Finaeon, Deutsche Bank



Since the early 1960s, 10 yr UST yields have typically tracked Nominal GDP apart from late 70s/early 80s when they soared above and in the 2010s when they were significantly below. The average gap over this period is around 70bps (NGDP>10yr USTs). 10yr average NGDP is c.5.5% so 4.8% seems broadly fair. However current NGDP is 6.6%. Do we need to factor this in?

10yr average nominal GDP growth and the 10yr Treasury yield

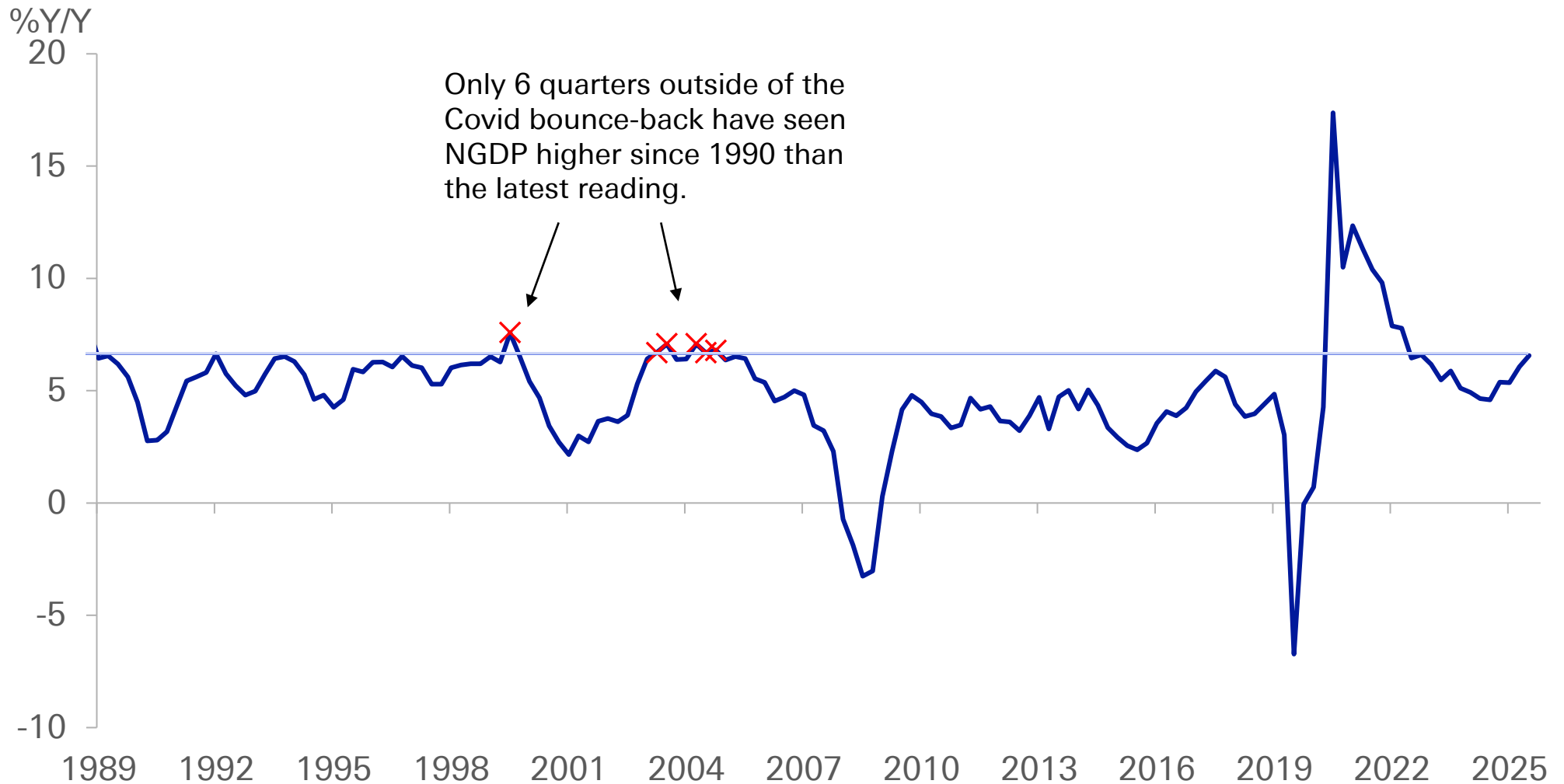


Source: BEA, FRB, Haver, Deutsche Bank

At 6.6%, current YoY US Nominal GDP is at the highest YoY rate since 2005 outside of the Covid bounce-back period. So even if not sustainable at these levels, this level of nominal growth has helped push UST yields higher....



US Nominal GDP growth with red crosses being the only time nominal GDP was higher than the current 6.6% outside of the post COVID rebound

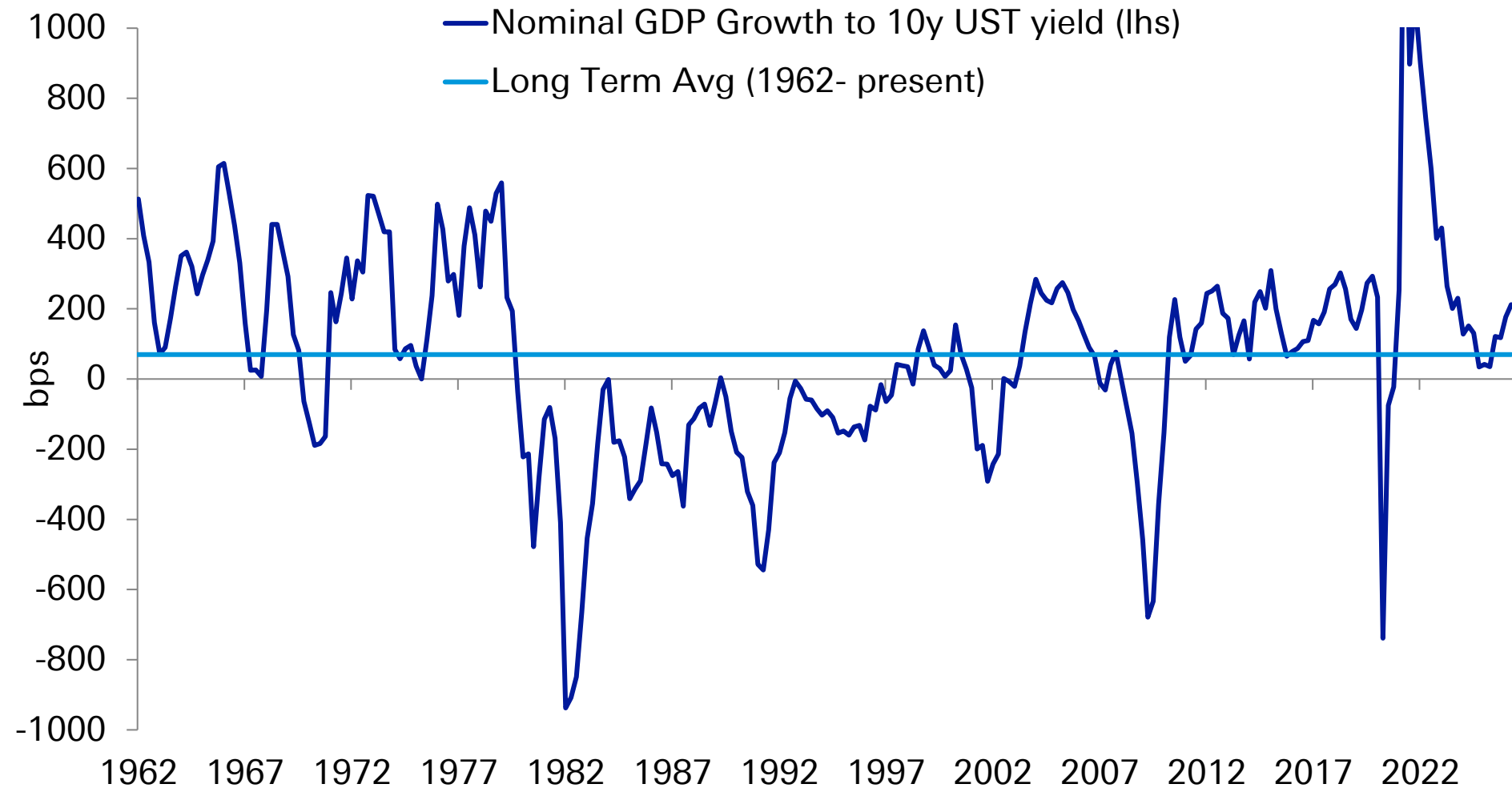


Source : BEA, Haver, Deutsche Bank



With spot Nominal GDP at 6.6% YoY, the current gap to 10yr yields is around 180bps. Clearly nominal GDP may be inflated by energy prices but overall it's hard to say US yields are too high based on this. If nominal GDP settles at 5.5% then 4.8% would be historically average. If we settle at 6% due to AI and inflation, then around 5.3% is reasonable before factoring in fiscal concerns.

US Nominal GDP minus 10yr UST yields

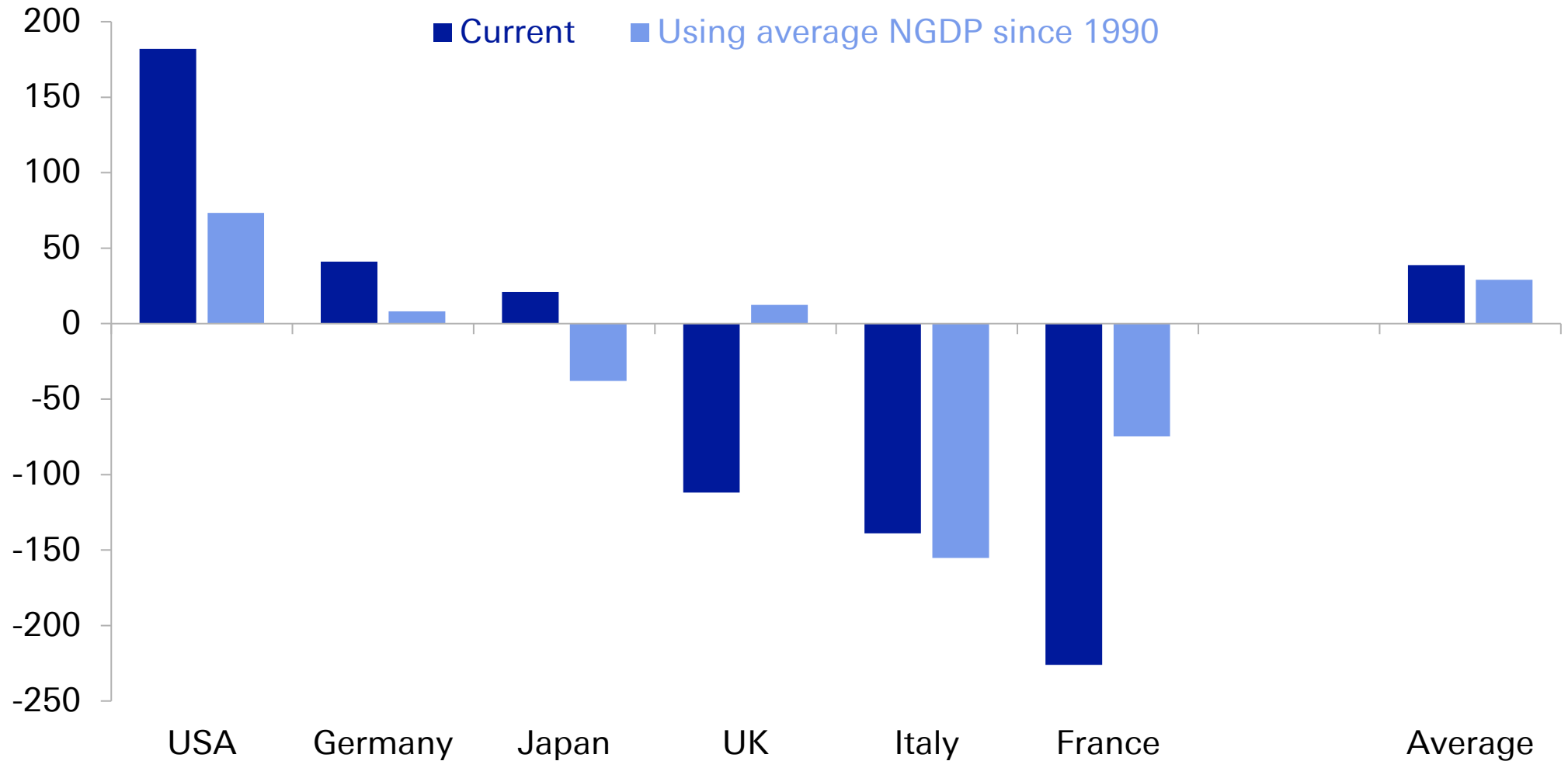


Source: Haver, Bloomberg Finance LP, Deutsche Bank



The US has the largest positive gap between current Nominal GDP and 10yr yields illustrating that growth is currently comfortably above funding costs. France, Italy and the UK are the most negative and have yields comfortably above current nominal growth.

Gap between Nominal GDP growth and 10y yields (bps)

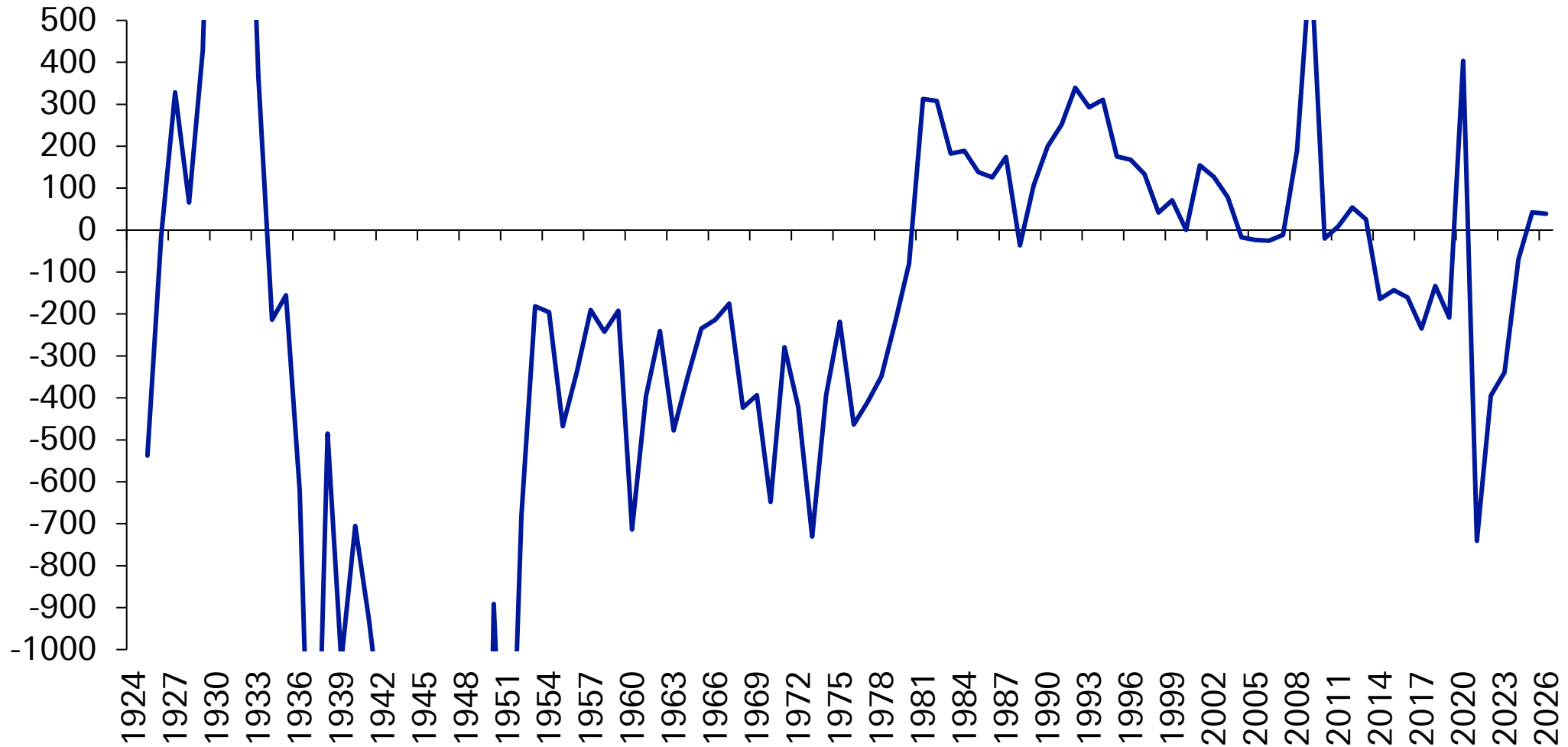


Source: Finaeon, Deutsche Bank

Across DMs, funding costs are now above nominal GDP on average. In the post GFC period of high debt and financial repression, yields were consistently below nominal GDP thus helping the funding situation....



Average difference between Nominal GDP and 10yr Govt. bonds for US, UK, Germany, France, Italy and Japan (in bps)

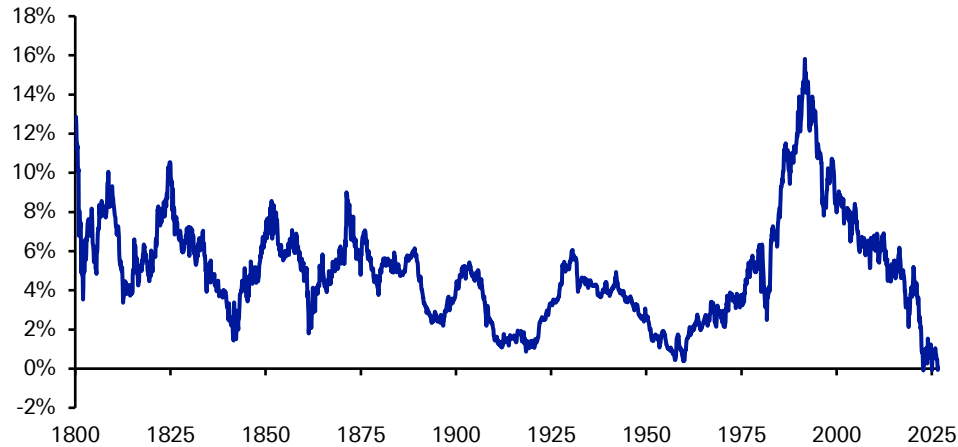


Source: Finaeon, Deutsche Bank

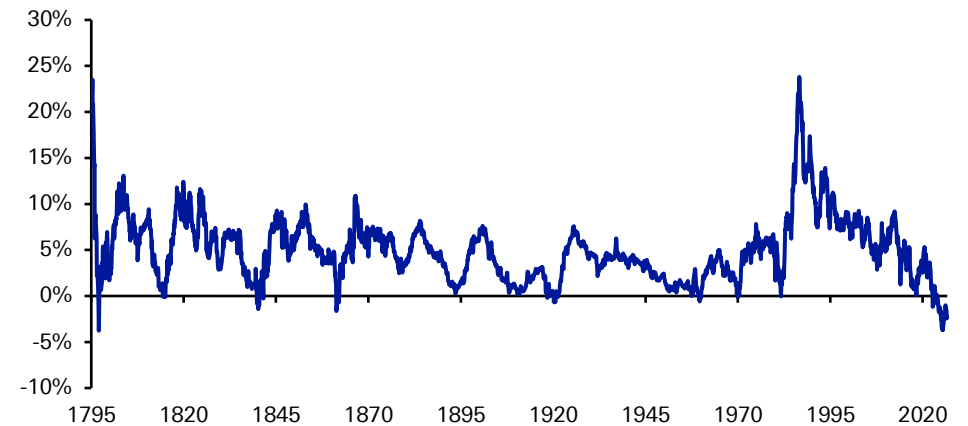


For 10yr Treasury yields, we are still hovering around the worst 5- and 10-year rolling nominal returns on record but it's clear that even with yields again approaching multi-year highs, returns have been stabilising more recently due to the higher coupons...

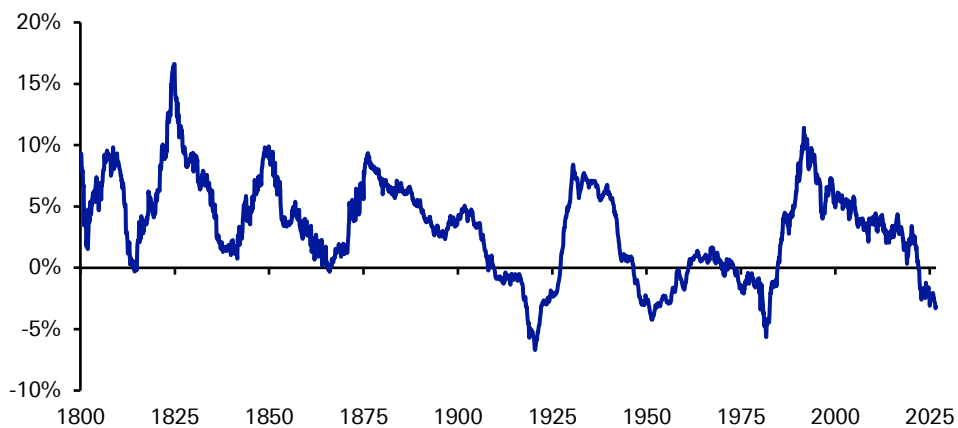
10yr Rolling Nominal Return (Annualised)



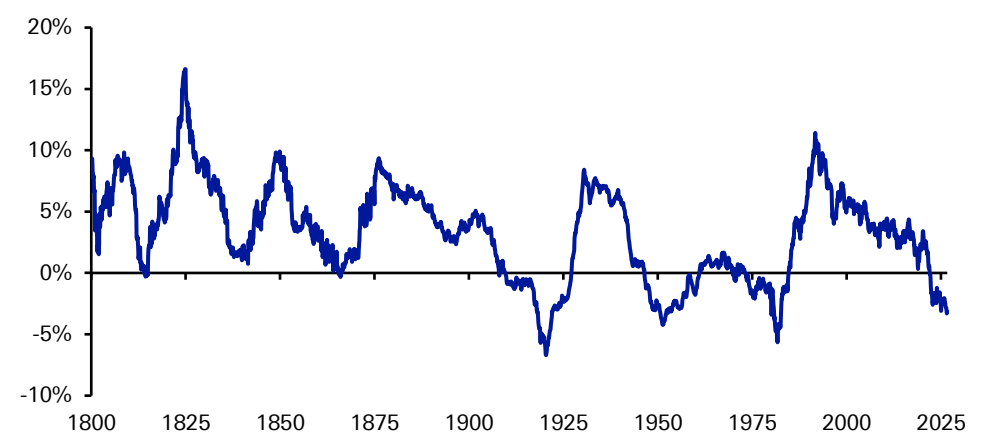
5yr Rolling Nominal Return (Annualised)



10yr Rolling Real Return (Annualised)



5yr Rolling Real Return (Annualised)

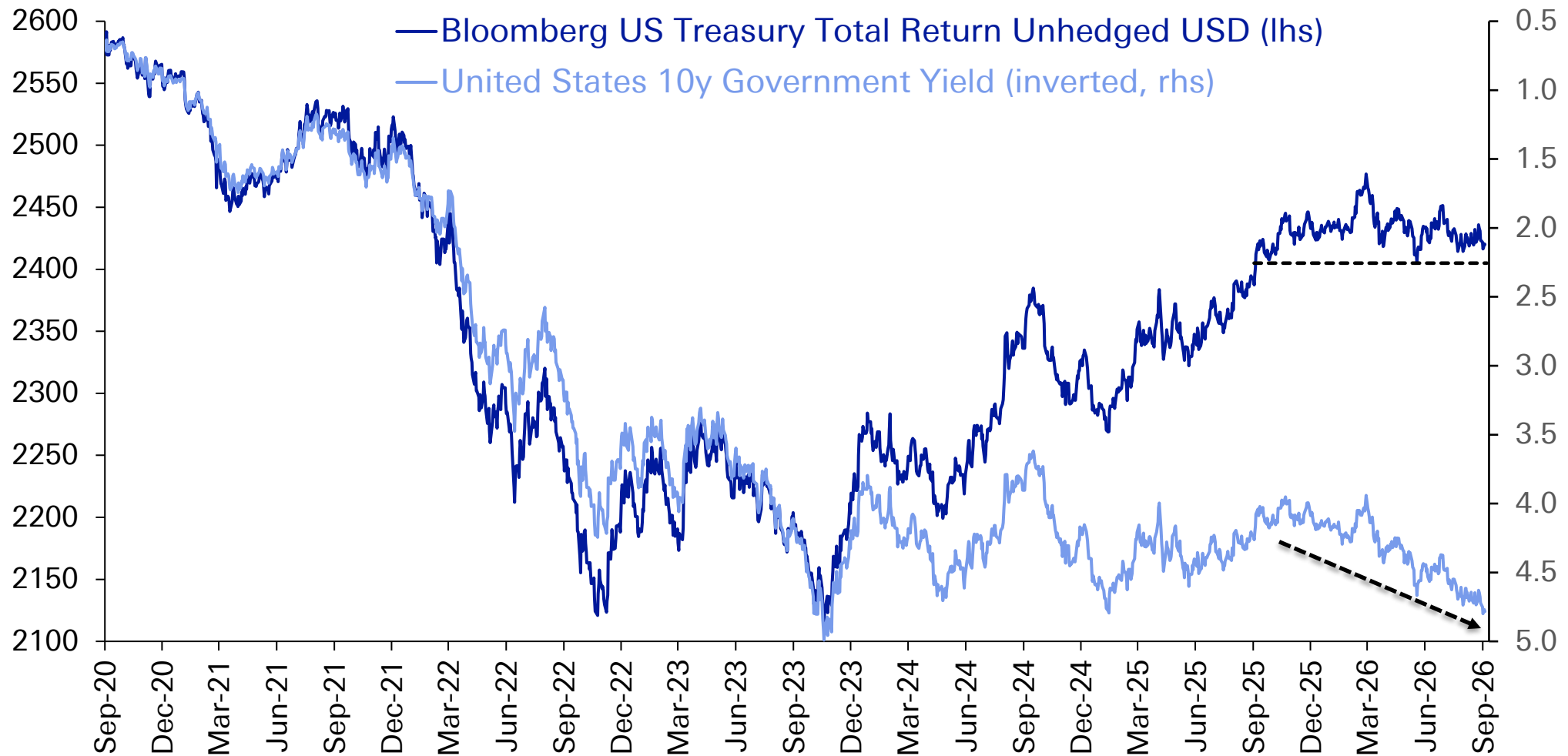


Source: Finaeon, Deutsche Bank



At least we're getting a decent compensatory yield now versus the 2010s and early 2020s. Over the last 12 months the US treasury total return index is higher even with a c.60bp rise in 10yr yields.. Since yields peaked in October 2023, the index has returned nearly 15%

US Treasury Index Returns vs. 10y Govt. Yields. Over the last year returns have been positive even with a notable rise in yields.



Source : Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7

Clearly it's a global phenomenon. For example, 30-year yields are hitting decade plus or multi decade highs in several countries.



30yr Gilt Yields (%)



30yr Treasury Yields (%)



30yr JGB Yields



30yr German Yields (%)

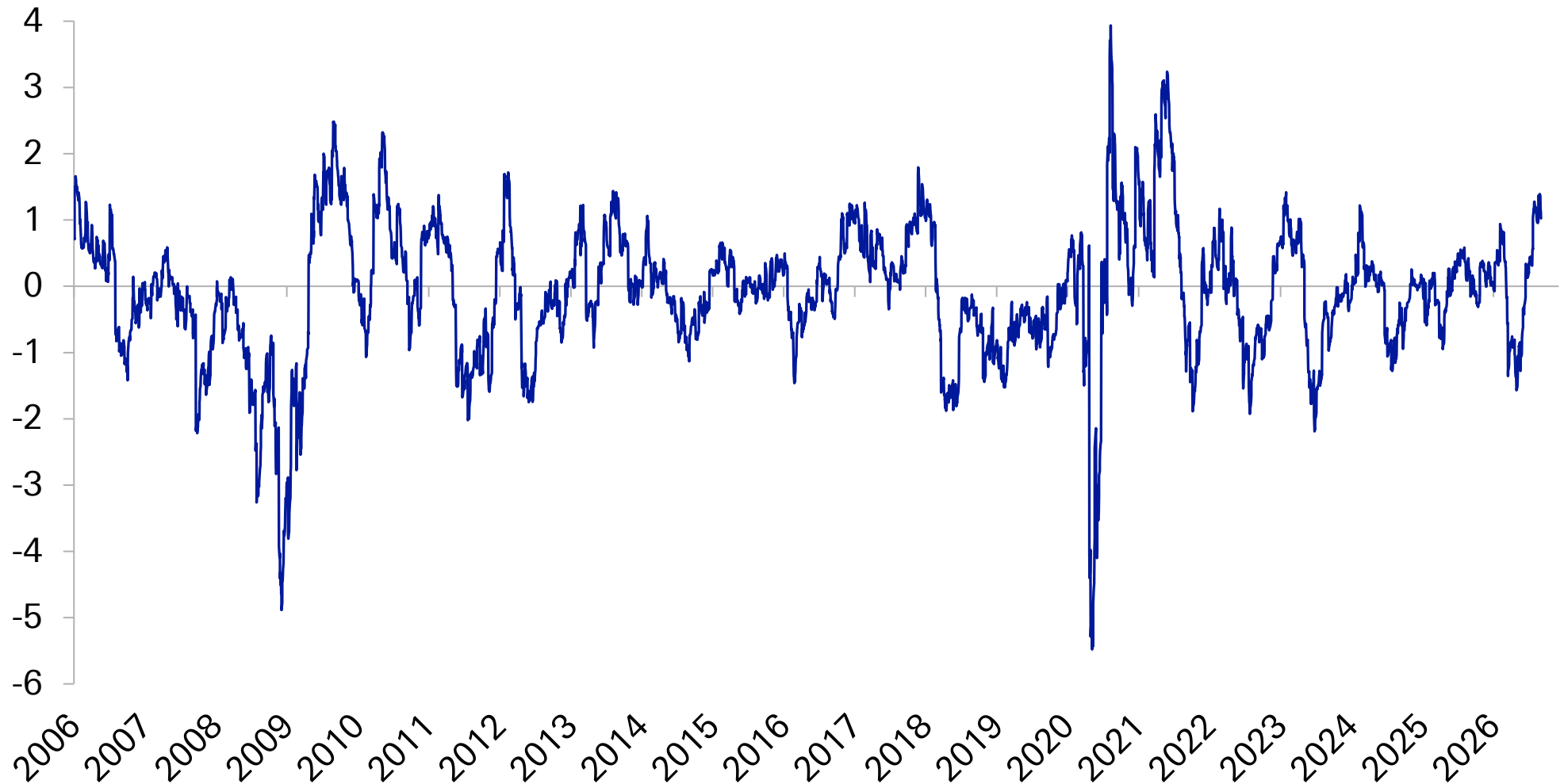


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

But some of this is justified by the data. For example, European growth surprises are nudging up against 20-year highs outside of the post Covid and GFC bounce-backs.



Deutsche Bank SIREN Surprise Index - Euroarea

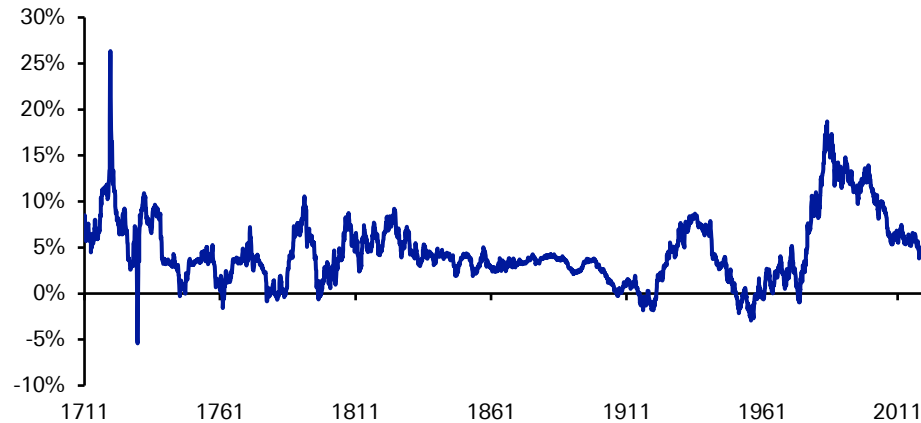


Source: Bloomberg Finance LP, Deutsche Bank; Last update : Sep 7

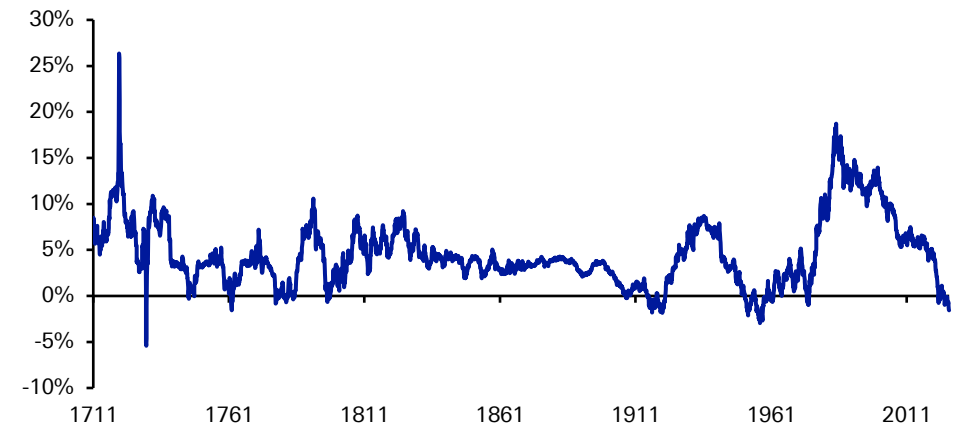
10yr UK gilts have also had a tough time in recent years. Rolling 5 and 10yr total returns are still near the lows..



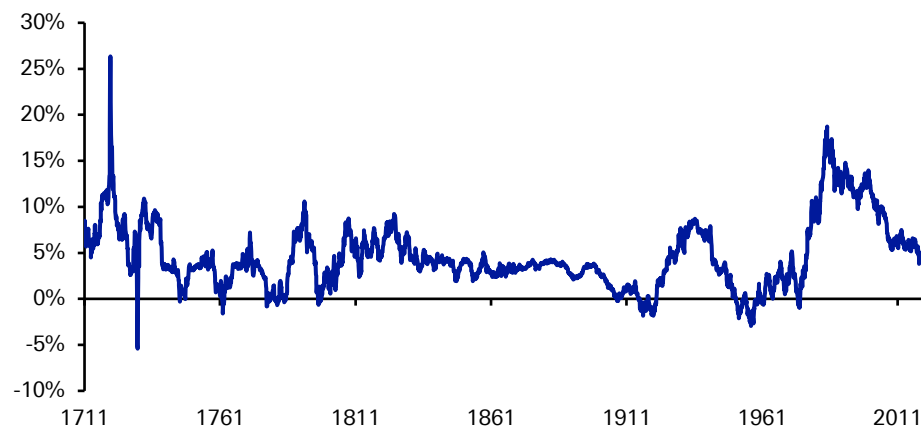
10yr Rolling Nominal Return (Annualised)



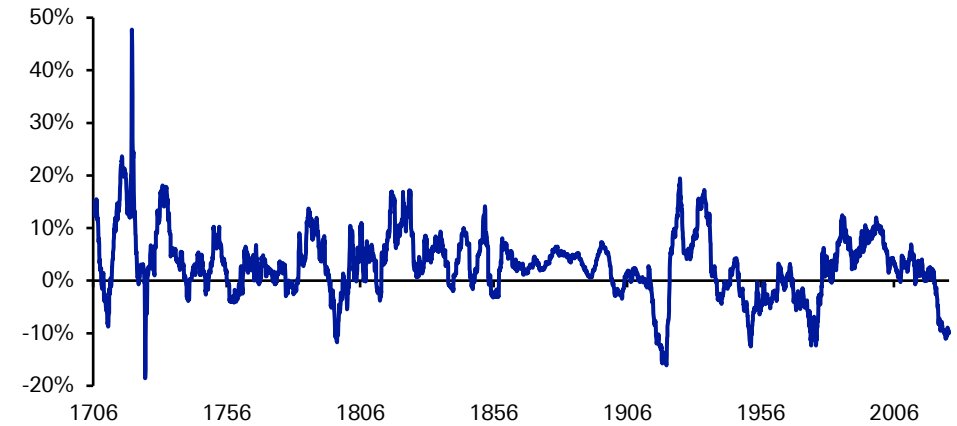
5yr Rolling Nominal Return (Annualised)



10yr Rolling Real Return (Annualised)



5yr Rolling Real Return (Annualised)

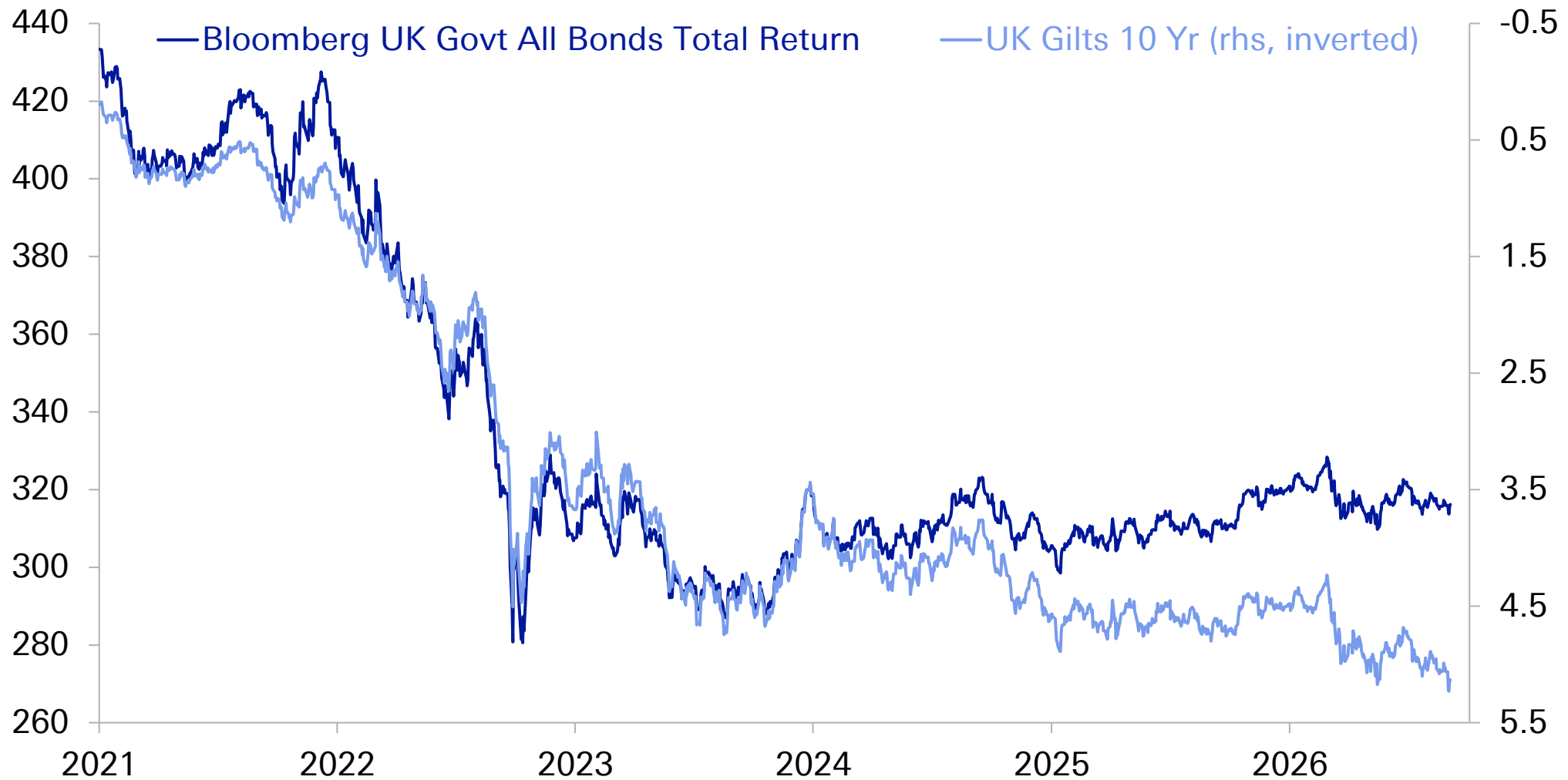


Source: Finaeon, Deutsche Bank

But even with all the negative headlines in the four years since the Truss mini-budget/LDI crisis, Gilt returns have generally crept higher even as yields have returned back to the highs. 10yr yields are currently around 65bps higher than their “crisis” peak in October 2022, yet the total return index is around 12% higher..



UK Gilt Index Returns vs. 10y Govt. Yields.

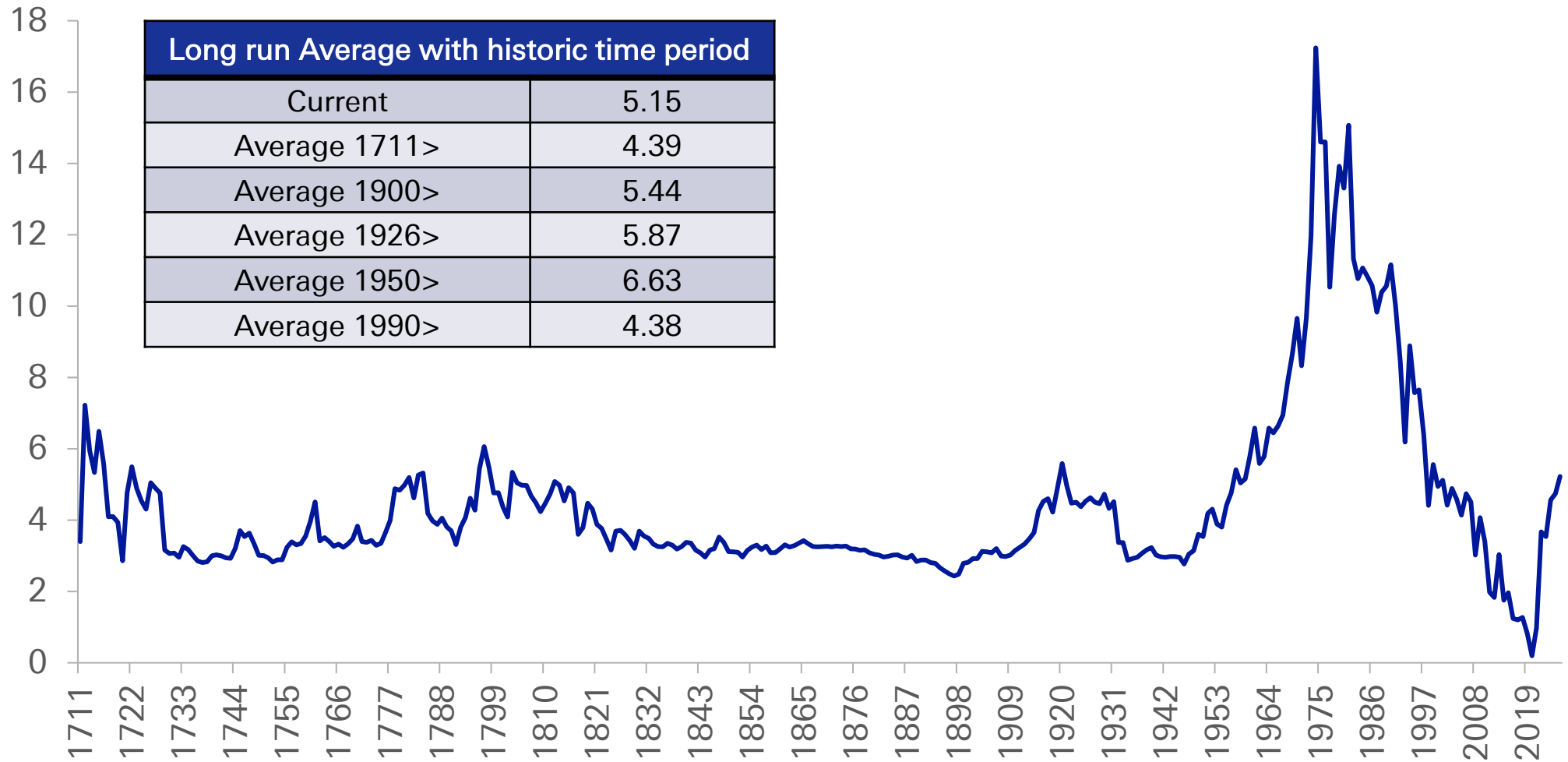


Source : Bloomberg Finance LP, Deutsche Bank, last update Sep 7

What average yield should you use in the UK? Over the long-run (since 1711- and since 1900-) the median is around one percentage point less than the average which is heavily influenced by 1960-1997 where 10yr yields were permanently above 5.5%. Outside of that period the current yield is at the top end of the historical range.



10yr govt yields in UK

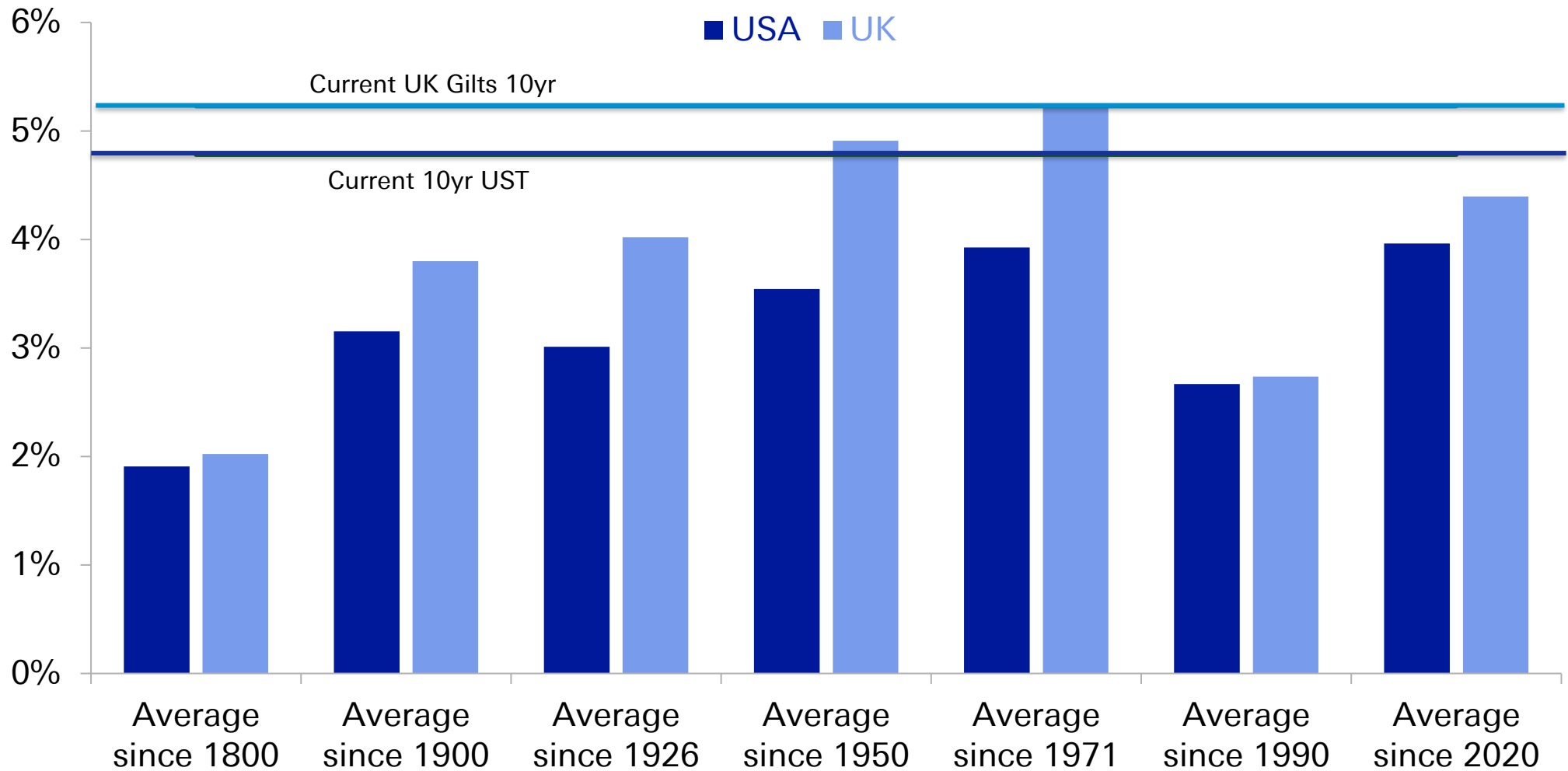


Source: Finaeon, Deutsche Bank

At least US and UK Bond yields are mostly above average inflation seen over virtually all extended periods through history...



Average inflation over various periods in the US and UK and the current 10 year bond yield

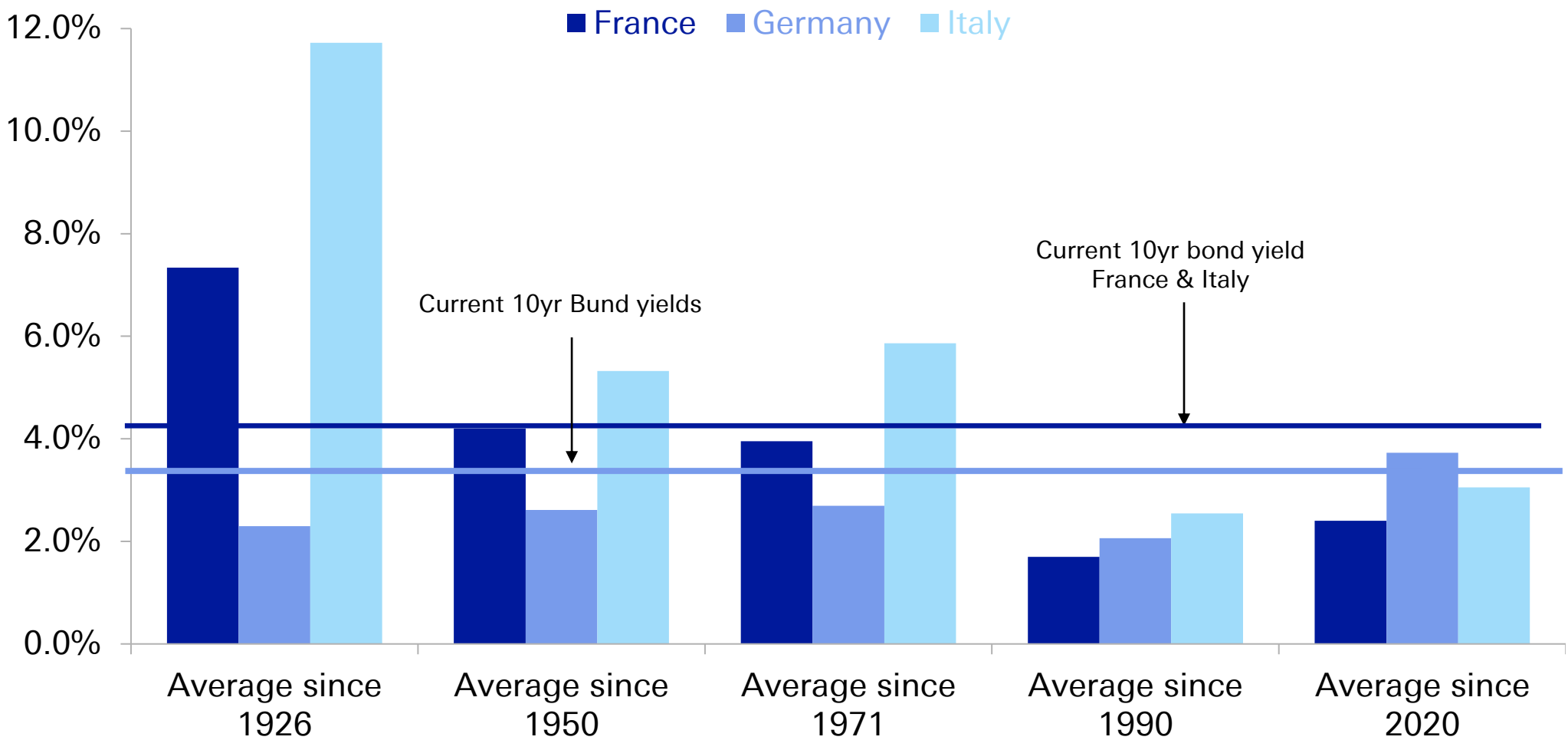


Source: Finaeon, Deutsche Bank

WWII makes this more complicated for European countries, as do lower current yields.



Average inflation over various periods in the France, Germany and Italy and the current 10-year bond yield

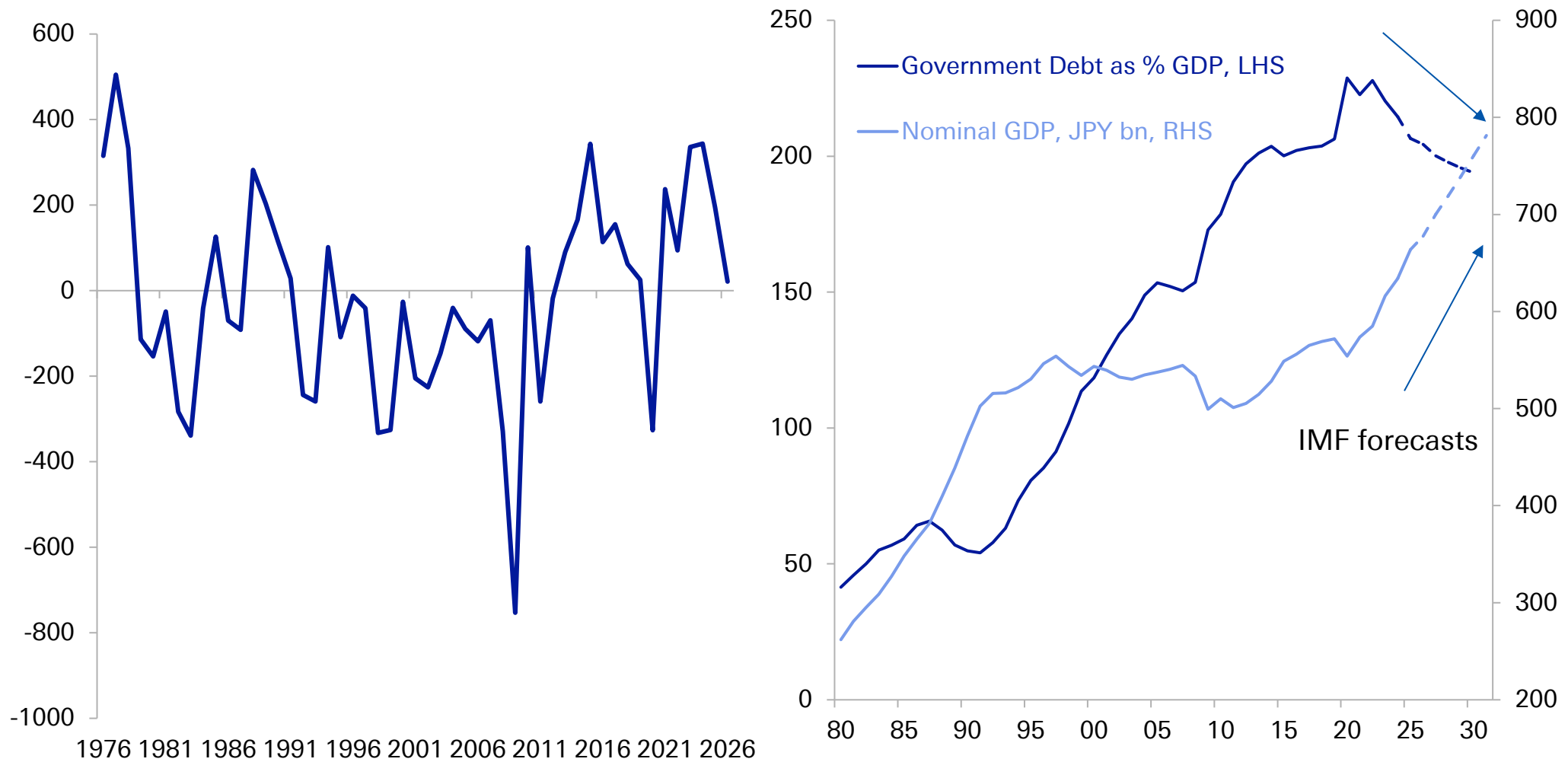


Source: Finaeon, Deutsche Bank

For Japan, rising yields and dipping nominal GDP do threaten what has been a falling debt/GDP since Covid when nominal GDP took off after being broadly flat for a couple of decades. Since 2021, Japanese nominal GDP has averaged over 220bps more than 10yr JGB yields. Today that stands at only around 40bps on a spot basis.



Japan: nominal GDP growth & 10yr JGB spread (left) & nominal GDP vs Govt debt (right)

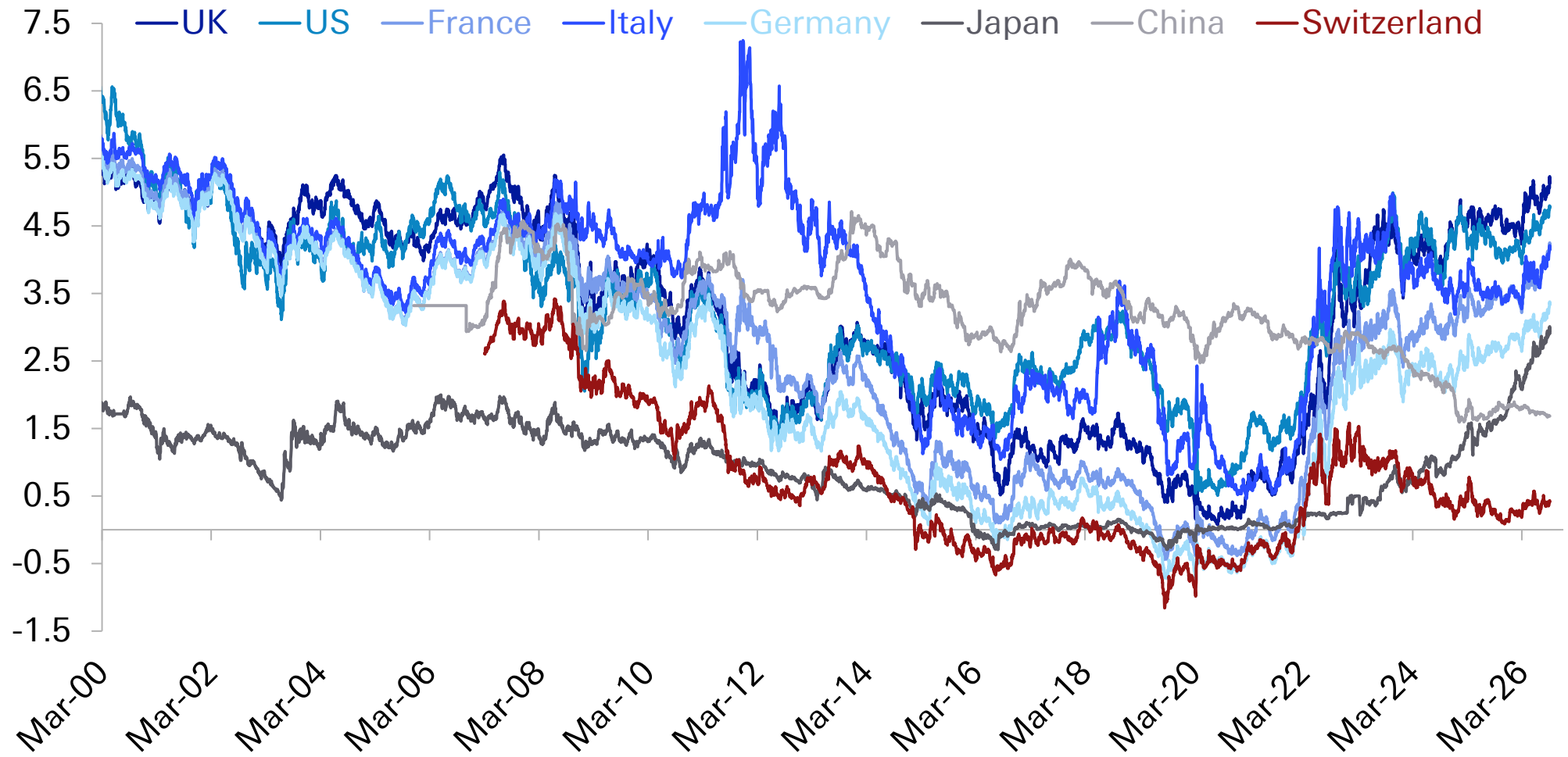


Source: Finaeon , Deutsche Bank Research, Haver Analytics, IMF

Interestingly, while most global yields hit multi-decade highs, Swiss and China yields have been flat or moving lower...



10y Government yields across major economies (%)

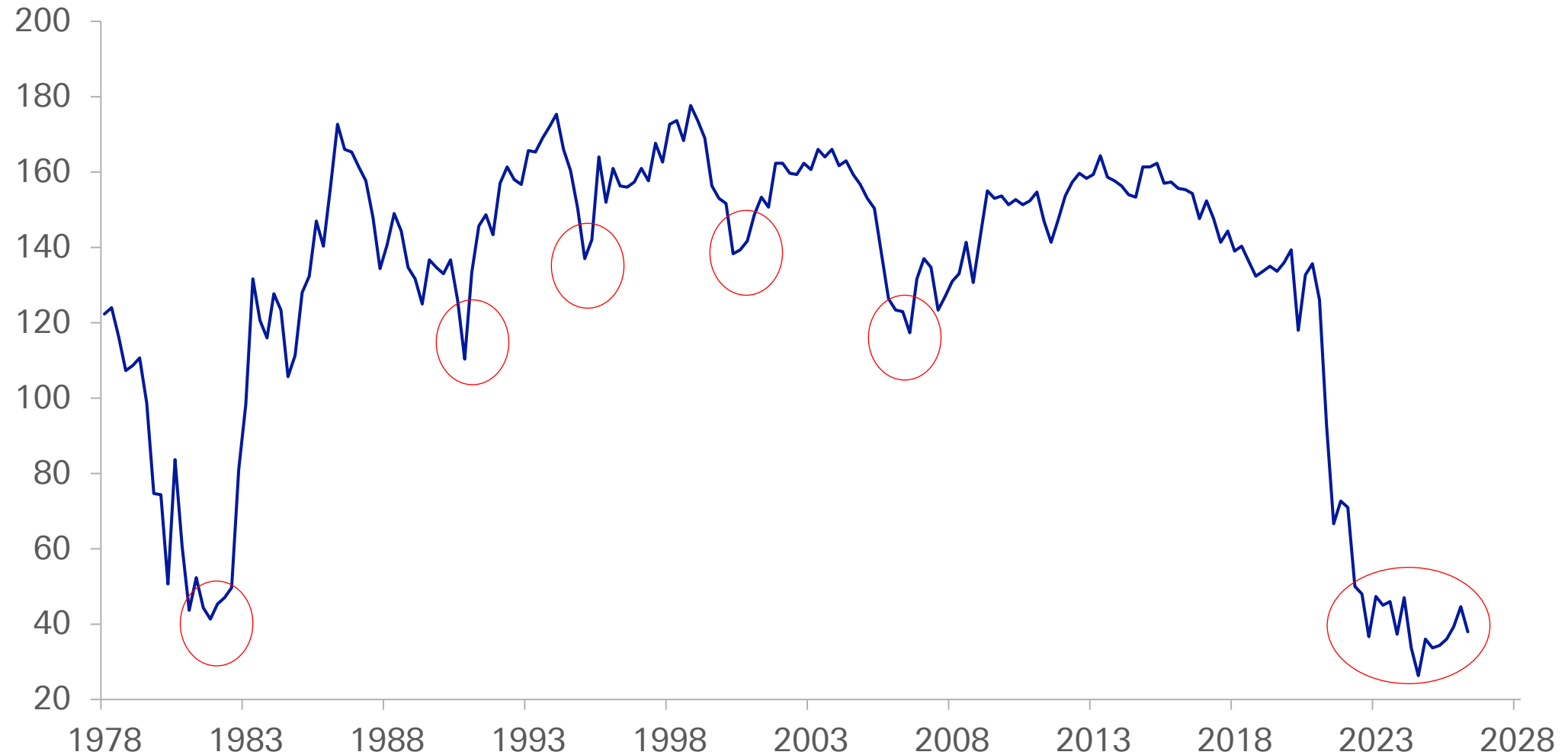


Source : Bloomberg Finance LP, Deutsche Bank, last update Sep 7, Note: Countries are ordered by current yield from high to low

Rising yields make the fiscal arithmetic more challenging but also continue to keep US housing affordability near record lows... Full employment and a lack of activity are the main thing holding prices up



UMICH: Current Conditions for Buying Houses (Index Score): sentiments hit rock bottom for buyers



Source: UMich, Haver, Deutsche Bank

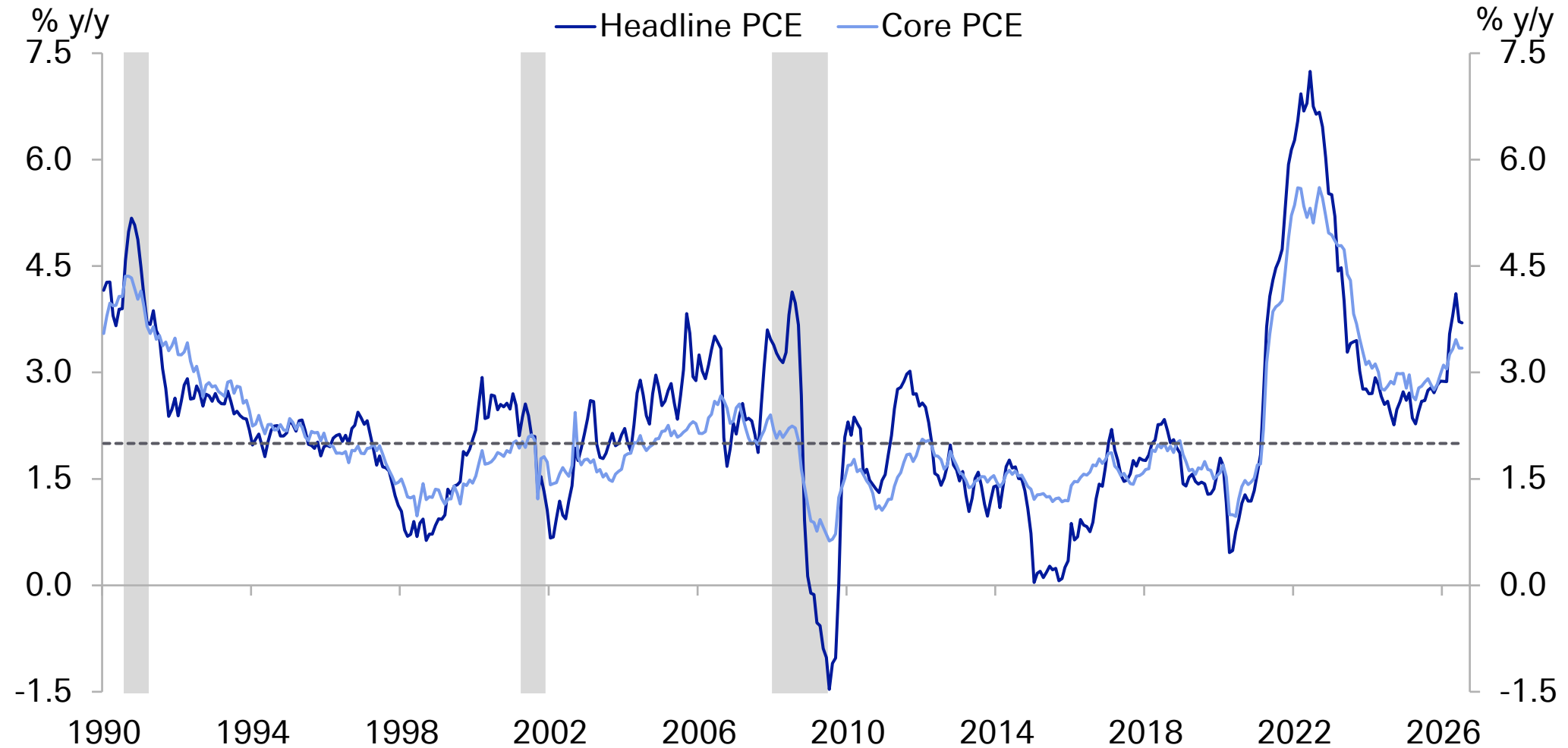


Should the Fed raise rates...?

PCE inflation remains elevated far above the Fed's 2% target. Excluding the pandemic anomaly, core PCE is only 15bps off its 34-year high seen in May



US PCE inflation

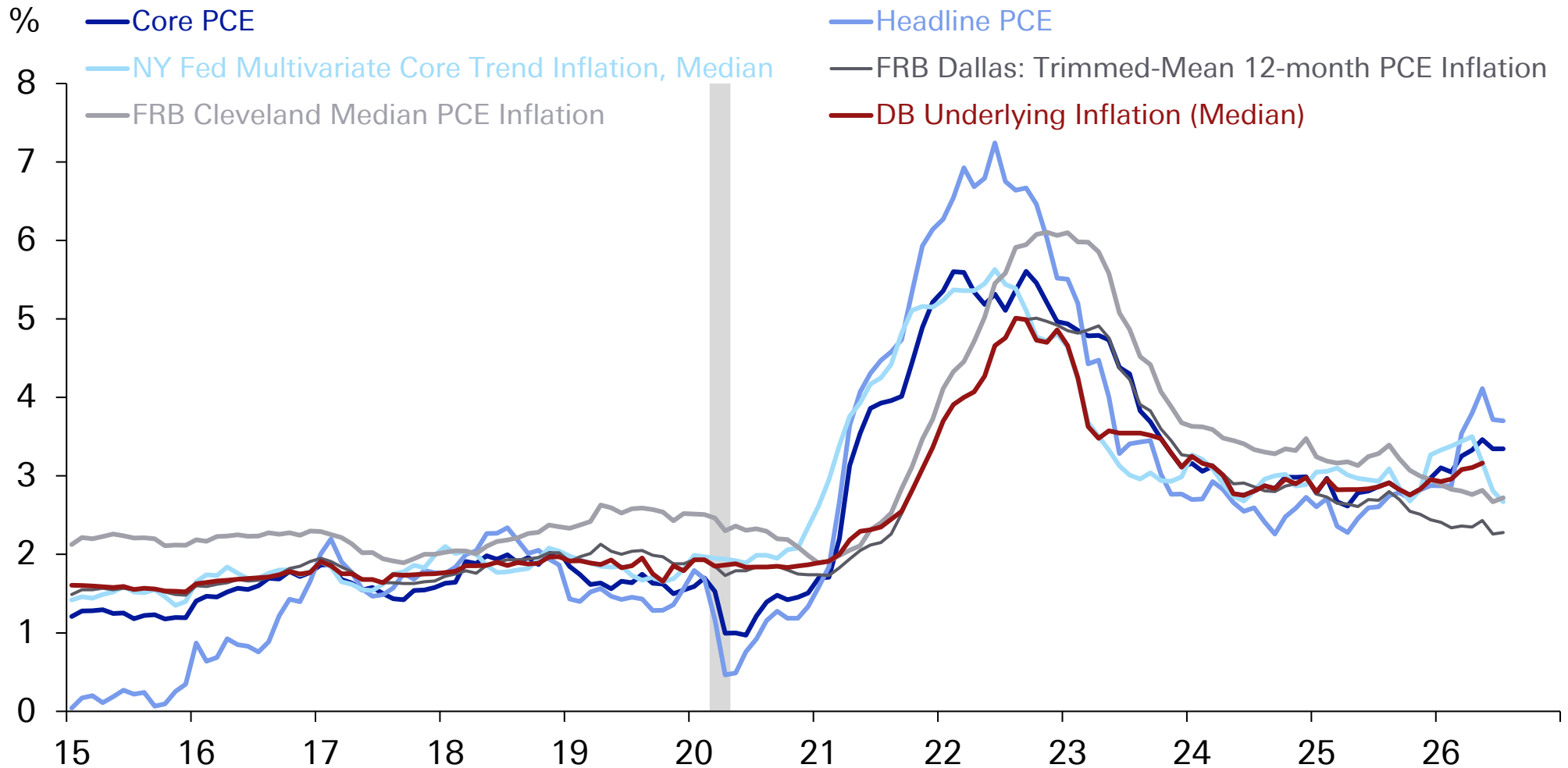


Source: BEA, Haver Analytics, Deutsche Bank

On most measures, trend inflation seems to be sticky at around 3%....



Alternative measures of trend PCE inflation

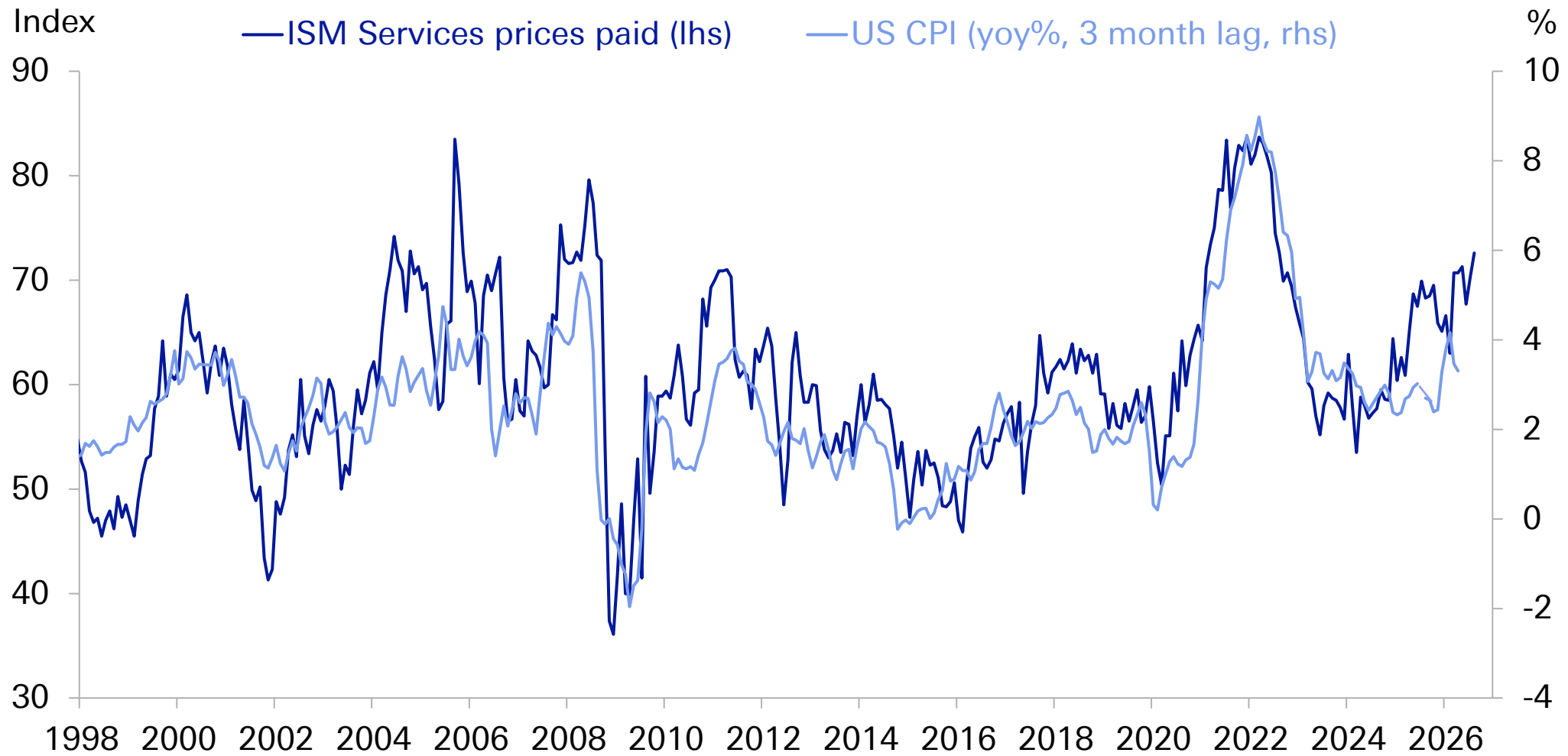


Source: BEA, FRB NY, FRB Dallas, FRB Cleveland, Haver, Deutsche Bank

Meanwhile, the ISM services prices paid component has been tightly correlated to inflation with a 3-month lag in the past... it's at a 4-year high and was consistent with CPI above 5% in the last inflation wave... While the correlation shouldn't be taken too literally, it highlights that price pressures remain in the pipeline...



US CPI Inflation and ISM Services Prices Paid

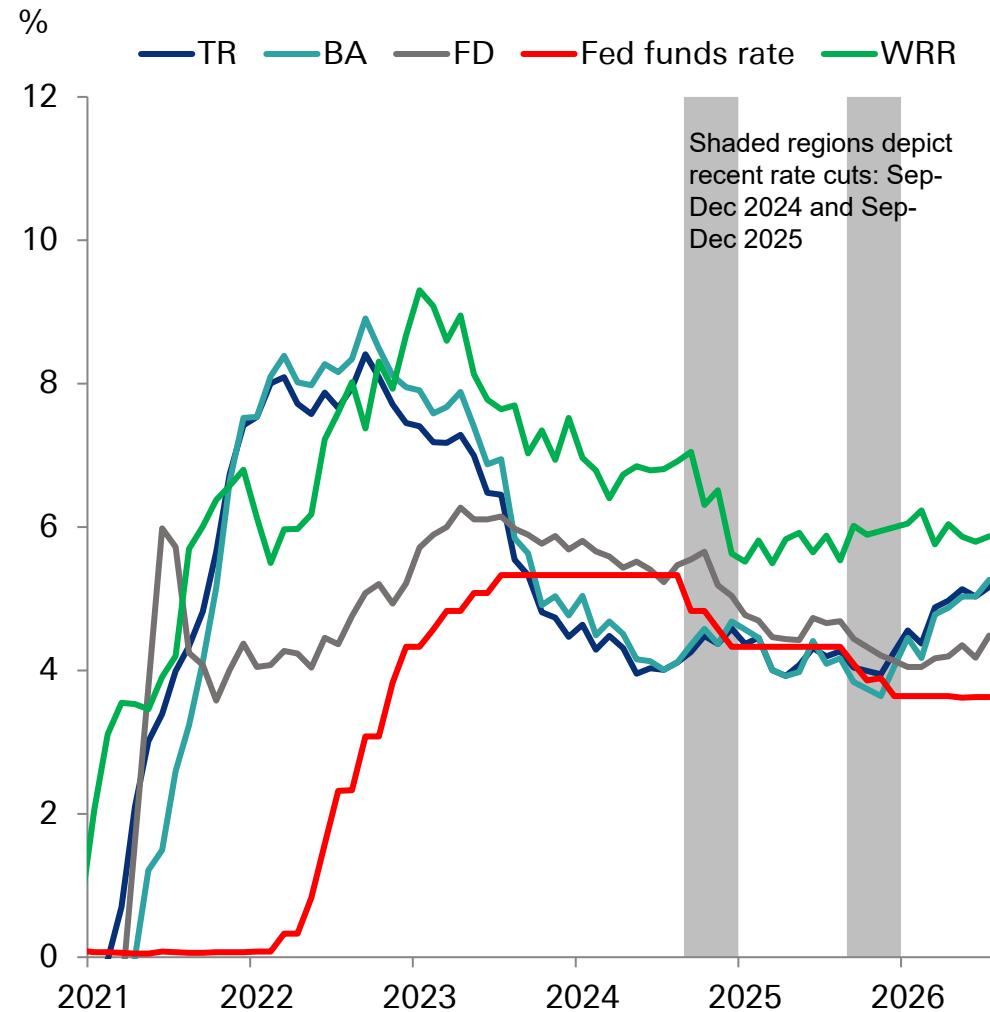


Source: ISM , BLS, Haver Analytics, Deutsche Bank

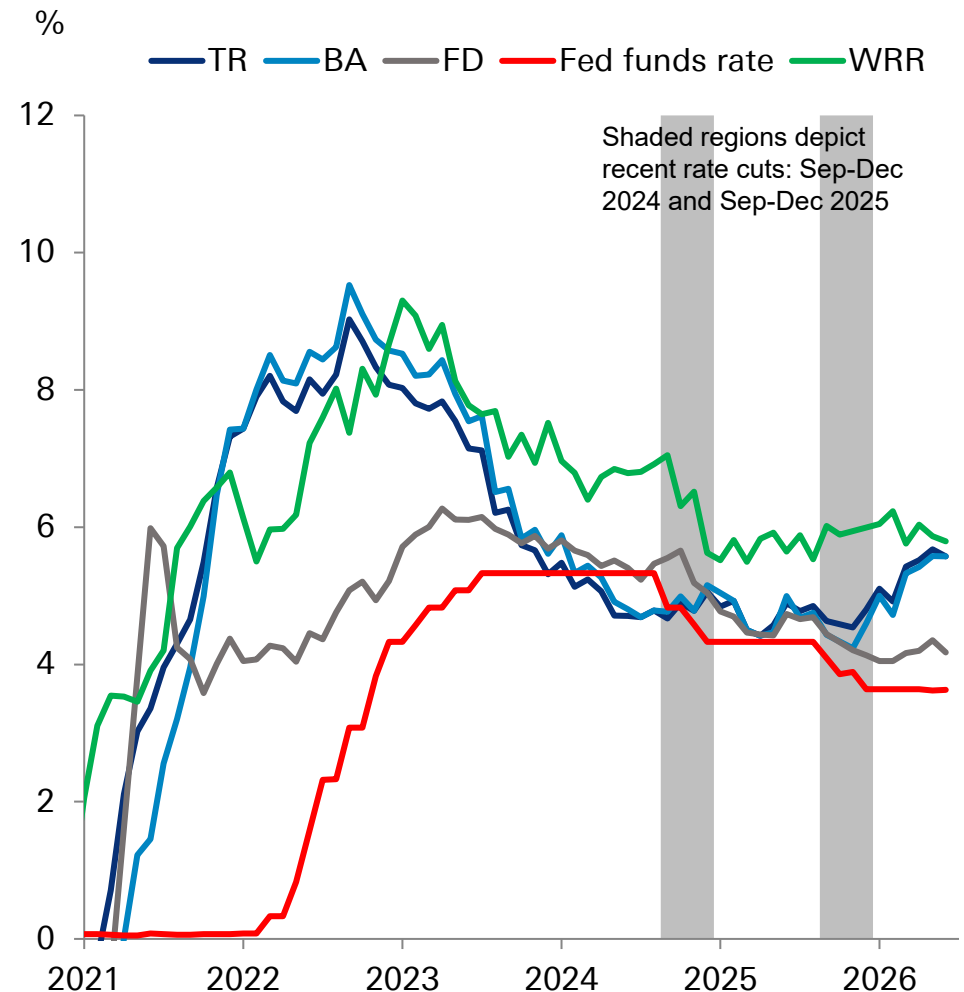
The policy rate has moved comfortably below policy rules after last year's "insurance" cuts



Fed funds rate versus rules w/ Fed r-star



Fed funds rate versus rules w/ DB r-star

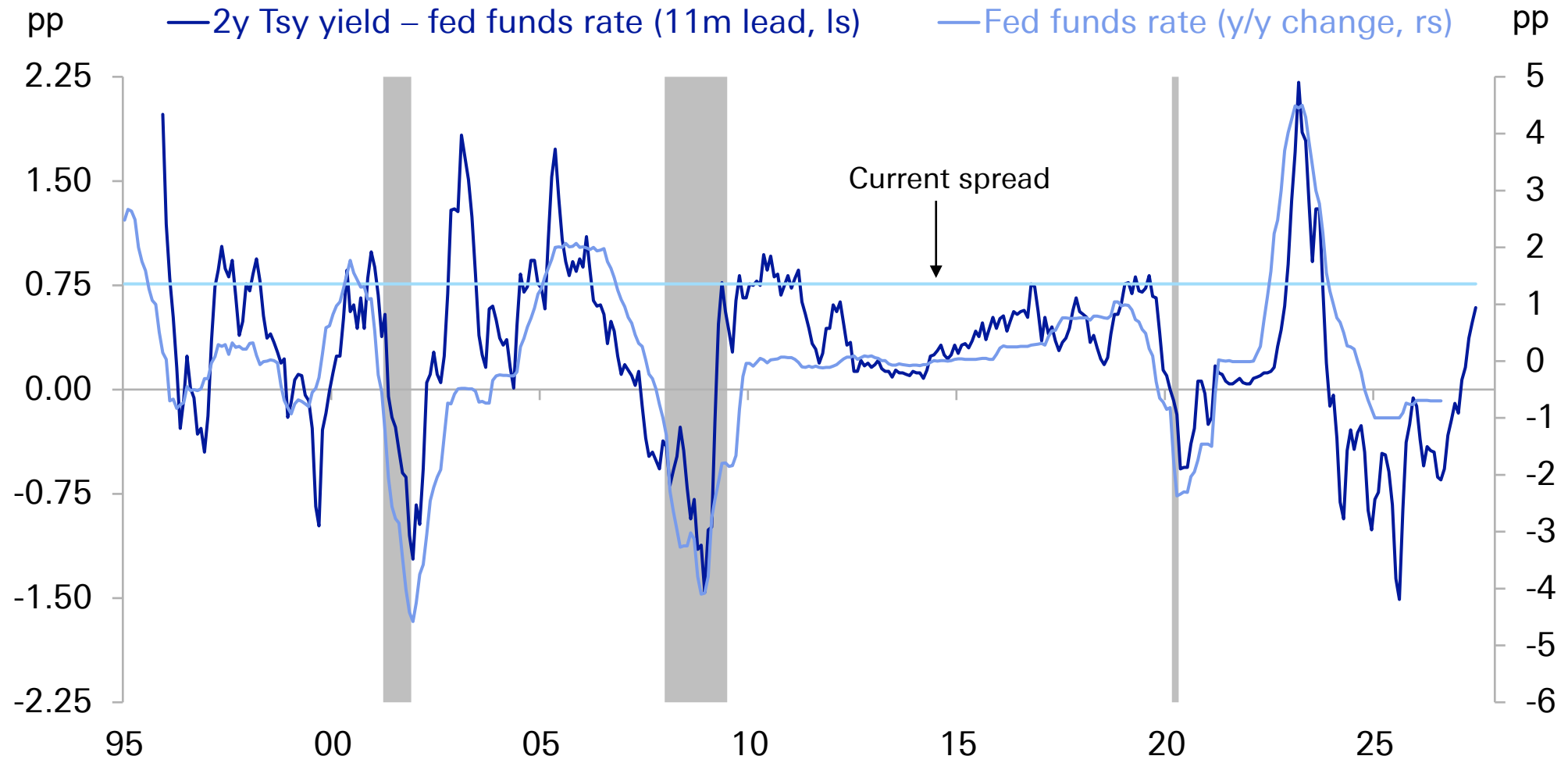


Source: FRB, BLS, CBO, Haver Analytics, Deutsche Bank

The wide gap between 2y yields and fed funds rate also predicts rate hikes



2y yield – fed funds rate and future rate changes

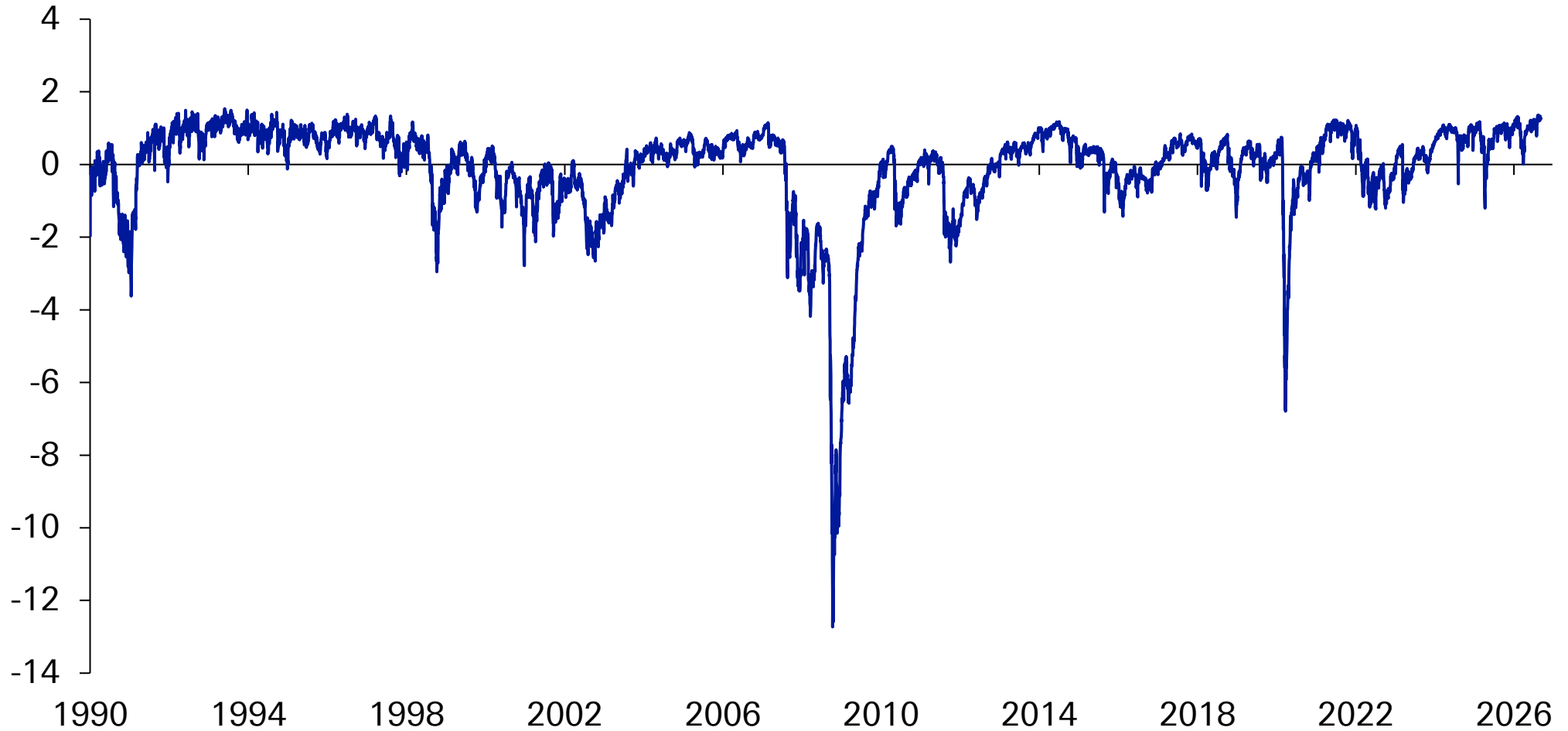


Source: FRB, Haver Analytics, Deutsche Bank

Financial conditions are broadly bumping up against the loosest they've been since 1990...



Bloomberg United States FCI

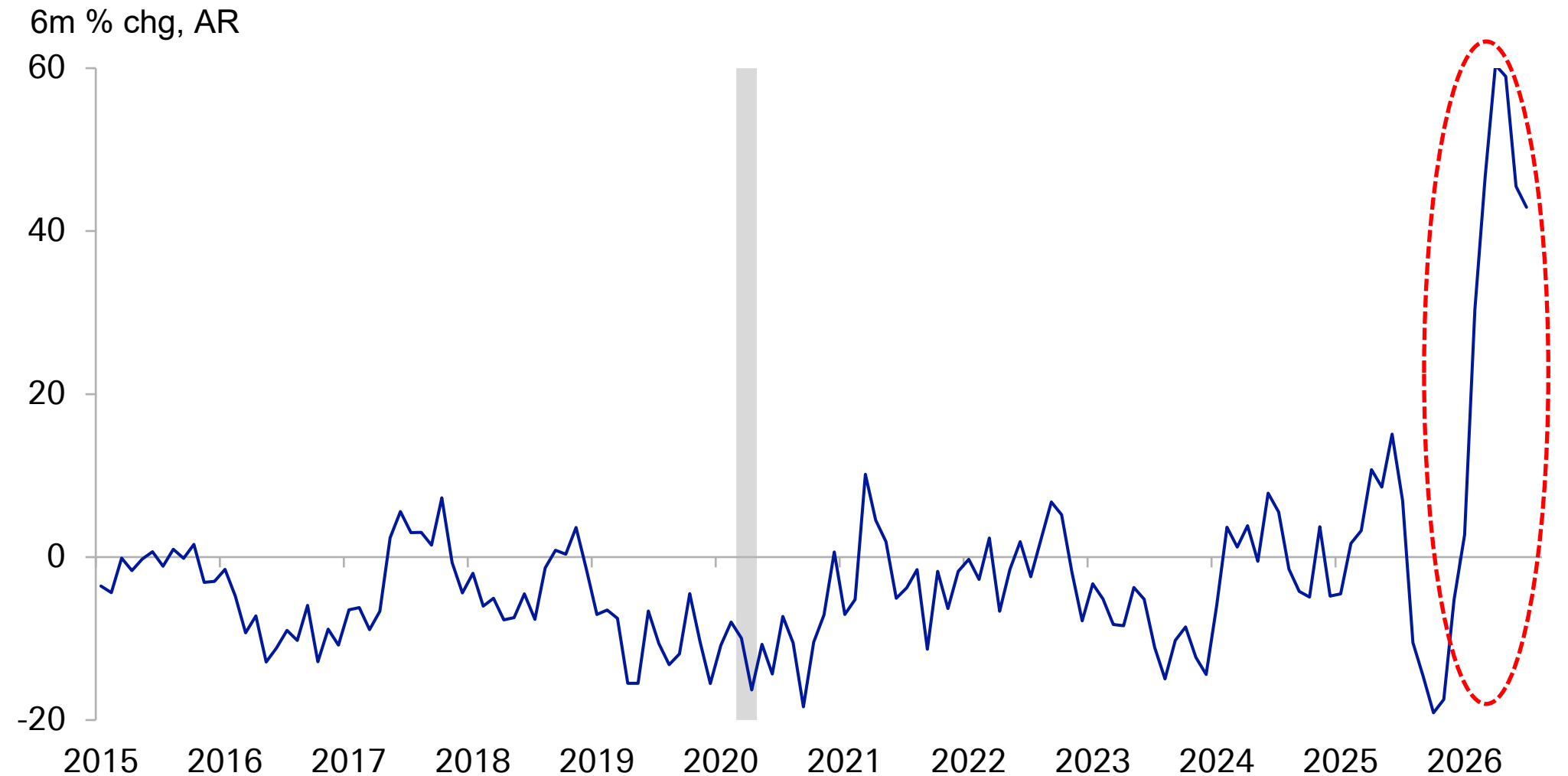


Source: Bloomberg Finance LP, Deutsche Bank; Last update : Sep 7; Note: Bloomberg U.S. Financial Conditions Index tracks the overall level of financial stress in the U.S. money, bond, and equity markets to help assess the availability and cost of credit. A positive value indicates accommodative financial conditions, while a negative value indicates tighter financial conditions relative to pre-crisis norms.

AI is an inflationary force today



PCE: software and accessories inflation

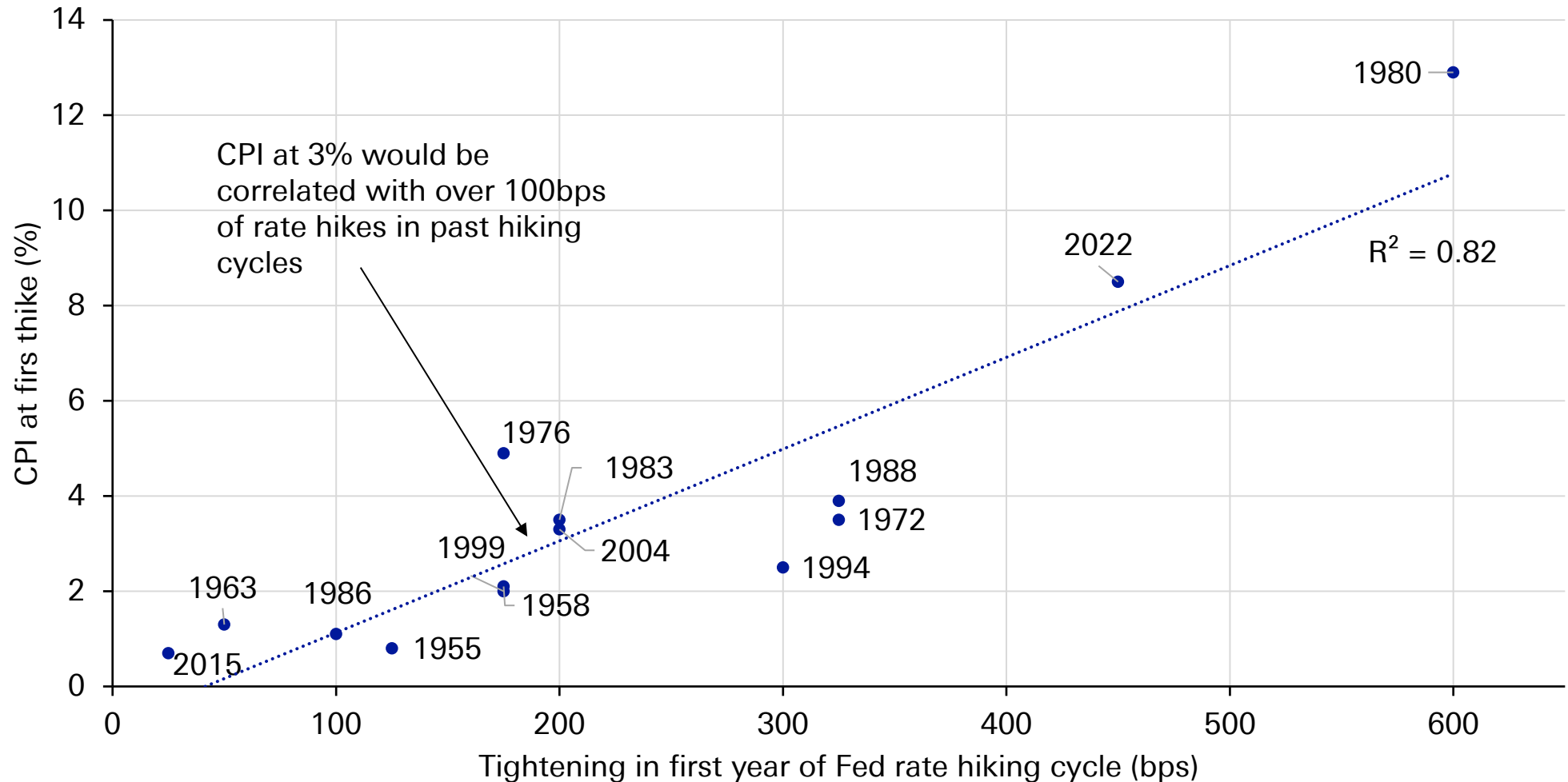


Source: BEA, Haver Analytics, Deutsche Bank



Historically, there's been a tight correlation between CPI inflation when the Fed starts hiking and how much they hike in the first year... with CPI at 3.4% in July, that would correlate to well over 100bps today... far above all expectations...

CPI and the amount of Fed tightening in the first year of a hiking cycle

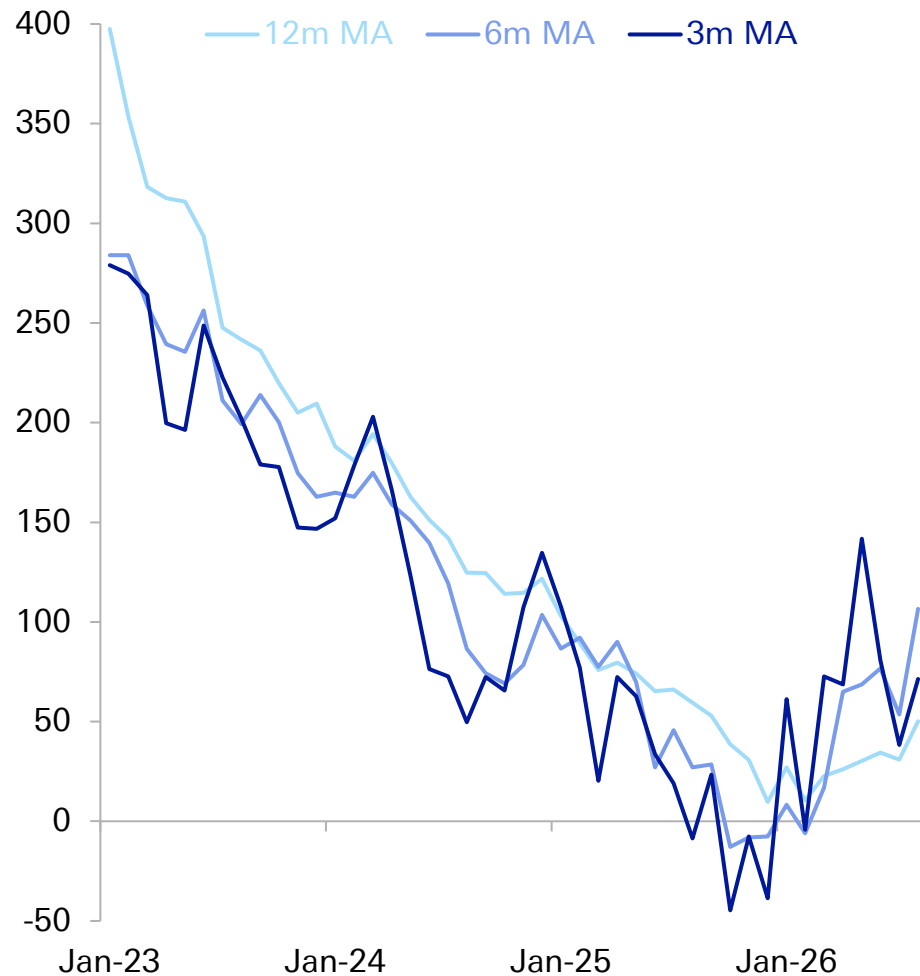


Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank

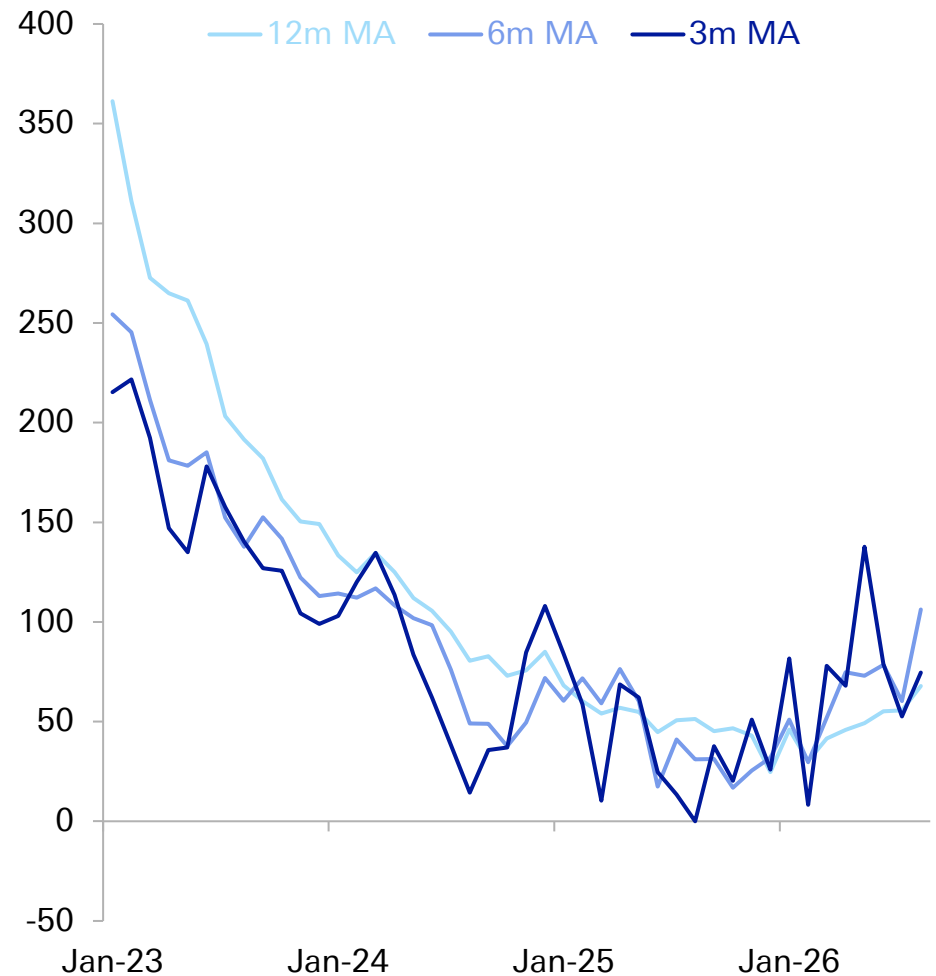
Payrolls did slow in 2025 which encouraged the insurance cuts but although volatile, there are clear signs that they have trended up in 2026.



Smoothed monthly gains in nonfarm headline payrolls



Smoothed monthly gains in private nonfarm headline payrolls

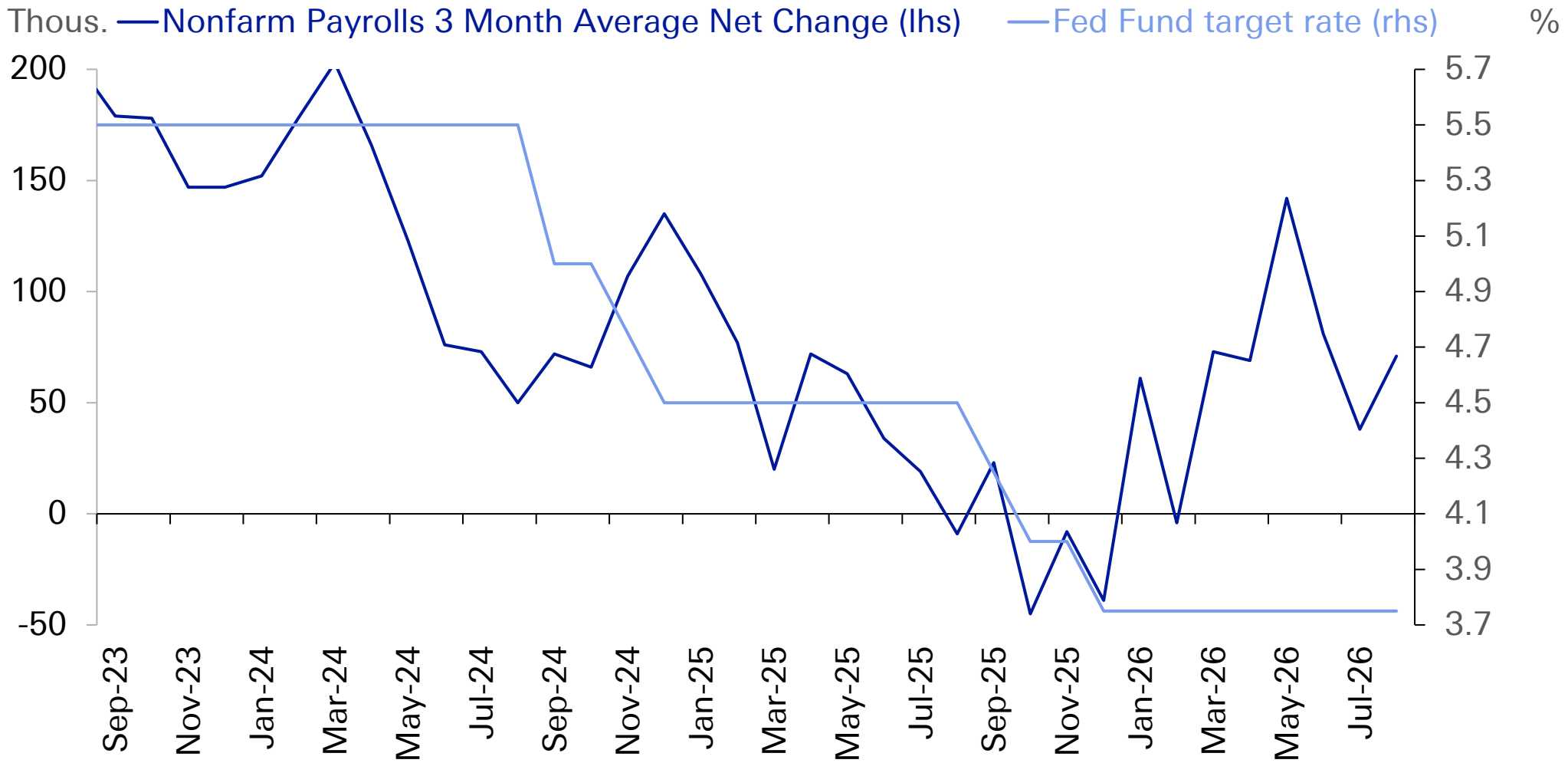


Source: BLS, Haver, Deutsche Bank

Indeed, with payrolls turning higher, those insurance cuts look ripe to be reversed...



Nonfarm payroll vs. Fed funds



Source: Bloomberg Finance LP, Deutsche Bank

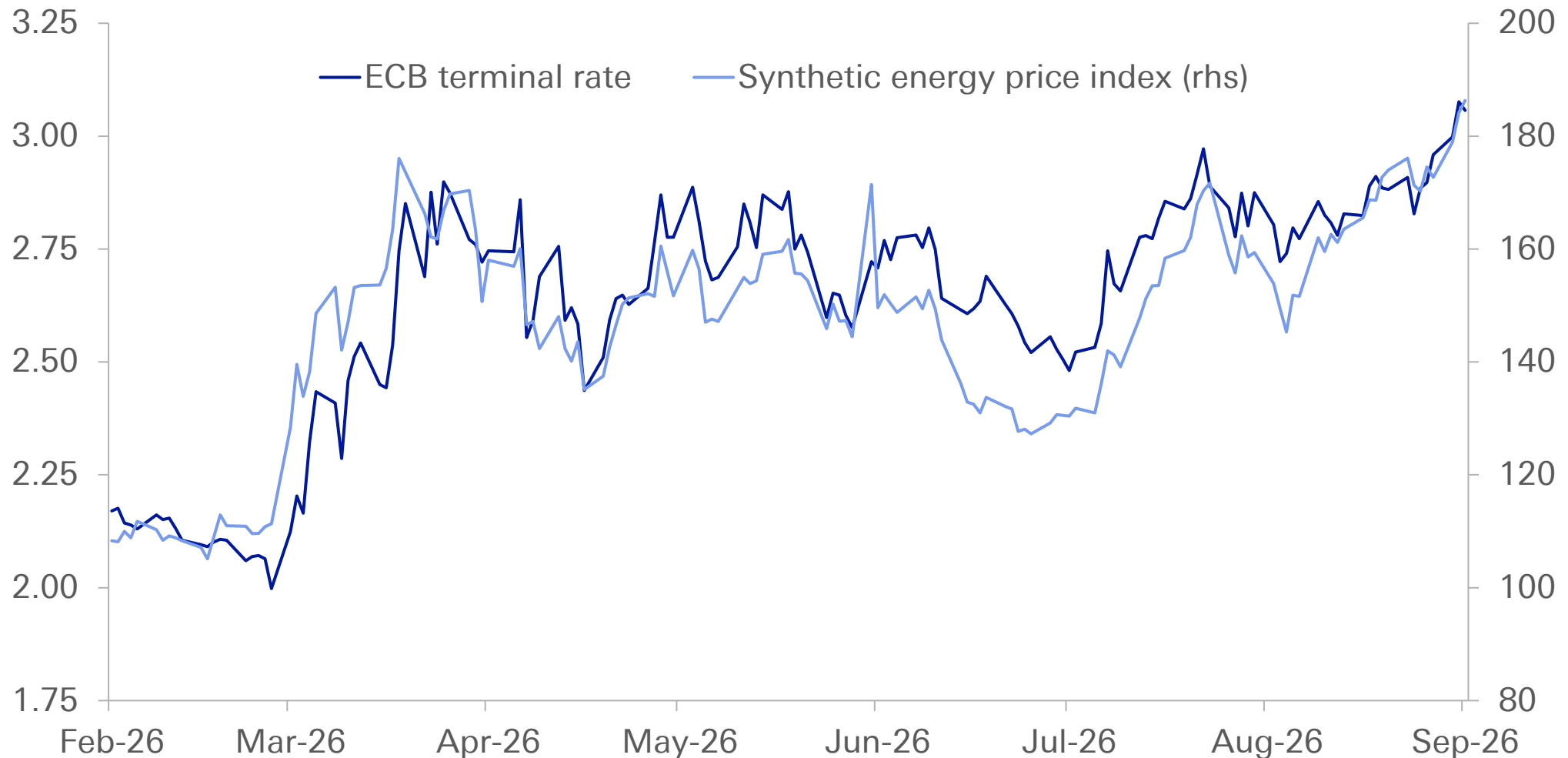


Should the ECB raise rates...?

ECB rate expectations (75bps more hikes) have tracked a synthetic energy price index.



ECB rate expectations & synthetic energy prices

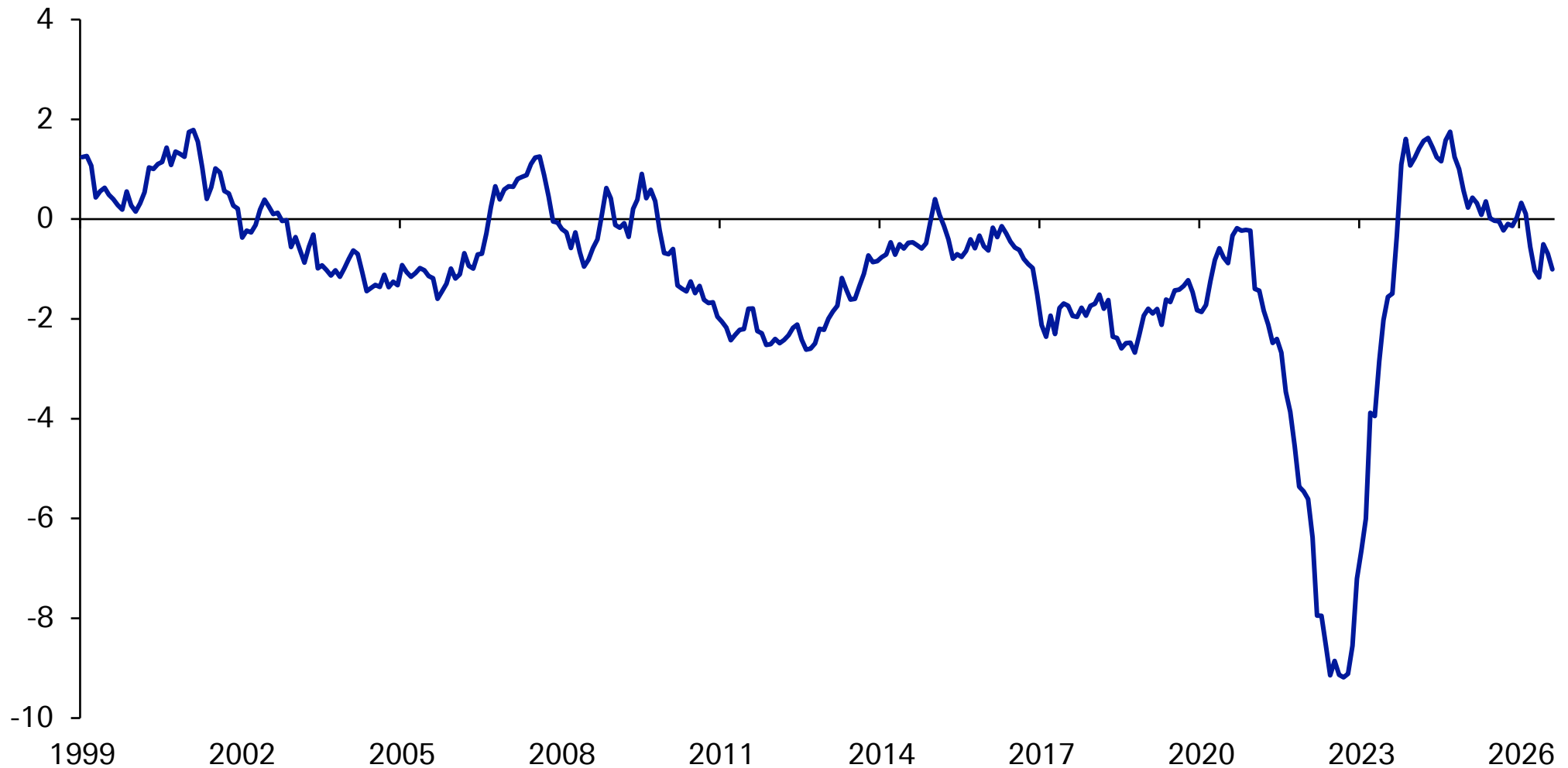


Source: Bloomberg Finance LP, Deutsche Bank. The index is an equal weighting of spot and 12M forward oil and gas prices

In February before the Iran conflict, the real adjusted ECB's deposit rate was still positive. But the energy shock pushed it back into negative territory again...



Real ECB Deposit Rate (the Deposit Rate minus year-on-year HICP inflation)

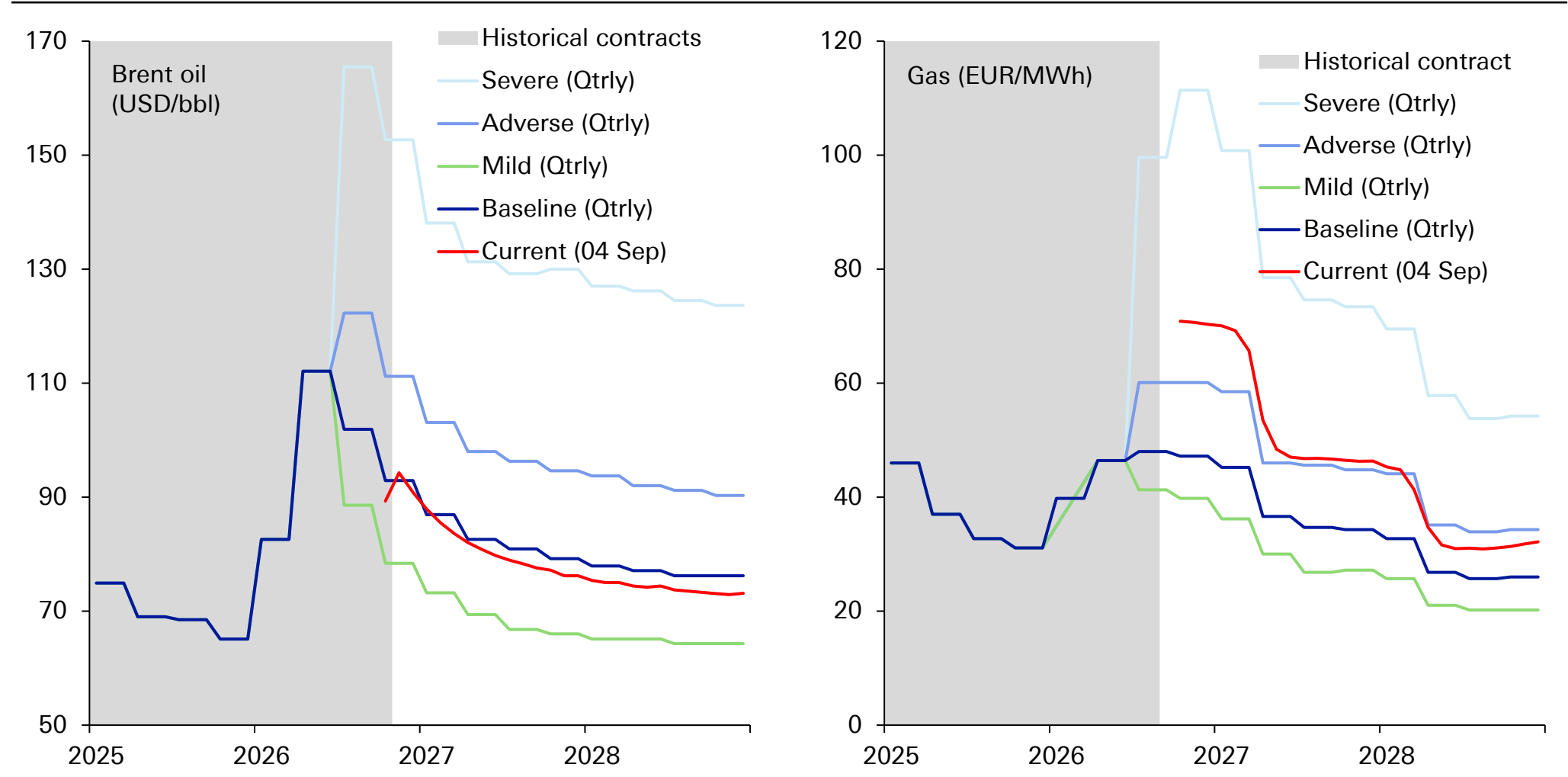


Source: Haver Analytics, Deutsche Bank



Brent is currently in-line with the ECB base case scenario from June but gas is now just above the adverse one. You can see our economics team's preview of the ECB meeting [here](#). They expect a hike this week and another to 2.75% in December. Note they have also upgraded '26 and '27 GDP to 0.8% and 1.2% (up 0.3pp and 0.1pp) respectively.

Oil prices (left) and Gas prices (right) vs. ECB baseline and alternative scenario



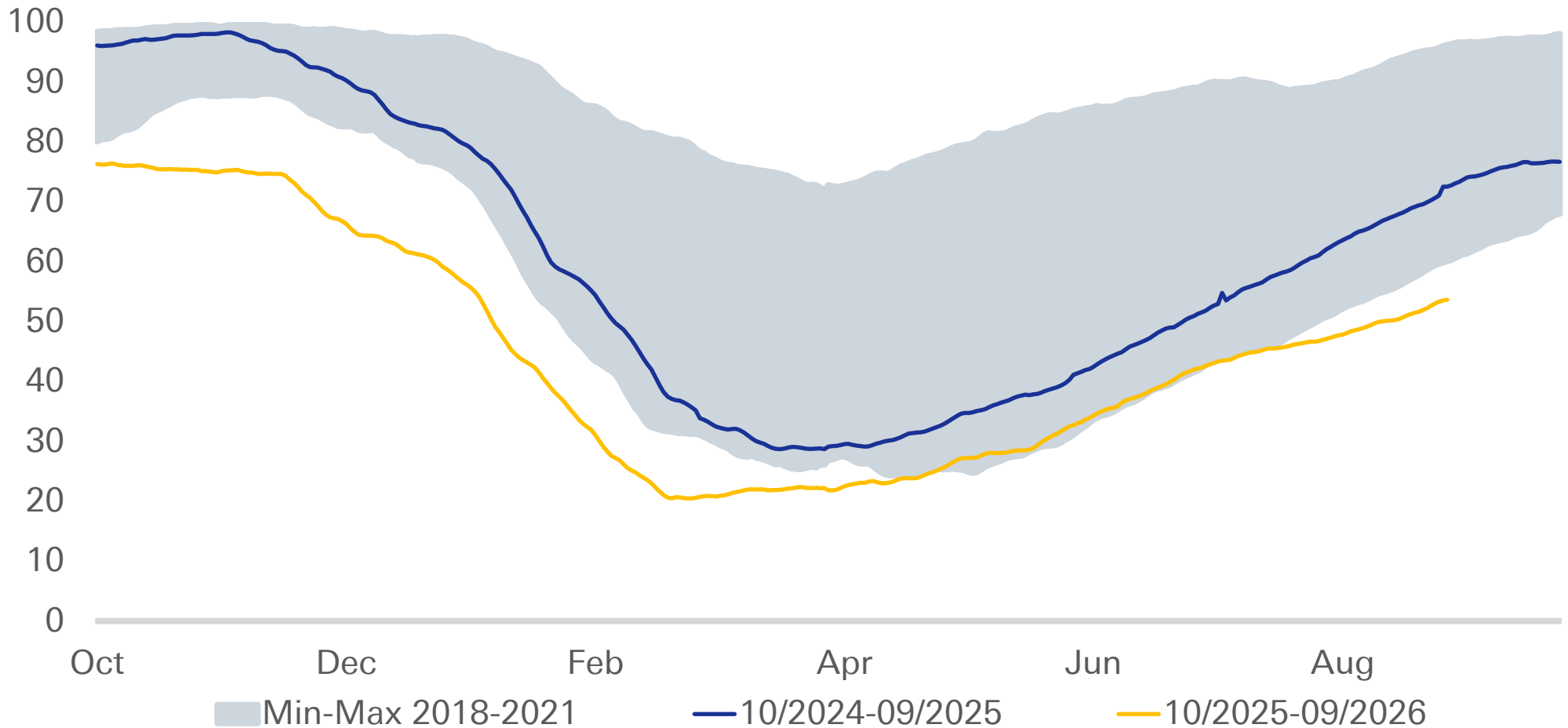
Source: ECB, Bloomberg Finance LP, Deutsche Bank Research

Gas storage levels haven't been an issue since the early days of the Russia-Ukraine conflict but we're starting to see low storage levels again ahead of the winter heating season. See Eric Heymann's blog [here](#) for more on this... Maybe El Nino will help prevent a cold winter!!



Gas Storage level in Germany

Gas storage level in Germany, %

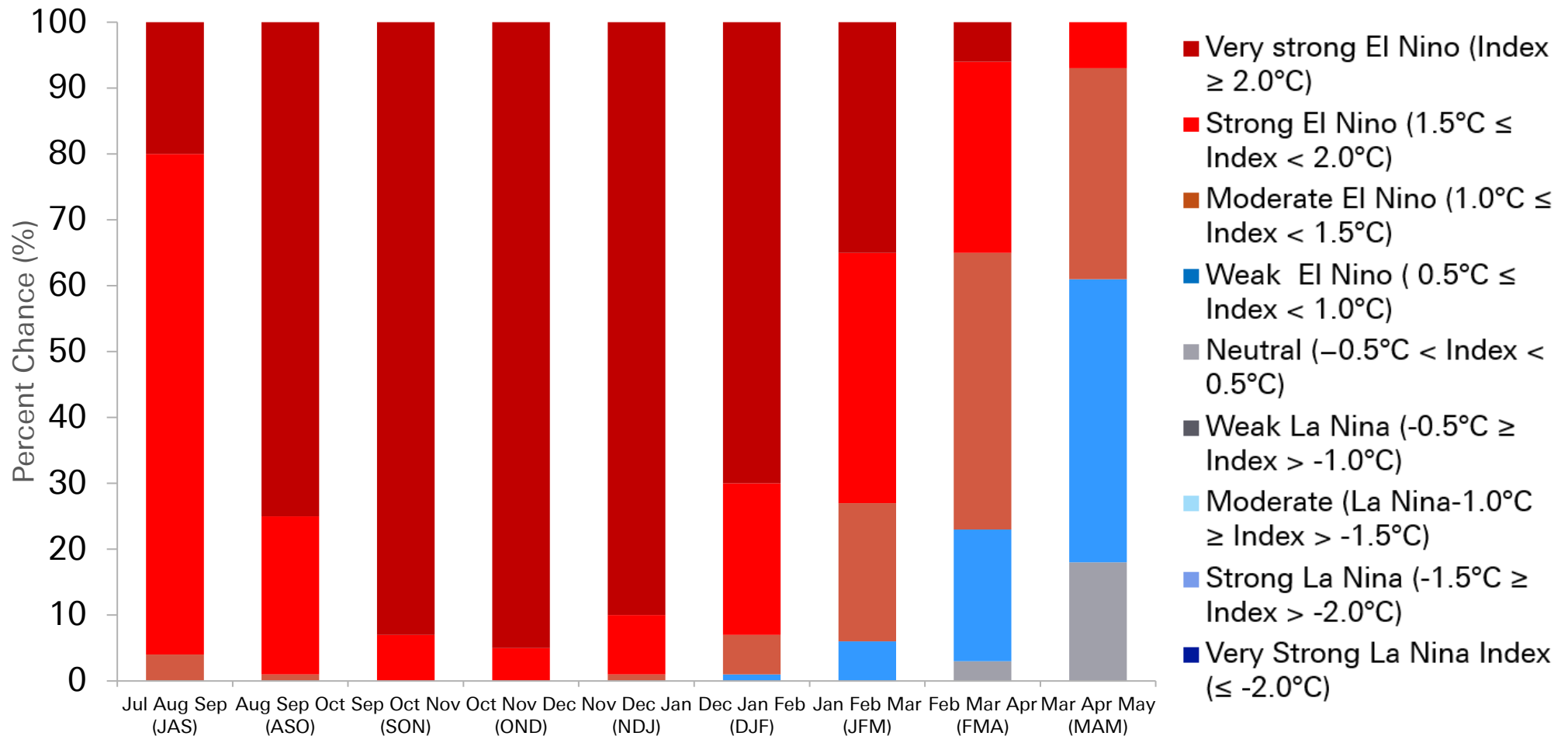


Source: Federal Network Agency, Deutsche Bank Research

Note that by the end of the year we could be seeing an El Nino unlike any we've seen before. Darkest red indicates the highest heat probabilities. This may have inflation and macro implications...



Official NOAA CPC ENSO Strength Probabilities (Issued August 2026)



Source: NOAA, Deutsche Bank ; Note: The bars show the chance of El Niño, ENSO-Neutral, and La Niña provided for 9 upcoming, overlapping 3-month seasons (each letter represents a month).

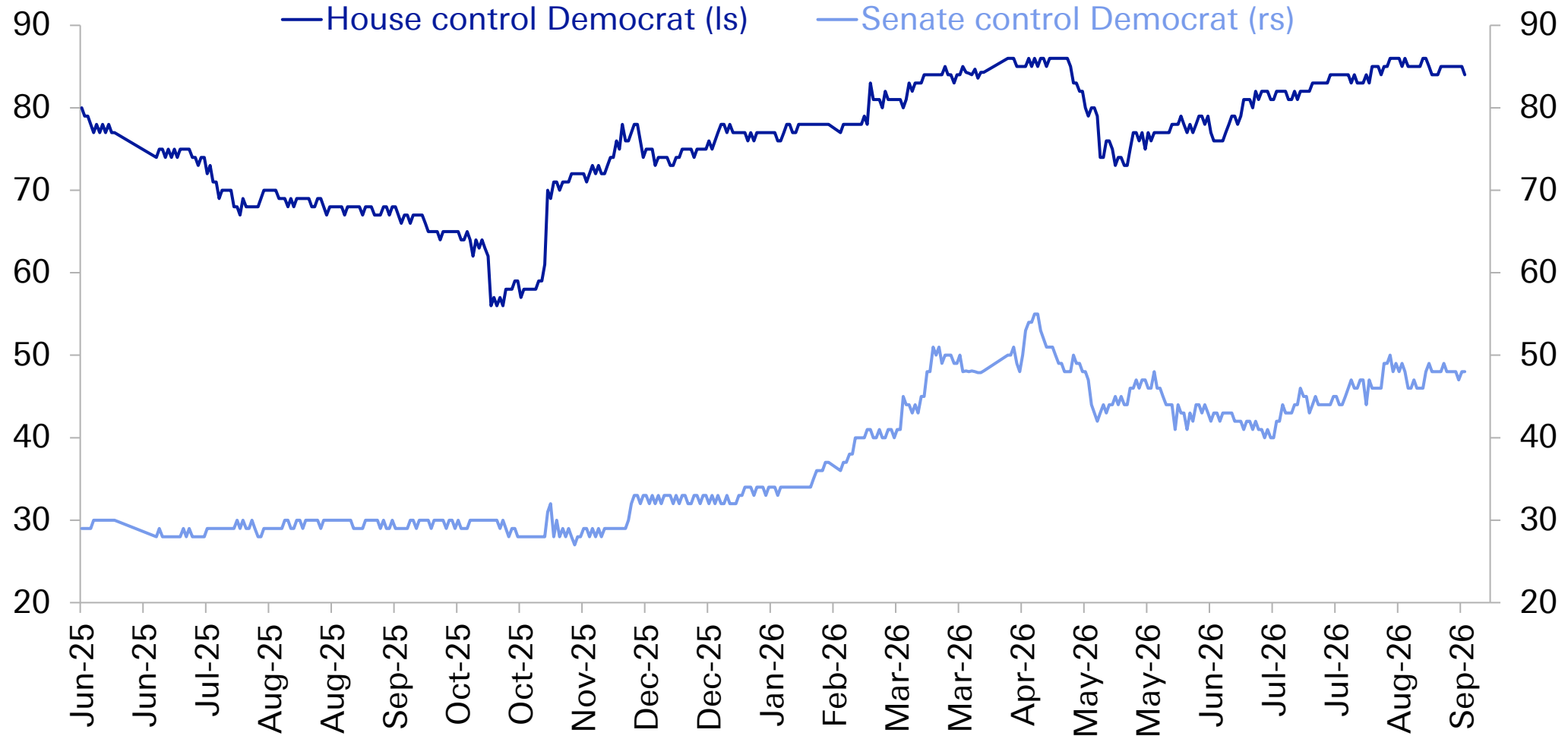


Midterms are fast coming into view...

While the House race has long been in favour of the Democrats, the Senate was firmly leaning GOP in prediction markets until this year and especially since the Iran conflict lifted inflation...



Predicted 2026 Mid-term election outcome

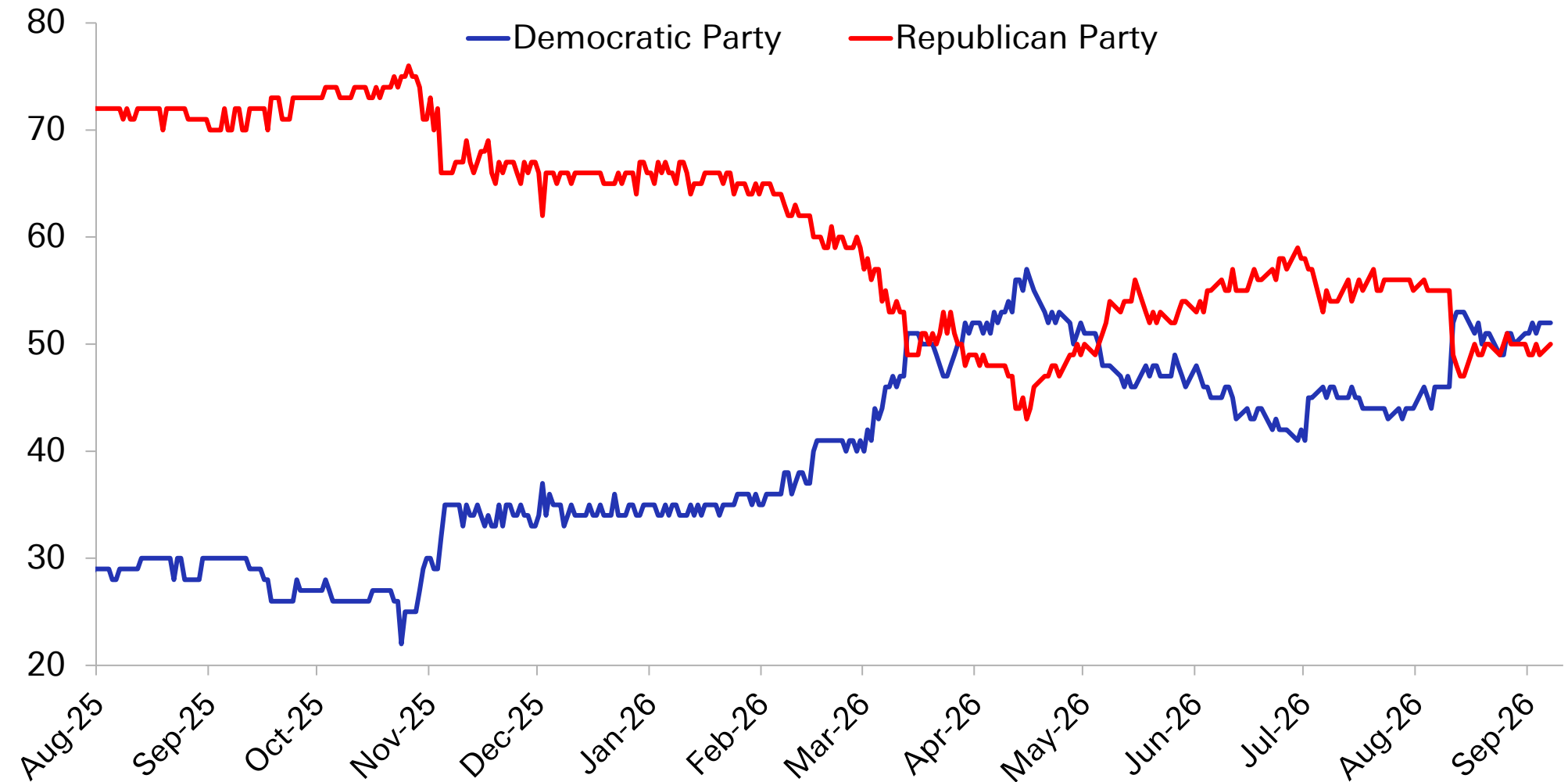


Source : Bloomberg Finance LP, Deutsche Bank, Polymarket, updated through Sep7

Since around the time the Iran strikes began, the Democrats have caught up in the race to retake control of the Senate... that would have a big impact on appointments like Fed Governors, cabinet members and Supreme Court justices, which all require Senate approval... Surely the Senate matters to Trump...



US Polymarket Party control Senate after 2026 Midterm election

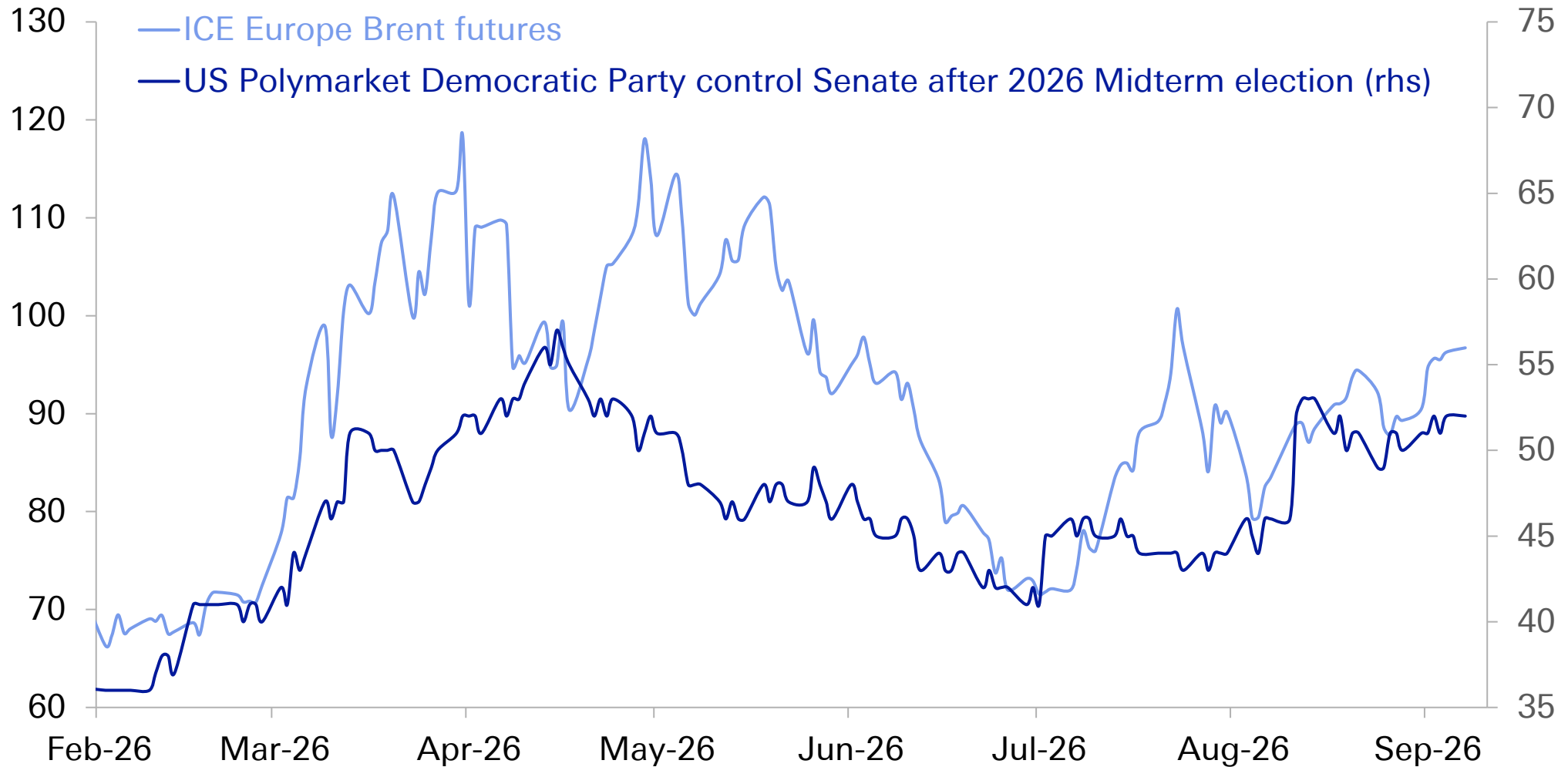


Source : Bloomberg Finance LP, Deutsche Bank, updated through Sep 7

There is a decent correlation between Brent and the probability of Senate control



Probability of Democrat control of the Senate after mid-terms and Brent price

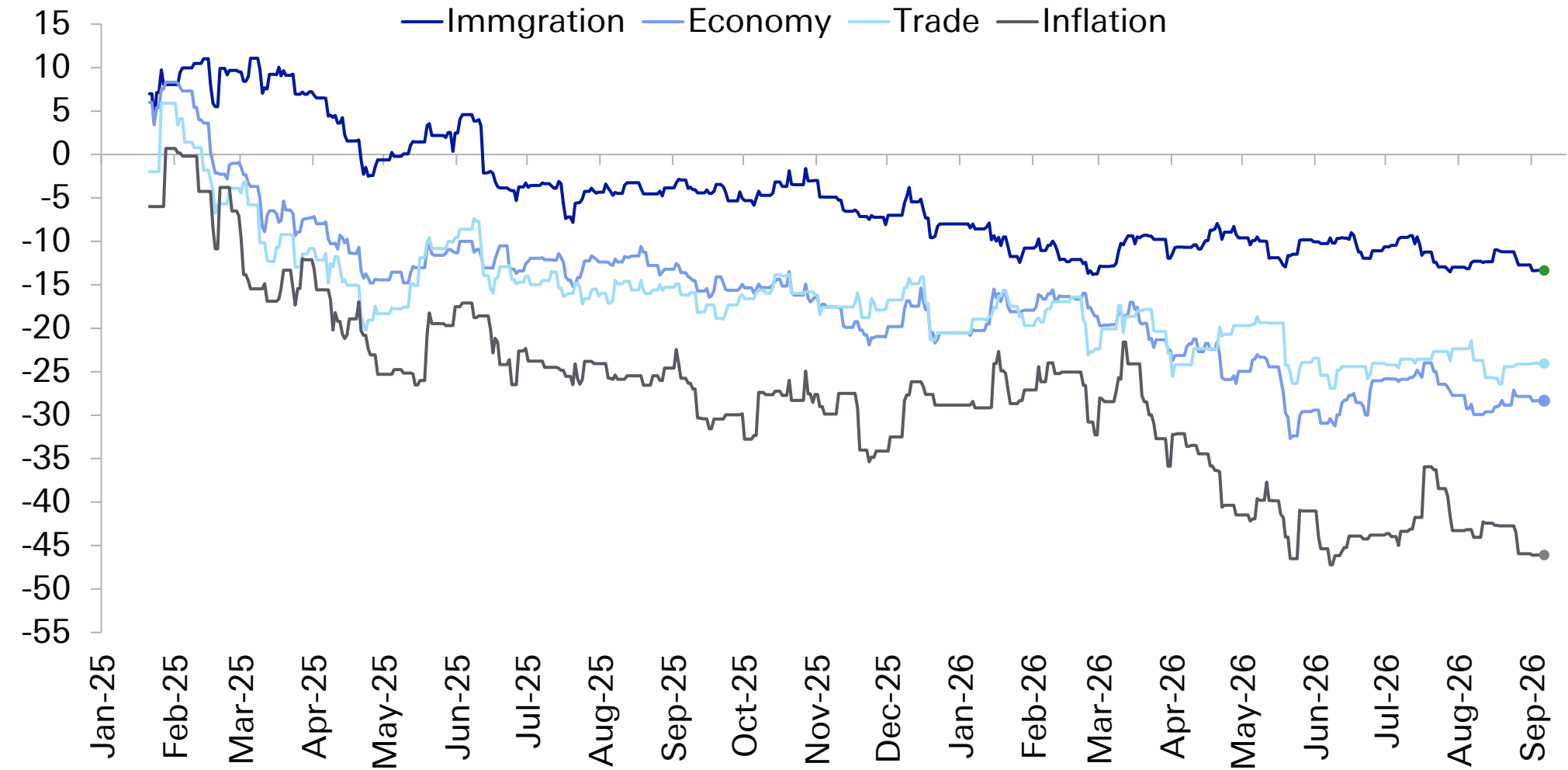


Source : Bloomberg Finance LP, Deutsche Bank, updated through Sep 7

Inflation is where Trump's approval rating seems most challenged and fell further after the Iran conflict began.



An updated polling average of Donald Trump's net approval rating on immigration, the economy, trade, and inflation, accounting for each poll's quality, recency, sample size, and partisan lean



Source: Nate Silver, Deutsche Bank, Last update : Sep 7



Mid-term years often see a weaker start to H2 for the S&P 500 and a fall into October as summer fades. After mid-terms the market tends to be strong regardless... 2026 initially followed the script but markets have been climbing since early August...

Average S&P 500 performance (first trading close on or after July 1 = 100)

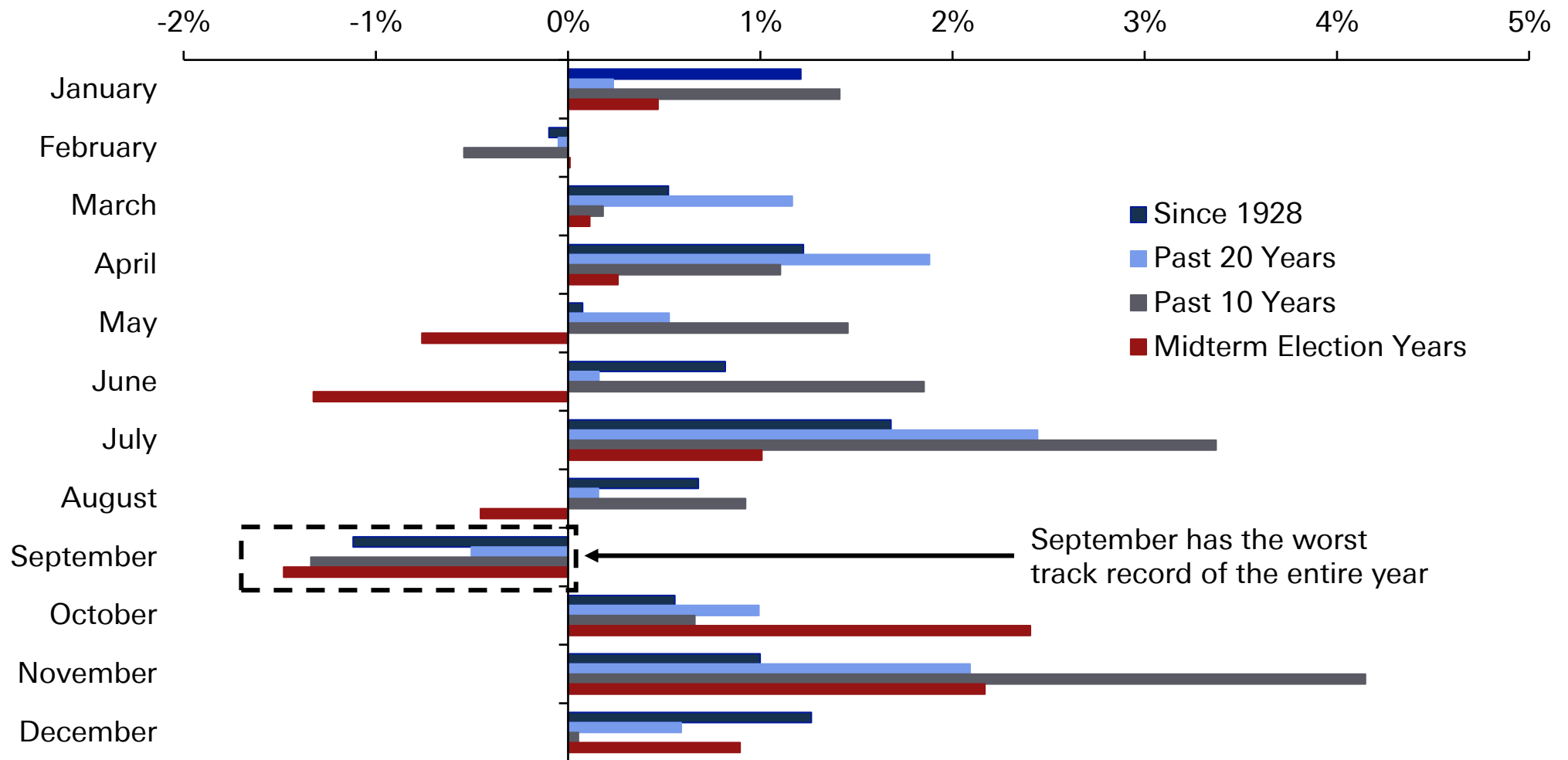


Source: Bloomberg Finance LP, Deutsche Bank

Throughout history and in recent years, the S&P 500 has consistently seen its worst month of the year in September... Midterm years are tougher but note that just before and after mid-terms, the market fully recovers so is this enough to justify lightening up?



Monthly Performance of the S&P 500 (1928-2025)

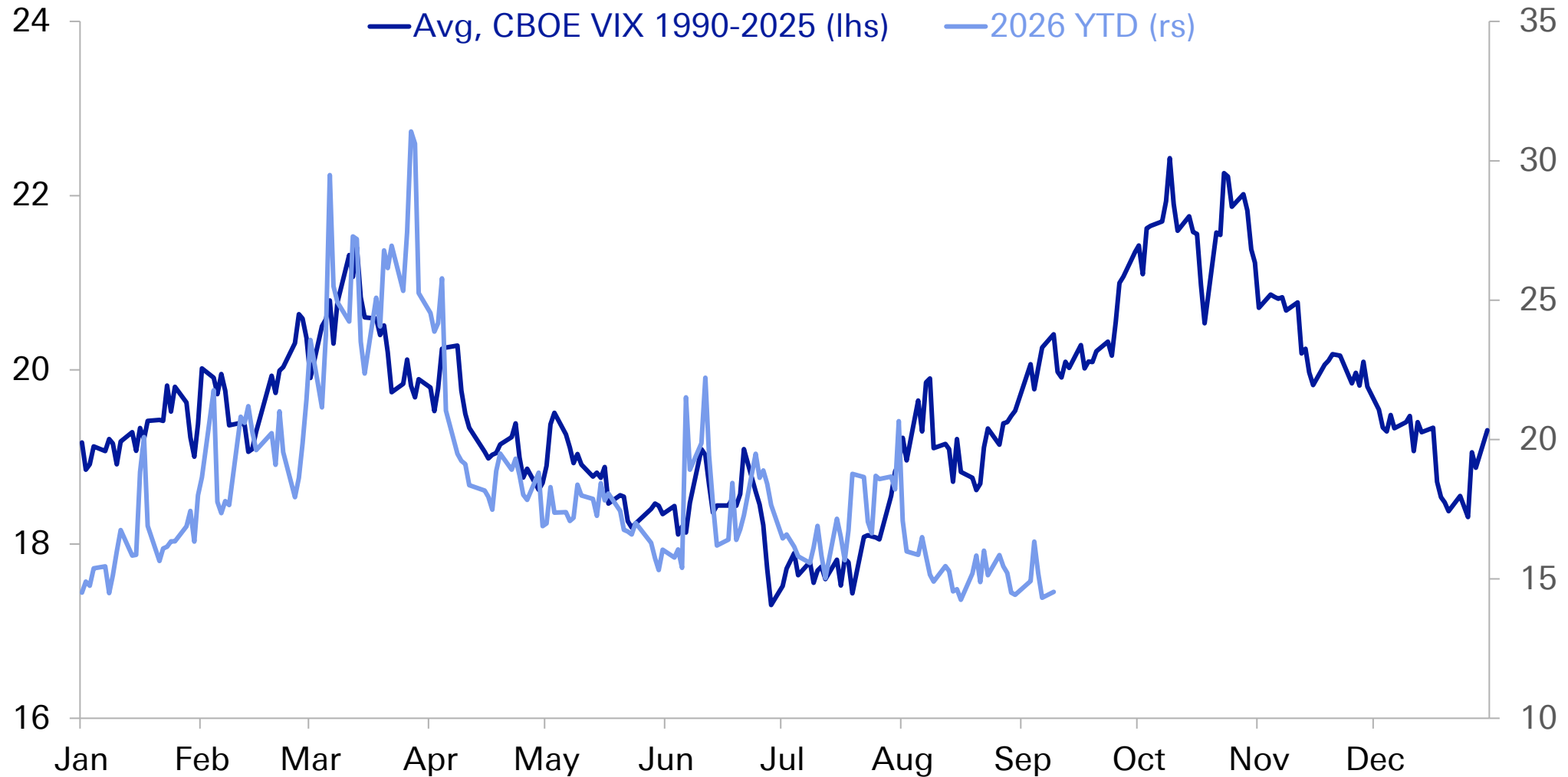


Source: Bloomberg Finance LP, Deutsche Bank

Even though there are a few ongoing concerns (e.g. Iran, oil, rising yields, mid-terms etc.), so far 2026 is resisting the seasonal pattern in terms of volatility...



Seasonality of the VIX index



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7



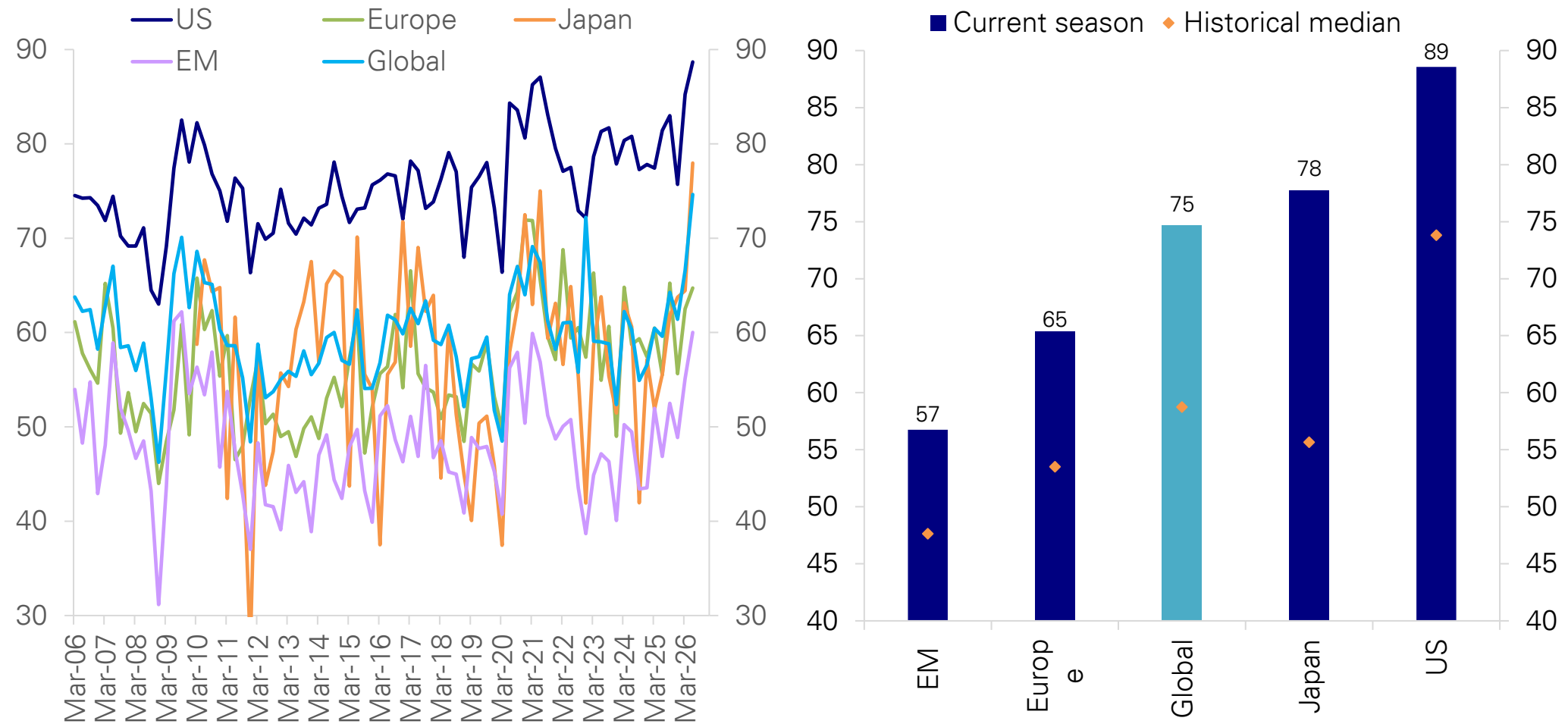
Global earnings may never have been stronger...

Please see Parag Thatte and Binky Chadha's excellent piece reviewing global earnings [here](#)



The breadth of companies beating expectations hit record peaks in Q2 in the US and Japan and remained well above historical averages in Europe and EM

Proportion of companies beating on earnings (%)



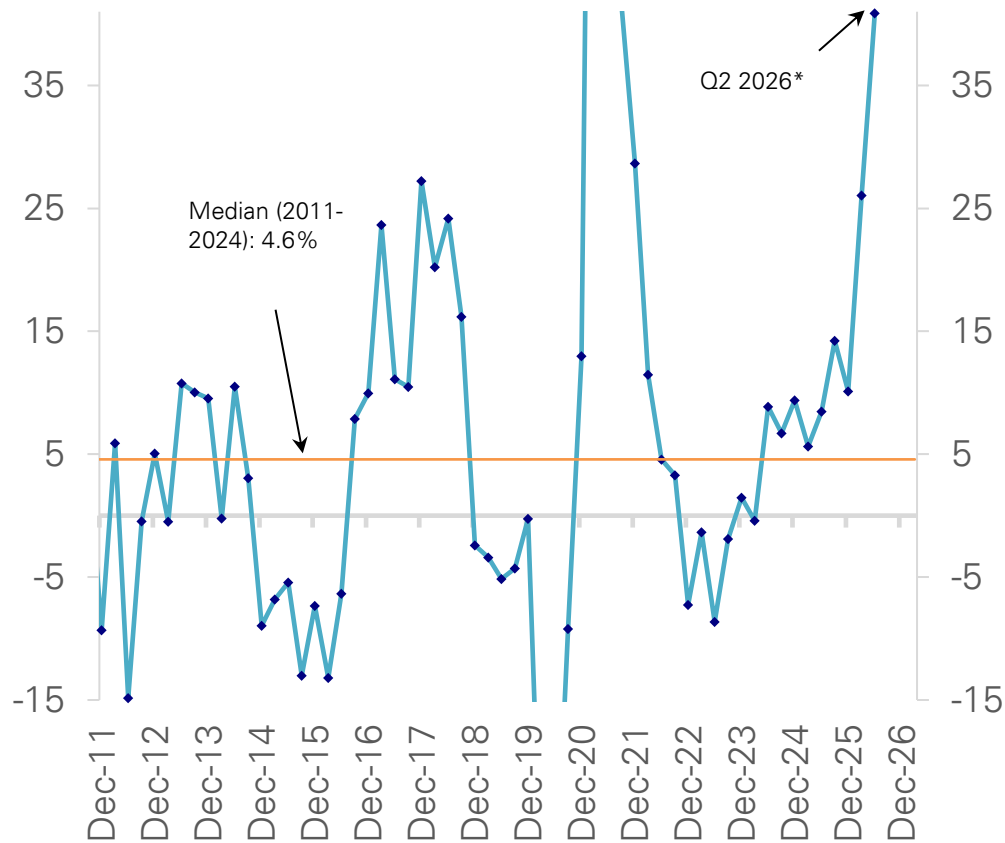
Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

And we haven't beaten low expectations. In Q2 we saw the strongest non-pandemic YoY global earnings expansion on record, driven by synchronised cyclical and commodity tailwinds.



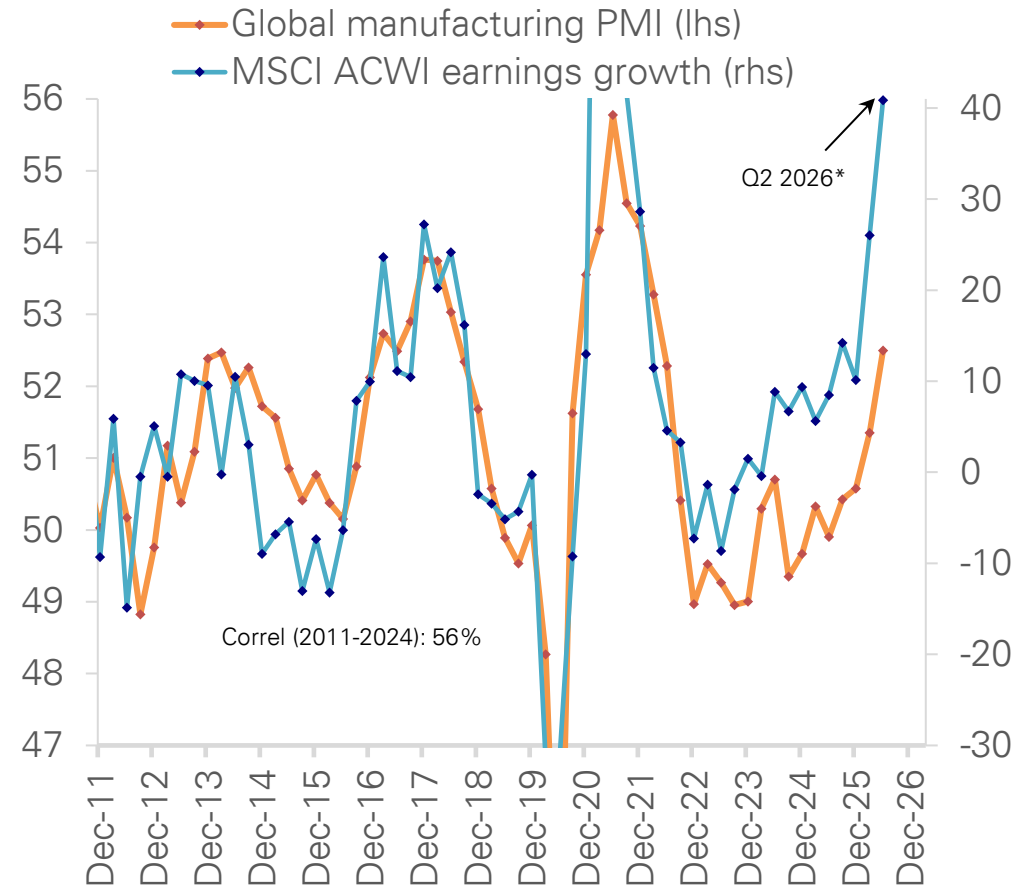
Global YoY Earnings growth

MSCI ACWI aggregate earnings growth (% yoy)



*Q2 is blended, assumes companies yet to report beat at current beat rate; growth in USD

Global earnings growth vs manufacturing PMI



*Q2 is blended, assumes companies yet to report beat at current beat rate

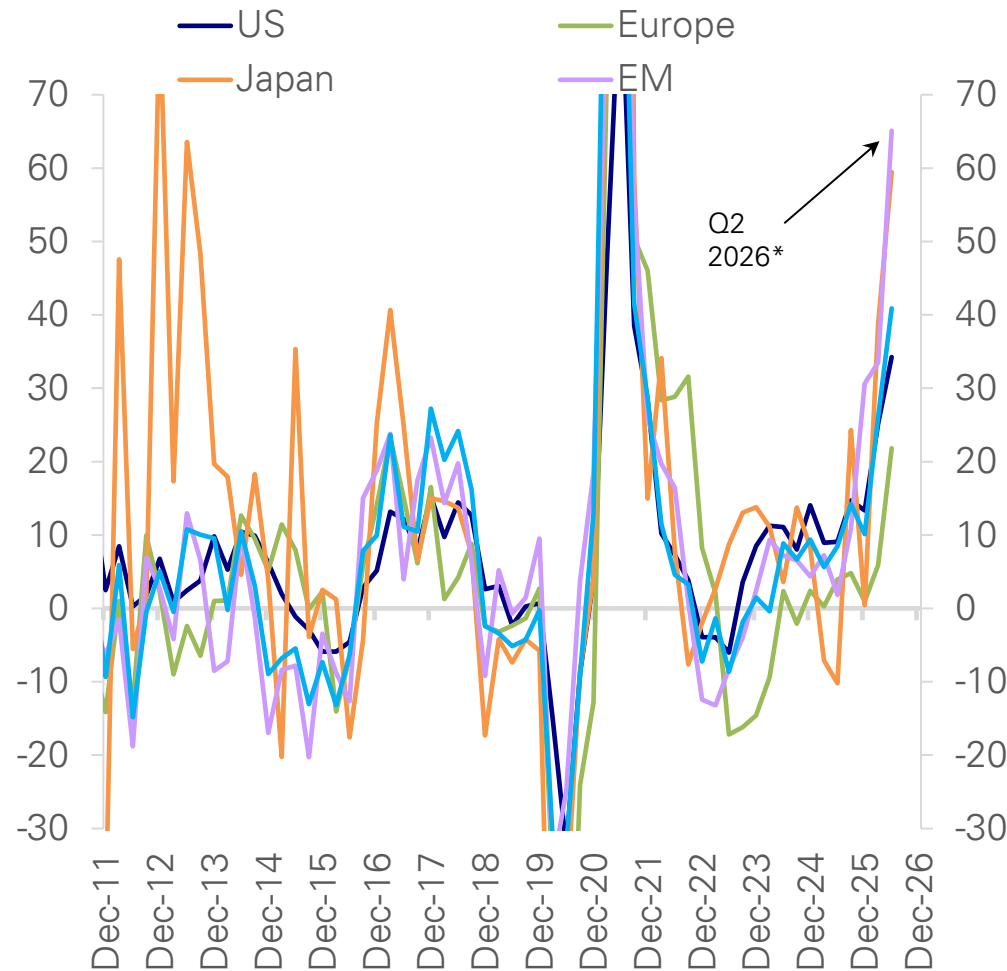
Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

Earnings growth has accelerated across all regions

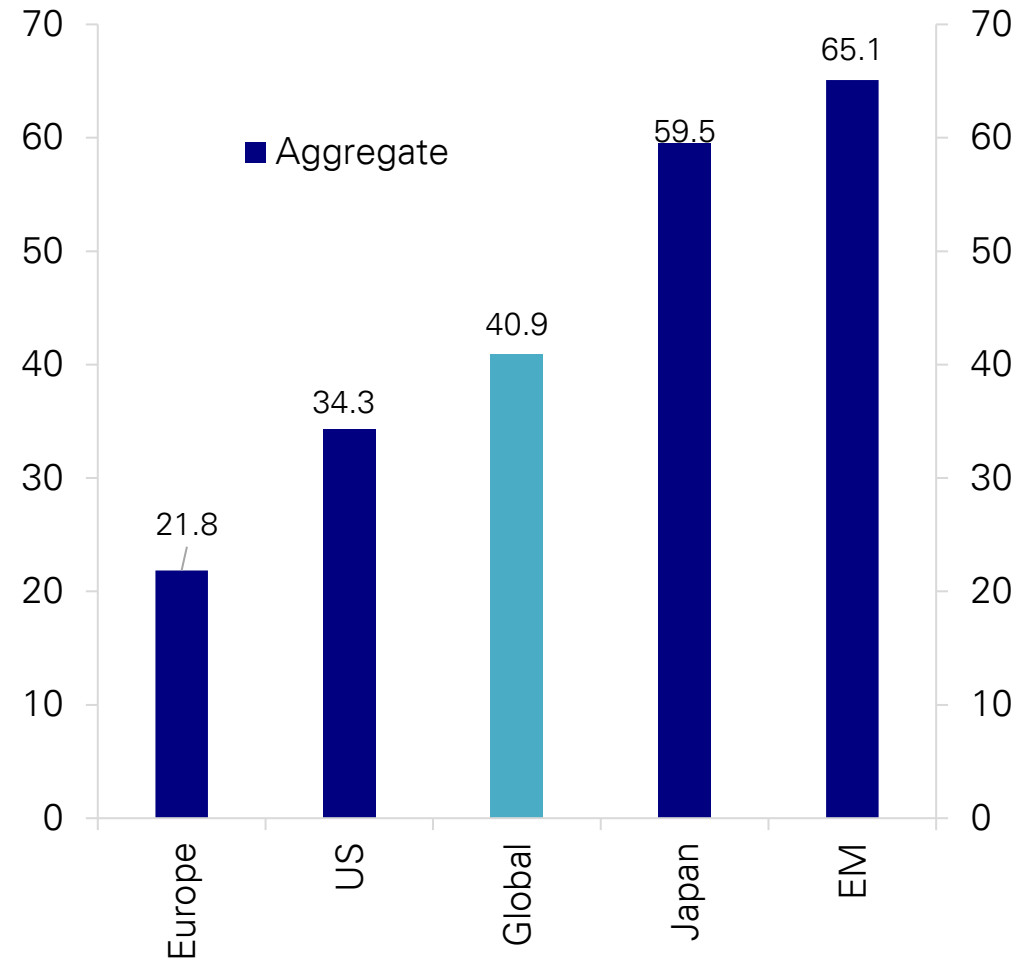


Quarterly aggregate earnings growth (% yoy)

Q2 2026 blended earnings growth (% yoy)



*Q2 is blended, assumes companies yet to report beat at current beat rate



*Q2 is blended, assumes companies yet to report beat at current beat rate

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

It's been a blockbuster Q2 for the US...

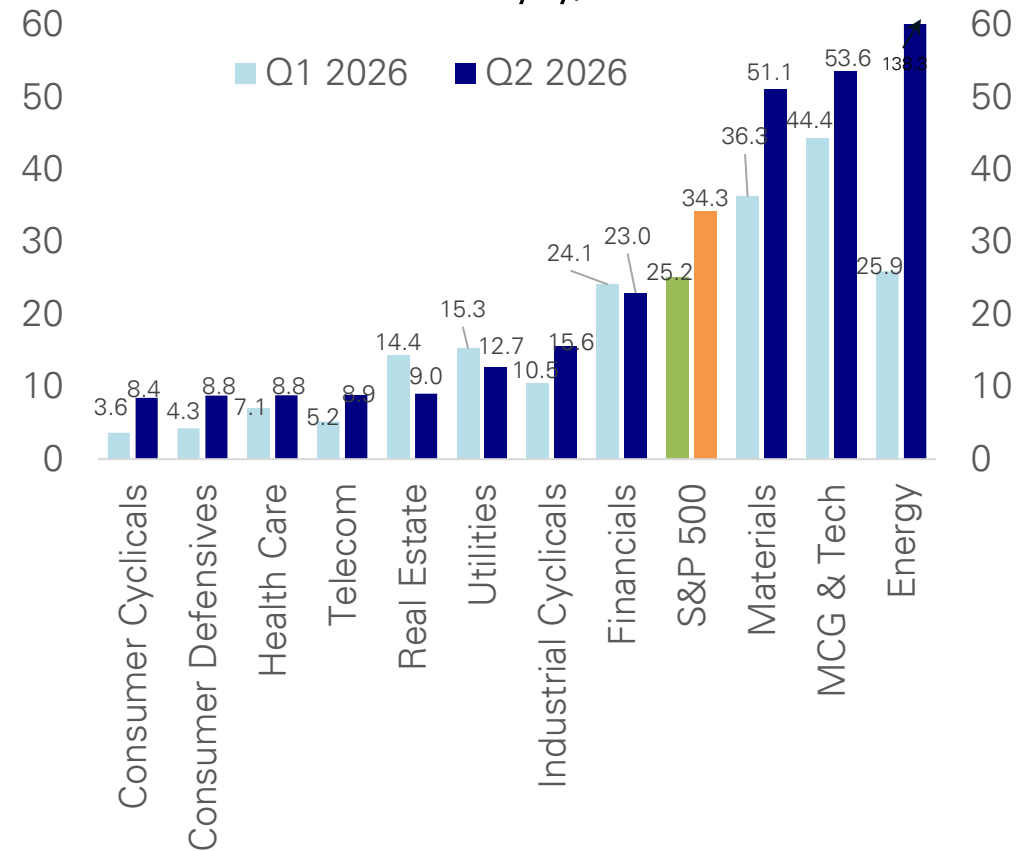


S&P 500 YoY earnings growth with sectors performance...



* Excluding 2018 tax cut

Earnings growth by S&P 500 sector group (% yoy)



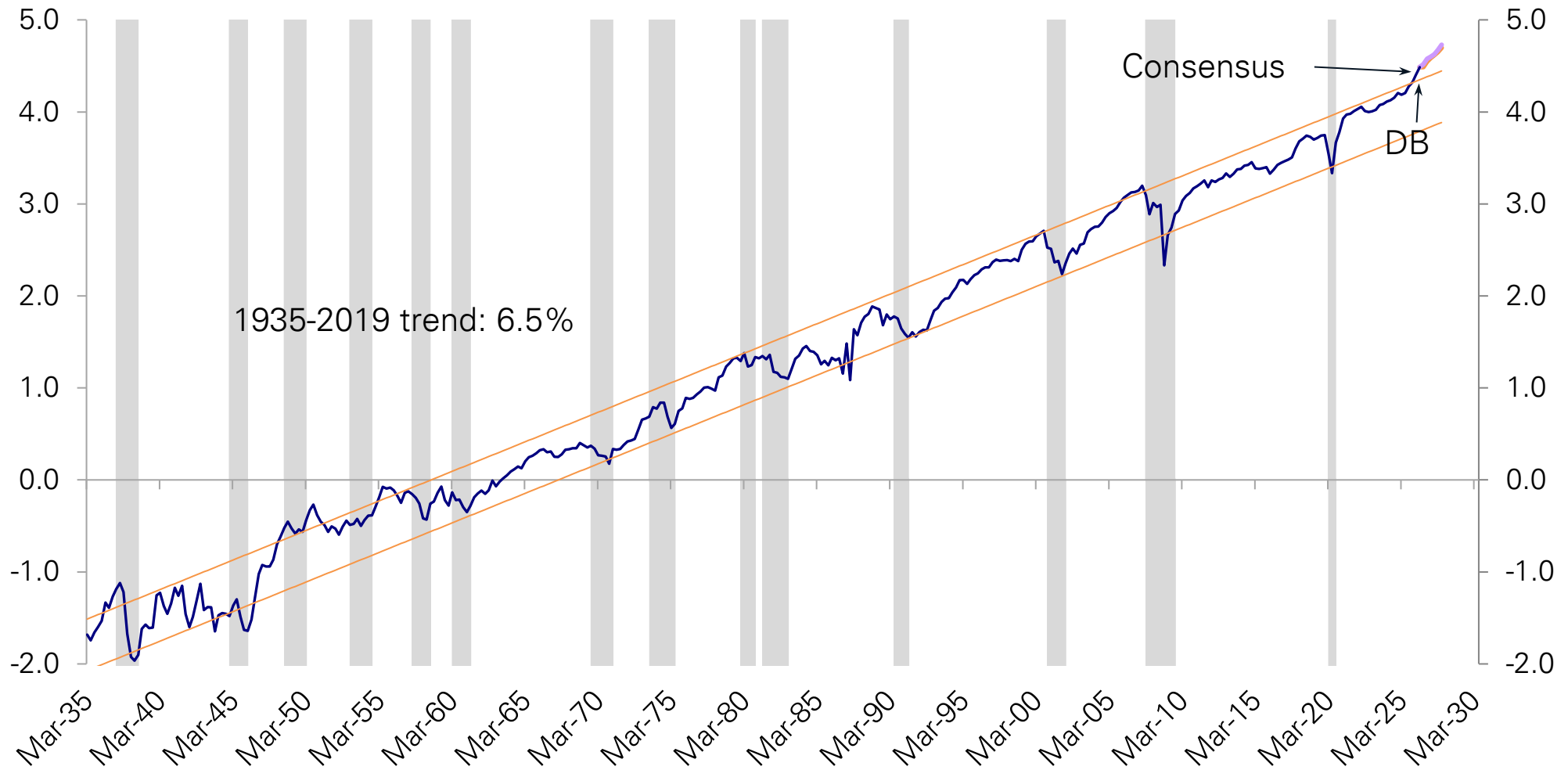
* Blended assumes companies that that report beat at the current beat rate

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation



US earnings are breaking out of a 90-year channel.... On the upside which is incredible when you consider the high growth periods of the past.... The debate will rage on as to whether this is a sustainable new plateau or an artificial period of AI related elevated earnings...

S&P 500 quarterly EPS seasonally adjusted (log)

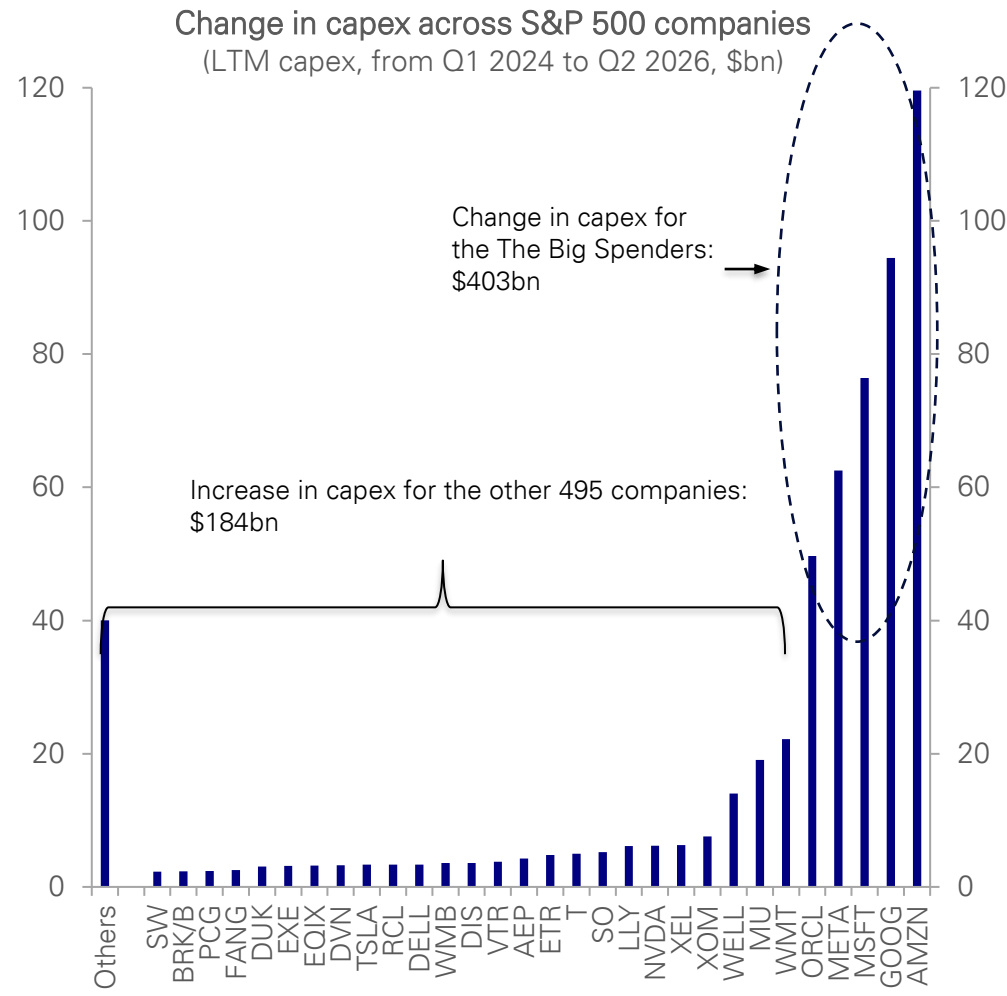
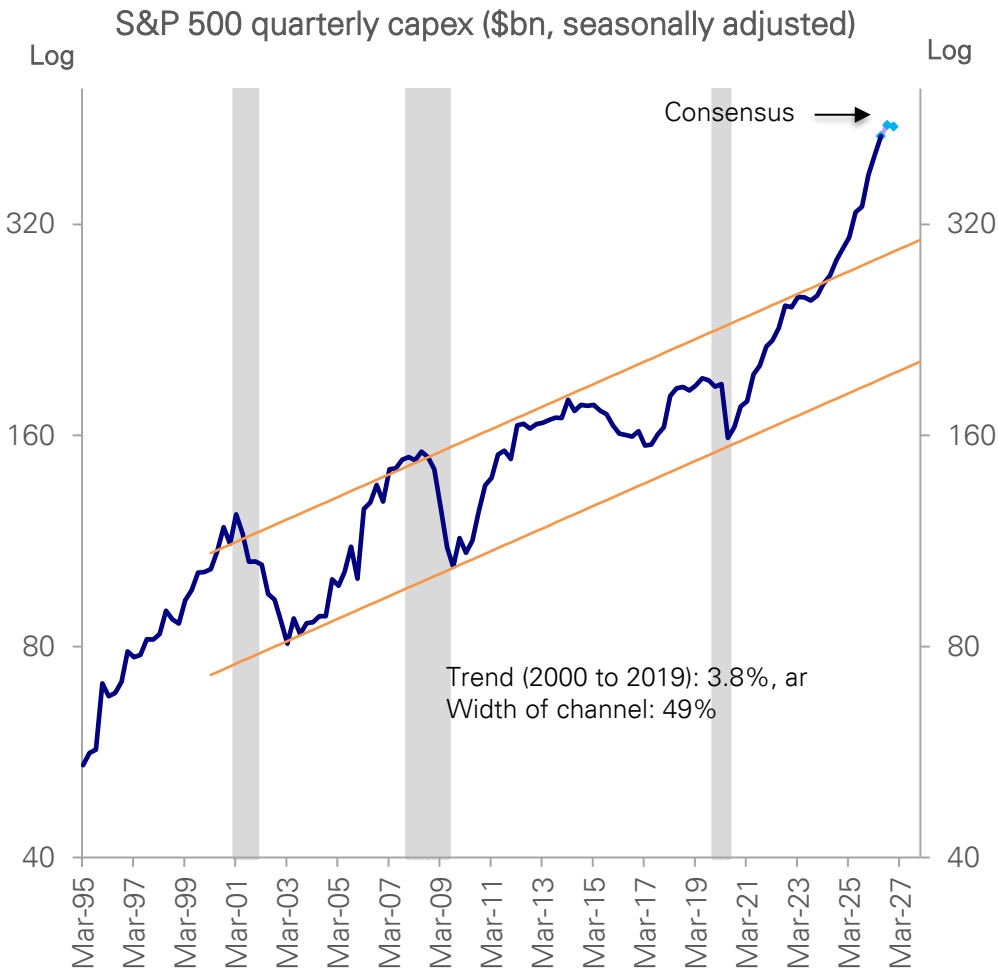


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

We are clearly in an AI Capex boom. The top 5 capex spenders are spending more than double that of the remaining 495 in the S&P 500



S&P 500 Capex spending...

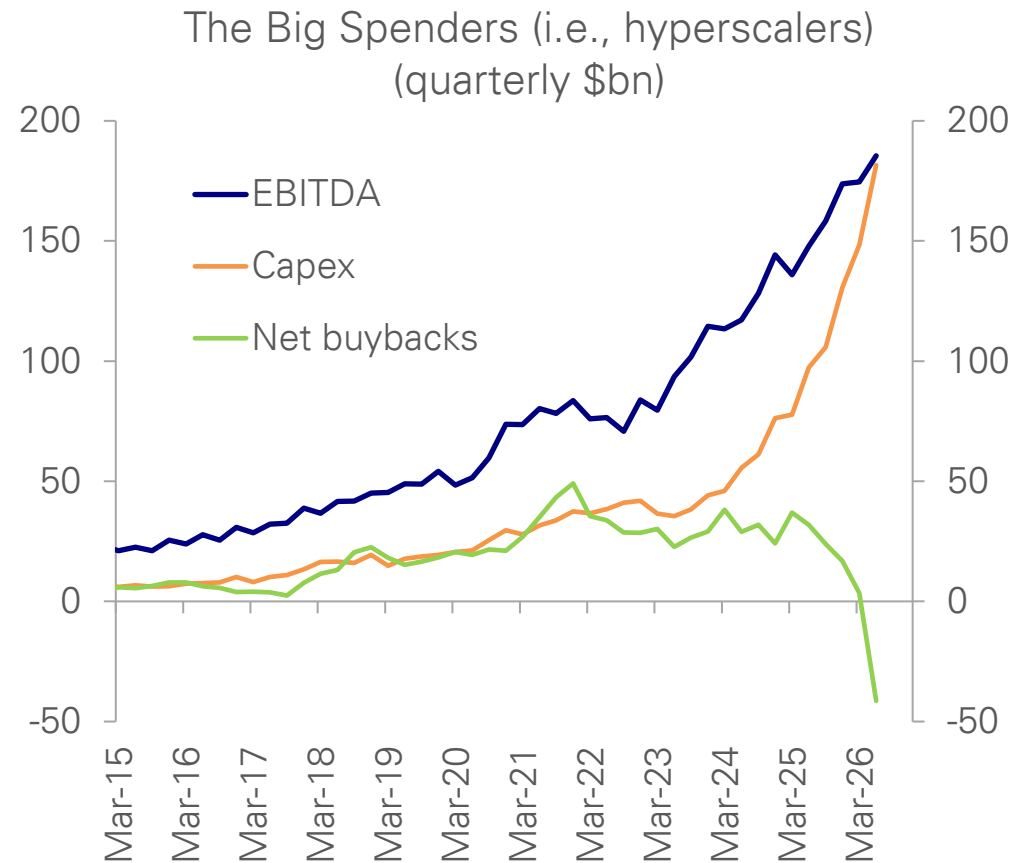


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

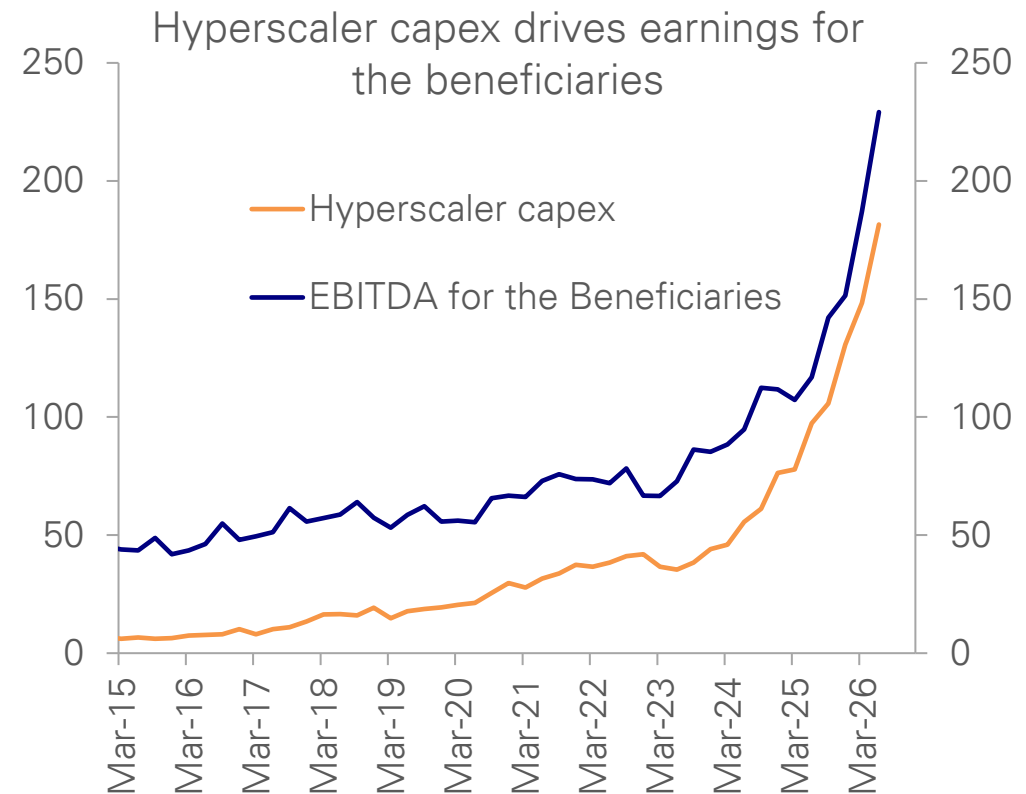
Buy-backs have turned into issuance for the hyperscalers but their spending is massively helping the AI food chain...



US Hyperscaler Capex, earnings and buybacks



Hyperscalers: AMZN, GOOG, MSFT, META and ORCL



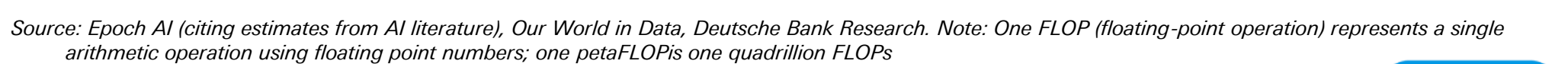
Hyperscalers AMZN, GOOG, MSFT, META and ORCL

Beneficiaries: Semis, Tech H/W & Equip., Construction & Engg., Electrical Equip., Electric Utilities, Independent Power Producers and Data Center REITs

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation



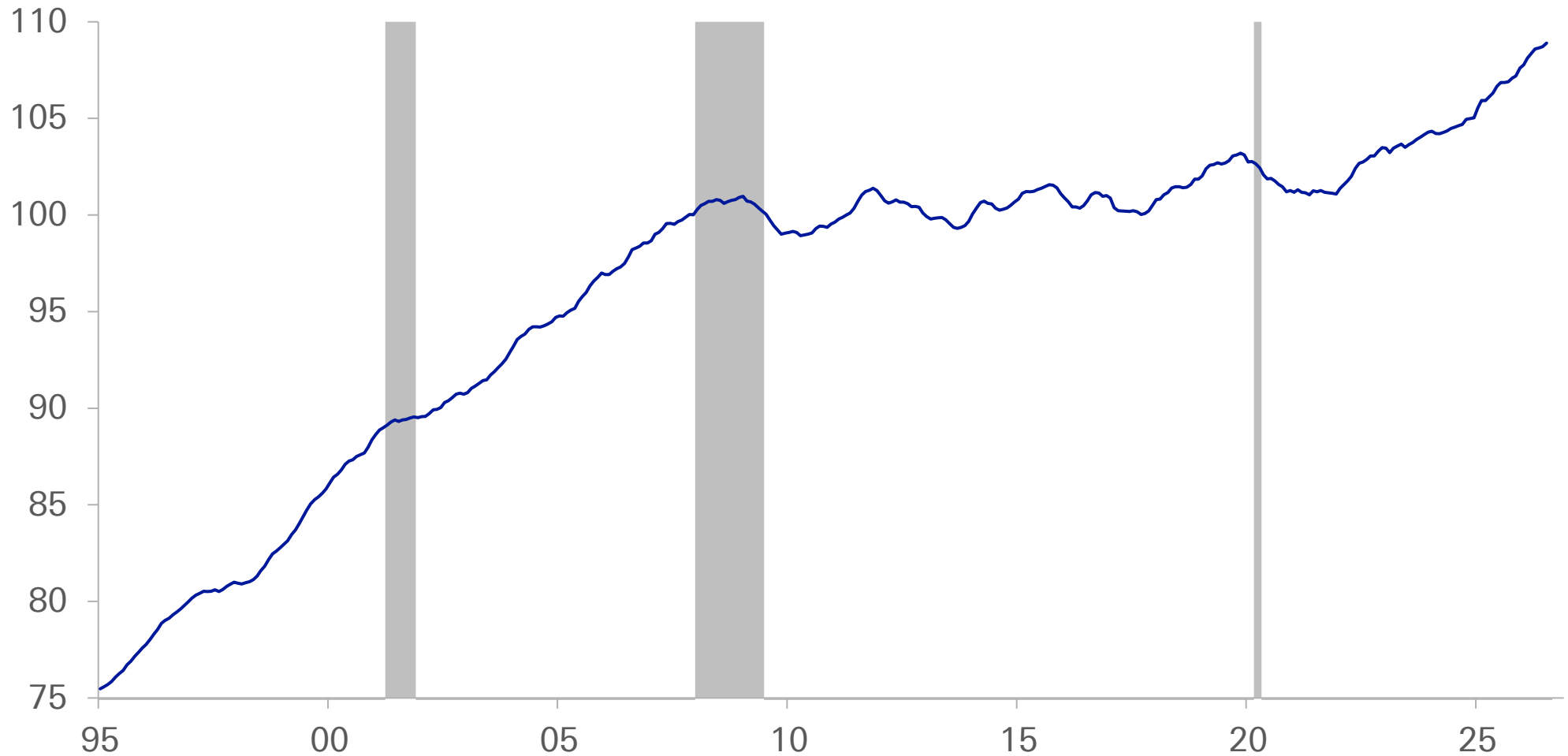
The AI train rolls on...



This exponential growth continues to drive demand for energy after a decade of stagnant demand...



IP: Electric Power Generation, Transmission and Distribution , 24-month Moving Average

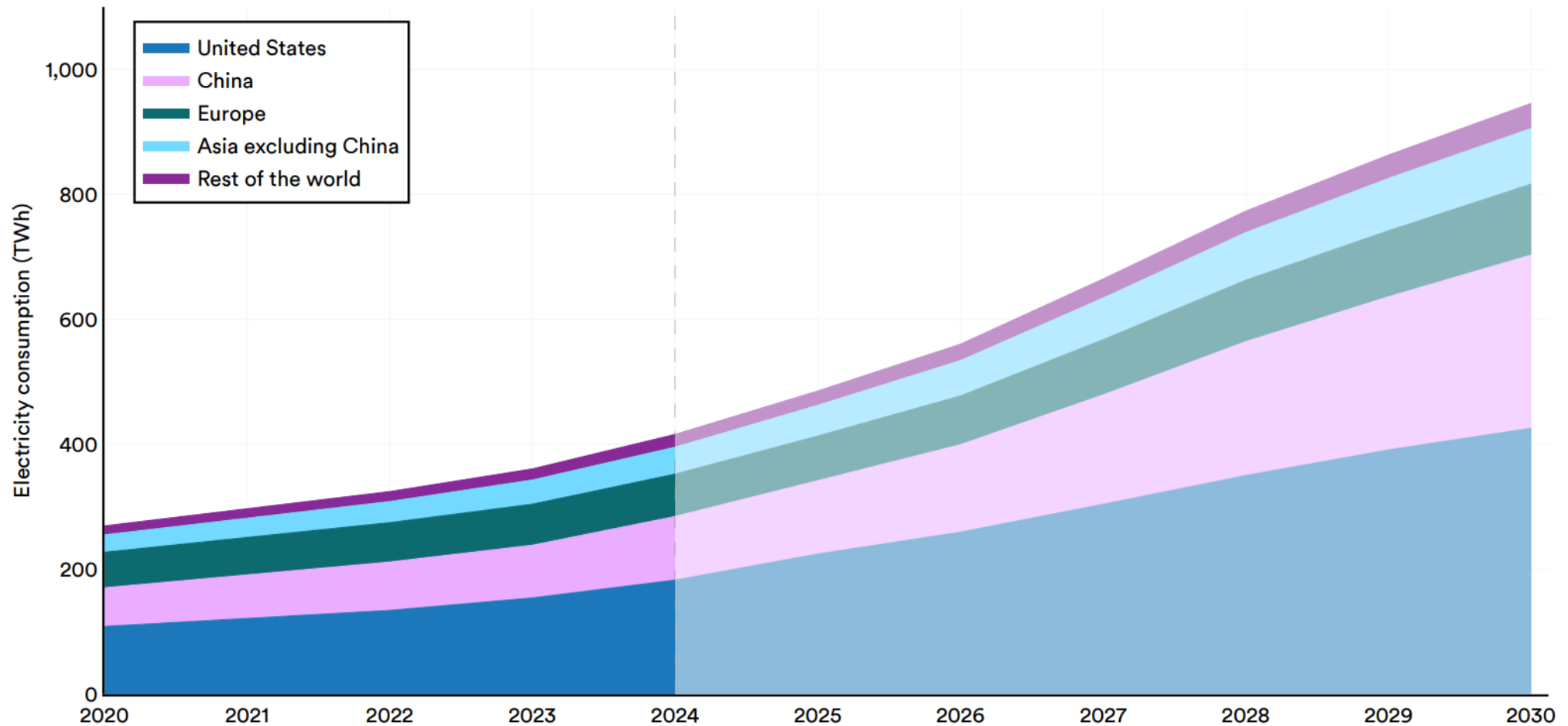


Source: FRB, Haver, Deutsche Bank

There is little sign that energy demand is going to wane...



Data center electricity consumption by region, 2020–30



Source: International Energy Agency (2025), Energy and AI, IEA, Paris | Chart: 2026 AI Index report, Deutsche Bank



Token costs are collapsing again amid cheaper compute, more efficient systems and pricing pressure from cheaper open-weight models. Lower prices and more sophisticated uses are fuelling adoption, even if the economics of some of the expensive frontier models may be tested

Silicon Data LLM Token Expenditure Index

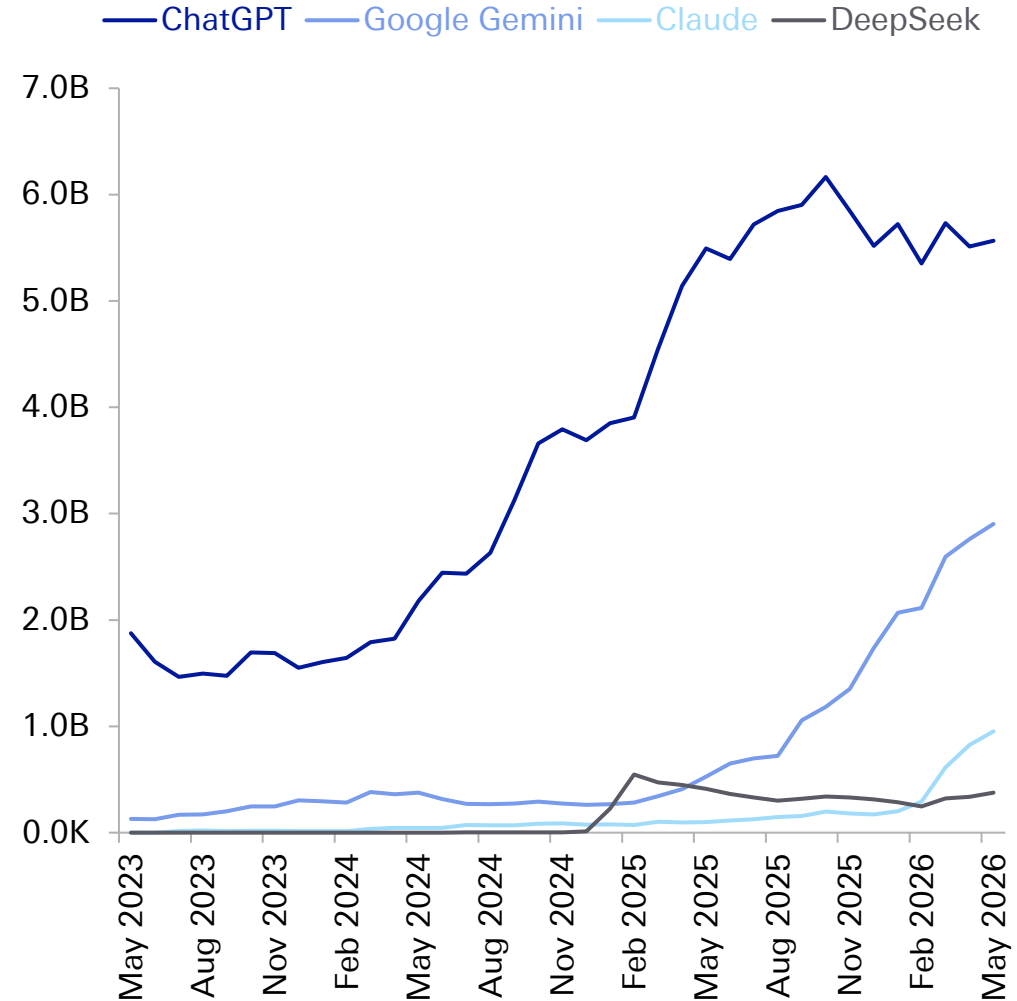
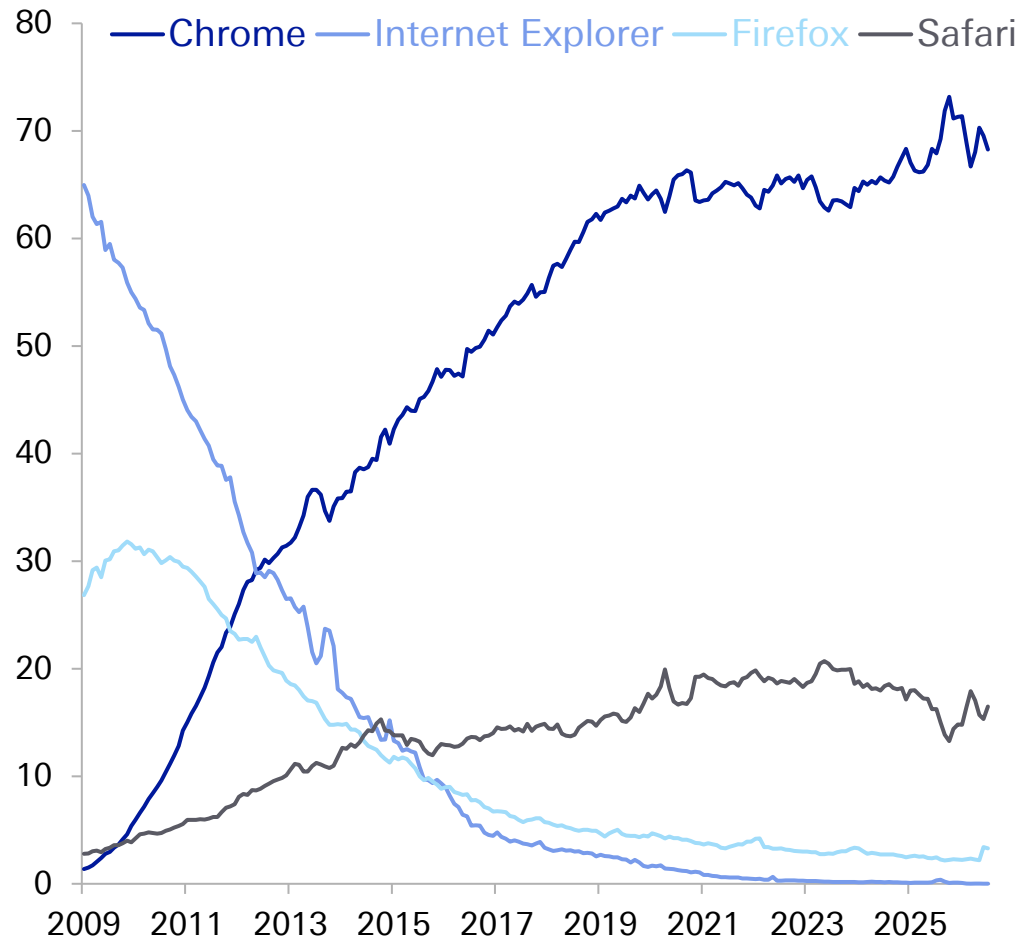


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

Remember the early winners don't always end up on top....



Browser market shares (%) (2009 to today) (right) and Top four generative AI platforms: monthly visits worldwide (desktop and mobile) (left)

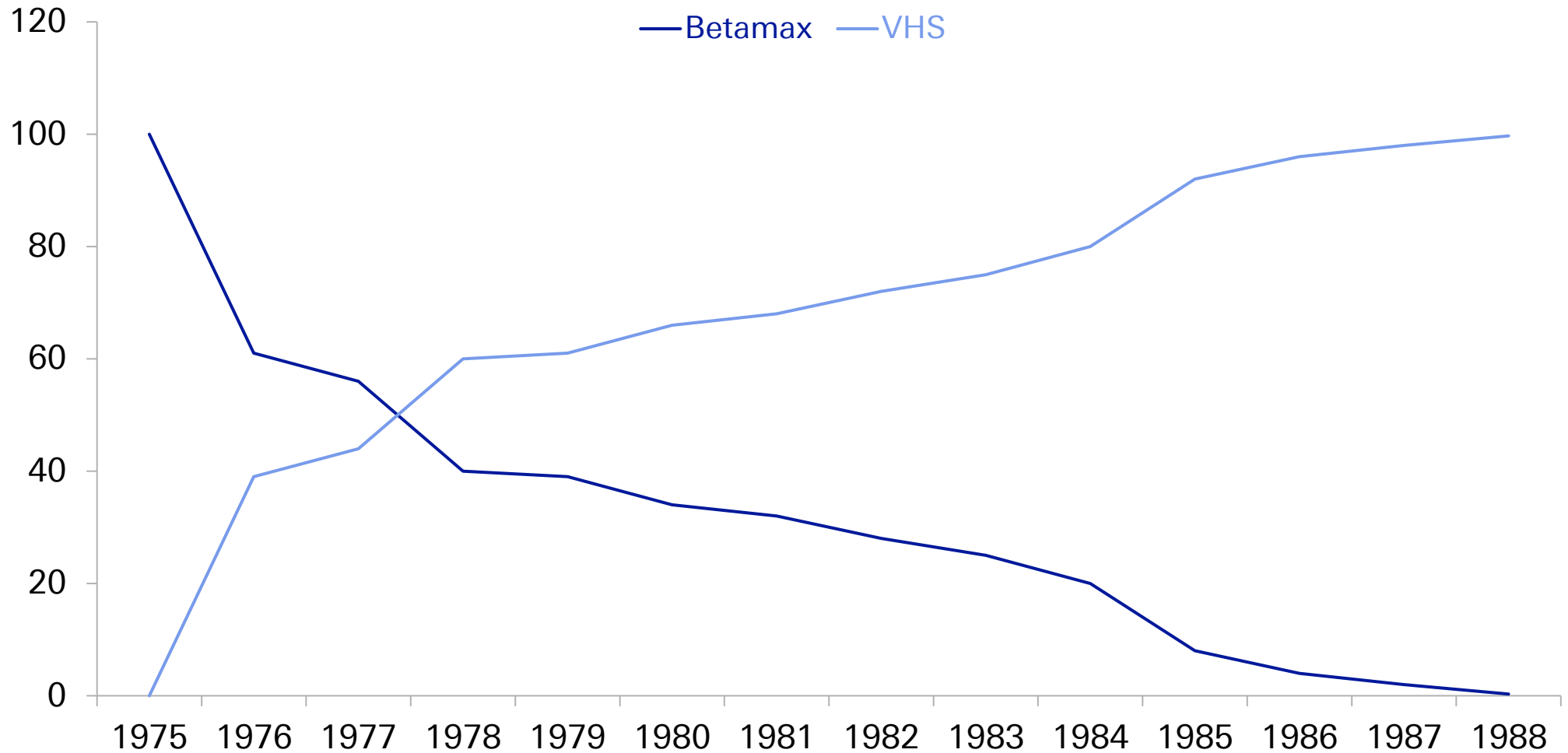


Source: Similarweb, Deutsche Bank

My favourite historical example of why winning early—and even having the better product— isn't always an advantage...



Betamax vs VHS market shares as proportion of annual production

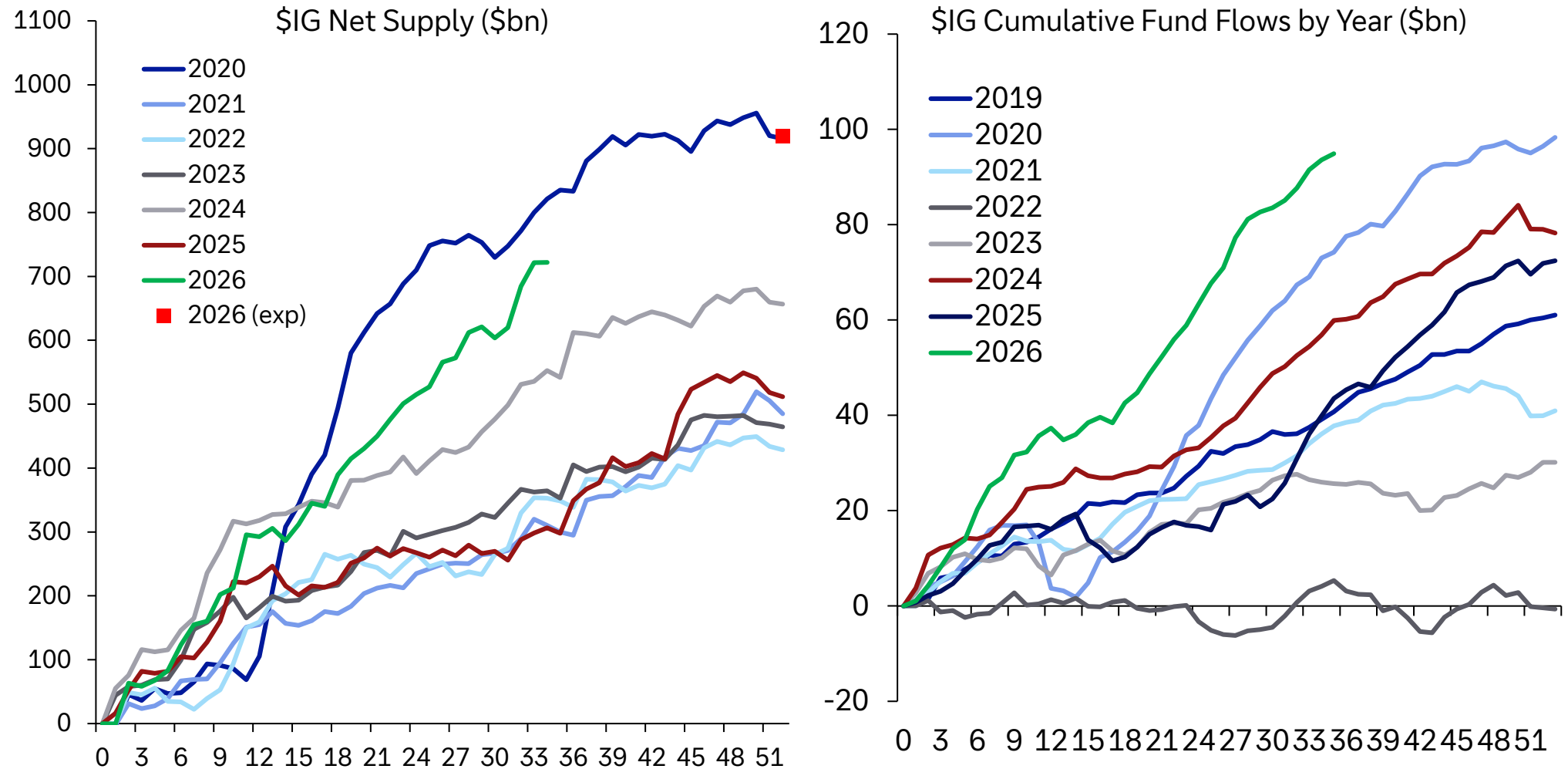


Source: Strategic Maneuvering and Mass-Market Dynamics (Cusumano, Mylonadis, Rosenbloom, 1992), Deutsche Bank Research

Technology—especially hyperscaler issuance—is helping put 2026 on course to surpass 2000 as the record year for US IG corporate net supply. However demand is also very high.



US IG Corporate Net Supply is likely to be the highest on record in 2026, but demand is also likely to be at a record...

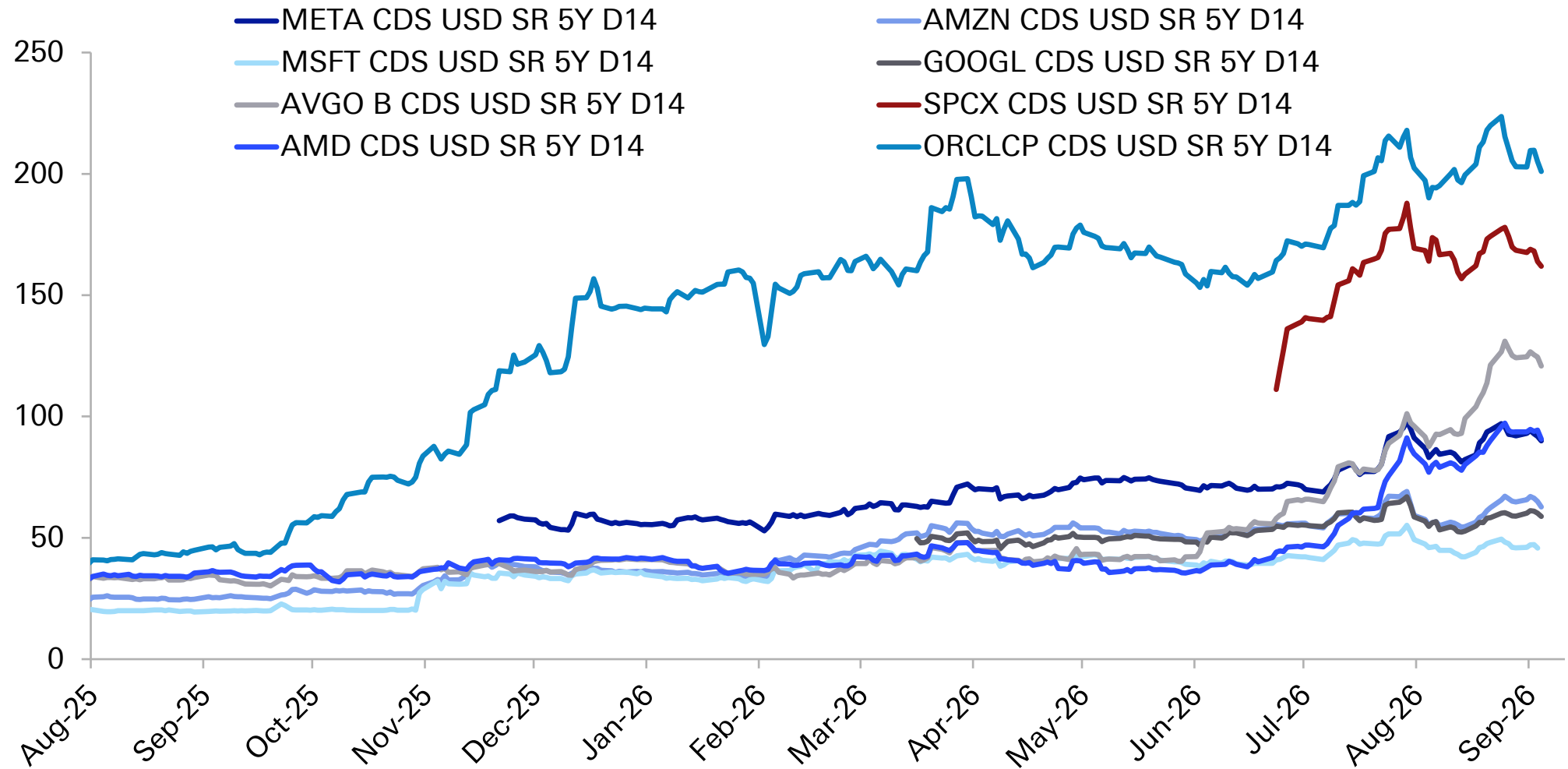


Source : EPFR, Bloomberg Finance LP, Deutsche Bank,

Hyperscaler (and related) CDS is climbing partly due to supply and increasing Capex but also as a hedge for all things AI-related so you have to be slightly careful in overinterpreting the moves...



Hyperscaler CDS

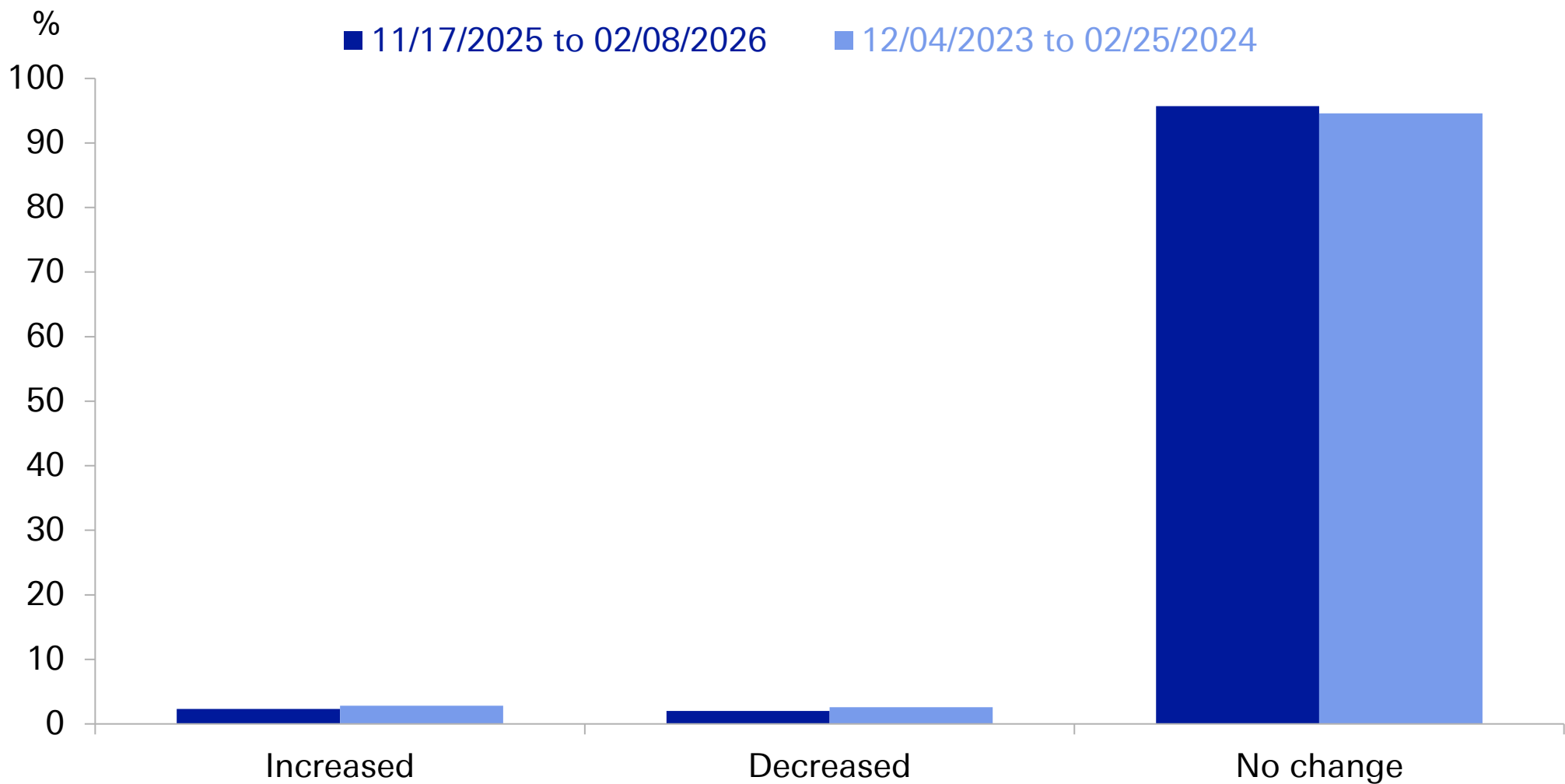


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

While AI adoption grows, a significant percentage of enterprises have not altered their workforce size as a direct result. We're still a long way from a revolution...



In last 6 months, how did use of AI affect business employment?

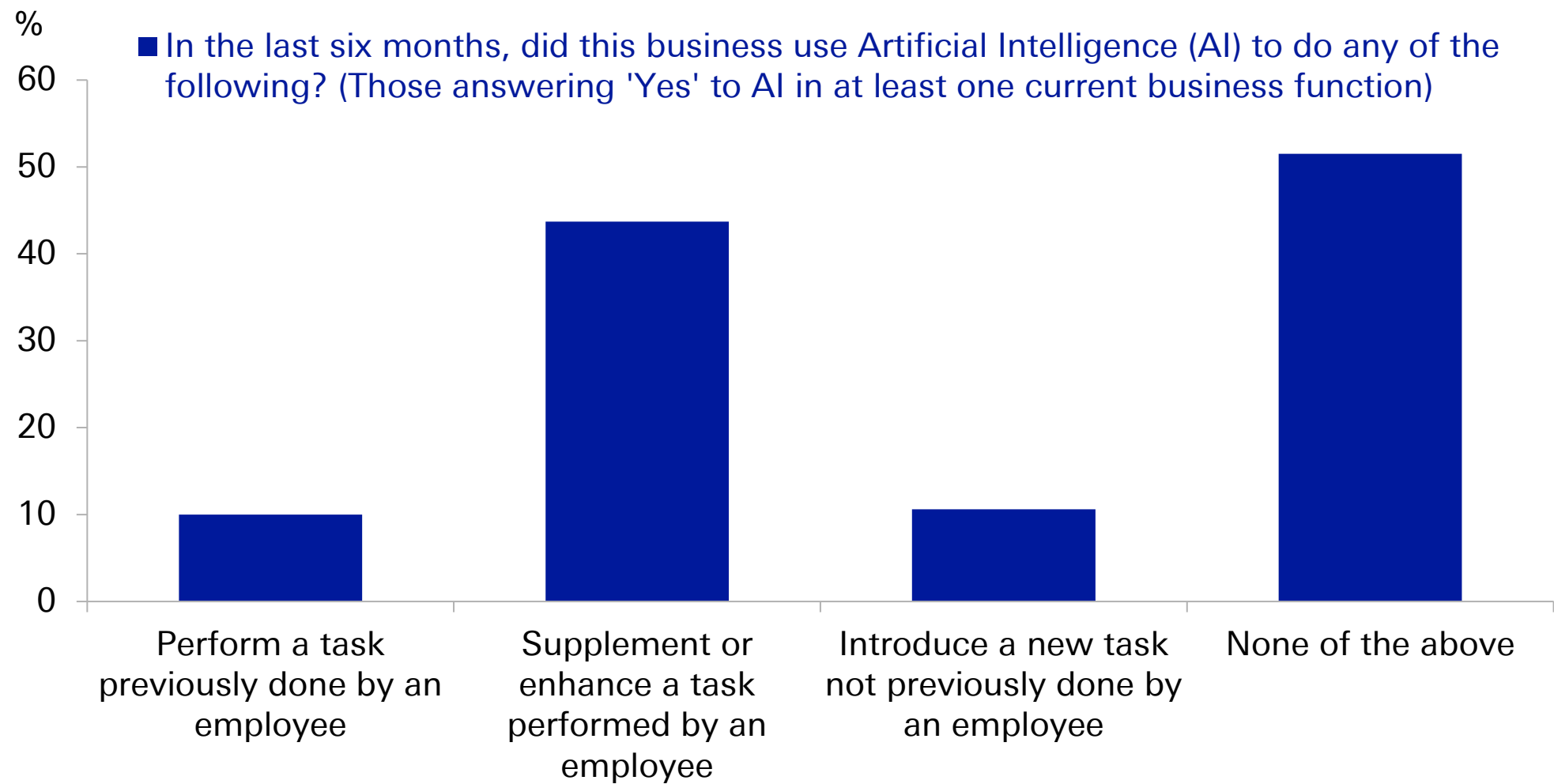


Source: BTOS, Census, Deutsche Bank

Companies adopting AI currently utilise it mainly to augment and enhance their employees' existing roles.



In the last 6 months, did this business use AI to do any of the following?

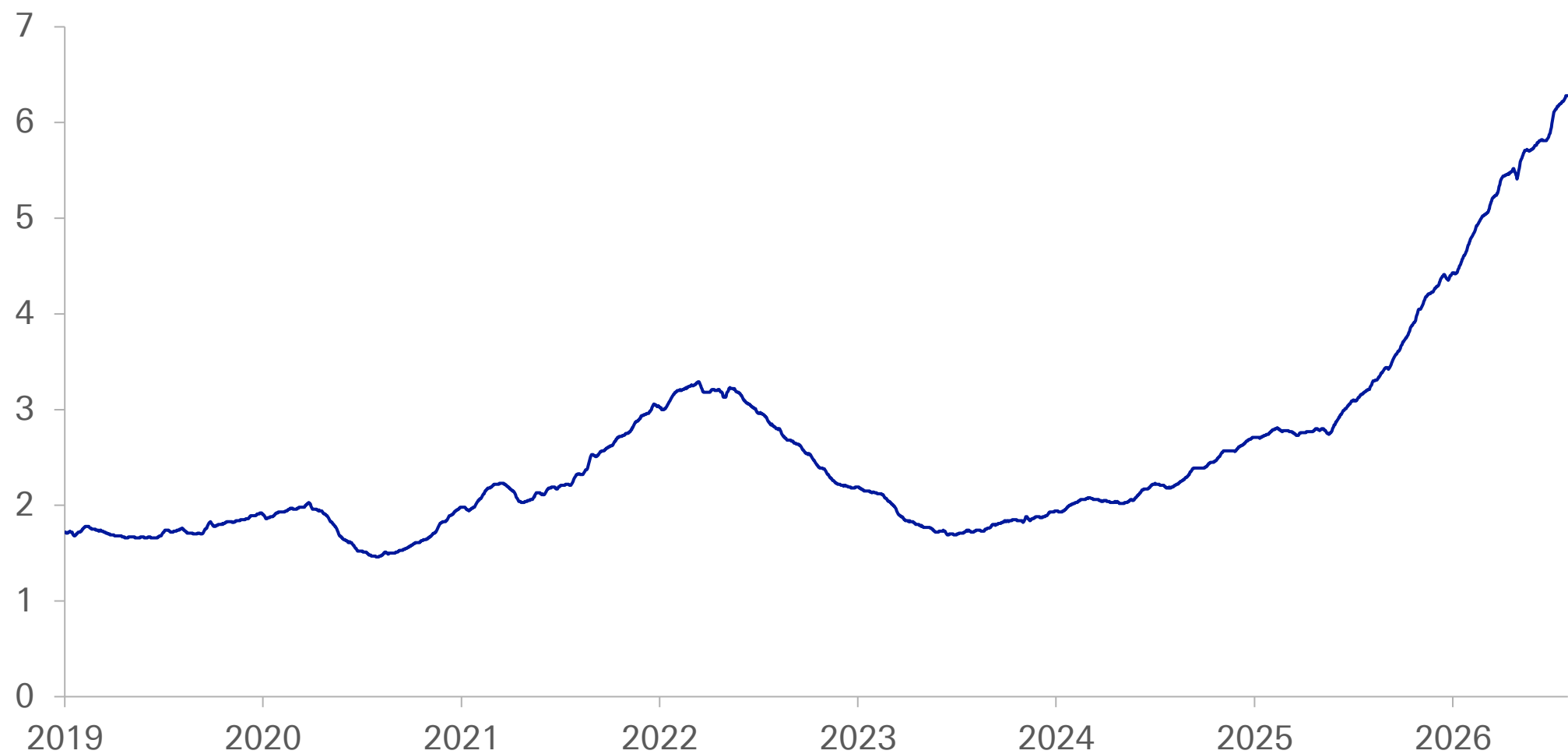


Source: BTOS, Census, Deutsche Bank

AI-related job postings have increased notably year-over-year in the US



Indeed AI Tracker: Share (%) of job postings on Indeed containing Artificial Intelligence (AI) terms in United States

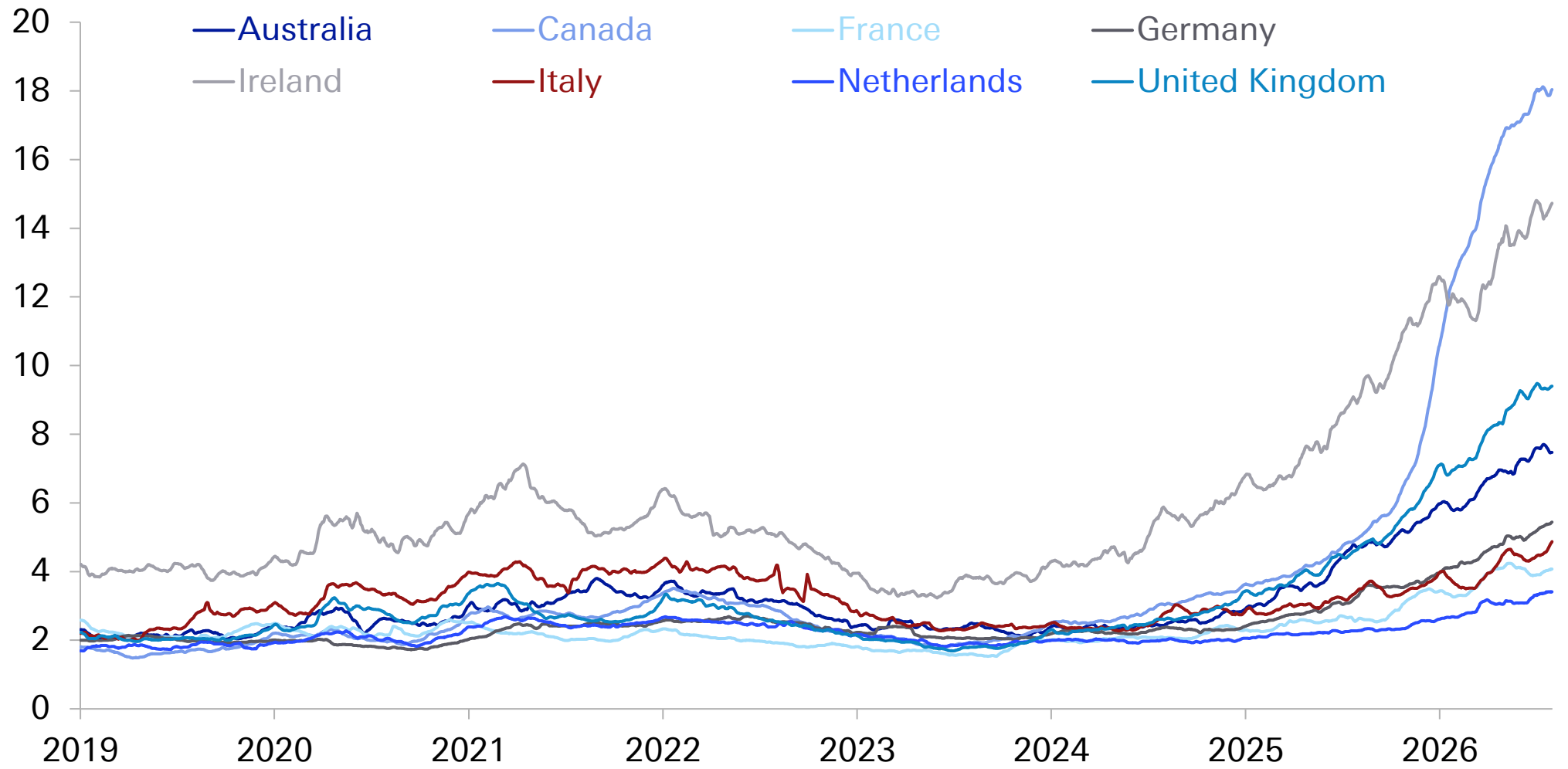


Source: Indeed Hiring Lab, Deutsche Bank

The share of job postings referencing AI is rising across countries. Is AI actually creating job openings rather than seeing them disappear?



Indeed AI Tracker: Share (%) of job postings on Indeed containing Artificial Intelligence (AI) terms

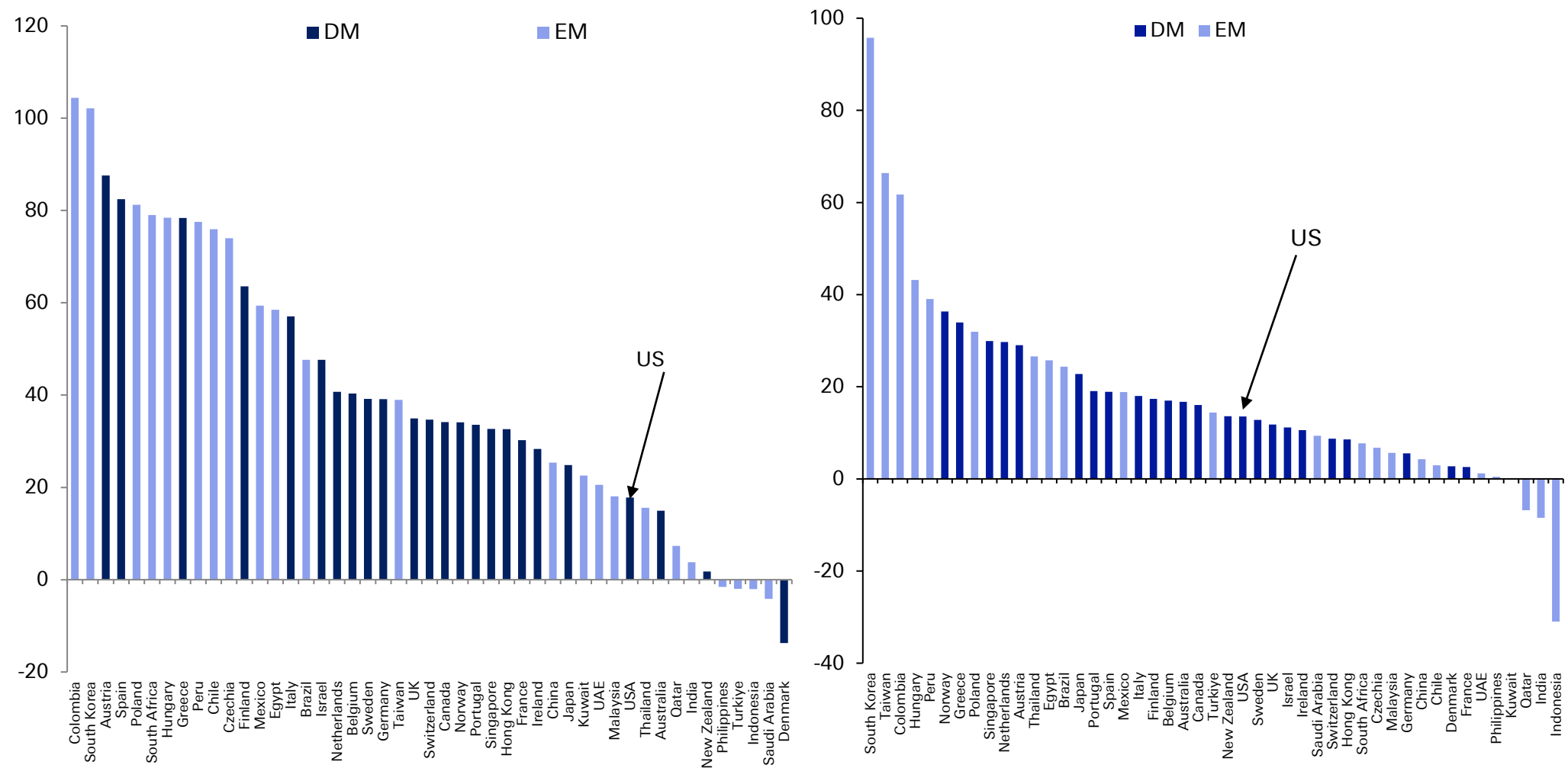


Source: Indeed Hiring Lab, Deutsche Bank

Global equity leadership has changed over the last two years as the AI cycle matures. The US has had two decent years but is in the middle of the global pack in 2026, after being around the bottom quartile in 2025...



Bloomberg large/mid-cap country stock market indices: 2025 total return in USD (%) (left) and 2026 YTD (right)

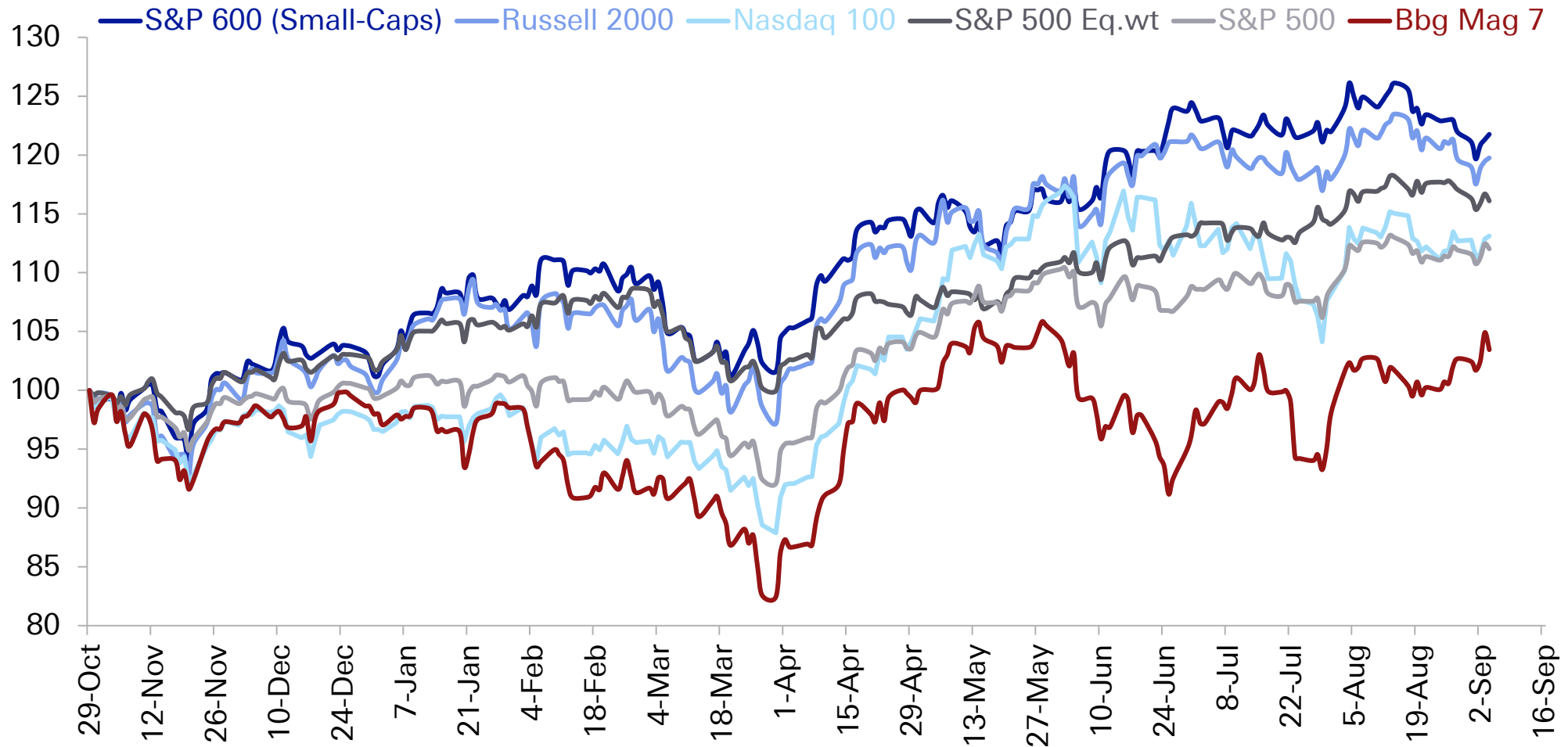


Source: Bloomberg Finance LP, Deutsche Bank, Note: 2026 YTD is updated through Sep 7

Since the end of last October, the Mag-7 have notably underperformed and are up less than 5%. However, other parts of the US market have performed strongly.



Rotation away from large cap tech. US Index performance (Oct 29, 2025=100)

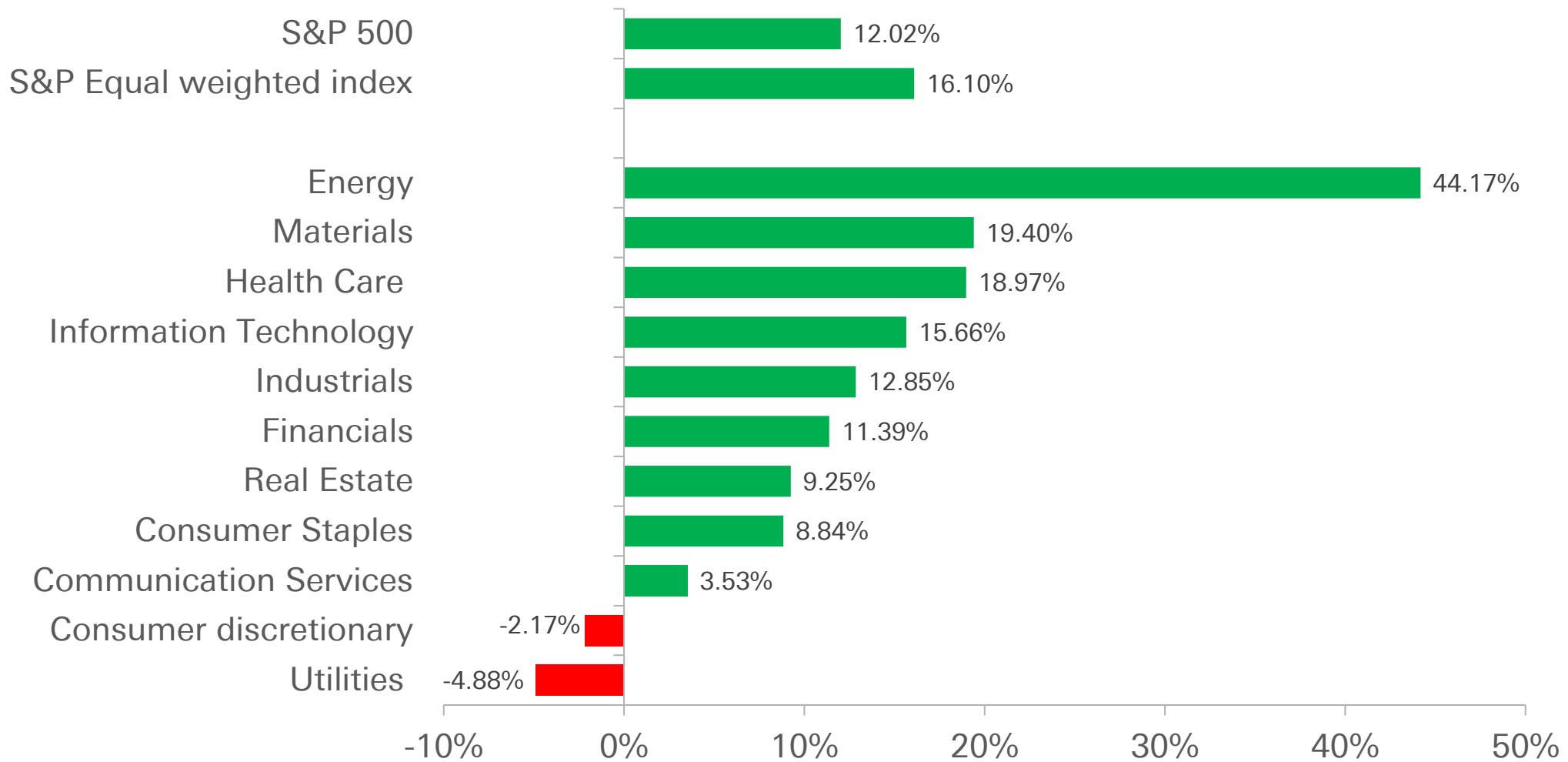


Source: Bloomberg Finance LP, Deutsche Bank., Last update Sep 7

Even with the Mag-7 underperforming, tech has performed in line with the market since tech started to see winners and losers emerge at the end of October last year. But tech has been outpaced by Energy, Materials and Health Care. The equal weight S&P 500 has outperformed.



S&P Sectors return since October 29, 2025

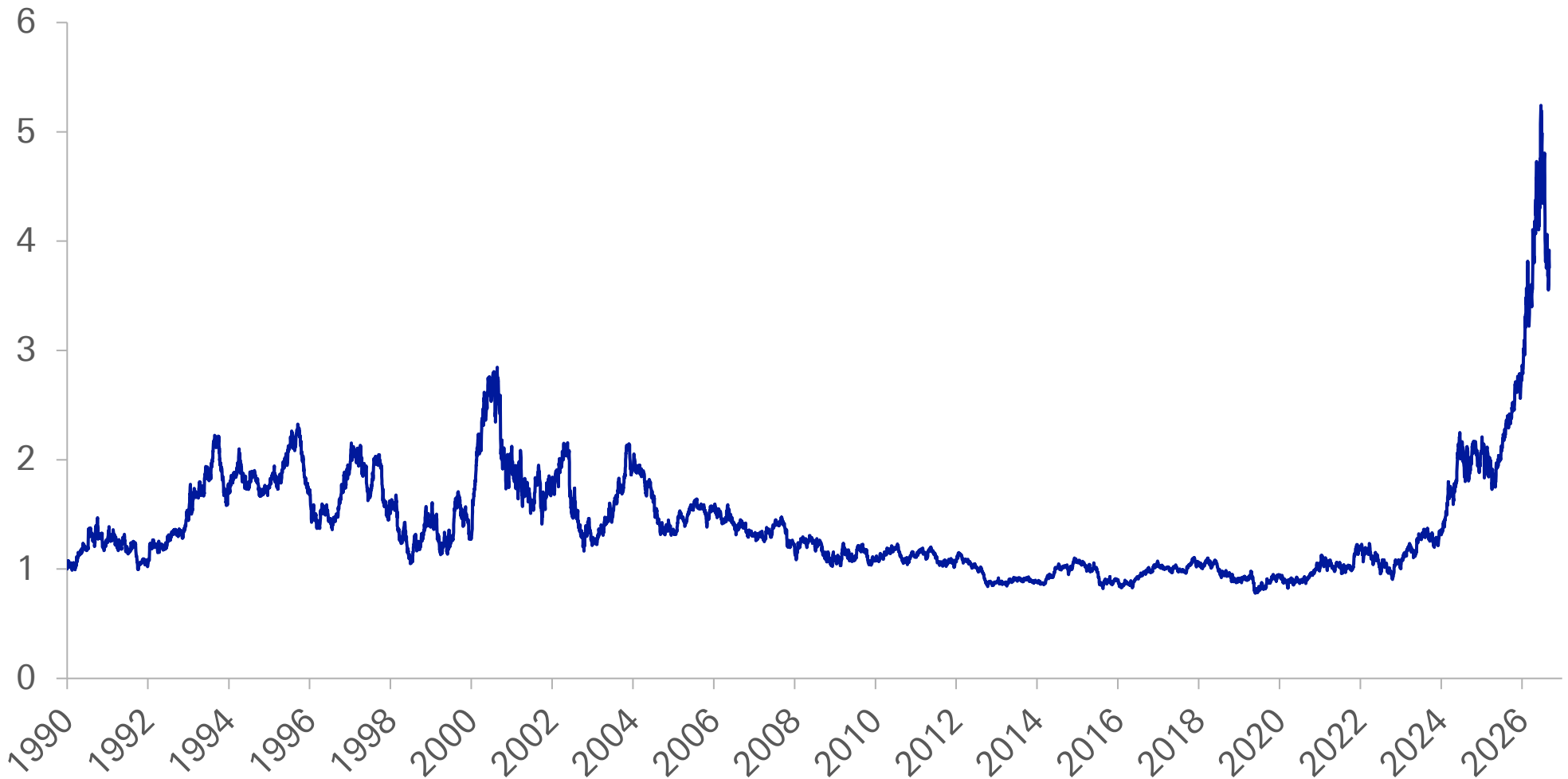


Source : Bloomberg Finance LP, Deutsche Bank., Last update : Sep 7

The tech winners and laggards theme can best be summed up in this chart... both are sizable sub-components within the tech index... Since the end of June software and services have made a comeback relative to semis but the major underperformance over the past 18 months is clear ...



S&P 500 semiconductors & equipment relative to software & services, total returns ratio, since 1990 (=1)



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7.

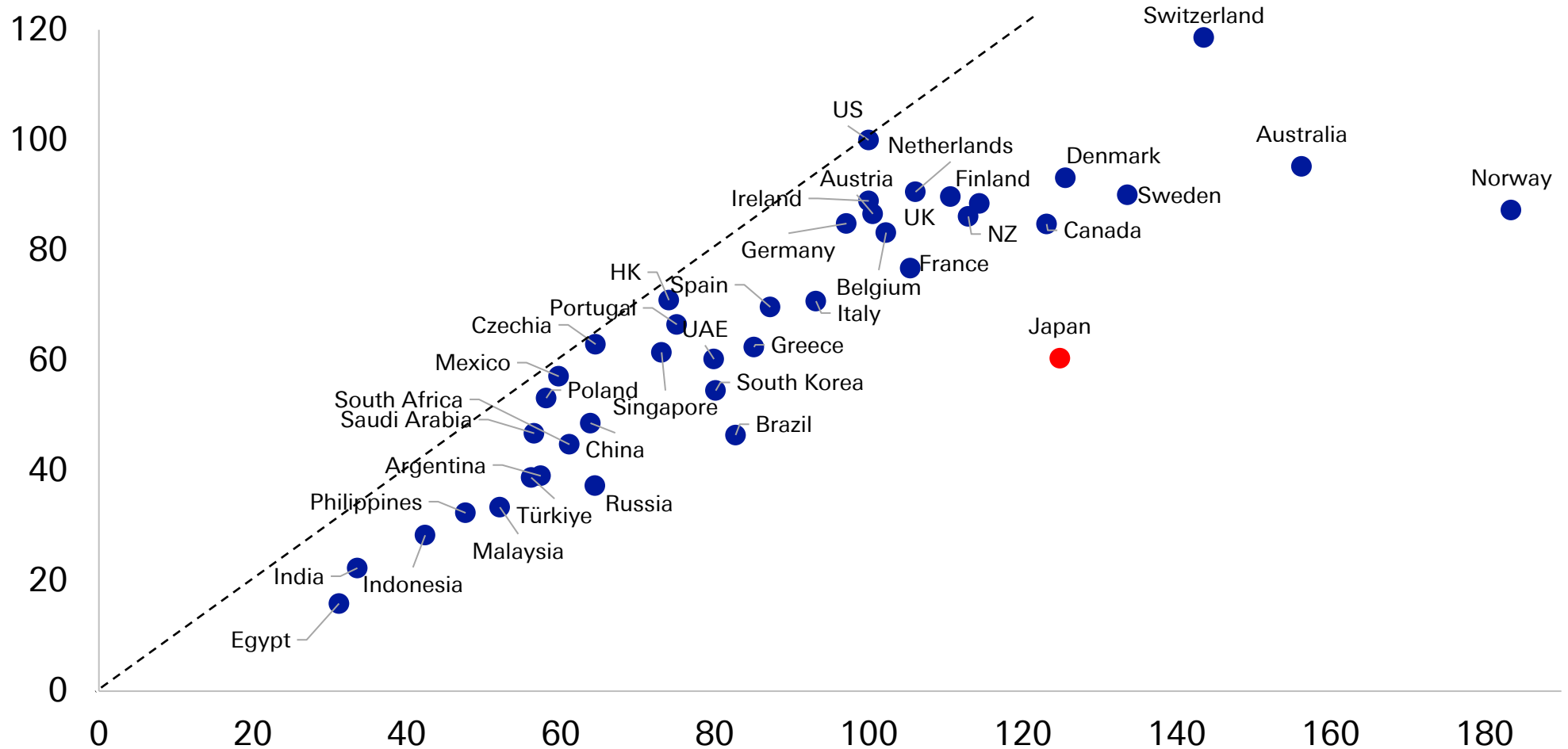


Japan is worth a watch...

Japan has cheapened up extraordinarily since the early 2010s....



Relative Price Levels as Implied by PPP (US = 100), 2012 (x-axis) and 2025 (y-axis)

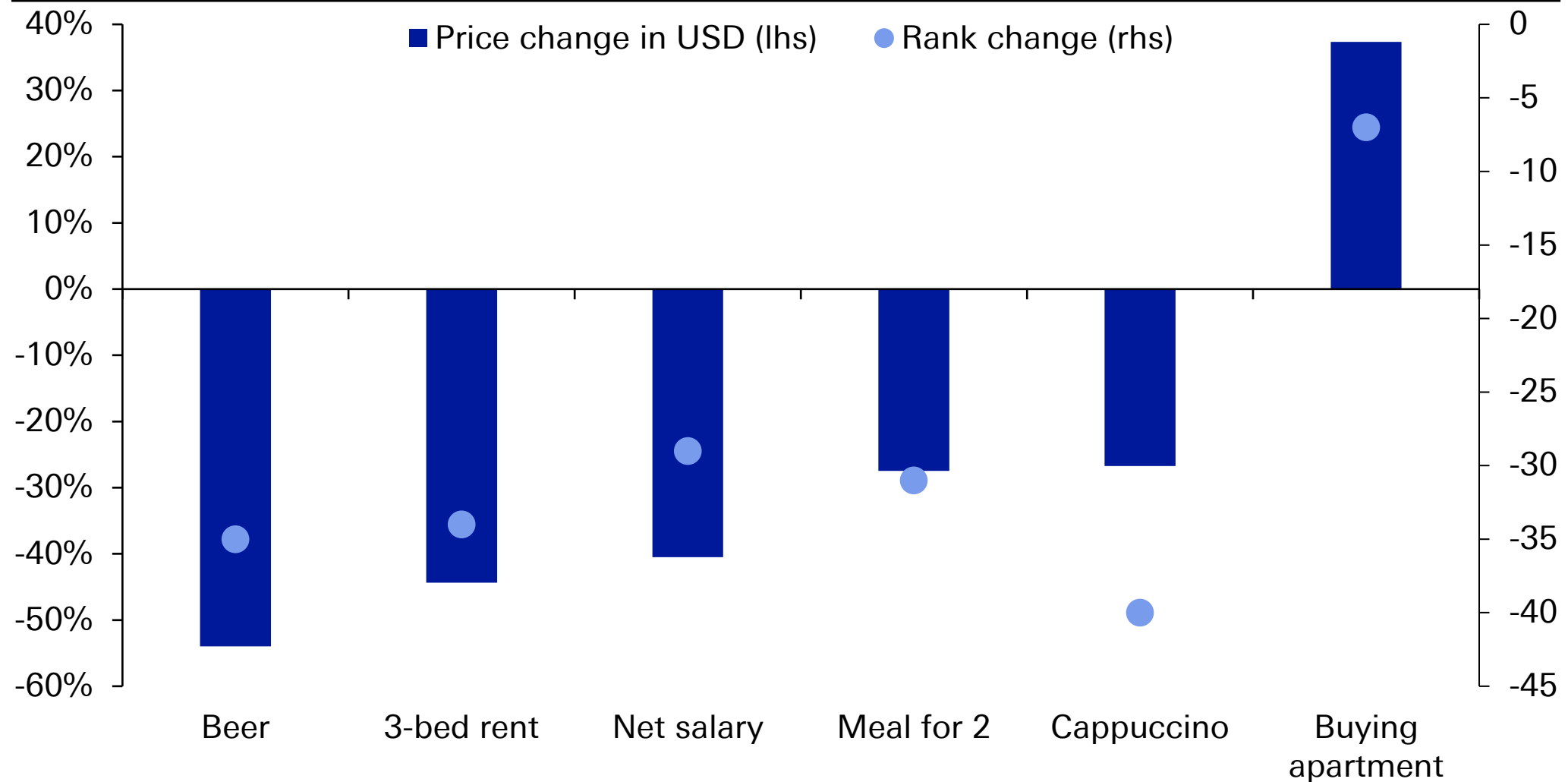


Source : IMF, WEO, Haver Analytics, Bloomberg Finance LP, Deutsche Bank calculations; Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on 31 December each year) to derive the implied price levels. Also note that some of the above numbers may not match those published last year exactly due to data revisions.

Is Tokyo the cheapest city in the world relative to its economic prowess? Maybe see you there for next year's holiday... its very cheap... See Mapping the World's Prices 2026 [here](#) for more evidence



Changes in prices of goods and services in Tokyo relative to 2012 in USDs



Source : www.numbeo.com, Deutsche Bank

The Yen is bouncing off 40-year lows...



USDJPY Spot Exchange Rate - Price of 1 USD in JPY

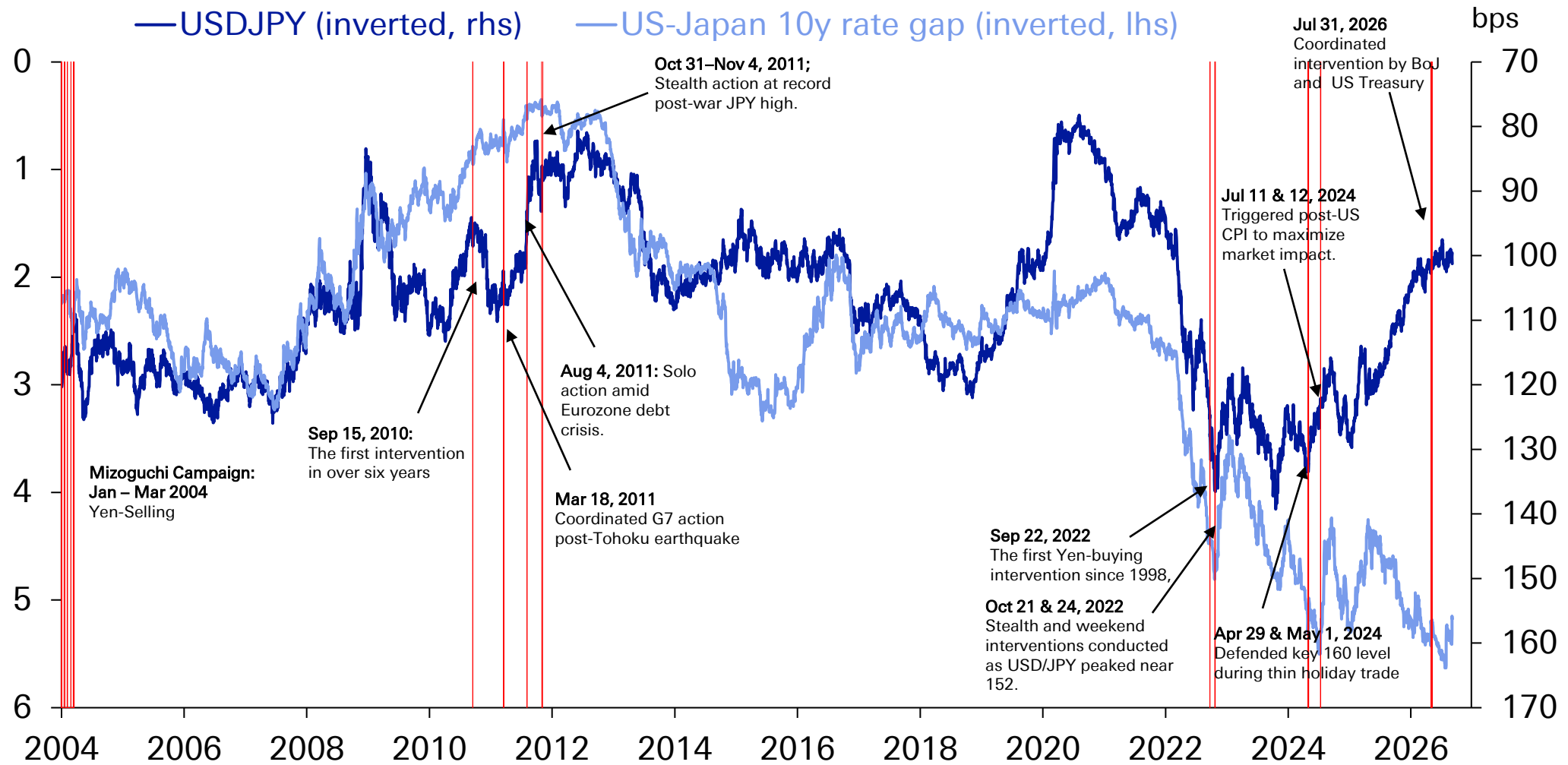


Source: Bloomberg Finance, Deutsche Bank; Last update: Sep 7

Since early 2025, the Yen has decoupled from its traditional anchor: interest rate differentials, compressing yield gaps have failed to spark JPY strength... The BoJ is the most likely source of a sustainable rally



USD/JPY and 10y spread with BoJ intervention episodes annotated (inverted axis)

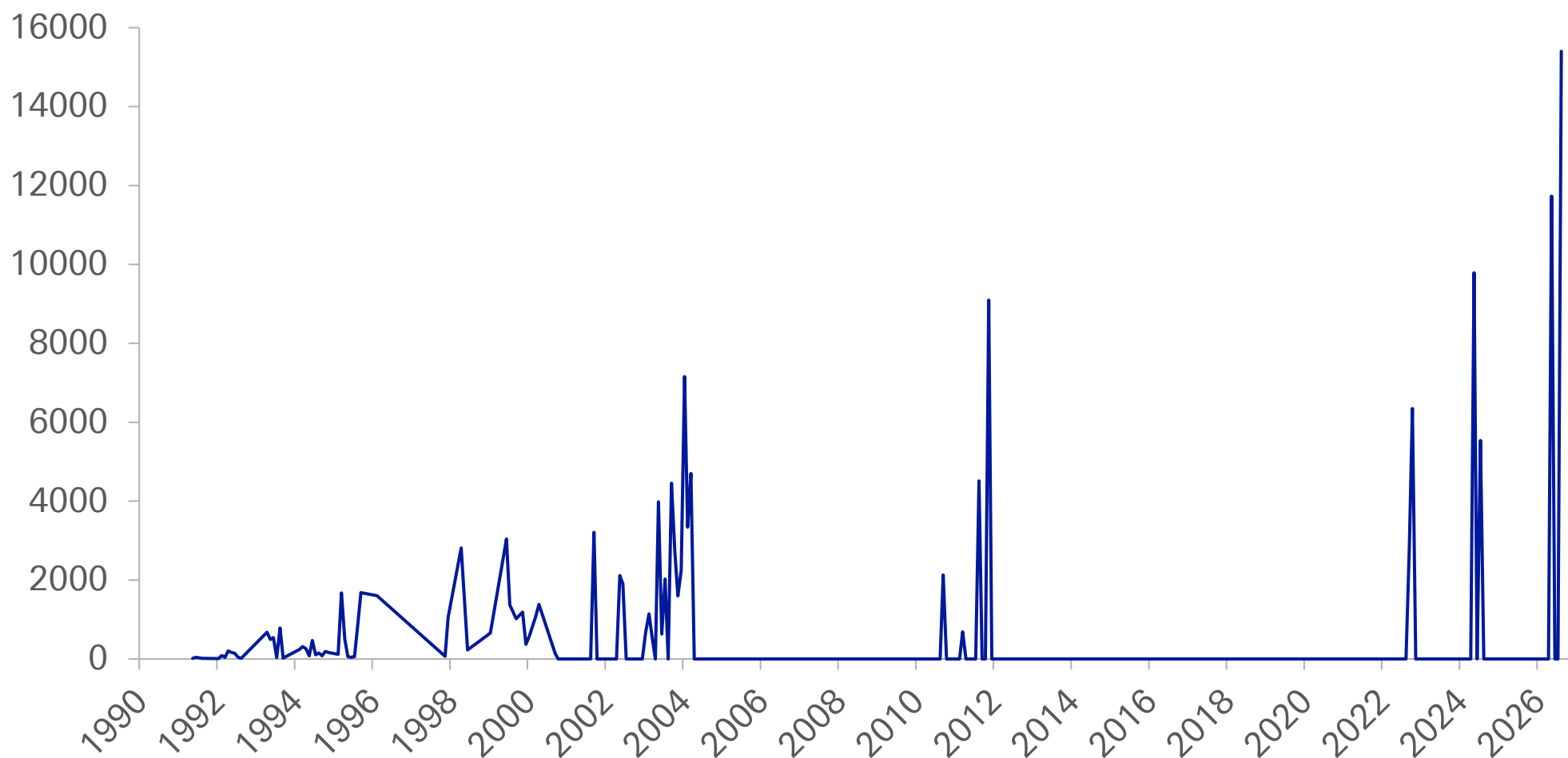


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

Japan spent ¥15.4tn on market intervention from April to August—the most since 1991



BOJ Japan Foreign Exchange Intervention Operations - Total Amount (JPY bil.)



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7

Appendix 1

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Capitalism and the dollar: the AI race, tokenization, and the dollar's future

September 03, 2026

A massive race for capital

The US is sprinting in its reach for capital. Private markets, debt, and public equities have been setting records, and the US is enjoying unprecedented amounts of foreign equity inflows. In this paper, we discuss how the US is leveraging the power of capitalism – supercharged by blockchain – to try and win the AI race. But the US reach for capital is not only about AI spending; it also reflects growing fiscal deficits and the need for foreign private funding. The US is increasing its reliance on private capital to offset public sector vulnerabilities. Meanwhile, China is running a very different race – given large trade surpluses, domestic excess savings, and a "low-cost" AI approach, China is raising less capital, with much less from abroad.

America embraces asset tokenization

The US is using blockchain technology to support US dollar and capital market dominance. We wrote last year about the crucial role of stablecoins. We are now at the next stage: asset tokenization – converting ownership rights to financial assets into digital tokens recorded on blockchain. Asset tokenization is not futuristic – it is happening now at the heart of US markets. Securities depositories and exchanges are building the infrastructure to tokenize US assets, with regulatory oversight.

TINA meets Tokenization: the future of the dollar

TINA already plays a key role attracting capital to the US. Tokenization could add to this by making US assets more attractive and accessible. The collateral velocity of tokenized assets could be a superpower. And if US assets are trading 24/7 with instant settlement, this could impact global behaviour, particularly for retail. But lowering the hurdles for inbound capital also exposes the US to capital going out. The dollar is becoming riskier, more equity sensitive, and leveraged to the AI race. If AI proves uneconomical, or the US starts to fall behind, the dollar is very exposed.

Open markets meet Open technology: the future of capitalism

The US is taking an "open markets-closed tech" approach to the AI race, while China is taking an "open tech-closed markets" one. The US is widening the doors for capital but keeping technology closed to preserve corporate pricing power. Meanwhile, China is pushing out low-cost models with open-weights. This is not just a contest between AI models, but economic models. If China's approach undercuts US profits, this could threaten the US model of shareholder capitalism itself. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation.

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Part 1: A massive reach for capital in the US

The US is sprinting at record speed in its reach for capital. Huge sums of money are being raised to fund the buildout of AI infrastructure and development of frontier AI models, with US companies expected to spend around \$800bn in AI capex this year.

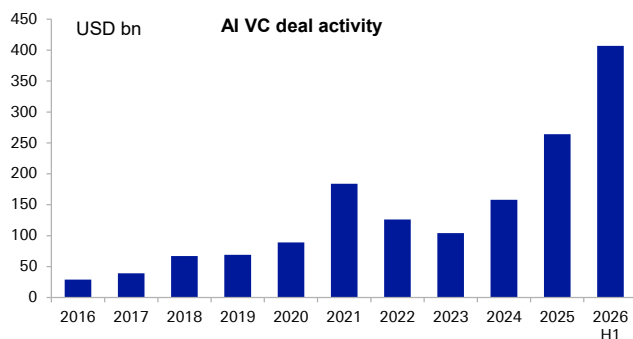
We argue in this paper that the US is leveraging the full power of capitalism to fund capital investments seen as crucial to winning the AI race

The US is raising enormous amounts of private capital

We simplify the key trends in the US' capital-raising landscape into a few observations. First, AI labs are turning to private markets to raise large sums of equity capital. Second, hyperscalers are turning to investment grade debt markets to issue meaningful amounts of debt for the first time. And third, public equity markets are back in focus for capital funding with some of the world's biggest companies returning to issue stock, and major IPOs in the pipeline.

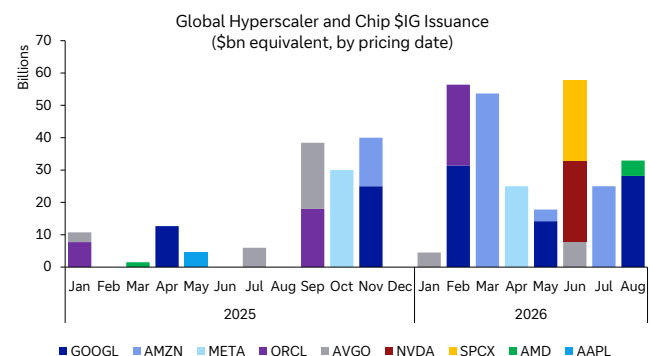
- The AI venture capital industry has raised over \$400bn this year alone, with over 90% of the biggest deal volumes in the US ([Pitchbook](#)). The two largest AI labs in the US have raised a combined \$217bn this year – valuing each company near \$1tn. We note that xAI had raised \$20bn earlier in the year, before being acquired by SpaceX in February. On an annualized basis, AI VC capital is running at three times last year's pace.
- In debt capital markets, the shift has also been staggering. While free cash flows remain an important source of funding for hyperscalers, the capital intensity of the infrastructure buildout is outrunning cash flows, forcing companies to turn to debt. Hyperscalers like Google, Meta, Oracle and Amazon raised an average of just USD25bn per year between them over 2020-24. In 2026 YTD, they have raised around ten times that amount through IG credit markets. Microsoft has yet to turn to debt markets.
- In public equity markets, Google also had its first primary equity raise since its 2004 IPO in June 2026, raising \$85bn. Returning to issue is a big reversal from years of buying back stock. This is also setting up to be a historic year for IPOs. SpaceX achieved the largest total market valuation of a newly listed company in history near \$2tn in its June 2026 IPO. The largest US AI labs are expected to come to the IPO market soon which could change the AI boom as we have written about [here](#).

Figure 1: US AI companies are attracting record amounts of private capital



Source : Deutsche Bank Research, Pitchbook

Figure 2: US hyperscalers have turned to public credit markets to issue huge amounts of debt



Source : Deutsche Bank Research, Bloomberg Finance LP



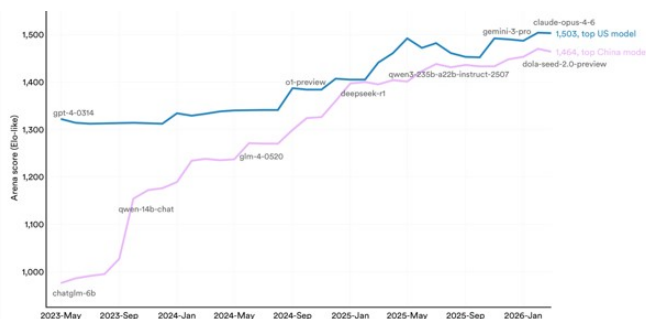
China is running a different race

On many metrics, Chinese models are closing the gap to the US in terms of performance and capabilities (Figure 3). While the US remains ahead in frontier development, Chinese models account for more than half of US company token usage on Open Router (Yahoo Finance).

But from a capital perspective, China appears to be running a different race - raising much less capital, less from private sources, and less from abroad. While US companies are expected to spend around \$800bn in AI capex this year, that number is closer to \$100bn in China. While the biggest US AI labs have raised nearly \$250bn in private funding rounds this year, Chinese labs have raised far less. The largest capital raise came from DeepSeek which raised \$7.4bn earlier in the year, and may be returning for a further \$8bn. China's National AI fund has participated in a number of capital raisings. Stanford University's [recent AI report](#) made a similar observation: noting that in 2025, US private AI investment was 23 times the \$12.4bn invested in China, although Chinese numbers are likely understated given government guided funds.

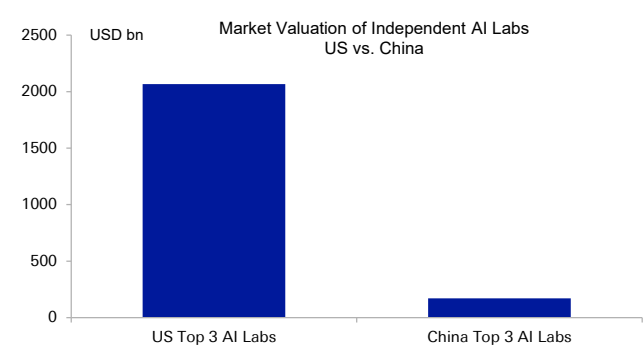
The stark difference in the approach to capital raising is also reflected in markets. With US AI labs raising far more capital for ownership stakes, the market valuations of US AI companies compared with their Chinese peers are many orders of magnitude apart (Figure 4). Furthermore, while Chinese hyperscalers like Tencent, Baidu and Alibaba have been tapping debt markets, they are doing so in smaller size, with a no major uptick compared with earlier years. They are, however, increasingly funding in dim sum bonds, suggestive of a desire to de-dollarize funding. See more on the Rise of the RMB Financing Ecosystem [here](#).

Figure 3: The gap in performance between US and Chinese models has narrowed a lot



Source : Deutsche Bank, Stanford University HAI

Figure 4: US AI labs have much larger implied market valuations than their Chinese peers



Source : Deutsche Bank, Bloomberg Finance LP, Various news sources; Note: Valuations are based on market cap for publicly listed companies and announced or intended funding rounds for private companies.

The US will likely want all the foreign capital it can get

The dramatic reach for capital in the US is not only reflective of an American economic model that prioritizes open capital markets, it also reflects the US' growing need for foreign financing given the US' large twin deficit position. The US is running a 6%+ fiscal deficit combined with a near 4% of GDP current account deficit. In the absence of fiscal savings, incremental private sector capex will require funding from abroad. At a time when private sector financing requirements for AI capex are rising, it is not a surprise the US is widening its reach for capital from abroad.

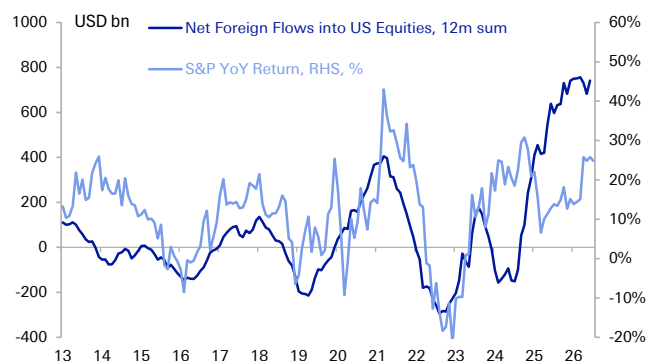


The US is in fact enjoying an unprecedented amount of equity capital inflow. In Q2 2026 alone, the US received over USD400bn of foreign equity capital, well above any previous quarter. The amount of capital coming into US equities is also well above what US equity performance alone would imply (Figure 5), and now well above debt flows (Figure 6) – the historical mainstay of US capital account financing.

We have written in the [past](#) about the big rotation underway in US financing. Geopolitical ruptures in long-standing alliances in defence and trade are hurting long-term appetite for USD government debt, particularly by the foreign official sector. But at the same time, technology is pulling enormous equity capital in from other types of foreign investors like retail. The gap between net equity flows to the US (increasing) and debt flows (falling) has never been wider, and the shift from official to private funding sources is becoming more noticeable. With the US on the cusp of a potentially significant IPO cycle, we find it notable that US companies are picking up on the funding shift. SpaceX reportedly initially earmarked 30% of its IPO for retail investors, more than three times as much as a traditional IPO offering.

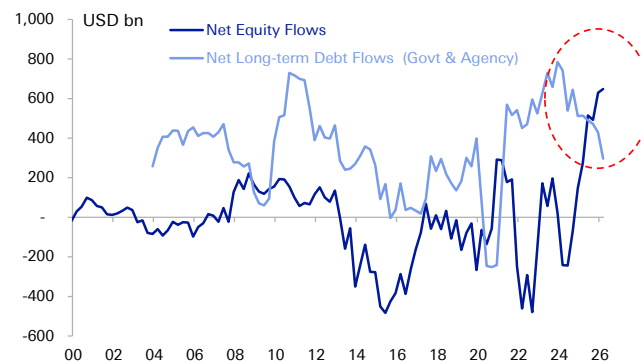
As we show in this paper, the US will likely remain very focused on attracting foreign private capital to fund both its AI ambitions and its deficits.

Figure 5: The US is attracting a record amount of foreign equity capital - more than market returns alone would suggest



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 6: Equity flows have replaced debt flows as the main source of foreign capital financing for the US



Source : Deutsche Bank, Haver Analytics

Where blockchain technology comes in

The US is taking a very capital-intensive approach to the AI race, raising record amounts of capital, and attracting huge amounts of it from overseas. Foreign capital not only helps support the dollar by providing deficit funding, it also improves the cost of financing for US companies. It is natural for the US to leverage technology that supports these capital flow trends.

We have [written](#) at length about how the US is using the power of blockchain technology and a new form of private money – stablecoins – to democratize access to the dollar. **We are now at the second stage of evolution, where the power of blockchain technology is being used to democratize access to US capital markets. This is called real world asset (RWA) tokenization.** While stablecoins aim put the US dollar "on chain" as a digital money token, asset market tokenization could look to put the entirety of US asset markets and their underlying financial



architecture on chain.

The two forces of tokenized money and tokenized assets are themselves highly symbiotic. The existence of tokenized assets should widen the appeal of holding tokenized money like stablecoins, that can then be more easily deployed for returns and yields "on chain", without needing to be moved off-chain back into fiat currency to invest in traditional assets. Both stablecoins and tokenized bank deposits could become important means of payment for tokenized assets, with a meaningful impact on the settlement process. We will explore the nuances of this in a forthcoming FX Special Report on Programmable Money.

From a macro perspective, we argue that US tokenization ambitions are deeply linked to supporting capital inflows into the US. Our earlier research on [stablecoins](#) discussed how they could support the diversification of dollar debt to private sector holders. Not only did the GENIUS Act introduce 1:1 reserve backing requirements for dollar stablecoin issuance, that helps power demand for front-end US T-bills; more importantly, if stablecoins help keep the world paying in dollars, they help retain the role of the dollar as a private sector savings instrument, which could help demand for assets like longer-dated Treasuries. If stablecoins are now combined with a wider universe of tokenized assets, this could encourage demand for many more US assets. This should help what we have termed a "geopolitics-to-technology funding swap". The US is moving away from attracting official, long-term, geopolitically motivated capital flows, towards private sector, shorter-term, technology motivated capital.

America's full-hearted embrace of asset tokenization – compared with China's turn away from it – can be understood in the context of both the AI race and the relative external funding position of both states. If the US is taking a capital-focused approach to the AI race, with a starting point of twin deficits, the US will naturally benefit from lowering the hurdles for foreign and private capital as much as possible. That is indeed what it is doing.

Part 2: America embraces Asset Tokenization

We begin with a definition for asset tokenization, borrowing from the [DTCC](#), the custodian for close to \$115 trillion in US assets and an organization that finds itself at forefront of this development. **Asset tokenization is the "process of converting ownership rights to a financial asset – such as stocks, bonds or real estate – into a digital token recorded on a blockchain. These tokens represent the same legal rights as the original asset but can be transferred, tracked and managed digitally."** They note that tokenization introduces new capabilities, from fractional ownership, faster cross-border transfers, transparent records and smart contracts. In a tokenized world, instead of waiting days for a U.S. Treasury bond to settle through multiple intermediaries, a tokenized version could settle in minutes. Also see "Asset Tokenization 101" from the DB Research Institute [here](#).

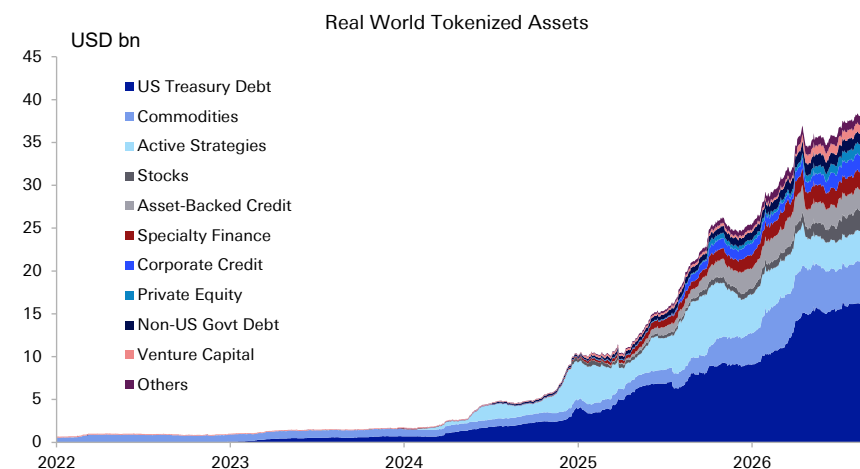
Only around \$40bn of real-world assets have been tokenized [to date](#), but the size of the tokenized asset universe is expected to grow much further and faster from here given adoption by traditional finance that is just getting underway.

In Part 1 of this paper we explored the *macro* motivations for America's embrace of tokenization: to reduce the hurdles for foreign capital to flow into the US and fund its AI ambitions in an increasingly competitive and existential AI race. In this



section, we look at some of the more micro motivations for tokenization, and how they come together.

Figure 7: Real world tokenized assets have been growing, but are still a fraction of the over USD100tn US asset universe - leaving big headroom



Source : Deutsche Bank, RWA.xyz

Some further motivating drivers for asset tokenization

The underlying technology that enables tokenization is distributed ledger technology (DLT) which allows the ownership rights to anything from money, to a stock, to a painting, to be represented as a digital token on a decentralized network like blockchain. But new technology alone cannot drive real world adoption and change, without the financial and economic incentives to use it.

So, why is DLT being used to tokenize real-world assets today and how have incentives aligned across economic agents – from retail investors, asset managers, corporates, financial firms, securities depositories and exchanges – to motivate this adoption? Beyond the macro, state-level motivation for attracting capital to fund the AI race and US deficits, we argue that private agents in the financial system are also likely to be deeply invested for their own reasons.

First, the revolution in overseas retail trading demonstrates the case for widening access to markets. Retail trading has been spurred on in recent years by a number of developments: from Covid lockdowns, the rise of zero-commission platforms, social media, fintech brokerages, improved access to information and charting, and even tax incentives to increase financialization of household savings (i.e., Japan's NISA program). Korea is arguably the poster child of the retail revolution, and offers a potential template for what might become possible at a global scale if technology is used to lower hurdles for entry. Korea represents 2% of global nominal GDP, but accounted for 10% of the \$740bn of foreign equity inflows to the US last year, a lot of which were powered by retail investors, or the "seohakaemi" (overseas stock ants). Access arguably played a big role. Korea began allowing for the fractional trading of foreign stocks by certain Korean brokerages as early as 2019, even earlier than for domestic stocks. Korean brokerages also signed up with a number of US alternative trading systems that offered daytime trading of US stocks in Korean hours. The combination of a highly digitized, connected population with lower hurdles for access was a dramatic flow of Korean capital into US equities. To us, the case of Korea demonstrates what is possible if US markets were to become easily



accessible 24/7 to everyone.

Second, corporates and institutions still face a number of inefficiencies around cash and capital in traditional off-chain markets. We focus on three examples here: trapped cash for settlement, inefficient collateral management, and manual corporate actions. While securities markets in the US have successfully moved from T+2 to T+1, this gap still involves the lock-up of capital to act as a backstop for trade settlement. If securities and payments were to settle atomically and instantly on blockchain, this could free up a huge amount of capital. Collateral markets are also still relatively inflexible today. Today, repo markets allow financial entities to borrow cash in exchange for lending collateral like US Treasuries. Repos are structured around overnight to longer-term financing and are constrained by end-of-day settlement cycles and operational windows. Digital assets held on a blockchain could arguably be posted instantly for much shorter periods, creating intraday repo possibilities and increasing the velocity of collateral. As a final example, NASDAQ highlights that processing corporate actions today alone "carries costs of more than \$58 billion across the industry, driven by fragmented systems, manual workflows, and duplicative record-keeping." The programmability offered by digital assets would help address this as corporate actions could be programmed into smart contracts attached to the asset. In sum, there are clear incentives for agents – from corporate issuers to institutional asset managers – to embrace a tokenized asset environment.

Third, the tokenization train had already left the station but was at risk of being mediated via third-party offerings and fragmenting market liquidity. In the traditional space, we note that fund managers like Blackrock and Templeton had already begun issuing some of their funds like BUIDL and FBOXX on public blockchains. This involved making the funds accessible as digital tokens, but importantly the underlying assets themselves remained off-chain in traditional financial instruments. More importantly, the presence of blockchain technology has allowed for entry and innovation by new private players that have begun offering US assets as digital tokens on blockchain rails. New tokenized equity frameworks offer *representations* of real US stocks and ETFs, backed 1:1 by underlying traditional equities, held in regulated custody, and issued as on-chain tokens on many different blockchains. They are available to non-US clients in 100+ countries, and allow fractional purchases with as little as \$1 in select US stocks and ETFs, some of which are available to trade 24/7. They do *not*, however, confer shareholder rights like voting or dividends in cash – instead, they offer price exposure to the underlying asset in an always-on format, via a *third-party* token. If tokenization was already underway, the incentive for traditional asset managers and exchanges to be part of the process was becoming more evident – to avoid losing funds and liquidity to third parties. Rather than simply tokenizing *claims* on US securities, developments underway are now building the infrastructure to allow underlying US securities *themselves* to be issued, traded, and held as tokens.

As we discuss in the next section, asset tokenization is not a futuristic or frontier concept – it is live and is now happening at the heart of traditional markets, led by core players in the US financial ecosystem.

What is happening?

In a nutshell: tokenization is going mainstream, with adoption by core, traditional financial players. Securities depositories and exchanges under the aegis of regulatory guidance are building the market infrastructure to tokenize the US asset universe, with participation from asset managers and banks.



Regulatory efforts arguably kicked off in force with the launch of [Project Crypto](#) by the SEC in July 2025, alongside the [White House's Digital Asset Report](#) that called for laying the groundwork for the US to become the deepest and most liquid digital asset market in the world. One of the first tasks was to develop a [taxonomy](#) for crypto assets, where tokenized securities were importantly defined as *remaining "securities"*. In January 2026, Project Crypto evolved into a [joint inter-agency framework](#) between the SEC and the CFTC that would allow for "coordination, coherence, and a unified approach to the federal oversight of crypto asset markets". Crucially, it is actions from regulatory agencies in the US that are driving progress, independent of pending efforts in Congress to pass the CLARITY Act.

The big green light for mainstream tokenization came in December 2025 when the SEC issued the DTC with a ["no-action" letter](#) which legitimized the idea that the same security could trade on two rails. DTC could therefore tokenize assets in its custody from US stocks, ETFs and bonds, and the digital version of these assets would have "the same entitlements, investor protections and ownership rights" as the traditional asset. This was followed up by a [ruling](#) in January 2026 which noted that the "format in which a security is issued or the methods by which holders are recorded (e.g., on-chain vs. off-chain) does not affect application of the federal securities laws". Assets can be held traditionally or in tokenized form (as a digital twin) and they will be treated the same, and will refer to the same security. In other words, **both traditional and tokenized architectures can now coexist, run side-by-side, and be interoperable, with the trust of DTC custody to back them.** While the passage of the CLARITY Act is still pending, the above SEC notifications and the commitment of Project Crypto to bringing US capital markets on chain means that the regulatory umbrella for tokenization has been sufficiently opened. This has been enabling of developments underway this year.

The progress at DTC remains crucial to track. DTC is the main central securities depository in the US – it is the venue where trades settle and clear, and the custodian where financial firms hold their assets. DTC has around \$115 trillion of US assets under its custody. Tokenization at DTC has now begun. In July 2026, the DTC converted the first asset into tokens that were used for real world trades. 40 different financial firms participated. Amongst many examples, the SPDR S&P500 ETF Trust (SPY) – the world's first ETF – was tokenized. In another test, JP Morgan successfully posted tokenized assets to satisfy a margin requirement at the CME. The full set of tests that day can be found [here](#). DTC intends to launch its tokenization service by October 2026, but full-scale tokenization will be a long process. The DTCC CEO has argued it could proceed over many years with markets "crawling, then walking, then running" and the pace to be set by market demand to hold assets as tokens. DTC is beginning by tokenizing the Russell 1000, ETFs, and US Treasuries before widening it out. Market [estimates](#) currently range from \$2tn to \$30tn for the size of the tokenized asset universe by the 2030s, but these estimates may well be revised higher as the infrastructure becomes operational.

US exchanges – NYSE and [NASDAQ](#) – are shifting to accommodate "always-on" trading. In current financial architecture, exchanges are the *trading* layer, where buyers and sellers meet and prices are discovered, while DTC is the *post-trade* infrastructure layer for settlement and custody, maintaining a central record of securities ownership after trades occur. The biggest development at US exchanges is that they are set to move towards round-the-clock trading. NASDAQ has also announced an intention to offer trading 23/5 trading by December 2025, while NYSE is aiming for 22 hours by later this year. As noted earlier, synthetic equities



have already been trading outside market hours on permissionless blockchains and alternative trading systems, so increasing exchange hours will allow that liquidity to come back on exchange, and prevent tokenized equities from becoming off-exchange products that fragment liquidity.

Exchanges are also building new platforms and tools to enable a tokenized trading experience. The NYSE's [new digital platform](#) – being built in conjunction with Securitize – aims to not just offer 24/7 operations, but also instant settlement, fractional share trading, and stablecoin based funding. NASDAQ has also announced the launch of an "[equity token design](#)" that will allow public companies to leverage tokenization to program shareholder engagement from corporate actions to proxy voting. This will address the current condition of third-party digital tokens that offer investors economic exposure but without the underlying shareholder rights and corporate actions. NASDAQ aims to have full tokenization capabilities by 2027.

The first mover advantage

Tokenization is going mainstream in the US and this is happening now: as we have described above, the biggest exchanges and securities depositories are making significant upgrades to allow US asset markets to be traded and held on chain. While tokenization of the US asset universe could proceed over many years, the ship has left the harbour.

With the US moving to provide rapid regulatory clarity and support for the development of tokenized securities, and building the settlement infrastructure to support it, it stands to benefit from a first mover advantage. Not only might the US be setting the standards for what on-chain financial trading and settlement looks like, it could benefit from future network effects. Just as the US benefits from agency over networks like SWIFT and USD correspondent banking, it may exercise them over the future on-chain infrastructure. **If the majority of on-chain private money (stablecoins) is dollarized, and the majority of on-chain assets are US listed, this could reinforce USD payments and capital markets dominance.** And if access to capital is increasingly concentrated in the US, global companies too may be increasingly interested in listing and issuing in the US to access them. As a potential harbinger, it is perhaps notable that SK Hynix made the largest foreign debut in US markets with its \$26.5bn ADR this summer, favouring US markets for their capital raising.

We look at some of the broader implications for markets from the interplay between the AI race and asset tokenization in Part 3.

See next page



Part 3: The future of the dollar, imbalances, and capitalism

In this final section, we look at how the AI race and approach to tokenization could impact the dollar, and how it might shape the future of global imbalances and capitalism itself.

Tokenization meets TINA

US equities already account for more than 50% of global equity market capitalization, and US Treasuries for more than half of OECD sovereign debt. For many investors, this gives US assets "There is No Alternative" status, popularly known as TINA: investors have no alternative but to hold assets which are such a large part of the global market portfolio. The position of US companies in the AI race thus far has further contributed to TINA, with investors compelled to have exposure to technology which could change the world. The ultimate outcome of the AI race will no doubt play an enormous role in whether the US' share of the global market portfolio grows or shrinks: it is not just how earnings and valuation multiples evolve, but also how terminal values of the corporate sector could change, and how debt sustainability could be impacted.

TINA is already playing a big role in attracting capital to the US, but tokenization could add to this: by making US assets more *attractive* and more *accessible*.

First, on attractiveness. The collateral implications of holding US assets on chain could be very attractive for investors. Fed Governor Cook noted in a [speech](#) that the most significant benefit of tokenization would be implications for capital and liquidity management for institutions. A tokenized US asset becomes more liquid and deployable, and capital and cash thus becomes more productive. In simple terms, you are more likely to hold a tokenized US equity or bond instead of cash, if you can easily repo your tokenized asset on chain when you need liquidity. If the value of US assets is going up, the cost of capital for US firms is going down. **The improved collateral velocity of tokenized assets could be another superpower for US assets, and another arrow in the quiver of US capital attractiveness.**

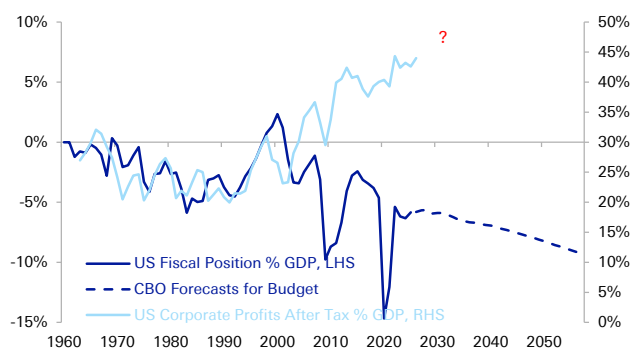
Second, on accessibility. We had argued that stablecoins could give anyone with an internet connection access to the dollar on chain. Asset tokenization could do the same for US assets. **In a world in which US equities and bonds are trading 24/7, with instant settlement and much lower costs and frictions, this could have dramatic knock-on effects on global behaviour, particularly for retail investors that could find access much easier.** We find it notable that [NASDAQ](#) openly acknowledged that "the investor base of public equities has the potential to expand as markets remain open across multiple time zones, making it increasingly important to have modern tools for public issuers to engage with investors worldwide." Tokenization need not also be limited to equities and bonds. One can imagine a world in which a person with even \$1 in a place 10,000 miles away could own a tiny fraction of a tokenized Times Square on chain.

From an FX perspective, if tokenization succeeds in making US assets more attractive and accessible, this could help attract private capital flows. It is perhaps not surprising that the US is leveraging the power of technology – both AI and blockchain – to attract more foreign private capital at a time when US geopolitical leadership could be under pressure (see important examples [here](#), [here](#), and [here](#)) and crucially when US fiscal deficits have been widening. Both increase the need for alternative sources of private sector funding. **The US is arguably going "all-in" on private capital to offset public sector vulnerabilities.**



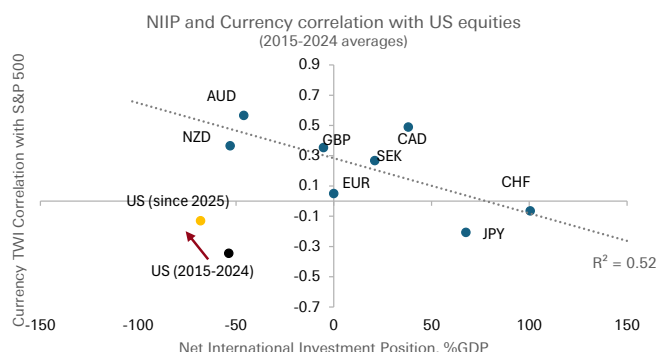
But lowering the hurdles for inbound capital, also exposes the US to capital going out. Tokenization would meaningfully increase capital mobility, in and out of US assets. The shift in US funding towards private sector, potentially shorter-term flows into equities – motivated largely by technology returns – will make the dollar riskier, more equity sensitive, and leveraged to the AI race. We have covered in earlier research how the dollar's historical properties of diversifying equity risk could be waning, with the dollar likely to become more positively correlated to risk. If for instance, AI proves uneconomical or the US starts to fall behind in the AI race, the dollar could be very exposed.

Figure 8: The US is increasing reliance on private capital - motivated by technology leadership - at a time of growing public sector vulnerability



Source : Deutsche Bank, Federal Reserve Bank of St Louis, CBO

Figure 9: The dollar is becoming a weaker diversifier for equity risk



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP

Europe is on board, but China is taking a very different approach

How does the highly enabling regulatory environment towards tokenization in the US compare with the other major economic blocs, namely Europe and China?

Europe has been receptive to the opportunity presented by tokenization.

Europe's DLT Pilot Regime introduced a regulatory sandbox for institutions to test the trading and settlement of tokenized securities on DLT as early as March 2023, even earlier than the US. ECB Governor [Lagarde](#) has noted that tokenization offers the prospect for Europe to "tackle the fragmentation that has held us back for decades." Note that Europe has over 30 central securities depositories across the union. An integrated market for tokenized assets could allow Europe to "leapfrog the fragmentation of existing legacy systems" (ECB). Europe has also embraced the role of tokenized securities as collateral, with DLT assets held in European CSD's eligible as collateral at [Euroclear](#) from March 2026. This puts digital assets on the same footing as traditional ones in Europe.

Where Europe sits apart from the US is perhaps in: the scale of its crypto ambitions, an explicit commitment to retaining a two-tier monetary system with central bank money as the final settlement asset, and the size of the AI economy itself. Europe is alive to the opportunities presented by tokenization for market unification – an important step to increasing Europe's capital market attractiveness, but the US arguably has more clearly stated ambitions to become "the crypto capital of the world". Moreover, Europe appears committed to keeping central bank money as the final settlement instrument between financial institutions even in a tokenized world, with initiatives like Pontes and Appia



underlying this. In the US, it remains to be seen whether private stablecoins become the primary instrument of settlement, disintermediating central bank money, or whether the existing system of bank deposits and reserves is tokenized. Ultimately, the biggest divide between Europe and the US is in the AI economy itself – Europe has a much smaller footprint in data centres and independent AI labs, and thus may not be where the appetite to raise or deploy capital – even in a tokenized asset world – is most compelling.

Meanwhile, China is taking a very different approach altogether; while the US is embracing stablecoins and RWA tokenization, China is tightly controlling – if not restricting – them. In February 2026, the PBOC and CSRC along with eight other national organization issued [a notice](#) that updated China's 2021 ban on crypto-related activities. The notice defined RWA tokenization and tightened regulation around it – noting that such activity or intermediation of it was prohibited in China, except with regulatory approval on specific financial infrastructure ([Xinhua](#)). The notice also banned entities from issuing stablecoins pegged to RMB abroad, without regulatory approval. We note that Chinese regulators prevented Ant Group and JD.Com from proceeding with plans for yuan stablecoins in Hong Kong in October 2025. Instead, in January this year, PBOC began to allow payment of interest on e-CNY, retail CBDC deposits suggesting relative encouragement for controlled forms of central bank digital money. We look further at some of the drivers of this very divergent approach between the US and China.

Open markets versus open technology

The US is pursuing a hyper-capitalist approach in the AI race: throwing open the doors for private capital to fund large capex, and using the power of asset tokenization to supercharge it. We call this the open markets approach. Under the US model of shareholder capitalism, US companies prioritize profit maximization, keep AI model weights closed to allow companies to control pricing via subscriptions, and offer the carrot of return participation to shareholders to attract in global capital. The US then uses those huge sums of capital to make large capital investments to innovate and stay ahead of the competition. Importantly, the open markets approach is paired with a closed technology one.

China is taking a very different approach in the AI race, pushing out lower-cost models with open-weights into the world. We call this the open technology approach, and it is paired with a closed markets one. Chinese companies are not only achieving similar model results, and charging clients less for tokens, they are offering their models with open-weights. Open weights allows users to download and customize models on their own servers, promoting global proliferation and usage among corporates. See more from the DB Research Institute on this [here](#). China may be counting on the same penetration of its AI models around the world as it has achieved with manufactured goods. One can perhaps start to look at China's approach to AI as an extension of China's broader involution approach to manufacturing where state-based alignment combined with fierce domestic competition achieves low pricing, improving quality, rising market share and ultimately industrial dominance. That China is now trying this in the technology space, where the US has thus far remained a leader, bears close watching. The US' open markets are meeting China's open weights.

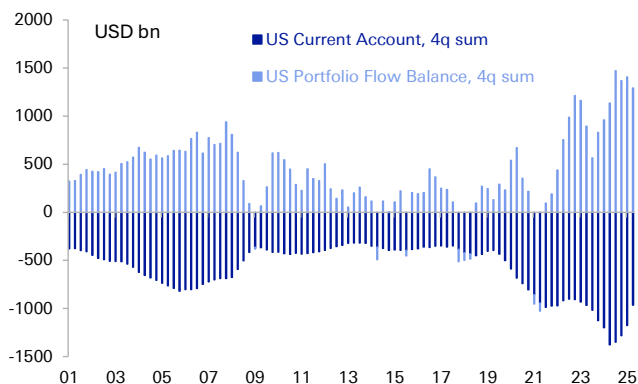
In sum, the US is taking an "open markets-closed technology" approach to the AI race, while China is taking an "open technology-closed markets" approach. This divergence is not just a reflection of different economic models, but underlying global imbalances, and very different needs for global funding.



A test for global imbalances and capitalism itself

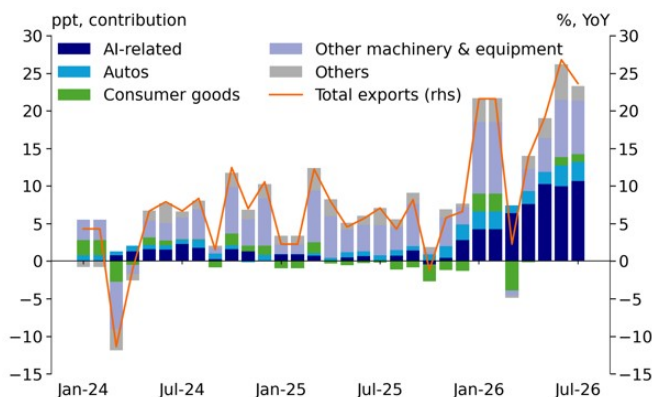
The US' open embrace of tokenization, compared to China's more closed approach to foreign capital is a reflection of underlying imbalances. The US' current account deficit combined with a fiscal deficit means the US requires foreign financing to fund incremental private capital expenditure, in the absence of domestic fiscal savings. In comparison, China is running a large and growing current account surplus, reflective of excess savings. China's trade surplus is in fact growing – with the AI cycle boosting AI-related exports. China does not need to turn to foreign capital because it can fund capital expenditures from abundant current account savings. This affords China the ability to maintain a more managed capital account, and explains the relatively restrictive approach to the high-speed blockchain rails that the US is so actively building.

Figure 10: The US is dependent on foreign capital flows to fund its deficits



Source : Deutsche Bank, Haver Analytics

Figure 11: Chinese exports are growing even faster as a result of AI



Source : Deutsche Bank, CEIC
Note: See "China Macro: Trade Imbalance and Rates Divergence" (31 Aug 2026)

The AI race could further reshape the state of global imbalances. As one potential path, imbalances could become more extreme. If China leverages AI to make industry even more efficient, adopting it for production and industrial automation, this could boost China's manufacturing dominance and good-driven surplus. One can imagine an alternative scenario in which the US current account improves, narrowing global imbalances. We have written in earlier research how AI revenue streams could help the US current account by around 1% over the coming decade via higher services exports. But this is crucially dependent on US AI labs retaining pricing power with private subscribers, and monetizing an international subscriber base. But China's low-cost, high-proliferation approach could cap the excess profits of US AI companies, and their ability to sell intellectual property for profit. How things evolve may come down to whether US and Chinese models become complements that coexist and serve different market segments, or whether they become substitutes in a winner-takes-all outcome. There is a lot at stake.

Ultimately, the AI race between US and China is more than a battle between AI models. It is a battle between economic models. The US is doubling down on capitalism: operating on the premise that increasing capital market openness will pull in the funding for large AI capital expenditures; in turn, this will help the US remain dominant in AI technology, allow corporates to maintain pricing power, and for the US to pull ahead in the race. Meanwhile, China does not need to leverage open markets to attract capital flows. China's industrial dominance and record



trade surpluses mean it has ample and cheap sources of domestic funding. This allows China to keep a more closed market approach that need not prioritize profits firsthand, and allows it to compete with low-cost, open-weight AI offerings. If China's approach starts to undercut US profits, this could threaten the US model of shareholder capitalism, which has become critical not only to attracting funding for AI, but for growing US deficits. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation. The AI race is a big test for capitalism itself.



Appendix 1

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The good, the bad and the ugly

September 03, 2026

- As the summer holidays come to an end, we take a look at the three questions that are likely to occupy our time in the months ahead: Is the euro area economy proving resilient to the energy shock? Will the ECB stop at 2.50% or continue hiking until policy is restrictive? And will public finances and elections prove a destabilising combination?
- The European economy is proving remarkably resilient, with some elements of positive demand shock offsetting the negative energy supply shock. Expectations are improving, but some caution is still needed: the energy shock is not over and the AI trade deficit is widening.
- With respect to the ECB, a rapid normalisation of energy prices may be needed to stop the hiking cycle at 2.50%. That said, the inflation outlook is not yet consistent with an outright restrictive stance either.
- Budget season is about to begin, debt dynamics are vulnerable and elections imply political risk and market volatility. The degree of tightening of financial conditions will be a function of political choices, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Q1: Is the euro area economy proving resilient to the energy shock?

Strictly speaking, as a net energy importer the euro area should be highly exposed to the negative energy supply shock triggered by the Iran war. In any case, HICP inflation jumped to 3.0% in Q2, 1.2pp higher than expected at the start of the year. The household real income shock alone should have knocked more than 0.5pp off GDP.

And yet euro area GDP was not weak in Q2, it was surprisingly strong. At 0.44% qoq, GDP growth exceeded expectations. We don't have detailed components yet, but on the basis of available data household consumption grew despite the inflation shock.

It is possible that elevated household savings buffered the real income shock on the assumption that it would be fleeting. There was also a targeted and temporary easing of fiscal policy (lower energy taxes), but this was not substantial and only moderated the rise in inflation slightly. That said, the euro area fiscal stance was already easy for 2026. Not nearly as easy as 2022, and skewed heavily towards Germany. But the stance is easy, not tight.

Most encouraging are the signals of rising investment spending: the BLS points to demand for loans for fixed asset investment, the flow of long-term loans to corporates has been a little volatile but hints at investment spending, and the rise in German industrial orders from elsewhere in the euro area has historically been consistent with a euro area investment cycle too.

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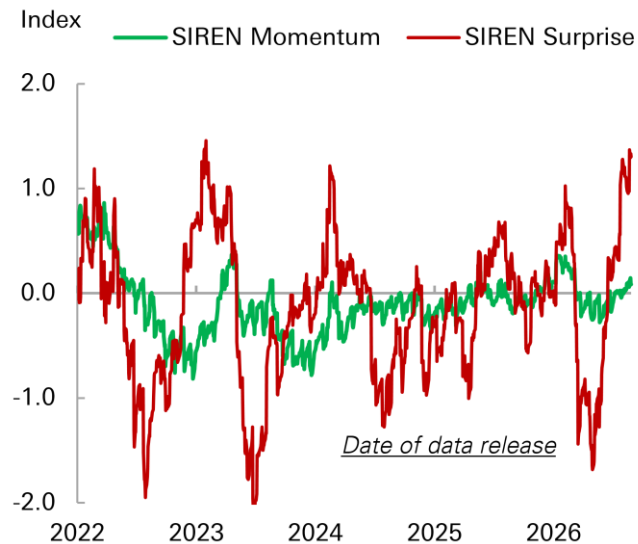
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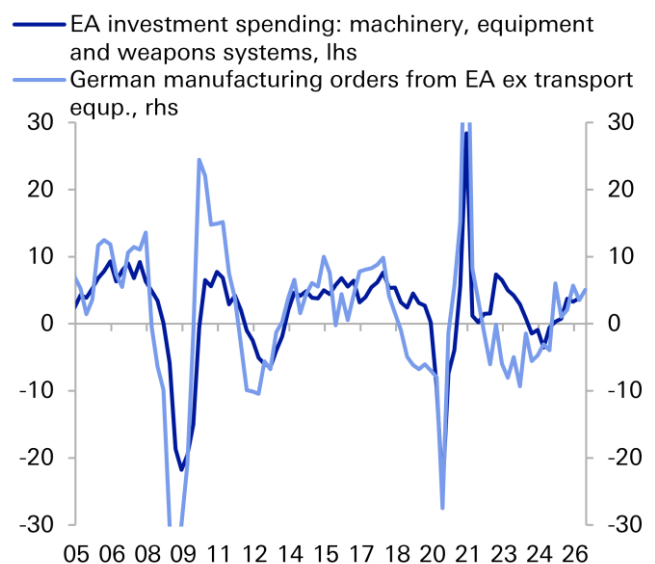


Figure 1: Euro area activity data showing strong surprises and building momentum



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 2: Encouraging signals of a capex cycle building



Source: Eurostat, German Federal Statistics Office, Haver Analytics LP, Deutsche Bank Research

We need to be careful. We might be seeing the lagged transmission of the earlier ECB rate cuts. Private sector balance sheets are healthy, which would aid transmission. Maybe structural demand from AI investment and defence spending is offsetting higher energy costs and geopolitical uncertainty.

In short, maybe what Europe is facing is a positive demand story, not just a negative supply story. Having quickly dropped to 48.8 in April, the composite PMI rose to 52.1 in August and might have been higher were it not for heat-related headwinds in services.

That said, there are reasons to be cautious about growth.

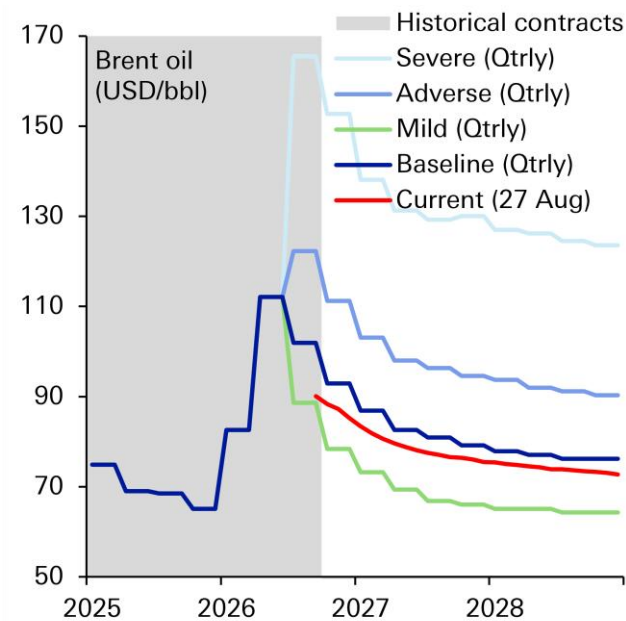
The energy shock is not over. Gas prices have reached a new high for the year, raising the prospect of an electricity-based production cost shock to add to the existing oil-based transport cost shock.

The energy shock might also be masked by temporary factors like stock-building. The shock in the Middle East went beyond energy into chemicals and metals. There was a threat to global supply chains. Fear of disruption might have triggered stock-building, which might explain why manufacturing has held up.

There is no evidence of domestic inventory building. But Europe might have benefited from temporary stock-building in its export partners. This might help explain export resilience.

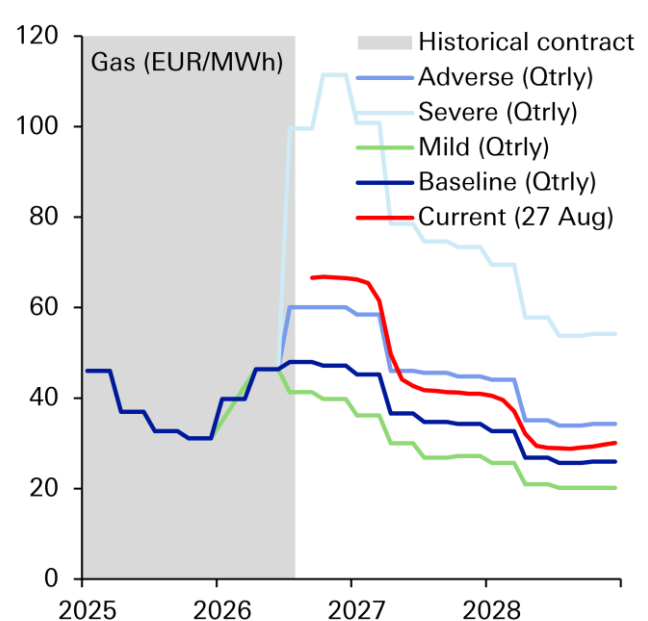


Figure 3: Oil prices vs. ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 4: Gas prices vs ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

AI investment spending should keep rising. But the AI spending boom might be less beneficial for GDP than it appears, however, as Europe's AI-related trade deficit is widening as both AI-related exports weaken and AI-related imports grow.

Monetary policy is tightening too, which will restrain growth with a lag (see our latest review of upstream and downstream financial conditions [here](#)). Concerns about sovereign debt sustainability could generate additional growth-sapping risk premia.

Finally, competitiveness remains weak and the Second China Shock and increasingly direct competition with China could weaken European production and employment.

The bottom line is the economy has been remarkably resilient to the energy shock so far and our euro area GDP forecasts are probably too low at 0.5% in 2026 and 1.1% in 2027 (our German team have raised their 2026 forecast from 0.5% to closer to 1.0%).

Q2: Will the ECB stop at 2.50% or continue hiking until policy is restrictive?

Since early March we have been expecting the ECB to hike twice (Jun and Sep), raising the policy rate to 2.50%, the upper end of the range of neutral. This felt like good risk management. It signals the ECB commitment to price stability on the one hand without overly burdening economic growth in the context of a negative supply shock on the other.

If growth was weakening amid a supply shock, inflation expectations were comprehensively anchored and data showed no risk of energy price pass-



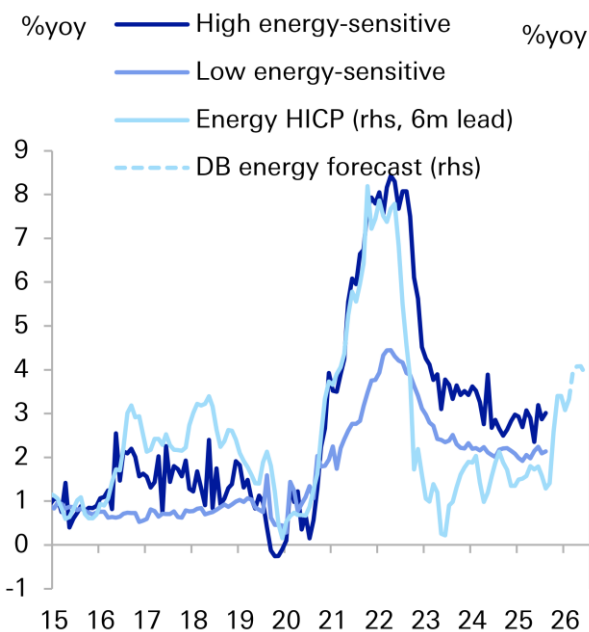
through, it would be easier to call into question the ECB's decision to hike rather than "look through" this energy supply shock.

In the round, the data, the ECB messaging (even this week Isabel Schnabel said the market "very well" understands the ECB reaction function, repeating what President Lagarde said in the July press conference) and the lack of resolution in the Middle East and still elevated energy prices mean a hike to 2.50% in September seems all but certain. Indeed, this is the impression from the July ECB Accounts. The question is, will the ECB stop there?

There are two scenarios in which the ECB hikes to 2.75%.

In one scenario, the ECB concludes that the upper end of the range of neutral is 2.75% rather than 2.50%. The persistent strength of credit growth in recent months despite higher lending rates and despite banks saying that lending standards are tightening might reveal that neutral is higher than expected. The scale of fiscal easing, even if led by Germany, could push r^* higher. If this is right, the hurdle to raising policy rates to 2.75% might be lower than assumed and might not need strong or clear evidence of indirect or second-round inflation.

Figure 5: Maybe it's too soon to expect indirect energy effect to show up



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 6: The credit impulse is consistent with a robust domestic economy



Source: ECB, Eurostat, Haver Analytics LP, Deutsche Bank Research

The argument is not one-sided. The strength of credit growth might reflect the lagged transmission of the earlier cuts rather than a higher neutral rate. In any case, our composite indicator of restrictiveness, based on a range of data that might reveal neutral, says that policy was still mildly restrictive in mid-2026 when the ECB started to hike. If that's right, it might be more difficult to argue 2.75% is still within the range of neutral.

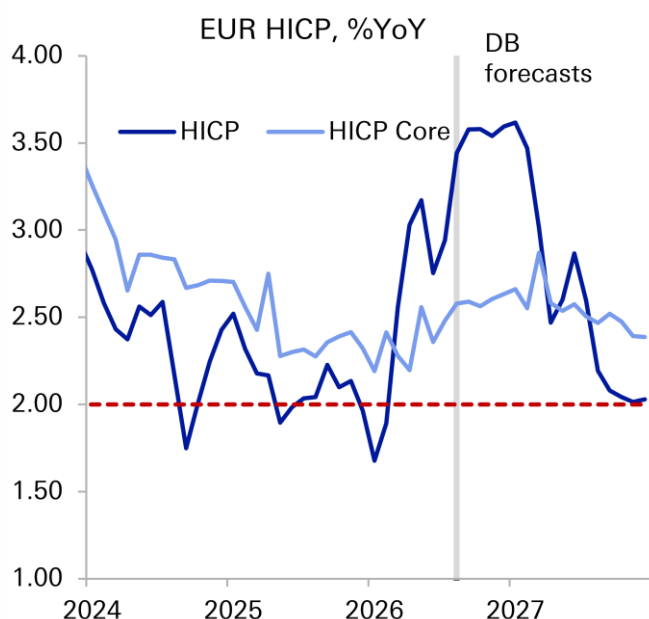


The other scenario for getting to 2.75% (or higher) is the ECB decides that a restrictive policy stance is appropriate. In the ECB's current framework, this means moving from a "measured" tightening to a "forceful" tightening. This requires the ECB to believe that the inflation shock is moving from a 'large but not too persistent' shock to a 'more significant and sustained overshoot of the target'.

There are different routes to believing in a more significant and sustained overshoot of the inflation target.

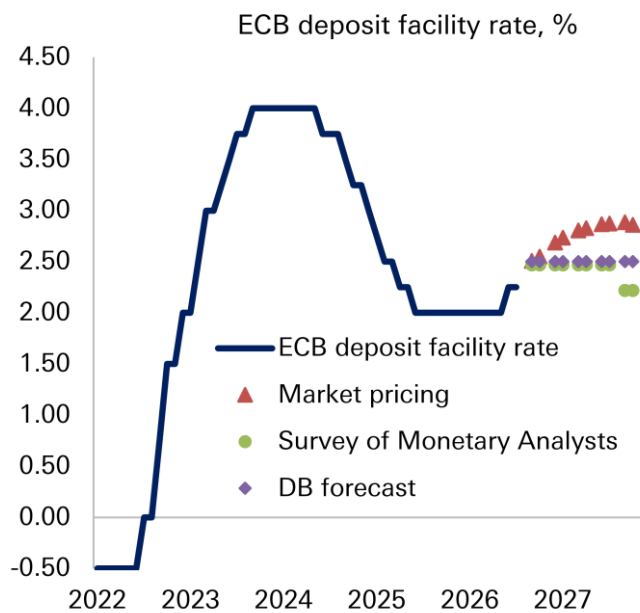
One would be if energy prices remain at elevated levels for longer than expected. Oil prices are in line with baseline, but gas prices are more consistent with the adverse scenario (at least in the near term). Given the lagged transmission of rising gas prices into inflation, this could add persistence to the headline HICP shock, lengthening the period during which energy inflation could spillover into core.

Figure 7: Headline and core HICP: DB forecasts



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 8: ECB policy rates: Market priced hikes to 2.75-3.00%



Source: ECB, Bloomberg Finance LP, Deutsche Bank Research

Another way would be if larger than expected indirect effects and in particular second-round effects materialised. So far, there is no compelling evidence of indirect effects and no evidence of second-round effects. However, both would only materialise with a lag. For example, it was six months before high energy-sensitive core HICP started rising in the 2022 shock.

A third way would be if stronger growth added persistence risk to inflation. The recent resilience of growth would add to this risk if it lasts. That said, if AI lifts productivity, the additional growth might not be as inflationary.

How the ECB staff forecasts change in September will be very important to the outlook for monetary policy. In June, the baseline forecast had core HICP



averaging 2.5% in 2027 and staying at least 30bp above target through the year. Importantly, however, headline HICP was expected to be back down to the 2% target from Q3-27 and stay there through to the end of 2028.

It is important to remember the baseline staff forecast in June was conditioned on policy rates rising to 2.75%. But despite sticky core inflation, it is difficult to imagine monetary policy tightening much further if headline inflation is sustainably returning to target within a year.

As such, the near-term path for policy could depend critically on whether the Governing Council believes core inflation will converge down to headline in the medium term or headline converges up to core. The former is less hawkish, the latter is more hawkish.

Which of these two outcomes seems more or less likely to the Governing Council will depend on how it judges the risk of inflation persistence. This is a function of pricing power, bargaining power and purchasing power. Our view is there is some risk of persistence, but not as much as 2022 and not enough to warrant an outright restrictive policy stance. Not yet, at least.

Our 2.50% baseline might be too optimistic, but 3.00% feels too restrictive. 2.75% is a grey zone: a non-negligible possibility unless the situation in the Middle East is resolved quickly and energy prices normalise rapidly. Our work on policy rules shows that the ECB 'baseline' scenario macro forecasts are consistent with 2.75%. The 'milder' scenario forecasts are consistent with 2.50%.

There may not be any new policy signals from the ECB at the Jackson Hole symposium. Unlike in the past, there is no global policymaker panel. Isabel Schnabel is the ECB member on the official agenda – speaking about tokenized finance. However, markets will be on the watch for press interviews by ECB attendees.

Q3: Will public finances and elections prove a destabilising combination?

The environment around European public finances is becoming more challenging for a number of reasons. Government bond markets are under pressure globally, driven currently by the US. The 'i minus g' gap at the heart of debt dynamics is becoming less favourable as the benefits of the low rates regime of the last decade expire, amplified by rising inflation risk, tighter monetary policy, rising term premia and generally low rates of economic growth.

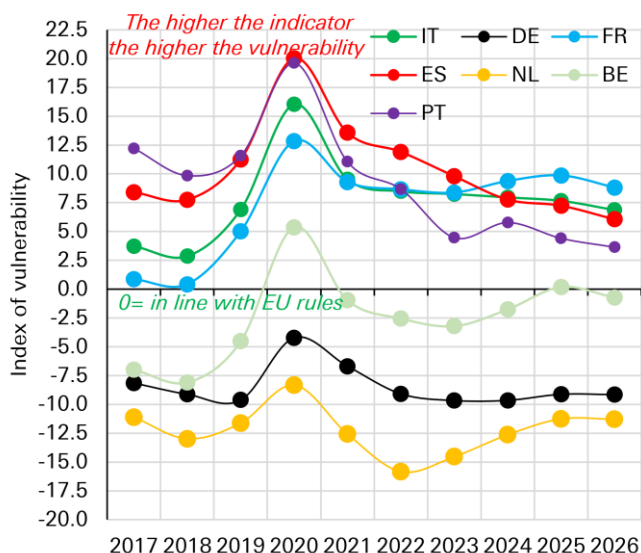
At the same time, public debt to GDP ratios have not returned to pre-pandemic levels, fiscal vulnerability is elevated even when accounting for improved external balances, and European governments are facing numerous spending demands, from defence and energy to climate change and technology.

The combination of more vulnerable public finances and elections in several significant member states over the next 12 months creates the potential for tighter financial conditions. National elections lie ahead in Italy, Spain, Greece, Finland, Estonia, Poland and, perhaps most consequentially, in France. Financial conditions are well-behaved so far (the ECB's state-of-the-art Macro-Finance FCI has hardly tightened). The degree of tightening of financial conditions going



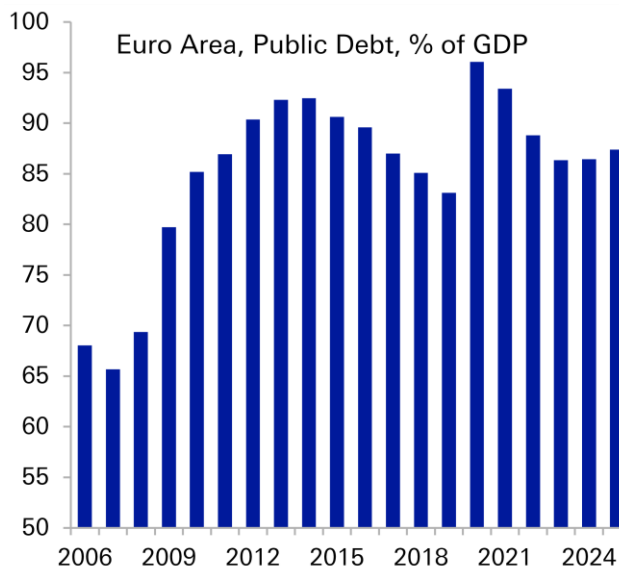
forward will be a function of the fiscal policy choices made by governments, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Figure 9: DB fiscal vulnerability indicator incl. net external position – France most vulnerable



Source: European Commission, Deutsche Bank Research

Figure 10: Public debt remains high and above pre-pandemic levels



Source: European Commission, Deutsche Bank Research

With regard to France, the opinion polls say that Rassemblement National (RN) has a credible path to winning the Presidency with Marine Le Pen. If that happens, there is a good chance an early legislative election will be called too.

OAT markets have been on edge since President Macron's ill-judged legislative election in 2024 resulted in a deeply divided parliament with little capacity to drive strong fiscal consolidation. On the basis of what is known at the moment, there is a lack of a credible consolidation trajectory in the RN fiscal plans. Even under moderately optimistic assumptions about consolidation, France's debt ratio will likely remain broadly flat in the range of 120-125% of GDP over the next decade. A flat trajectory means vulnerability to any new negative surprises on growth, funding costs and/or deficits – and as we have said before, the deficit is under pressure from inflation (indexed spending), legacy policy decisions (stalling the pension reform), defence, ageing, etc.

Much will depend on the course of French fiscal policy, particularly RN's fiscal policy. This could be revealed in three distinct stages over the year ahead.

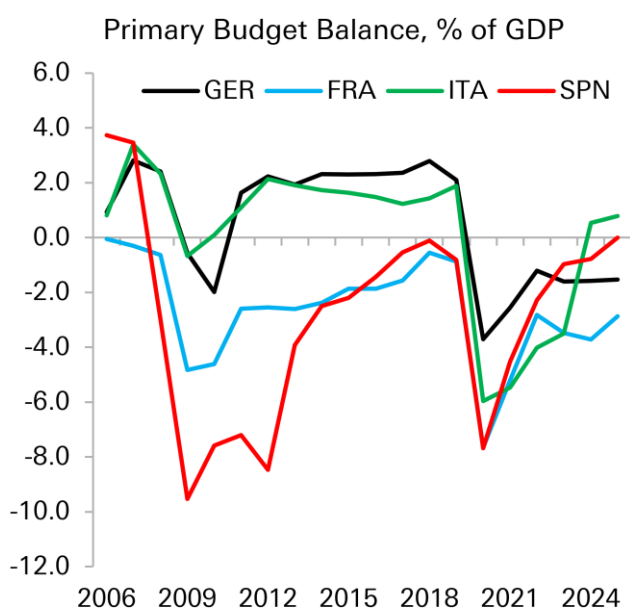
- **The 2027 budget.** The incumbent Lecornu government needs to have a budget agreed by year-end. A deeply divided parliament and approaching election will make strong consolidation difficult. Importantly, however, RN is signaling that it won't oppose a budget. The alternative ("special law") would mean a less stable financial position in 2027. RN would prefer to have some breathing space if it is elected.
- **The election campaign.** This will reveal more about RN's fiscal plans. The limited details available so far imply a wider deficit. However, RN's fiscal



identity is unclear and RN President Bardella's admiration for Italy's market-friendly fiscal stance under PM Meloni suggests an understanding of the benefits of a stable bond market. The extent of the RN's intention to challenge Brussels will be important too. This could influence market confidence in the availability of European backstops (ECB TPI, ESM, etc).

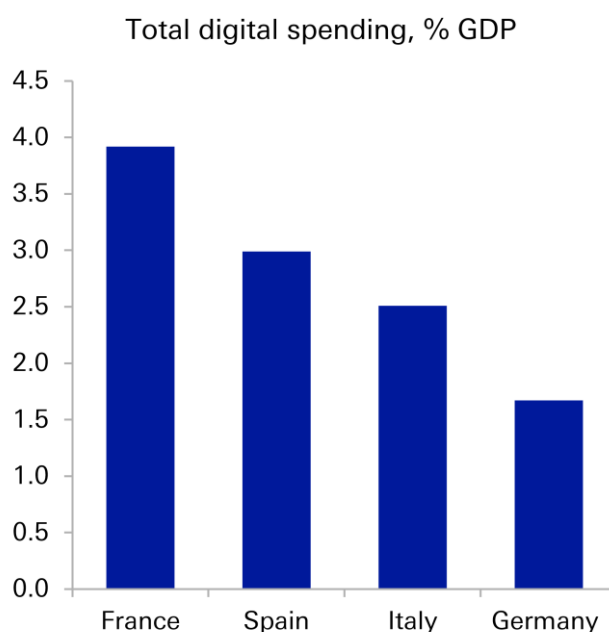
- **Post-election decisions.** Assuming an RN President would dissolve parliament in the spring to trigger legislative elections by the summer, the first task for the new parliament would be the preparation of the 2028 budget. This assumes a budget for 2027 has already been passed. If France is operating under a special law, a 2027 budget would take immediate precedence. These would reveal RN's fiscal plans and their consistency with EU rules.

Figure 11: France has the largest primary deficit among the Big4...



Source: European Commission, Deutsche Bank Research

Figure 12: ...but France might have better growth prospects, helped by AI (digital investment spending)



Source: Eurostat, Deutsche Bank Research

Counterbalances. France will be the focal point for the market. As we have argued (here, here and here), the key theme for investors is not outright regime change, but a constant, volatility-inducing push and pull between RN's political objectives – shifting in places – and the fiscal, legal, and market constraints that would ultimately shape its capacity to govern. The push for political change will generate friction.

There are several buffers that could help keep the country off the more challenging paths.

- **The market is already pricing weak fundamentals.** French yields have been gradually adjusting to these weak fundamentals over the last couple



of years. For example, the current yield already anticipated further credit downgrades.

- **Robust banking.** French banks are well-capitalised and unlike the crisis-hit periphery a decade and a half ago hold relatively little direct exposure to OATs. As a global safe asset, OATs are widely held abroad.
- **Backstops.** Unlike the sovereign debt crisis 15 years ago, there are backstops in place. For example, the ESM can lend to crisis-hit member states and the ECB can intervene in the market for eligible member states with TPI. If foreign investors offload OATs in scale, the French banks have some capacity. However, as bank exposure increases, the potential for a more destabilising sovereign/banking doom-loop would increase.
- **Economic growth.** There are several reasons why French economic growth could outperform its peers and help stabilise debt. One is AI. Among the larger member states, France is spending the most on digital investment and was the only EU member state to launch a notable AI model in 2025 (50 in the US). France is also relatively more exposed to defence production, and among the larger EU member states has relatively favourable demographics.



Appendix 1

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The Great Rebalancing Act IV: Connectivity

September 2, 2026

The connectivity chokepoint

- Digital connectivity—the transmission of data through terrestrial fibre, mobile networks, subsea cables and satellites—has become a critical arena of geopolitical competition.
- The Internet's physical infrastructure is increasingly viewed as a source of strategic leverage, resilience and vulnerability.
- In the era of “infrastructure realism,” the contest increasingly involves both states and hyperscalers such as Google, Meta, Microsoft and Amazon, whose AI and cloud ambitions are reshaping global connectivity infrastructure.

Digital connectivity has become a central arena of great-power competition. While states have long used communications infrastructure to project influence and control, in the age of AI, the contest to dominate digital networks has become a defining geopolitical frontier as countries and – increasingly – private actors look to win the AI race.¹

This chapter in our Infrastructure Realism series examines the physical systems that enable digital connectivity: undersea cables and communications satellites. We assess their role on the Internet—the largest and most consequential connectivity network—and the strategic advantages and vulnerabilities they create for state actors.

Subsea cables are indispensable to the global economy. They carry more than 99% of intercontinental data traffic, making sabotage or prolonged disruption a direct threat to global Internet connectivity.² Satellites carry only a small share of international data, but they are crucial for last-mile access where fiber cannot reach, particularly for governments and defense users. As we go on to explore, the rise of communications-based satellites is also becoming integral to the global AI race.

Several incidents have underscored the criticality of connectivity infrastructure. In April, reports of Russian surveillance near UK cables and pipelines raised concerns.³ In May, amid conflict in the Middle East, Iran threatened to sever undersea cables in the Red Sea. Iran's IRGC, also floated the idea, through state-affiliated media, of charging foreign operators infrastructure fees and licensing royalties for routes through Iranian territorial waters.⁴ This would have significant implications for hyperscalers such as Meta, Google and Microsoft, which rely on these corridors for portions of their international traffic and cloud connectivity.

¹ The first instance of communications sabotage was during WW1, when the British cut Germany's undersea telegraph cables in the English Channel, forcing German telegraph traffic onto British-controlled networks that could be intercepted. And in the 21st century, autocratic states like Iran and Russia have imposed internet shutdowns to control their population / information access.

² The 99% does not account for total global internet traffic, given the vast majority of internet traffic is domestic and does not cross an ocean at all. Instead, the 99% applies to internet traffic that goes between continents. Mauldin, Alan. 2023. 'Do Submarine Cables Account for Over 99% of Intercontinental Data Traffic?'. In *Telegeography.Com*.

³ Olivia Ireland, Thomas Copeland, and Frank Gardner, "UK Says Russia Ran Submarine Operation over Cables and Pipelines," BBC News, April 9, 2026, <https://www.bbc.co.uk/news/articles/cre13qn9z7do>.

⁴ "IRGC Media Float Plan to Levy Fees for Subsea Internet Cables," Amwaj. Media, 2020, <https://amwaj.media/en/media-monitor/irgc-media-floats-plan-to-levy-fees-for-subsea-internet-cables>.

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Section 1 explains how the Internet works and traces the physical journey of data, highlighting the points at which state disruption can occur. We give less emphasis to 5G, due to our definition of what constitutes the Internet's infrastructure. This is because while states—most notably China—have used 5G equipment to project influence, 5G is primarily an access network: it links mobile devices to the wider Internet but does not replace the role of subsea cables and satellites.

Sections 2 and 3 analyse subsea cables and communications satellites, detailing the strategic advantages and vulnerabilities of each ecosystem and how states are seeking to consolidate their influence. Finally, Section 4 examines the rise of private actors such as Google and Meta, which are reshaping global connectivity by building their own data corridors and controlling more of the Internet stack to meet rising demand.



1. How connectivity works

1.1 What is the Internet?

The Internet enables global communication, data exchange and access to information through a system of cables, satellites and wireless technologies. It is not a single entity but a common set of protocols that allows independently owned networks to communicate. In this sense, it is a “network of networks” through which data moves across billions of devices.

To move across the Internet, data is broken into small “packets” and routed across different systems using standard protocols. The primary ones are the Transmission Control Protocol (TCP), which ensures the reliability of the connection and data, and the Internet Protocol (IP), which acts as an addressing system to find and route packets to their destination.⁵ Using these protocols, data travels through a global web of fiber-optic cables that connect major network hubs, forming what is known as the Internet Backbone. The process is as follows:

- A packet leaves the user’s device and enters a local access network.⁶ Over fixed broadband, the data packet passes through a Wi-Fi router and then to an Internet service provider (ISP). Over mobile broadband, 5G carries it to a radio base station and the mobile operator’s core network before it reaches the wider Internet.
- From the ISP, the packet is passed from one junction (or router) to the next, crossing multiple networks toward its destination server.
- If the destination server is nearby, the packet may be exchanged at a local Internet Exchange Point (IXP). If it is distant, the ISP routes it onto the backbone, across a country or ocean, and into an ISP at the other end.
- The response takes a similar path back, with each router along the way making forwarding decisions based on the destination IP address.

How one network gets its traffic to the rest of the Internet, and whether it must pay for it, is where key corporations come in. There are two ways networks connect: 1) through transit, where a smaller network pays a larger one to carry its data across the globe, and 2) through peering, where two networks of similar size agree to exchange traffic directly at no cost. Within this system, a hierarchy of companies has emerged.

At the top are Tier 1 network operators, which can reach the entire Internet through settlement-free peering. This small group includes major telecom operators such as Lumen, AT&T, Tata Communications, NTT, Arelion, GTT and Cogent.

Below these are Tier 2 network operators, such as regional carriers and internet service providers like Comcast and Vodafone. These entities primarily purchase transit from Tier 1 providers to reach parts of the internet they cannot access via peering. Finally, Tier 3 network operators rely exclusively on paying for transit from Tier 2s to connect their customers. These are typically the local ISPs that provide Internet service to homes or offices. Therefore, a data packet from a user’s laptop typically travels from a Tier 3 provider to a Tier 2, onto a Tier 1 backbone, across an undersea cable to another Tier 1 network, and then down through the tiers to its destination server.

1.2 How data physically travels

The Internet ultimately depends on physical infrastructure. As we said in step 1 above, the first leg is not always a cable, with mobile technologies like 5G carrying data wirelessly between the user’s device and a nearby base station. Satellite systems can also provide a

⁵ Rus Shuler, “How Does the Internet Work?,” Stanford.Edu, 2002, <https://web.stanford.edu/class/msande91si/www-spr04/readings/week1/InternetWhitepaper.htm>.

⁶ Access networks connect the user’s device or premises to the first point in an Internet or telecom provider’s network. They can be wired or wireless, and in addition to 5G and Wi-Fi mentioned above, other examples are fiber-to-home, cable broadband, and satellite broadband.



wireless connection over much longer distances. Beyond that access layer, however, both systems typically rely on terrestrial fibre and the wider Internet backbone. Most long-distance traffic travels as pulses of light through fibre-optic cables buried on land or laid across the seabed, and each glass strand is thinner than a human hair.

Terrestrial cables handle domestic and regional internet traffic, while undersea fiber-optic cables carry intercontinental internet and telecom traffic. When a user sends data across the globe, it travels through local terrestrial cables to a coastal landing station. This station then transmits the data across the ocean via a subsea cable to a corresponding landing station on another continent, which receives the data and sends it through local terrestrial cables to the destination server.

Satellites complement cables, providing connectivity in remote or disrupted environments where fiber-optic cables cannot reach. There are two different types: traditional satellites, which sit in a fixed, high-altitude orbit, offering broad coverage but with a noticeable delay due to the long signal path. A more recent approach, exemplified by services like Starlink, uses large constellations of thousands of satellites flying much lower to the earth to reduce latency and improve service.

1.3 How the Internet transitioned from a public good to a private advantage

Similar to crypto's arc discussed in this series ([here](#)), the Internet came to symbolise decentralisation during the globalisation and commercialisation of the 1990s and 2000s. Its ability to reroute traffic reinforced the perception that it was neutral and disintermediated. Yet the network depends on physical assets concentrated among a limited number of carriers—and increasingly hyperscalers—that control key digital corridors and possess extensive visibility over network traffic.

Conceived as a “network of networks,” the Internet began as a US government initiative in the 1960s to facilitate information sharing among research universities.⁷ Its predecessor, ARPANET (Advanced Research Projects Agency Network), became the first operational “packet-switching network” in 1969. Famously, its first attempted message – “LOGIN” – crashed the system after only the letters “LO” appeared, an event now widely regarded as the moment the Internet was born. Over the following decades, public researchers cultivated the Internet ecosystem by developing concepts like email, data packet switching, the World Wide Web service (WWW), and the protocols needed to create the Internet ecosystem, all with the goal of keeping the Internet open for all.

By the 1990s, the emergence of major internet exchange providers and the release of WWW software into the public domain enabled millions of people to use the Internet freely.⁸ As private interest in commercializing the Internet grew, the US Congress passed legislation in 1993 that opened access to Internet applications like computing and networking to private enterprise, which had previously only been available to universities and government agencies. This led to thousands of firms entering the market to pursue commercial ventures using TCP/IP technologies.

By 1995, the government agency responsible for overseeing the Internet, NSFNET, was decommissioned and replaced by four privately operated Network Access Points in Washington, New York, Chicago, and California. This marked the Internet's transition from a government-funded academic experiment to a fully private enterprise. The dot-com bubble in the late 1990s and eventual collapse of Internet stocks in the 2000s cleared the way for the companies that today define the modern Internet, including Amazon (1994), eBay (1995), Google (1998) and Facebook (2004). The more recent roll-out of broadband,

⁷ The Department of Defense created ARPA (the Advanced Research Projects Agency) in 1958 in an attempt to lead technological breakthroughs for the military. It was replaced by NSFNET as the main US research network for the Internet in the 1980s.

⁸ The Internet is the global infrastructure of interconnected computers and networks, while the WWW is a service running on that infrastructure, using web browsers to access linked hypertext documents.



fiber-to-the-home, and 3G/4G/5G cellular networks has also helped make the Internet ubiquitous and increasingly mobile.

1.4 Why a decentralized system still creates concentration

No single entity owns the Internet, but a small number of companies and countries control the infrastructure on which it depends on. As a result, the Internet's ecosystem remains highly concentrated across few backbone carriers and transit providers, telecom-equipment suppliers, cable landing stations, and internet exchange points to name a few, thereby creating numerous chokepoints that different countries have raced to ensure a foothold in.⁹

Beyond telecom equipment, the physical layer of digital connectivity is also concentrated. Subsea-cable ownership spans telecom operators, consortia, state-backed entities and technology companies, yet control over capacity, construction, strategic routes and landing points is narrowing. Cable ownership is distinct from Tier 1 backbone status, but the boundary is blurred because operators such as AT&T and Tata also own subsea infrastructure or capacity.

Today, this landscape is also shifting as large content providers have become the dominant users of international bandwidth and increasingly fund, co-own or purchase large allocations of new subsea cable capacity. As we will explain more in section 4, this is flattening the traditional hierarchy, since under these private backbones, hyperscalers are reducing their reliance on traditional Tier 1 transit hierarchy by peering directly with consumer ISPs and even placing their own servers inside those ISP networks.

Figure 1: Notable chokepoints in the connectivity corridor

	Selected key players
Subsea cables (suppliers)	SubCom, Nippon Electric Company (NEC), HMN Technologies, Alcatel Submarine Networks (ASN)
Subsea cables (owners)	Hyperscalers (Google, Meta, Microsoft, Amazon) and Telecom carriers (Orange, Tata, Vodafone, Telstra, China Telecom, China Mobile)
Tier 1 Backbone Operators	Lumen, Arelion, AT&T, Tata Communications, Orange, NTT, GTT and Cogent.
Satellite LEO Constellations	Starlink, OneWeb/Eutelsat, Amazon Kuiper, China SatNet
Telecom Equipment / 5G	Huawei, Ericsson, Nokia, ZTE
Data centres / cloud	AWS, Microsoft Azure, Google Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud, Baidu AI Cloud, Oracle Cloud

Source: Deutsche Bank Research, CAIDA.

2. Undersea cables: The invisible backbone of the internet

2.1 How undersea cables work

Subsea cables are long, fiber-optic conduits laid on the seabed that, in total, stretch for more than 1.8 million kilometers. There are approximately 500 active submarine cables around the world, and while many are barely thicker than a garden hose, yet they carry over 99% of all intercontinental data traffic. Their dominance is due to their high capacity, low latency and reliability, which make them ideal for transmitting vast volumes of data and information quickly. While most of the world's cables are laid by just four companies – SubCom (US), Alcatel Submarine Networks (France), Nippon Electric Company (Japan),

⁹The geopolitical case of China's Huawei is one example. As the company that provided many of the physical and software components needed for 5G, Huawei rapidly ascended to become a significant part of the Internet's underlying infrastructure, and thus a critical player in global digital connectivity. This later proved to be a problem during accelerating tensions between China and Western countries, as many nations sought to disentangle themselves from Huawei's supply chains in their telecom networks.

and HMN Technologies (formerly Huawei Marine Networks; China) – it is US hyperscalers who now finance and control most of the new cable capacity.¹⁰

Today, subsea cables are the foundation of the global telecommunications industry because they transfer data and voice traffic more cheaply and quickly than any alternative. Yet for much of the 20th century, the bulk of transoceanic traffic was primarily carried by satellites. The transition only began in 1988 with the installation of TAT-8, the first trans-oceanic fibre-optic cable, which linked the US, the UK and France. This marked the point at which submarine cables began to outperform satellites in volume, speed and cost-effectiveness of data and voice communications.¹¹ Consequently, almost all transoceanic telecommunications are now routed via the submarine cable network, with satellites handling only around 3% of total traffic now, according to ESCAEU.¹²

Figure 2: Global submarine cables, with the most geopolitically contentious corridors



Source: TeleGeography, www.submarinecablemap.com, Deutsche Bank Research.

2.2 Why undersea cables are strategic

Beyond serving as the Internet's backbone, these underwater cables are essential for global trade, finance, cloud computing, and government communications. For example, submarine cables can carry around 375mn simultaneous telephone conversations on 1 fiber optic strand.¹³ The SWIFT network also relies on cables for transmitting financial data to more than 8,300 member financial institutions in 195 countries.¹⁴ It is also estimated that \$10trn worth of financial transactions travel through these subsea cables every day.¹⁵

A 2026 study found that undersea cable disruption also had massive macroeconomic consequences for global economies.¹⁶ Specifically, a typical cable disruption often triggers an immediate 2ppt decline in GDP capita growth for an economy. There is also evidence to show that the impact continues to accumulate over time; within three to six

¹⁰ The number of systems by supplier is ranked by ASN (23), Subom (9), NEC (8), and HMN Tech (6). Data taken between 2021 – 2025. Kieran Clark, "Submarine Telecoms Industry Report," SubTel Forum, October 30, 2025, <https://subtelforum.com/submarine-telecoms-industry-report/>.

¹¹ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

¹² "Submarine Telecommunication Cables." n.d. www.Escaeu.Org. Accessed July 7, 2026. <https://www.escaeu.org/articles/submarine-telecommunications-cables/>.

¹³ Submarine Telecommunication Cables." n.d. www.Escaeu.Org. Accessed July 7, 2026. <https://www.escaeu.org/articles/submarine-telecommunications-cables/>.

¹⁴ Swift data

¹⁵ "Securing Europe's Subsea Data Cables." 2024. Carnegie Endowment for International Peace. 2024. <https://carnegieendowment.org/research/2024/12/securing-europes-subsea-data-cables?lang=en>.

¹⁶ Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. (hal-05544990)



years, the affected country's economy diverges, falling 9 ppts below its non-disrupted neighbors. These disruptions also reduce overall economic efficiency, innovation and labor productivity, followed by a tightening of financial conditions as domestic credit for non-financial private agents' contracts and cross-border banking positions weaken. The study also noted that in countries with fewer international cable relations and/or without internet exchange points, disruptions widen the GDP per capita gap immediately and substantially over the medium run, compared to nearby non-disrupted economies that instead experience offsetting gains consistent with reallocations.¹⁷

These findings highlight a broader point: the resilience of subsea cable infrastructure is not merely a telecommunications issue, but a broader precondition for economic stability and the functioning of modern financial systems.

The strategic importance of these networks is further amplified by the geography of the routes themselves. Because international data traffic is concentrated within a limited number of physical corridors, damage to key cable systems could have magnanimous consequences. One example that illustrates this point is to consider a high-frequency trader in London pulling live market data from a server on Wall Street in New York. The request would be transmitted as packets across terrestrial fibre networks in the UK, where it enters a low-latency transatlantic cable system. Examples of such routes include EXA Infrastructure's Express system, which forms part of the low-latency route between New York and London or Aqua Comm's AEC-1 linking New York and Ireland, and then on to the UK.

Many of these transatlantic cables pass through the GIUK Gap in the Atlantic, a geopolitically sensitive maritime corridor between Greenland, Iceland, and the UK. The passage is contentious because it is a critical chokepoint for NATO maritime surveillance and because it is also increasingly a primary route for Russia's Northern Fleet to access the Atlantic. So this concentration of critical data infrastructure within a militarily significant area creates a potential vulnerability.

As such, subsea cable infrastructure security has become a national priority for several countries, including the EU, US, Japan, India and Australia.

2.3 Undersea cable vulnerabilities

The criticality of subsea cables is heightened by the fact that they can be vulnerable to damage, outages and cyber risks, such as natural or man-made hazards, and hazards to other infrastructure including the landing stations and IT systems that support the cables. This systemic risk is amplified because many global networks rely on a limited number of routers, hubs and suppliers.

Crucially, there is often no "Plan B" when a major cable is disrupted, and there are a reported 150-200 cable faults recorded annually.¹⁸ The challenge is threefold: first, repairs are immensely costly, estimated at \$500,000-\$1mn per incident for a subsea fibre optic cable, and \$10 - \$100mn per incident for a subsea power cable.¹⁹ Second, there are only ~40 specialist cable repair vessels in the entire world, owned by a handful of companies like SubCom, Orange Marine, and HMN Technologies. These vessels operate on a first come first served basis within the consortium's zone agreement. And while the repair ships can mobilize within a 24-hour period, repair time frames vary according to fault location, while the requirement to secure repair permits within territorial or exclusive economic zone waters can be burdensome and significantly delay starting operations.²⁰ Finally, these cables are often concentrated in physical chokepoints – such as along

¹⁷ Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. <hal-05544990>

¹⁸ International Cable Protection Committee.

¹⁹ Qiu, Winston. 2025. "Statistics on Subsea Cable Fault and Repair - Submarine Networks." Submarinenetworks.Com. June 28, 2025. <https://www.submarinenetworks.com/en/nv/insights/statistics-on-subsea-cable-fault-and-repair>.

²⁰ Constable, Mike, Burdette, Lane, Mauldin, Alan. "The Future of Submarinen Cable Maintenance: Trends Challenges, and Strategies." June 2025.



shallow straits or continental shelves – that are also areas of geopolitical strife, making conducting any reparations difficult in this environment.

While direct evidence of deliberate cable sabotage has been hard to prove thus far, several incidents of damage to critical undersea infrastructure sites have caught the attention of governments.²¹ The US Congress, in its proposed Strategic Subsea Cables Act of 2026, noted damage to cables connecting the Norwegian mainland with the Norwegian archipelago of Svalbard (2022), multiple cuts to cables in the Taiwan Strait (2023, 2025), and disruptions to cables in the Baltic Sea (2024, 2026) as key examples of this growing threat.²²

Perhaps the more immediate concern for individual users was the threat that emerged earlier this year, when Iran threatened to cut its undersea cables in the Red Sea. Since these routes carry around 17-25% of global internet traffic, the ramifications of such an act would be severe.²³ Even before this threat, subsea cable operators in the Red Sea have been plagued with numerous problems. Yemeni territorial disputes and Houthi attacks on shipping lanes have obstructed the deployment of several new cables – including the 2Africa cable, which is almost complete, and other cables such as the IEX, Africa-1, and Raman, which are still awaiting deployment. The geopolitical strife in the corridor has led to recent reports that all cable projects there have been “delayed indefinitely.”²⁴

Geopolitical risk is rising, but most cable damage remains accidental. The International Cable Protection Committee estimates that fishing gear and ship anchors account for roughly 70–80% of faults.

2.4 State actors responding

Governments have long regarded submarine telecommunications networks as critical infrastructure requiring strong protection (for example, ACMA, 2007).²⁵ More recently, they have also recognized the geopolitical value of controlling these corridors. In 2015, China set a goal of capturing 60% of the fibre-optic cable market through state-backed firms under its Digital Silk Road initiative.²⁶ HMN Technologies (formerly Huawei Marine) is one of four companies that collectively dominate global cable manufacturing.

However, China’s influence is primarily concentrated in the Global South, as Chinese firms have been largely excluded from most Western-connected systems since the US launched its 2020 Clean Network program to counter China’s influence across telecoms and data. Under the Clean Network, Chinese telecom and cable firms were shut out of lucrative bids for cable-laying projects, most notably when HMN Technologies lost the SEA-ME-WE 6 project (running from France to Singapore via the Suez) in 2023 despite its competitive pricing. aChina has pivoted to financing new cable projects connecting Africa, Asia and the Middle East, such as its flagship PEACE cable (Pakistan-East Africa-Europe) and EMA (Europe-Middle East-Asia) project, which is backed by three state-owned telcos: China Telecom, China Mobile, and China Unicom in partnership with Egypt and Saudi Arabia.

Over the last few years, the US has consolidated its toolkit to protect its subsea cable security. In February of this year, the US Congress introduced the bipartisan Strategic

²¹ In 2025 Western media discovered that engineers at China’s Lishui University in Zhejiang had filed a patent for the creation of a submarine cable cutting device back in 2020. While it was argued it was needed for emergency situations, the announcement raised suspicion amongst Western media, which uncovered the patent applications in January 2025.²² And in April 2026, the South China Morning Post reported China had successfully tested cable cutting 11,483 feet (3,500m) undersea.

²² Joe Wilson. 2026. US Strategic Subsea Cables Act. https://joewilson.house.gov/sites/evo-subsites/joewilson.house.gov/files/evo-media-document/undersea-cables-signed_0.pdf.

²³ [Iran Signals a New Threat to the Gulf’s Internet | WIRED Middle East](#)

²⁴ “Stalled Subsea Cable Projects Threaten Middle East Digital Ambitions.” 2026. FinancialTimes. Financial Times. June 25, 2026. <https://www.ft.com/content/5e6ce736-3a48-406a-933a-ed215370bd16?syn-25a6b1a6=1>.

²⁵ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

²⁶ “The Digital Silk Road: A Growing Priority for Beijing as Its Tech Champions Expand Overseas.” 2024. Merics. March 27, 2024. <https://merics.org/en/tracker/digital-silk-road-growing-priority-beijing-its-tech-champions-expand-overseas>.



Subsea Cables Act. A core proposal of the act requires the President to impose sanctions on foreign actors who intentionally damage critical undersea infrastructure, including subsea fiber optic cables.²⁷ If passed, this act will build upon the US's 2020 Undersea Cable Control Act, which focused more on preventing adversaries from accessing technology to build and operate undersea cables. Together, the two bills aim to deny countries like Russia and China the ability to conduct what a sponsoring senator referred to as “shadow warfare” against US critical infrastructure in subsea cables.²⁸

Finally, a landmark development to watch is the recent US Federal Communications Commission (FCC) vote to extend cable surveillance to terrestrial endpoints, such as cable landing stations and the hardware housed within them that converts the subsea optical fiber signals. The FCC has identified these endpoints as chokepoints vulnerable to data interception, physical sabotage, and cyber espionage. Under the new rules, hyperscalers and tech firms employing AI infrastructure could receive expedited licensing for their terrestrial hardware, while foreign rivals would need to apply for individual licenses and face the possibility of denial. This is a potential win for the hyperscalers, although the pool of eligible hardware suppliers may significantly shrink, creating short-term bottlenecks in cable landing equipment manufacturing.²⁹

Europe has historically had a strong presence in the subsea cable industry, but this position is being challenged by competing US and Chinese interests and the rise of US hyperscaler control. France remains a leading global supplier through its ownership of Alcatel Submarine Networks and Orange Marine, while Italy's Elletra holds a strong mid-tier position.

The EU is attempting to carry out a coordinated approach towards cable security via its 2025 EU Action Plan on Cable Security, but implementation has been slow. In June of this year, however, the European Commission finalized a €5.8mn grant to establish the first two regional cable hubs in the Baltic Sea and the Mediterranean Sea, and a separate €40mn fund to increase its submarine cable repair capacity.³⁰

Finally, US-allied attempts to find a geopolitically secure “Fourth Route” away from volatile transits in the Red Sea and South China Sea have led to the creation of the Far North Fiber express route. This will be the first long-haul submarine cable connecting Europe and Asia exclusively through Allied-sovereign waters. It is expected to cost \$1.17bn and should be ready for service by the end of 2026.³¹ The project is owned by US Alaskan company Far North Digital, in conjunction with its Canadian affiliate True North Global Networks, but it has since expanded into a consortium with companies from Japan and Finland as well. Alcatel Submarine Networks will supply the cables.

All of this strategic interest has translated into unprecedented subsea cable infrastructure investment, estimated to have averaged over \$2bn per year in new construction over the last 9 years.³² Most new investment is forecast to be directed toward trans-pacific routes.

The most consequential response, however, is coming from the private sector, which controls most of the infrastructure. AI is sharply increasing bandwidth demand and prompting content and cloud providers—notably Google through its Pacific Connect

²⁷ Qiu, Winston. 2026. “US Introduces Strategic Subsea Cables Act of 2026 — Senior Industry Analyst Briefing.” Submarine Networks. April 14, 2026. <https://www.submarinenetworks.com/en/nv/insights/us-introduces-strategic-subsea-cables-act-of-2026>.

²⁸ “Wilson and Meeks Introduce ‘Strategic Subsea Cables Act Of 2026.’” 2026. Representative Joe Wilson. March 27, 2026. <https://joewilson.house.gov/media/press-releases/wilson-and-meeks-introduce-strategic-subsea-cables-act-2026>.

²⁹ Qiu, Winston. 2026. “FCC Overhauls Subsea Cable Licensing: SLTE Licensing and Fast-Track Process for US Tech Giants.” Submarine Networks. June 26, 2026. <https://www.submarinenetworks.com/en/nv/insights/fcc-subsea-cable-licensing-slte-fast-track-process>.

³⁰ Ernst, Lauren. 2026. “EU Launches Regional Cable Hubs To Protect Critical Undersea Infrastructure.” Ocean News & Technology. June 23, 2026. <https://oceannews.com/news/subsea-cable/eu-launches-regional-cable-hubs-to-protect-critical-undersea-infrastructure/>.

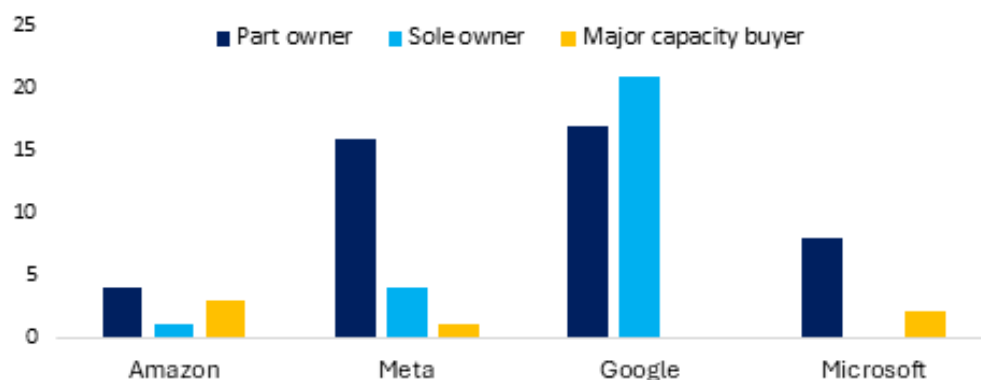
³¹ “Far North Fiber - Submarine Networks.” 2026. Submarinenetworks.Com. 2026. <https://www.submarinenetworks.com/en/systems/trans-arctic/far-north-fiber>.

³² Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.



Initiative and Meta through Project Waterworth—to accelerate investment.³³ This private dominance means resilience will depend as much on international coordination with corporate owners as on national policy.

Figure 3: Number of submarine cables owned by content providers



Source: Telegeography, Deutsche Bank Research..

3. Satellites: Beyond terrestrial networks

Like subsea cables, satellites form a part of the Internet's physical layer. Throughout much of the 20th century, satellites carried the bulk of all transoceanic digital connectivity. The launch of Telstar, the first operational telecommunications satellite in 1962, was soon followed by other satellites that helped commercialize the international communications and broadcast industry.

Today, however, traditional satellite communications (satcom) complement rather than replace cables. Satellites provide broadcast and communication services to sparsely populated regions not served by terrestrial infrastructure and act as a strategic back-up for disaster-prone regions. While subsea cables deliver more consistent, high-capacity links between major population centres, satellites are a cost-effective solution for governments seeking to connect regions with low population densities.³⁴

The renewed focus on satellite communications is driven less by their current share of global traffic than by their strategic potential. Two forces are decisive. First, the war in Ukraine and US-China technology competition are prompting sovereign investment in space-based communications. Second, rising AI workloads are intensifying demand for bandwidth and encouraging the deployment of satellite mega-constellations. We examine the implications in greater detail in the next chapter.

For these reasons, communications-based satellite networks and space-based connectivity represent the next major frontier in the contest for control over digital connectivity.

3.1 How satellites support connectivity

There are three main types of satellites in operation: communications, Earth-observation, and radiofrequency- based. This section focuses on communication satellites, which provide services ranging from radio, TV, telecommunications, and Internet by receiving signals from Earth and retransmitting them to a ground station elsewhere, creating a communications channel between distant regions that were previously unable to communicate with one another.

³³ The State of the Network: 2026 Edition. 2026. Telegeography.

³⁴ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.



Traditional satellite communications (also referred to as “satcom”) rely on large geostationary (GEO) satellites positioned about 35,800 kilometres above Earth. Systems such as the early Intelsat 1 (1965) and the more recent Jupiter 3 (2023), provide broad, fixed coverage with high availability and favourable ground-terminal geometry.³⁵ GEO assets have historically supported predictable business models because of their long operating lives, typically around 15 years, and high launch costs. Their strategic value is clear, but so is their limitation: distance creates latency. The farther a satellite is from Earth, the longer data takes to travel, which is one reason subsea cables continue to carry most global connectivity.

In addition, GEO satellites also suffer from increasingly longer development cycles and higher integration complexity. As a result, both state and private actors are turning their interest towards low-earth orbit (LEO) satellites instead. LEO satellites orbit between 160 and 1,500km above the Earth’s surface, enabling a more seamless user experience for applications like live streaming, video calls, remote desktops, etc. Because they circle the planet up to 16 times a day, LEO constellations can provide ubiquitous coverage to nearly any point on Earth. SpaceX’s Starlink was the harbinger of the LEO satellite model, but several LEO constellations have emerged in response to Starlink since then, including Amazon LEO, OneWeb (now part of Eutelsat), Telesat Lightspeed, and China’s Guowang and Qianfan projects. In fact, in July 2026 Amazon announced it had filed an application with the US FCC to deploy a constellation of more than 5,100 satellites that would provide direct-to-device connectivity by 2028.³⁶ Doing so will directly challenge Starlink Mobile, a subset of Starlink comprised of around 650 satellites that beam mobile data to phones in remote regions on earth, and which has recently also announced plans to launch “thousands” more satellites by end of 2028.³⁷

3.2 Why satellites are strategic: the arrival Starlink

Although subsea cables carry the bulk of internet traffic, communications-based satellites are also integral to the global economy. They provide connectivity to remote and underserved regions that undersea cables cannot reach, while also enabling communications for mobile platforms such as ships, aircraft and military assets. Indeed, industries such as aviation, rail, shipping and logistics depend on satellites for navigation, route optimization, and cargo tracking.

Perhaps to an even greater extent than cables, satcom has become strategic to a country’s military infrastructure. Satcom companies are developing partnerships to cater to civilian and government forces. For example, Canada’s Telesat (c. 1969) announced a strategic partnership with the Canadian Armed Forces to develop and deliver a multifrequency, Arctic military satellite communications capability via its forthcoming Lightspeed constellation, which the company targets to launch in late 2026.³⁸ Meanwhile, established satcom companies with significant government -related revenue streams, like Viasat, SES and Eutelsat, have seen strong growth over the past year. Our DB equity analysts estimate that existing GEO communications contracts alone account for around \$50bn-60bn in defence-related spending (excluding non-US allies).³⁹

The greatest harbinger has been SpaceX’s Starlink, which has provided encrypted communications, imagery and geolocation services for the Ukrainian military against Russia since 2022. In addition, Starlink’s “direct-to-device” business model that connects smartphones directly to satellites threatens to collapse the traditional, hierarchical

³⁵ Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. “State of the Private Markets: Rise of Micro-GEO Satellites.” Deutsche Bank Research.

³⁶ “Amazon Targets Musk’s Starlink with Satellite Constellation for Mobile Services.” 2026. FinancialTimes. Financial Times. July 27, 2026. <https://www.ft.com/content/83e93ce0-1ebd-4b04-a70c-82ff7b64a5f4?syn-25a6b1a6=1>.

³⁷ Kan, Michael. 2026. “Starlink Mobile Will Span ‘Thousands’ of Satellites By 2028.” PCMag UK. August 3, 2026. <https://uk.pcmag.com/networking/166530/starlink-mobile-will-span-thousands-of-satellites-by-2028>.

³⁸ Yu, Edison, and Laura Li. 2026. “Telesat: Focus Increasingly Shifts toward Defense.” Deutsche Bank Research.

³⁹ Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. “State of the Private Markets: Rise of Micro-GEO Satellites.” Deutsche Bank Research.



Internet infrastructure, where cables traverse terrestrial data and comms-based satellites take it on its last leg.

Starlink's ability to bypass the conventional hierarchy of connectivity provided vital communications for Ukrainian forces and civilians at a time when conventional communications like fiber lines, cellular towers and power grids were damaged early in the war. This highlighted how a private company could become essential to a nation's military capability during a war. The gravity of this power was demonstrated again in February 2026, when SpaceX and Ukraine introduced a terminal verification system that effectively blocked Russia from using Starlink military terminals, after it was revealed that Russia had acquired Starlink terminals via third-party countries and grey markets to conduct strike drones.

The entry of Starlink in 2019 has shifted the goalposts and urgency for more communications-based satellites. Bringing affordable, high-speed internet to those that lack access to traditional broadband, Starlink's true purpose is to help deploy Musk's goal of AI compute at a massive scale. In fact, Musk confirmed that Starlink is also designed to support a project called Starwind, which aims to place massive data centers in space. He recently filed applications with the FCC for a constellation of space-based AI data center satellites through this project. These space AI data centers will compute in space but then use lasers to offload the processed data directly into the Starlink constellation, which will beam the results back to Earth with low latency.

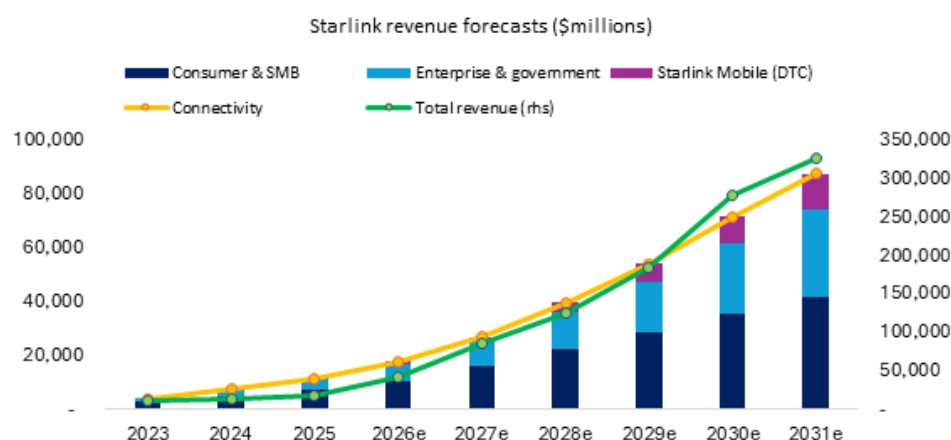
By using Starlink to create a reliable and robust orbital network used for Earth, Musk hopes to eventually extend the satellite internet service to the Moon, Mars, and beyond. Our equity analysts place Musk's aims for orbital data centers within a 5-year horizon and human voyage and colonization in Mars and off-world within 10-year horizon.

Starlink has upped the game for the rest of the world considerably. First, by its sheer dominance in the number of satellites it operates and in the numbers of subscribers it now has on its network. Starlink now accounts for 54% of all operational satellites in orbit,⁴⁰ and our equity analysts estimate Starlink's consumer broadband subscribers will grow to around 17mn by the end of 2026.⁴¹ Second, its ability to simultaneously increase speeds while cutting hardware prices has made it very hard for anyone else to compete. Finally, Starlink's entry into LEO also made it harder for pure-play GEO industry players like Viasat and Intelsat to focus solely on consumer broadband, although as mentioned above there is a growing market opportunity for these players to enter the government services market. We will go more in depth on Starlink and its parent company SpaceX in our final chapter on Cloud and Data Centers.

⁴⁰ "10,400+ Starlink Satellites in Orbit — Live Count (2026) | Orbital Radar." 2026. Orbital Radar. May 31, 2026. <https://orbitalradar.com/how-many-starlink-satellites>.

⁴¹ Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.

Figure 4: Starlink connectivity revenue to reach \$18bn in 2026 and \$88bn by 2031



Source: Deutsche Bank Research equity research, company filings.

3.3 Satellite vulnerabilities

Because communications-based satellites are so critical, their vulnerability to deliberate attacks is a matter of strategic concern. Satellites are susceptible to attacks such as jamming (when the uplink or downlink frequency is filled with noise so the signal cannot get through) and spoofing (transmitting fake signals) of their radio waves. Countries like Finland, Estonia and Poland have all reported [GNSS](#) disruptions associated with alleged Russian activities in the Baltic region, which has affected numerous services such as commercial aircraft and maritime navigation. Other examples of disruption include cyberattacks on satellites via their ground stations and control networks, as well as physical and cyberattacks on the orbiting satellites themselves. Notable incidents include evidence that Russia was conducting widespread [jamming](#) of Starlink GPS signals soon after their invasion of Ukraine in 2022. In 2021, Russia also conducted an anti-satellite weapon test that destroyed an inactive satellite and generated a massive field of orbital debris.

Figure 5: Major satellite networks in orbit

Constellation	First launched	In orbit	Total planned
Starlink (US)	May 2019	10,877	42,000
Amazon Leo (US)	Oct 2023 (prototype satellites); production deployment began Apr 2025, initial commercial service expected in late 2026	390+	7,700
OneWeb (UK/Eutelsat)	Feb 2019, merged with Eutelsat in 2023.	654	Gen-1 complete, additional 440 satellites planned with Airbus
Guowang (China)	December 2024	177	12,992 (~13,000) filed with the ITU
Qianfan (China)	August 2024	238	Over 15,000 satellites approved for the final system

Source: Satellitemap.space, ChinainSpace, CNBC.

3.4 How state actors are responding

A rapid, high-stakes military space race looks to be underway between the US, China, and Russia, focused on establishing, securing, and defending satellite communication (satcom) constellations, particularly those in Low Earth Orbit (LEO).

While US companies currently dominate the satcom landscape, they face a growing challenge from other countries, especially China. China is developing its own satcom



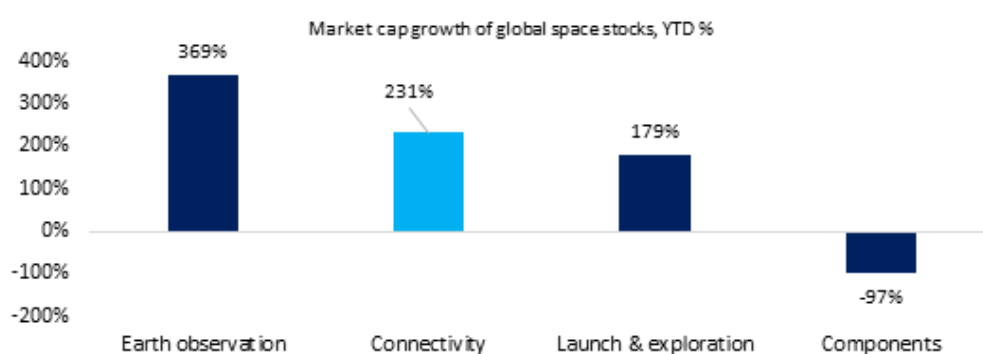
ecosystem through two key initiatives: the state-run Guowang network, aimed towards China's domestic telecom and national security, and the Shanghai-backed Qianfen constellation, which is aimed at the commercial and export markets. Together, these projects aim to deploy between 13,000 and 15,000 satellites.⁴² Guowang is designed primarily to connect rural areas and government institutions, while Qianfan is more commercially focused, aiming to provide in-flight connectivity (IFC) and direct-to-device (D2D) communications, with active pilots underway in Brazil, Malaysia, and Thailand.

Both constellations are in their early stages, with fewer than 200 satellites each currently in orbit. In December 2025, China also submitted filings with the International Telecommunication Union (ITU) to reserve orbital slots and spectrum for nearly 200,000 additional satellites – a move to secure future resources, even though China does not yet have the launch capacity to deploy such a number.⁴³ To facilitate more launch activity, China has also implemented new regulatory guidelines to facilitate IPOs for firms developing reusable rockets.

The EU is also attempting to reduce its dependence on US private operators by building IRIS, a government-controlled constellation of approximately 290 satellites, with services targeted to begin in 2029. However, progress on the project remains slow. Other notable sovereign interests include the UK's stake in OneWeb and Canada's investment in Telesat's Lightspeed project.

Driven by these strategic imperatives, the market for communications-based satellites is growing, as they become critical infrastructure for consumers, enterprises, and governments. Deutsche Bank equity analysts estimate the satcom market grew by over 35% YoY to nearly \$26bn in 2025, mostly driven by consumer broadband and government contracts.⁴⁴ This growth is expected to continue, with market revenue projected to reach \$65-70bn by 2030.

Figure 6: Oversized demand for comms-based satellites has led to a rally in space connectivity stocks



Source: Bloomberg Terminal, Deutsche Bank Research. Returns up to August 2026. Connectivity stocks: AST SpaceMobile, APT, Beijing BDStar, China Satcom, Echostar, Eutelsat, Globalstar, Gogo, Iridium, JSAT, SES, Space42, Telesat, Thaicom, Viasat.

⁴² Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.

⁴³ "China ITU Filing to Put ~200K Satellites in Low Earth Orbit While FCC Authorizes 7.5K Additional Starlink LEO Satellites – IEEE ComSoc Technology Blog." 2026. Comsoc.Org. January 13, 2026. <https://techblog.comsoc.org/2026/01/13/china-itu-filing-to-put-200k-satellites-in-orbit-fcc-authorizes-7-5k-additional-starlink-leo-satellites/>.

⁴⁴ Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.



4. The new power map of connectivity

4.1 The rise of private control

While cloud and space data centers will be the focus of our next chapter, this section zooms in how connectivity supply chains are being affected by the AI build-out.

In the connectivity model set out in Section 1, hyperscalers own the destination of data: the cloud, where applications, websites and AI models run. Historically, these companies have relied on telecom operators for transmission and concentrated investment on data centers and cloud services. That division is now eroding, as hyperscalers are increasingly acquiring, funding or building the underlying infrastructure, including subsea cables and communications satellites—that carry data to and from their platforms.

AI's data intensity is accelerating this vertical integration. In 2016, traditional backbone providers carried most international bandwidth; today, content and cloud networks account for nearly three-quarters of demand.⁴⁵ Owning connectivity allows hyperscalers to lower long-run unit costs, scale capacity with cloud demand and reduce dependence on third-party networks. It also gives them greater control over the economics and governance of the data chain.

This trend is evident in new cable construction, where private companies are starting to swallow up market share. For example, of the \$14bn projected investment in new submarine cables entering service between 2025 and 2027, driven largely by content providers such as Google and Meta rather than the traditional internet backbone providers.⁴⁶ While traditional telecom operators still account for the majority of existing builds, this new investment demonstrates the steady and growing influence of large-scale cloud and content companies in shaping global connectivity.

The same dynamic is beginning to emerge in the satellite ecosystem, where hyperscalers and space technology companies are exploring space-based data centers as a potential long-term solution to the terrestrial constraints that increasingly accompany the buildout of AI compute capacity.

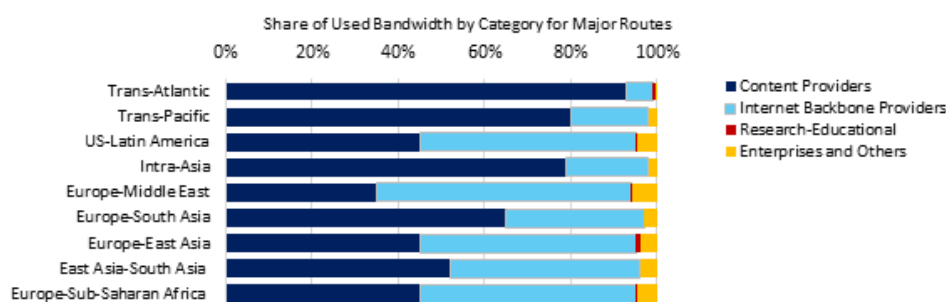
To make orbital computing viable, companies need communications infrastructure capable of moving massive amounts of data between space and Earth, creating a strategic rationale for greater ownership of satellite networks and related connectivity assets. A notable entrant is Jeff Bezos' Blue Origin, which announced TeraWave in January 2026, a multi-orbit communications network comprising 5,280 satellites in LEO and 128 satellites in MEO. Blue Origin plans to begin deploying the constellation in 4Q27 and is targeting enterprise, data-center, and government customers. DB analysts estimate that the constellation could require ~\$18bn of capex.

Although Blue Origin has not explicitly positioned TeraWave as infrastructure for orbital data centers, its high-capacity, optically interconnected network could eventually provide part of the communications layer required to connect space-based compute systems with terrestrial networks. In addition to SpaceX and Blue Origin, Amazon is building its LEO communications constellation, while Nvidia-backed Starcloud has filed for regulatory approval to deploy orbital data-center satellites. Starcloud also plans to launch AWS Outposts hardware aboard its second satellite in October 2026, providing an early test of AWS infrastructure operating in space.

⁴⁵ The State of the Network: 2026 Edition. 2026. Telegeography.

⁴⁶ Ibid.

Figure 7: Content networks represent around three-quarters of all global bandwidth demand



Source: Deutsche Bank Research. Data inferred from Telegraphy's 2026 State of the Network Report. Represents used bandwidth as of year-end 2024.

4.2 Effects of the AI disruption

The result is a sharp shift in control over global connectivity. Traditional carriers are increasingly becoming minority consortium members or implementation partners, while hyperscalers lead funding and technical design. The Internet's decentralised architecture increasingly coexists with concentrated private ownership of critical connectivity infrastructure.

The consequences are substantial. By owning subsea cables and satellite networks, hyperscalers are increasingly able to build private connectivity corridors linking their data centres, reducing dependence on traditional telecom operators. This vertical integration bypasses—and may ultimately compress—the tiered, multi-provider structure that has defined telecommunications for decades.

This disruption has forced traditional telecom operators to either merge to survive or retreat into more niche services. By shrinking the telecommunications market, the remaining players can better afford the massive capex required for increasing broadband demand. Notable deals in recent years include \$7.3bn merger between US-based Viasat and British satellite firm Inmarsat in 2023, SES's acquisition of Intelsat for \$3.1bn in 2024, and the \$8bn Rocket Lab/Iridium⁴⁷ consolidation in 2026. By creating these multi-orbit groups, they have been able to concentrate on key segments of mass-market connectivity.

Europe is perhaps at the greatest disadvantage in this new connectivity map, finding itself crowded out by US-led private innovators and state-backed Chinese companies. The European Centre for Development Policy Management argues that the growing concentration of subsea-cable ownership and operational control among US hyperscalers is creating structural dependencies that could erode Europe's digital sovereignty.⁴⁸ This concern is particularly acute for Europe, where ownership and operational control of critical subsea infrastructure is increasingly concentrated among non-EU actors. Unlike the United States, which exercises oversight through institutions such as the FCC, Europe lacks a unified governance framework for the subsea cable ecosystem, raising concerns about long-term digital sovereignty.⁴⁹

The same risk exists in space, where Starlink dominates the global satellite network. As the CEO of the French telecoms operator, Bouygues Telecom, warned in an interview earlier this year, Europe needs to "get some sovereignty" or else it risks having its

⁴⁷ Deutsche Bank and/or its affiliates is acting as advisor to Rocket Lab Corporation on the potential acquisition of Iridium Communications, Inc..

⁴⁸ Pearson, Sasha. n.d. "Troubled Waters: Europe's Subsea Telecommunications Network." Ecdpm. <https://ecdpm.org/work/troubled-waters-europes-subsea-telecommunications-network>.

⁴⁹ Ibid.



connectivity controlled by a non-state actor like Starlink.⁵⁰ Reflecting these pressures, in June 2026, Bouygues itself partnered with Orange and Free-illiad Group to acquire and break up French telecommunications company SFR for €20.35bn.

In the space arena, the EU has attempted to address this problem with its proposed Space Act, which aims to cultivate European champions. However, this has created friction with the US, whose FCC has publicly warned that Washington would retaliate if Europe imposed protectionist measures that favored its own satellite companies over American ones.⁵¹ In the meantime, the formidable presence of Starlink continues to drive a new wave of private communications-based satellite initiatives seeking to compete.⁵²

States therefore face a strategic dilemma. AI and cloud adoption are directing more traffic through hyperscaler-owned networks, increasing dependence on private technology companies even as governments must guarantee resilience and control in a crisis. The US is best positioned because its firms dominate the industry. European governments have responded by buying stakes in, or taking control of, strategic telecom assets, including Italy's acquisition of TIM Sparkle and Spain's 10% stake in Telefónica.

Figure 8: Aside from 2022, where activity dropped sharply amidst the backdrop of Russia Ukraine, the telecoms M&A industry has seen massive growth



Source: Deutsche Bank Research, Dealogic. Data only draws upon publicly available deal value at announcement data. Data up to July 2026.

5. Conclusion

Digital connectivity emerged from an ideal of neutrality and disintermediation. The Internet was designed to share knowledge across research institutions, while cables and communications satellites expanded international exchange. Today, connectivity is an arena of state and corporate power. Control over data flows is becoming as consequential as control over payment infrastructure, and the race for AI leadership is intensifying the strategic value of the networks that carry information.

Subsea cables are uniquely fragile because their criticality is matched by limited substitutability. The parallel with payments is instructive: excluding a country from SWIFT can amount to financial siege; severing its cable links can impose a digital one. Yet payments have emerging alternatives—including CIPS, SPFS and stablecoins—whereas subsea cables have few practical substitutes. A replacement cable cannot be laid during a crisis, so a major disruption could simultaneously impair payments, markets, military command-and-control and basic communications.

⁵⁰ Ohlen, Elsa. 2026. "Europe Doesn't Realize How Dangerous It Is': Telecoms CEO Warns of U.S. Dominance in Satellites, AI." CNBC. May 21, 2026. <https://www.cnbc.com/2026/05/21/starlink-spacex-satellites-telecoms-europe-dangerous-bouygues-ceo.html>.

⁵¹ Ranjit, Roshan, Robert Grindle, and Keval Khiroya. 2026. "European Satellites: Levelling Up." Deutsche Bank Research.

⁵² Yu, Edison, and Laura Li. 2026. "The Final Frontier: Space 101." Deutsche Bank Research.



The most important actor to watch may not be a state but a company: SpaceX. Starlink has reshaped satellite communications on cost, scale and performance, compressing the traditional hierarchy of space connectivity. Its ambitions now extend into markets historically served by terrestrial telecom and cable providers, in both advanced and emerging economies.⁵³ The implications for governments, defence and incumbent telecom operators will be significant.

In the era of infrastructure realism, power lies less in the number of cables or satellites built than in control of the network's junction points and architecture—from the ocean floor to low-Earth orbit. As with dollar-based payments and Taiwan's semiconductor capacity, concentration in cable landing rights and orbital slots creates a new class of systemic risk. The central question is no longer who wins the cable race, but who controls the rails through which global value flows—and on whose terms.

The final part of our series turns to data centres, moving the chokepoint further up the technology stack—from minerals and cables at the physical layer, through chips at the fabrication layer, to the infrastructure that stores and processes data at scale.

We would like to thank Edison Yu, Laura Li, Benjamin Black, Raymond Wong and Yi Xiong for their valuable insights and comments to this paper.

⁵³ Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.



Appendix 1

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The Next 90: Your Geopolitical Risk Radar

Sep–Nov 2026

September 2026

Dr. Helen Belopolsky | Miha Hribernik

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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The Next 90: Your Geopolitical Risk Radar | Sep–Nov 2026



A geopolitical calendar mapping the next 90 days

SEPTEMBER

For a detailed breakdown of key events, issues and themes in September, please see: [Latest Geopolitical Monthly](#)

Early	US/Mexico	Next bilateral round of USMCA review talks
1	G20	Meeting of Finance Ministers and Central Bank Governors
1	SCO	Shanghai Cooperation Organisation (SCO) Heads of State Summit in Bishkek
6-20	Germany	Regional elections
8	UN	Opening of the 81 st UN General Assembly
8	Canada/US	Canadian retaliatory tariffs on the US are set to enter into force
12-13	BRICS+	18 th BRICS+ summit in India
13	Sweden	General election
15	EU	European Parliament plenary session
18-20	Russia	Legislative and regional elections
22-28	UN	UN General Assembly debate
24	US/China	Xi-Trump Summit in the US
27	France	Senate election
30	US	Key government funding deadline

OCTOBER

3	Latvia	General election
4	Brazil	General election
5-28	Canada	General election in Quebec, municipal elections in British Columbia, Ontario and Manitoba
9-17	Czech Republic	Senate and municipal elections
12-18	IMF/World Bank	Annual meetings of the IMF and WB in Bangkok
15-16	EU	European Council meeting
19	Canada	Alberta independence referendum
23	Morocco	General election
24	Slovakia	Municipal and regional elections
25	Bulgaria	Presidential election
27	Israel	Legislative election
30-31	G20	G20 Foreign Ministers' Meeting in Atlanta
TBD	EU/China	Ministerial meeting in Beijing; deadline for "tangible results" on trade talks
TBD	China	Fifth Plenum of China's Central Committee

NOVEMBER

3	US	Midterm elections (House, Senate, gubernatorial and state legislatures)
4	South Africa	Municipal elections
7	New Zealand	General election
9	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
9-20	COP31	COP31 in Türkiye
10	US/China	Expiry of the 2025 Busan agreement
10-12	ASEAN	49 th ASEAN Summit in Metro Manila
18-19	APEC	APEC Economic Leaders' Meeting in Shenzhen
28	Taiwan	Local elections
28	Palestinian National Authority	Legislative election
TBC	US/North Korea	Potential meeting between Donald Trump and Kim Jong-un

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

Appendix 1



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This Month in Geopolitics: September 2026

Key events, themes and trends to watch in the month ahead

September 2026

Dr. Helen Belopolsky | Miha Hribernik

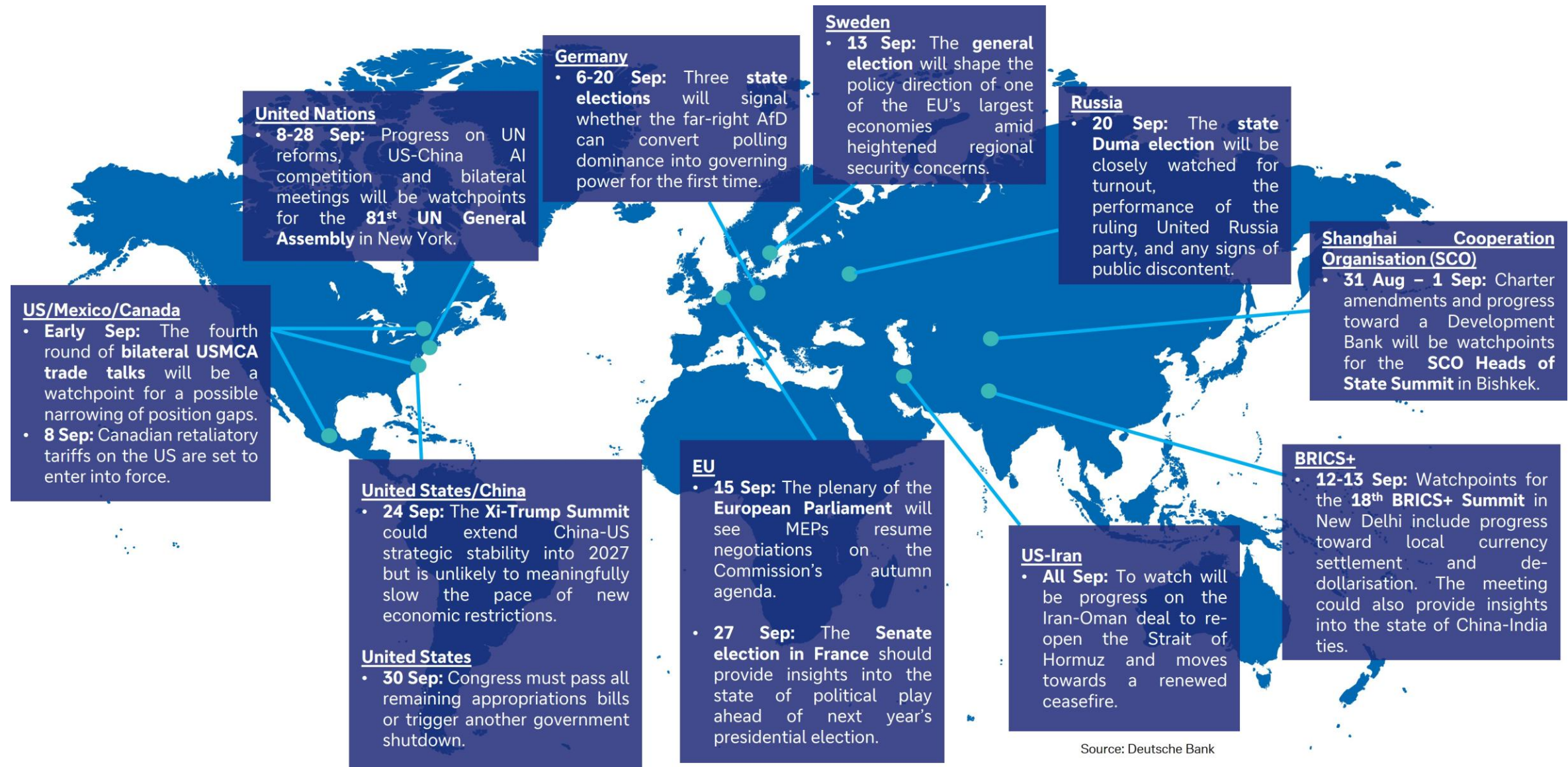
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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in September 2026

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



In the spotlight for September 2026: US-Iran war and China-US Summit



US-Iran war (all of September)

August brought a mix of escalation and small progress in the conflict: US strikes on Iranian oil infrastructure and an ongoing blockade pushed fuel shortages inside Iran as President Trump ramped up economic pressure. Iran and Oman outlined a proposal for a temporary shipping corridor through the Strait of Hormuz, with talks ongoing toward a permanent arrangement.

It appears both the US and Iran are waiting for the other side to blink though a new framework agreement isn't beyond the realm of possibility as backchanneling continues. Economic pressure on Iran is ramping up with President Trump's "economic D Day" threat and fuel shortages within Iran adding to the regime's challenges. Political pressure on the Trump administration is likely to ramp up moving closer to the mid-terms. To watch will be whether the Iran-Oman shipping corridor talks move from proposal to an actual operating channel, which could signal a genuine de-escalation amid a war that seems to keep grinding on with no military victory able to drive resolution. In addition, the US' latest round of sanctions targeting networks moving Iranian oil through China, Hong Kong and the UAE, will be a point of friction at the Trump-Xi summit. We should prepare for more volatility in the coming months as neither side is prepared to lose face.

Trump-Xi Summit (24 September)

The summit could extend China-US strategic stability into 2027 but is unlikely to meaningfully slow the pace of new economic restrictions. During the run-up to the summit, investors should watch for any sign of progress in three areas: the Board of Trade to support non-sensitive commerce, AI dialogue and a possible extension of the Busan agreement beyond November. The meeting could also deliver new purchase commitments. **While relations remain stable, there is little evidence to indicate a major breakthrough.** [Recent reporting](#) suggests stalled progress in some areas, such as on a new Board of Investment, in line with our [assessment](#) that national security concerns will limit progress in this area.

Investors should also be prepared for both sides to unveil new restrictions or retaliatory measures ahead of the summit, as Beijing and Washington work to strengthen their negotiating positions. US economic curbs against [Iran's](#) economic partners will be a watchpoint. New restrictions would add to a growing list of measures adopted this year (see table), which will likely expand further in the coming years. There is little prospect that the September summit can resolve structural competition in areas such as critical minerals, semiconductors or AI.

Beyond summitry: China-US relations remain competitive in 2026

January	Washington approved limited Nvidia H200 chip sales to China , signaling a more transactional approach to tech controls.
February	The US launched a major critical minerals effort , including new bilateral frameworks, financing tools and a 56-country coalition, aimed at reducing dependence on China.
April/May	China codified new anti-derisking regulations , which allow Beijing to penalize companies that cut Chinese suppliers or comply with foreign sanctions/export controls. China separately ordered the unwinding of the Meta-Manus AI deal, signaling scrutiny of US investments in sensitive China-linked tech firms. New US Congressional bills , such as the MATCH Act, seek to tighten semiconductor equipment controls, close loopholes and pressure allies to align with US restrictions. US sanctions on Chinese refiners tied to Iranian crude triggered China's first formal use of blocking measures.
June	The Pentagon included several major Chinese tech companies on its Chinese Military Companies list. In response, China introduced export controls on 10 US companies, and restricted procurement from another 46 corporates. The US banned China-linked Polestar from selling cars in the US from model year 2027 onward due to national security concerns.
July	China's new outbound investment regime covers the overseas deployment of Chinese technology, data, talent and capital, especially in AI, chips, batteries, EVs and solar. The US imposed tariffs under a Section 301 forced labour investigation on over 60 markets, including China (12.5%). Washington also banned the import of new Chinese humanoid and quadruped robots and power inverters.
August	The US announced tariffs of up to 100% on imported drones and components , primarily impacting Chinese exporters. A White House report alleged the transshipment of China-linked goods into the US through more than 40 countries. The US also added a record 43 Chinese companies to the UFLPA Entity List ; China retaliated by imposing countermeasures on 7 US compliance, audit and testing firms. US sanctioned dozens of mainland and HK entities over Iran China tightened drone-related exports to the US , potentially constraining access to critical minerals, rare earths, advanced electronics and other dual-use components. In late August, Beijing approved limited sales of H200 chips in China.
November	The 2025 Busan agreement is set to expire but will likely be extended, delaying the introduction of new bilateral economic restrictions.
December	The US Congress will adopt annual must-pass defence legislation , which frequently includes provisions aimed at China.

Note: The table provides a non-exhaustive list of selected key developments.
Source: Deutsche Bank Research

Key events, issues and themes to watch in September 2026

Listed in chronological order



Shanghai Cooperation Organisation (SCO) Heads of State Summit (31 August – 1 September)

Charter amendments and development bank progress will be key watchpoints. The summit in Bishkek will bring together leaders from the SCO's 10 member states: China, Russia, India, Pakistan, Iran, Belarus, Kazakhstan, Kyrgyzstan, Tajikistan and Uzbekistan. Founded in 2001, the SCO aims to promote security and economic cooperation across Eurasia, but its practical impact has been limited given its relatively loose structure and member states' diverging geopolitical priorities.

However, the Bishkek summit could offer significant watchpoints, including amendments to the SCO's Charter and possible progress toward the establishment of an SCO Development Bank. The latter would give the SCO a dedicated mechanism to fund regional infrastructure. A dedicated development bank could also support local currency settlement – potentially boosting efforts by member states such as China and Russia to reduce their reliance on the US dollar.

Next bilateral round of US-Mexico USMCA review talks (Early September)

The fourth round of talks is unlikely to deliver a breakthrough but should show whether US-Mexico position gaps are narrowing. 1 July marked the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process whereby the three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode. The US has declined to confirm the renewal of USMCA "in its current form".

We expect USMCA negotiations to remain difficult and protracted. In early September, the US and Mexico will hold further negotiations focused on resolving outstanding issues (e.g., automotive rules of origin, steel/aluminum, economic security, and issues around agriculture, labour, environment and alleged [China-linked transshipment](#)). Auto talks will be a key watchpoint. Washington is reportedly requesting 50% US-made content in North American vehicles, while Mexico's [counter-proposal](#) would exempt Canadian and Mexican parts and levy only non-North American content – potentially lowering US vehicle prices.



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German state elections: Saxony-Anhalt (6th), Mecklenburg-Vorpommern (20th) and Berlin (20th)

The three state elections in September will close out Germany's 2026 "super election year" and are together being read as a stress test of whether the far right AfD can convert polling dominance into governing power for the first time. In Saxony-Anhalt, the AfD is polling around 41%, well ahead of the Chancellor Merz's CDU's roughly 23-27%. The polling puts the AfD within reach of an absolute majority – a level of power no AfD branch has held in any German state. In Mecklenburg-Vorpommern, the AfD leads at 34-38% support, well ahead of the SPD (19-27%), which has led the state since the late 1990s. Berlin, by contrast, remains genuinely fragmented; the CDU sits around 19-22%, with the AfD, Greens, and Die Linke behind. These elections are seen as a test of the AfD's rise and whether that puts strain on Chancellor Merz's federal coalition.

Canadian retaliatory tariffs on the US (8 September)

On 8 September, Canada will enact retaliatory tariffs matching US levies (15-50%) across roughly 700-900 tariff lines, roughly C\$27.6bn of US imports (including steel, dairy, appliances, electronics, etc.). Aluminum duties will double to 50%. Despite Prime Minister Carney suggesting "we are at war," Ottawa says it remains open to a deal. But September could still see further escalation as both sides seek to gain leverage before renewed negotiations.

UN General Assembly's 81st Session (8-28 September)

The event will bring together heads of state and government from more than 190 countries at a time of significant geopolitical volatility. The session – officially opening on 8 September and followed by a General Debate during 22-28 September – will oversee the final stages of appointing the next UN Secretary-General. The 81st UNGA will be crucial for advancing the "Pact for the Future" (the political vision) and the UN80 initiative (the institutional delivery mechanism), the aim of which is to make the UN more efficient and "fit-for-purpose" by reviewing mandates, improving efficiency, and undertaking structural reforms. Within the reform agenda, the Global Digital Compact has become a flashpoint for US-China competition over AI governance. UNGA debates will be critical in shaping global norms for AI, pitting the US vision of open, values-based innovation against China's model of state-centric control. Numerous bilateral meetings between world leaders will take place alongside formal UN proceedings.



Source: Bernd Dittrich on [Unsplash](#)

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18th BRICS+ Summit in India (12-13 September)

Watchpoints include signs of a deepening China-India diplomatic thaw and BRICS+ progress toward local currency settlement. India's official agenda includes cooperation and resilience, development finance, trade finance and global institutional reform. To watch will be whether President Xi's expected attendance – marking his first visit to India in seven years – signals an acceleration in the ongoing China-India diplomatic thaw.

The summit could also revive discussion of local-currency settlement, BRICS Pay (an initiative to facilitate cross-border payments in BRICS currencies) and the BRICS 'Unit', a prototype gold-backed settlement instrument. Nevertheless, progress toward cooperation on de-dollarisation will likely remain gradual given member states' diverging geopolitical priorities.

Sweden's general election (13 September)

The election will shape the policy direction of one of the EU's largest economies and a key NATO member amid heightened security concerns in the Baltic region. The governing Moderate Party-led centre-right coalition, supported by the Sweden Democrats will compete against the opposition Social Democrats and their centre-left allies. The campaign is expected to focus on crime, immigration, cost of living, economic competitiveness, energy security and defence following Sweden's accession to NATO. The outcome will influence Sweden's approach to defence spending, support for Ukraine, migration, energy policy and industrial competitiveness, while providing an important indicator of broader political trends across Northern Europe.

European Parliament plenary session (15 September)

The first plenary after the summer recess is expected to be a key legislative session as MEPs resume negotiations on the Commission's autumn agenda. Particular attention will focus on implementation of the European Defence Readiness 2030 agenda; follow up to the Competitiveness Compass, including measures to reduce regulatory burdens and strengthen the Single Market; and progress on the Economic Security Strategy, notably outbound investment screening and export controls for sensitive technologies. The Parliament is also expected to debate additional support for Ukraine, including use of frozen Russian assets, the next package of sanctions, and progress on Ukraine's EU accession.



Source: Frederic Köberl on [Unsplash](#)

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Russian state Duma elections (18-20 September)

While the elections are not expected to be either free or competitive, they remain an important political event as the Kremlin seeks to reinforce domestic legitimacy. The vote will be closely watched for turnout, the performance of the ruling United Russia party, and any signs of public discontent despite tight political controls. Internationally, the results will be scrutinized for indications of the regimes confidence and willingness to sustain military operations in Ukraine, while any voting in occupied Ukrainian territories is likely to draw strong international condemnation. While the results of the elections are a forgone conclusion for President Putin's United Russia party, Mr. Putin appears to be avoiding a possible mass mobilization announcement in light of battlefield losses in Ukraine until after the elections.

France Senate election (27 September)

The vote should provide insights into the state of political play ahead of next year's presidential election. 178 out of the Senate's 348 seats will be contested in the indirect election to the Upper House of France's parliament. The electoral college mainly consists of local officials. While the election's outcome on state policy will likely be relatively limited, the outcome could provide signals ahead of the French presidential election in April 2027.

A key development to watch is whether the far-right National Rally Party can secure enough seats to form its first parliamentary group in the Senate. Marine Le Pen, the party's presidential candidate, continues to hold a significant lead in recent opinion polls.

Key US government funding deadline (before 30 September)

30 September marks the end of the US federal fiscal year, and it's the "hard" deadline by which Congress must pass all remaining appropriations bills or trigger another shutdown. This is a real risk given that 2026 has already seen two shutdowns tied to disputes over immigration enforcement funding, including a record 76-day partial DHS shutdown.

Congress still has yet to reconcile FY2027 appropriations work before the fiscal year closes. It's also political loaded timing – landing right before the 3 November midterms. How Congress handles this deadline (smooth passage versus another standoff) could shape voter sentiment, especially on issues like immigration enforcement that have already driven this year's funding fights.



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Appendix 1



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